Counting fluctuations: discrete distributions and cumulants

A counter registers a mean of four events per observation window. That mean is consistent with independent rare arrivals, occupancy of one thermal mode, or a finite number of independent opportunities. These models predict probabilities of ten or more events of 0.008132, 0.107374, and 0.002595, respectively, for the examples developed here. A threshold alarm designed from the mean alone can therefore be badly miscalibrated. The task is to infer what a count distribution says about its generating mechanism, and what it leaves undetermined.

> Based on lecture 1 and lecture 2 of V. Balakrishnan's IIT Madras course. Their order supplies the sequence of models. The equal-mean comparison, counterexamples, numerical laboratory, and exercises here are original teaching constructions. Cumulant identities complete the definition introduced at the end of lecture 2; some are developed further in lecture 3. Two caption tracks were read and fifteen timed still frames inspected; this is not continuous audiovisual transcription.

1.1 Counts, states, and the rule for assigning probability L1 1:31

Let X be the random count in one specified observation window, and let n denote one possible integer value. A *probability mass function* assigns $p_n = \Pr(X = n)$, with $p_n \geq 0$ and $\sum_n p_n = 1$. The set of values with positive mass is the support. For an ordinary event counter it lies among the nonnegative integers; a difference of two counters can instead be any integer. A probability model specifies these masses as well as the support. Listing possible outcomes does not make them equally probable.

Two independent fair dice give a small example of the distinction. A detailed outcome is the ordered pair of face values. There are thirty-six such pairs with probability $1/36$ each. The measured score S discards that detail and retains their sum. For scores s from two through seven, the first face can be any of $1, \ldots, s-1$, giving $s-1$ pairs. For scores from eight through twelve, there are $13-s$ pairs. Thus a direct counting derivation gives

$$\Pr(S = s) = \begin{cases} (s-1)/36, & 2 \leq s \leq 7, \\ (13-s)/36, & 8 \leq s \leq 12. \end{cases} \tag{1.1}$$

The score seven has six contributing pairs and probability $1/6$, although no individual pair is preferred. A *microstate* is the detailed configuration; a *macrostate* records only the chosen coarse observable. Equal probabilities of microstates can produce sharply unequal probabilities of macrostates. Conversely, assuming equal macrostate probabilities would define a different experiment.

Making the dice look identical does not change the result: an unordered pair of distinct faces still collects two ordered outcomes. Quantum particle counting changes the permitted state space through symmetrization or antisymmetrization and also needs an ensemble assigning probabilities to those states. Merely concealing a

classical label does not establish that ensemble. Entanglement language alone cannot substitute for the counting rule.

Definition 1.1 — Mean and scatter. For a count with finite second moment, define its mean $\mu = \mathbb{E}[X] = \sum_n n p_n$ and variance $\sigma^2 = \mathbb{E}[(X-\mu)^2]$. The standard deviation is σ. When $\mu > 0$, the relative fluctuation is σ/μ, and the variance-to-mean ratio is $F = \sigma^2/\mu$, commonly called the Fano factor. Counts and these ratios are dimensionless.

Expanding the square and using $\mathbb{E}[X] = \mu$ gives $\sigma^2 = \mathbb{E}[X^2] - \mu^2$. Nonnegativity follows from the expectation of a nonnegative square; zero variance means $X = \mu$ with probability one. The two-dice example gives mean seven and variance $35/6$, checked by finite enumeration in the laboratory. Neither the mean nor variance determines all the masses. Our equal-mean counting problem will keep $\mu = 4$ fixed while changing the law and its fluctuation mechanism.

1.2 Independent opportunities: Bernoulli and binomial counts L1 13:38

A fixed set of opportunities differs from an unconstrained arrival stream. Let N be a positive integer counting opportunities, and let B_j, for $j = 1, \dots, N$, be independent random variables taking the value one with probability p and zero with probability $q = 1 - p$. Each B_j is a *Bernoulli trial*. Assume the same p for every trial and set $X = \sum_{j=1}^{N} B_j$. Independence specifies the joint law; equality of the success probabilities specifies identical marginal laws. Both assumptions enter the binomial formula.

Proposition 1.1 — Binomial count. Under these assumptions,

$$p_n = \binom{N}{n} p^n q^{N-n}, \qquad n = 0, \dots, N, \qquad \mu = Np, \quad \sigma^2 = Npq. \tag{1.2}$$

The combinatorial coefficient is $\binom{N}{n} = N!/[n!(N-n)!]$, where $n!$ is the product of the positive integers through n, with $0! = 1$.

Derivation.

A specified sequence with n successes and $N - n$ failures has probability $p^n q^{N-n}$, by independence. Choosing which n positions succeeded gives $\binom{N}{n}$ mutually exclusive sequences. Adding their probabilities proves the mass formula. The binomial expansion gives $\sum_n p_n = (p+q)^N = 1$, so it is normalized.

For the mean, linearity of expectation gives $\mathbb{E}[X] = \sum_j \mathbb{E}[B_j] = Np$. Since $B_j^2 = B_j$, each trial has variance $p - p^2 = pq$. Expanding the variance of the sum gives the sum of variances plus twice the sum of pairwise covariances. For two trials, the covariance is $\mathrm{Cov}(B_i, B_j) = \mathbb{E}[(B_i - p)(B_j - p)]$. Independence makes each covariance between distinct trials zero, leaving Npq. This proves the moment statements. ∎

The factor $1 - p$ reduces the variance below the mean: the count cannot exceed its fixed number of opportunities. This is a statement about this model, not a universal test of the physical origin of a count. For

$N = 20$ and $p = 0.2$, the mean is four, the variance is 3.2, and $F = 0.8$. Its probability of an empty window is $0.8^{20} = 0.01152922$. These exact values will be compared with other four-count models.

Generating the factorial moments

Define the *probability-generating function* of a nonnegative integer-valued X by $G_X(z) = \mathbb{E}[z^X] = \sum_{n=0}^{\infty} p_n z^n$. The argument z may be complex. Absolute convergence is guaranteed for $|z| \le 1$; extension outside that disk requires the particular law's tail. For a finite-support law the sum is a polynomial. Its coefficient of z^n is the probability of count n.

Whenever the corresponding factorial moment is finite, differentiation at one from inside the convergence region gives

$$G_X(1) = 1, \quad G'_X(1) = \mathbb{E}[X], \quad G''_X(1) = \mathbb{E}[X(X-1)], \quad \sigma^2 = G''_X(1) + G'_X(1) - G'_X(1)^2. \tag{1.3}$$

The second derivative generates $X(X-1)$, rather than X^2, because differentiating z^n twice brings down $n(n-1)$. Adding the first moment restores the missing X. At order k, the derivative generates the falling product $X(X-1)\cdots(X-k+1)$, called the kth factorial moment. Derivatives and sums can be interchanged directly for finite support; for the nonnegative infinite series, finite factorial moments justify the limit at one.

For the binomial law, summing the binomial expansion gives $G_X(z) = (q + pz)^N$. Differentiating gives $G'_X(1) = Np$ and $G''_X(1) = N(N-1)p^2$; substitution into (1.3) again gives $Np(1-p)$. The two calculations check different parts of the argument: the sum-of-trials calculation exposes independence, while the generating function exposes how the whole law encodes its moments.

At $p = 0$ or $p = 1$, the count is deterministic and the variance vanishes, as (1.2) predicts by continuity. If success probabilities differ, the generating function becomes $\prod_j (1 - p_j + p_j z)$, not a single binomial power. If trials are correlated, even this product fails. A physical application must identify why its trials have the assumed probabilities and joint law.

1.3 A gas: local fluctuations and the thermodynamic limit L1 31:22

Consider N noninteracting classical particles in a vessel of volume V, at a fixed time in a spatially uniform equilibrium state. Choose a subvolume v, with $0 < v < V$, and let X count particles inside it. For each particle, the indicator of being inside has probability $p = v/V$. Independence of the particle positions makes their sum binomial. This is a statement about one equilibrium snapshot; it does not assert that snapshots taken arbitrarily close in time are independent.

Write the mean number density as $\rho_{\text{gas}} = N/V$, so its units are inverse volume. Substituting $p = v/V$ into (1.2) gives the derived relations

$$\mu = \rho_{\text{gas}} v, \qquad \sigma^2 = \rho_{\text{gas}} v \left(1 - \frac{v}{V}\right), \qquad \frac{\sigma}{\mu} = \sqrt{\frac{1 - v/V}{\rho_{\text{gas}} v}}. \tag{1.4}$$

The mean has no residual volume units. The variance is reduced by the finite-reservoir factor $1 - v/V$: when the observation volume becomes the entire vessel, its count is the fixed total N, so the variance must vanish. If the subvolume occupies one fifth of a vessel holding twenty independent particles, its count reproduces the running binomial model, with mean four and variance 3.2.

There are two different large-system operations. At fixed fraction v/V, increasing the total particle count increases the mean local count and suppresses its relative fluctuation in proportion to $N^{-1/2}$. In the *thermodynamic limit*, let N, V increase with $N/V = \rho_{\text{gas}}$ fixed. If the observed v is held fixed, its mean $\rho_{\text{gas}} v$ remains finite and its relative fluctuation approaches $1/\sqrt{\rho_{\text{gas}} v}$. It does not vanish merely because the reservoir becomes infinite. The Poisson limit in Section 1.5 makes that distinction explicit.

Interactions can invalidate independent particle positions. For any sum of indicators, the exact variance still equals the sum of individual variances plus twice the pairwise covariance sum. Positive spatial correlations can increase fluctuations; a fixed-total constraint can reduce them. An inverse-square-root scaling argument therefore needs its correlation assumptions, as well as a declaration of which volume grows. Time correlations and dynamics require the process descriptions introduced in chapter 5.

1.4 Waiting failures and thermal occupation L1 37:05

Keep the Bernoulli success probability p, but change the experiment: repeat independent trials until the first success. Let X be the number of failures before that success. Its support starts at zero, because the first trial may succeed. The event $X = n$ requires n failures followed by one success, so independence gives $p_n = pq^n$, with $q = 1 - p$. This is the *geometric distribution* in the failures-before-success convention. Counting the trial on which success occurs would instead give $X + 1$, shifting the mean by one without changing the variance.

Proposition 1.2 — Geometric fluctuations. For $0 < p \le 1$,

$$G_X(z) = \frac{p}{1 - qz}, \qquad \mu = \frac{q}{p}, \qquad \sigma^2 = \frac{q}{p^2} = \mu(1 + \mu). \tag{1.5}$$

For $0 < q < 1$, the generating series converges when $|qz| < 1$.

Derivation.

Summing $\sum_{n \ge 0} p(qz)^n$ gives the geometric series in (1.5). At $z = 1$, its value is $p/(1 - q) = 1$. Differentiation gives $G'_X(z) = pq/(1 - qz)^2$ and $G''_X(z) = 2pq^2/(1 - qz)^3$. Hence $G'_X(1) = q/p$ and $G''_X(1) = 2q^2/p^2$. Substitution into (1.3) yields $q^2/p^2 + q/p = q/p^2$. Solving $\mu = q/p$ for p gives $p = 1/(1 + \mu)$, proving the last equality. ∎

The mass function can therefore be parameterized by its mean alone:

$$p_n = \frac{1}{1 + \mu}\left(\frac{\mu}{1 + \mu}\right)^n, \qquad F = 1 + \mu, \qquad \frac{\sigma}{\mu} = \sqrt{1 + \frac{1}{\mu}} \quad (\mu > 0). \tag{1.6}$$

The variance grows quadratically at large mean, and the relative fluctuation remains greater than one. For mean four, $p = 1/5$, the empty-window probability is 0.2, and the variance is twenty. The probability of ten or more counts is the tail $q^{10} = 0.8^{10} = 0.10737418$. A broad law can have both many empty windows and a substantial high-count tail while retaining a modest mean.

A single radiation mode

The same law describes a different physical object. A single electromagnetic *mode* is one independent oscillator degree of freedom of the field, including its spatial pattern and polarization. It is not the collection of every mode sharing one frequency. Let $\varepsilon = \hbar\omega > 0$ be the energy of one photon in the selected mode, where ω is angular frequency and $\hbar$ is the reduced Planck constant. The state with n photons has excitation energy $n\varepsilon$. Its zero-point energy is common to all relative probabilities and cancels during normalization.

Assume thermal equilibrium with a bath at absolute temperature $T > 0$. Write $\beta = 1/(k_B T)$, where k_B is Boltzmann's constant. The canonical weight for the number state is $\exp(-\beta n\varepsilon)$. This Boltzmann weighting is the statistical-mechanical model input; the distribution and moments below are derived from it. Set $q_T = \exp(-\beta\varepsilon)$, a dimensionless number between zero and one. Normalization sums a geometric series:

$$p_n = (1 - q_T) q_T^n, \qquad \mu = \frac{q_T}{1 - q_T} = \frac{1}{e^{\beta\varepsilon} - 1}, \qquad \sigma^2 = \mu + \mu^2. \tag{1.7}$$

The occupation of this one thermal mode is thus geometrically distributed. The Bose occupation factor is its mean; it is not the probability of finding exactly that number of photons. The broad fluctuations follow from the full sequence of number-state weights. No direct interaction between photons is needed to obtain the excess variance.

For ordinary blackbody equilibrium, matter can absorb and emit photons, so photon number is not a conserved thermodynamic constraint and no independent photon-number chemical potential appears in (1.7). Zero rest mass alone is not a sufficient argument for omitting such a constraint in every model. The thermal model also differs from a waiting-time experiment: sharing a geometric law does not mean photons are produced by successive failed coin tosses.

The single-mode qualification matters quantitatively. Combining independent thermal modes adds their counts and variances. Equal-mean modes produce the negative-binomial law in Section 1.8; unequal occupations generally produce another law. A single-frequency label or a large collection aperture does not establish the required number or equality of independent modes.

1.5 The rare-event limit L1 53:59

Increase the number of independent opportunities while decreasing each success probability. Let N tend to infinity and set $p = \mu/N$, holding a positive mean μ fixed. For each fixed nonnegative count n, rewrite the binomial mass as

$$\Pr(X = n) = \frac{\mu^n}{n!} \left[\prod_{j=0}^{n-1} \left(1 - \frac{j}{N}\right)\right] \left(1 - \frac{\mu}{N}\right)^N \left(1 - \frac{\mu}{N}\right)^{-n}. \tag{1.8}$$

The finite product tends to one, the next factor tends to $e^{-\mu}$, and the final factor tends to one. These are limits at fixed n; no large-count Stirling approximation is needed. The resulting masses sum to one by the exponential series. They define the *Poisson distribution*:

Proposition 1.3 — Poisson limit and moments.

$$p_n = e^{-\mu}\frac{\mu^n}{n!}, \qquad G_X(z) = e^{\mu(z-1)}, \qquad \mathbb{E}[X] = \operatorname{Var}(X) = \mu. \tag{1.9}$$

The binomial laws just constructed converge to these masses as $N \to \infty$.

Derivation.

Equation (1.8) proves the pointwise mass limit. Normalization of the limiting sequence ensures that probability has not escaped to arbitrarily large counts. For the generating function, sum $e^{-\mu}(\mu z)^n/n!$ to obtain $e^{\mu(z-1)}$. Its first two derivatives at one are μ and μ^2. Formula (1.3) then gives variance $\mu^2 + \mu - \mu^2 = \mu$. ∎

In the running case $\mu = 4$, the Poisson empty-window probability is $e^{-4} = 0.01831564$, and the variance is four. A finite binomial with twenty opportunities has the same mean but a noticeably narrower law. To quantify the approximation, let a_n and b_n be the masses of two integer-valued probability laws, and define their *total-variation distance* as $d_{\mathrm{TV}} = \frac{1}{2}\sum_n |a_n - b_n|$.

This distance bounds the absolute error assigned to any event. To prove the bound, write $\delta_n = a_n - b_n$. Normalization gives $\sum_n \delta_n = 0$, so the sum of the positive δ_n equals the absolute sum of the negative δ_n, and each equals d_{TV}. For any set of outcomes A, the sum $\sum_{n\in A}\delta_n$ lies between $-d_{\mathrm{TV}}$ and d_{TV}. Thus the two probabilities assigned to A differ by at most d_{TV}. Choosing A to contain exactly the outcomes with $\delta_n > 0$ attains the bound.

At mean four, the computed distances from the Poisson law are 0.05600005 for twenty trials, 0.01024234 for one hundred trials, and 0.00100498 for one thousand trials. These are numerical comparisons, not a universal convergence-rate theorem.

The ideal-gas thermodynamic limit with fixed observed subvolume satisfies exactly this scaling: $p = v/V \to 0$, while $Np = \rho_{\mathrm{gas}} v$ remains fixed. Its local count becomes Poisson, with finite variance equal to that mean. By contrast, a finite radioactive sample with independent decays in a fixed interval gives a binomial law exactly; a Poisson approximation needs many nuclei and small individual decay probabilities. A constant-rate Poisson process is a further temporal model, not a consequence of recognizing one Poisson histogram. Event timing and independence of increments belong to chapter 6.

Equality of mean and variance follows from a Poisson law, but the converse is false. For example, a count taking zero and two with equal probabilities has mean and variance both one yet cannot be Poisson. Inferring a mechanism requires more of the distribution and, when the mechanism concerns time, more than a one-time distribution.

1.6 Coherent-state photon counts L2 3:27

The thermal-mode construction assigns probabilities directly to number states. A coherent quantum state instead starts from complex amplitudes. Only a small amount of quantum notation is needed to derive its count law. Write $|n\rangle$ for a normalized state with exactly n photons in the selected mode. These states form an orthonormal basis: $\langle m|n\rangle$ is one when $m = n$ and zero otherwise. A superposition $\sum_n c_n|n\rangle$ has normalization $\sum_n |c_n|^2 = 1$; an ideal number measurement has probability $|c_n|^2$ for outcome n. This last rule is the quantum-mechanical input.

Define the lowering operator a by $a|n\rangle = \sqrt{n}\,|n-1\rangle$ for positive n, and $a|0\rangle = 0$. A *coherent state* $|\alpha\rangle$ is a normalized eigenstate satisfying $a|\alpha\rangle = \alpha|\alpha\rangle$, with complex dimensionless eigenvalue α. Substituting its expansion in number states and comparing the coefficient of $|n\rangle$ gives $c_{n+1}\sqrt{n+1} = \alpha c_n$. Iteration gives $c_n = c_0\alpha^n/\sqrt{n!}$. Normalization fixes $|c_0|^2 e^{|\alpha|^2} = 1$. Choosing an irrelevant common phase makes c_0 real and nonnegative, so

$$|\alpha\rangle = e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}|n\rangle, \qquad p_n = e^{-|\alpha|^2}\frac{|\alpha|^{2n}}{n!}. \tag{1.10}$$

This derives a Poisson number distribution with mean $\mu = |\alpha|^2$. The same construction appears in Glauber's original treatment, printed page 2769, equations (3.1)–(3.9). The optical model and its measurement rule differ from independent rare classical trials, although their resulting count masses agree. An ideal coherent mode of mean four therefore has the Poisson variance four, rather than the thermal variance twenty.

Number counting alone loses phase information. Replace α by $|\alpha|e^{i\theta}$, where $i^2 = -1$ and θ is a real phase. Every mass in (1.10) remains unchanged because it contains only $|\alpha|$. Moreover, an ensemble formed by randomly selecting such a phase gives the same count probabilities as a fixed coherent state: averaging identical probabilities changes nothing. The phase-averaged ensemble has no fixed phase, so the count law cannot by itself certify the full state. Phase-sensitive measurements would distinguish additional properties.

Real laser output can have extra noise, multiple modes, and a measurement chain that changes observed counts. Equation (1.10) is a statement about the specified ideal state and ideal number measurement. It does not identify every laser with that state or establish coherence from a measured variance-to-mean ratio. The remaining count models provide useful alternatives before any such inference is attempted.

1.7 Adding sources and subtracting backgrounds L2 14:06

A detector may count two unresolved sources together or subtract a reference channel from a signal channel. These operations change the support as well as the moments. Let X and Y be independent nonnegative counts with generating functions G_X and G_Y. Their sum has generating function

$$G_{X+Y}(z) = \mathbb{E}[z^{X+Y}] = \mathbb{E}[z^X z^Y] = G_X(z)G_Y(z). \tag{1.11}$$

The factorization is independence expressed at the level of the whole law. At the mass-function level, a total s can be assembled as $(n, s-n)$ for $n = 0, \ldots, s$, giving the *convolution*

$\Pr(X+Y=s)=\sum_{n=0}^{s}\Pr(X=n)\Pr(Y=s-n)$. The finite upper limit matters because neither count can be negative. The product in (1.11) automatically incorporates this constraint through its coefficients.

For independent Poisson counts with means λ_+ and λ_-, multiplication gives $\exp[(\lambda_+ + \lambda_-)(z-1)]$. Their sum is therefore Poisson with the sum of the means. Here the lambdas are dimensionless mean counts in the chosen window, not rates; a constant-rate temporal model would multiply each rate by the window duration.

A difference has negative outcomes

Define the signed count $D_c = X - Y$. It can take any integer d, so its generating expression is a *Laurent series*, allowing negative powers of z. For nonzero complex z, the Poisson expressions give an absolutely convergent series and

$$\mathbb{E}[z^{D_c}] = G_X(z)G_Y(z^{-1}) = e^{-(\lambda_+ + \lambda_-)}e^{\lambda_+ z + \lambda_- z^{-1}}. \tag{1.12}$$

For a fixed $d \geq 0$, write $X = Y + d$ and sum over $Y = j \geq 0$. The probability of $D_c = d$ is

$$e^{-(\lambda_+ + \lambda_-)} \sum_{j=0}^{\infty} \frac{\lambda_+^{j+d}\lambda_-^{j}}{(j+d)!\,j!}. \tag{1.13}$$

To name this series, define the integer-order *modified Bessel function* by $I_d(x) = \sum_{j\geq 0}(x/2)^{2j+d}/[j!(j+d)!]$ for nonnegative integer d, and $I_{-d} = I_d$. Factoring $(\lambda_+/\lambda_-)^{d/2}$ out of (1.13) identifies its argument as $2\sqrt{\lambda_+\lambda_-}$. Negative d follows by exchanging the two counts. Thus, for both means strictly positive,

Proposition 1.4 — Skellam law.

$$\Pr(D_c = d) = e^{-(\lambda_+ + \lambda_-)}\left(\frac{\lambda_+}{\lambda_-}\right)^{d/2} I_{|d|}(2\sqrt{\lambda_+\lambda_-}), \quad d \in \mathbb{Z}. \tag{1.14}$$

Its mean is $\lambda_+ - \lambda_-$, and its variance is $\lambda_+ + \lambda_-$.

Derivation.

The convolution sum (1.13) and its exchanged version prove the probability formula. Linearity gives the difference of means without needing independence. For the variance, expand $\operatorname{Var}(X-Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) - 2\operatorname{Cov}(X,Y)$, where $\operatorname{Cov}(X,Y) = \mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])]$. Independence sets this covariance to zero; (1.9) supplies each variance. ∎

The special-function formula compresses an elementary convolution; no knowledge of its differential equation is needed here. The related generating identity is independently tabulated in NIST DLMF 10.35.1. When one mean vanishes, use the direct difference definition rather than evaluating a ratio with zero denominator. The result becomes a one-sided Poisson law, possibly reflected about zero.

Taking $\lambda_+ = 7$ and $\lambda_- = 3$ produces the running mean four, but variance ten and probability 0.05801782 of a zero difference. Subtracting a background removes its mean and adds its independent noise. In particular, a

signed count with zero mean can still fluctuate substantially; dividing its variance by that mean is then undefined. The Fano factor is not a useful universal summary for signed counts.

For arbitrary real constants a_c, b_c, the exact identity is $\mathbf{Var}(a_cX + b_cY) = a_c^2\,\mathbf{Var}(X) + b_c^2\,\mathbf{Var}(Y) + 2a_cb_c\,\mathbf{Cov}(X,Y)$. The mean is always linear, but variance has this covariance term. Scaled Poisson variables generally have a different support and are not ordinary Poisson counts. The independence assumption, not simply the appearance of a plus or minus sign, licenses dropping the covariance.

1.8 From one thermal component to many L2 40:26

Collect r independent geometric counts with the same success parameter p, where r is a positive integer. Their sum X has generating function $[p/(1-qz)]^r$, by (1.11). For a total n, the individual nonnegative counts can be arranged in $\binom{n+r-1}{r-1}$ ways: place $r-1$ separators among $n+r-1$ positions to divide n objects into r possibly empty groups. Every arrangement has product probability $p^r q^n$. This proves the *negative-binomial distribution*:

$$p_n = \binom{n+r-1}{n} p^r q^n, \quad n \geq 0, \qquad G_X(z) = \left(\frac{p}{1-qz}\right)^r. \tag{1.15}$$

The name refers to the negative power in the binomial series, not to negative counts. Its support remains the nonnegative integers. In a sequential-trial interpretation it counts the failures before the rth success. In the thermal interpretation it counts the total occupation of r independent equal-mean modes. The same law again admits different mechanisms.

Adding the geometric means and variances gives $\mu = rq/p$ and $\sigma^2 = rq/p^2$. Solving for $p = r/(r+\mu)$ puts the comparison at fixed total mean:

Proposition 1.5 — Equal-mean components.

$$G_X(z) = \left[1 + \frac{\mu}{r}(1-z)\right]^{-r}, \qquad \sigma^2 = \mu + \frac{\mu^2}{r}, \qquad F = 1 + \frac{\mu}{r}. \tag{1.16}$$

At $r = 1$ this is geometric; as $r \to \infty$ at fixed μ, it approaches Poisson.

Derivation.

Each component has mean μ/r and variance $\mu/r + (\mu/r)^2$, by (1.5). Independence adds the r variances, proving (1.16). For the limit, take the logarithm of its generating function: $-r\log[1 + \mu(1-z)/r] \to \mu(z-1)$ for fixed z near one. The limiting function is the Poisson generating function (1.9); equivalently, substitution in (1.15) gives the Poisson mass limit at each fixed count. ∎

At total mean four, one mode gives variance twenty, two equal modes give twelve, and four equal modes give eight. One hundred modes give variance 4.16, close to the Poisson value four. Figure 1.1 compares the laws and the fluctuation trend. The limit holds total mean fixed: each mode becomes more weakly occupied as more

modes are collected. If each mode's occupation is instead held fixed, both total mean and variance grow with mode count and their ratio remains constant. These are different experiments.

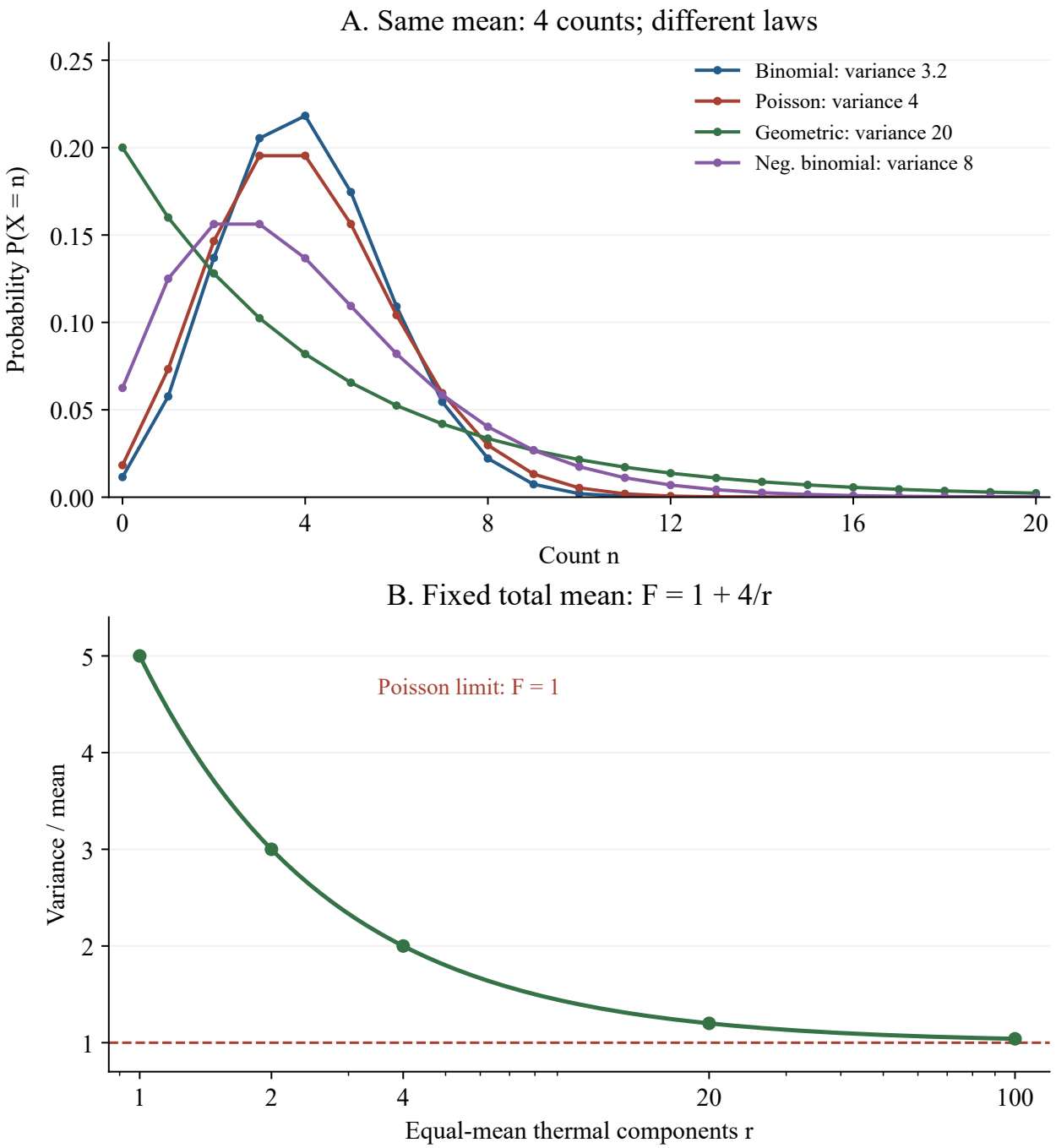


Figure 1.1. Exact count laws at equal mean. Panel A joins discrete masses as a visual guide; it displays counts zero through twenty without renormalizing the omitted tail. Panel B plots $F = 1 + 4/r$; markers at integer component counts are physical equal-mode examples. The intervening curve is the positive-shape negative-binomial extension. Computed by the retained generator from the laboratory data.

Positive real shape and a different construction

A positive noninteger r can parameterize a valid negative-binomial law without being a literal mode count. To establish this, let the conditional count be Poisson with random positive mean Λ. Give Λ a gamma density $f(\lambda) = (r/\mu)^r \lambda^{r-1} e^{-r\lambda/\mu}/\Gamma(r)$, for $\lambda > 0$, where $\Gamma(r) = \int_0^\infty y^{r-1} e^{-y}\, dy$. For any real $r > 0$, integration of $e^{-\lambda}\lambda^n/n!$ against that density yields

$$p_n = \frac{\Gamma(n+r)}{\Gamma(r)n!}\left(\frac{r}{r+\mu}\right)^r\left(\frac{\mu}{r+\mu}\right)^n. \tag{1.17}$$

The integral uses the substitution $y = (1 + r/\mu)\lambda$; its remaining power integral is $\Gamma(n + r)$. As a mixture of normalized Poisson laws with a normalized nonnegative density, (1.17) is normalized. For positive integer r, integration by parts gives $\Gamma(n + r)/\Gamma(r) = (n + r - 1)!/(r - 1)!$, recovering (1.15). A fitted real shape is consequently not evidence that a detector collected that many physical modes.

An arbitrary combination of coherent and thermal contributions need not be negative binomial. For independent Poisson mean μ_c and geometric mean μ_t, the total generating function is $e^{\mu_c(z-1)}/[1 + \mu_t(1 - z)]$. This is generally different from (1.16). Equal first two moments can conceal the difference, as Exercise 1.2 shows. Unequal thermal mode means likewise give a product of distinct denominators. Inferring an admixture from an effective negative-binomial fit therefore requires a separate model of how the contributions combine.

1.9 Cumulants organize independent contributions L2 49:03

Generating functions multiply when independent contributions add. Taking a logarithm converts that product into a sum. This produces quantities that retain the additivity of mean and variance at higher orders. Assume for this section that the *moment-generating function* $\mathcal{M}_X(u) = \mathbb{E}[e^{uX}]$ is finite on an open real interval containing zero. For the dimensionless counts here, u is dimensionless. For a dimensional variable it must have inverse-variable units.

Since $\mathcal{M}_X(0) = 1$, its real logarithm is defined near zero. Define the *cumulant-generating function* $\mathcal{K}_X(u) = \log \mathcal{M}_X(u)$, and the kth cumulant $\kappa_k(X) = \mathcal{K}_X^{(k)}(0)$, for positive integer k. For nonnegative counts, $\mathcal{M}_X(u) = G_X(e^u)$ wherever the expression is finite. Substituting e^u rather than differentiating G directly makes derivatives generate ordinary powers of X.

Proposition 1.6 — Additivity, scaling, and shifts. For independent X, Y with moment-generating functions finite near zero, and real constants a_c, b_c,

$$\kappa_k(X + Y) = \kappa_k(X) + \kappa_k(Y), \qquad \kappa_k(a_c X) = a_c^k \kappa_k(X). \tag{1.18}$$

Adding b_c changes only κ_1, by b_c. The first cumulant is the mean and the second is the variance.

Derivation.

Independence gives $\mathcal{M}_{X+Y}(u) = \mathcal{M}_X(u)\mathcal{M}_Y(u)$. Logarithms add, and differentiation at zero proves the first identity. Also $\mathcal{K}_{a_c X + b_c}(u) = b_c u + \mathcal{K}_X(a_c u)$; differentiating proves both the scaling and shift claims. Finally, $\mathcal{K}'_X = \mathcal{M}'_X/\mathcal{M}_X$ and $\mathcal{K}''_X = \mathcal{M}''_X/\mathcal{M}_X - (\mathcal{M}'_X/\mathcal{M}_X)^2$. At zero, these are $\mathbb{E}[X]$ and $\mathbb{E}[X^2] - \mathbb{E}[X]^2$. ∎

The fourth cumulant illustrates why central moments are insufficient. Put $Y = X - \mu$, and let $m_{3,c} = \mathbb{E}[Y^3]$ and $m_{4,c} = \mathbb{E}[Y^4]$. Expanding at zero gives $\mathcal{M}_Y(u) = 1 + \sigma^2 u^2/2 + m_{3,c} u^3/6 + m_{4,c} u^4/24 + \cdots$. In $\log(1 + w) = w - w^2/2 + \cdots$, the leading contribution of $w^2/2$ is $\sigma^4 u^4/8$. Comparing the coefficients of u^3 and u^4 therefore proves

$$\kappa_3 = m_{3,c}, \qquad \kappa_4 = m_{4,c} - 3\sigma^4. \tag{1.19}$$

The subtraction removes a contribution made solely by two second-order fluctuations. For two independent centered variables, expanding the fourth power of their sum gives the sum of fourth central moments plus $6\sigma_X^2\sigma_Y^2$. Thus fourth central moments fail to add; the subtraction in (1.19) cancels exactly that cross term. This is the higher-order analogue of subtracting the squared mean to form the variance.

For a Poisson count, $\mathcal{K}_X(u) = \mu(e^u - 1)$, so every cumulant equals μ. For the independent Poisson difference, $\mathcal{K}_{D_c}(u) = \lambda_+(e^u - 1) + \lambda_-(e^{-u} - 1)$, giving $\kappa_k(D_c) = \lambda_+ + (-1)^k\lambda_-$. Odd cumulants retain the signed imbalance; even cumulants add the two sources. This provides a systematic explanation of the different mean and variance in (1.14).

These identities require the stated existence conditions. For a concrete counterexample, let X have masses $p_n = C/n^6$ for integers $n \geq 1$, with $C = (\sum_{n=1}^{\infty} n^{-6})^{-1}$. Its first four moments are finite: for $1 \leq k \leq 4$, the defining sum is $C\sum_{n\geq 1} n^{k-6}$, which converges since $k - 6 \leq -2$. For every real $u > 0$, however, the terms Ce^{un}/n^6 do not even tend to zero. Therefore $\mathbb{E}[e^{uX}] = \infty$, and its moment-generating function is not finite on any open interval containing zero.

The same law has a characteristic function $\phi_X(t) = \mathbb{E}[e^{itX}]$ for every real t: its defining series converges absolutely because $|p_n e^{itn}| = p_n$ and $\sum_n p_n = 1$. Here $i^2 = -1$. Finite moments alone therefore do not justify the moment-generating-function method used in this section. Chapter 2 develops characteristic functions; chapter 4 studies stable laws for which familiar moments fail.

1.10 Computational laboratory: distinguish equal-mean sources

The running problem now has explicit alternatives. All four laws in the table have mean four, but their empty-window and high-count probabilities differ. The binomial uses twenty opportunities with success probability 0.2. The geometric uses the nonnegative-count convention of (1.6). The negative binomial uses shape four and total mean four. The entries are exact-law numerical evaluations; the final column is a separate Monte Carlo check with 200,000 draws per law and seed 20261108.

Law	Variance	F	Empty window	Ten or more	Sample F
Binomial	3.2	0.8	0.01152922	0.00259483	0.797928
Poisson	4	1	0.01831564	0.00813224	0.998945
Geometric	20	5	0.20000000	0.10737418	5.007600
Negative binomial	8	2	0.06250000	0.04614258	2.011407

The table identifies why both zero counts and rare large counts are informative. The geometric distribution concentrates more probability at zero than the narrower laws while retaining a much heavier upper tail. Matching a mean estimate alone does not choose among them. Even matching mean and variance leaves the counterexample in Exercise 1.2 unresolved.

The executable laboratory evaluates these laws on integer grids, checks missing tail mass, verifies normalization and moments, compares a direct factorial binomial sum with library probabilities, and checks

Skellam probabilities by an independent Poisson convolution. It also measures both binomial and negative-binomial convergence toward Poisson at fixed mean. Its outputs are machine-readable results and the numerical report, written beside the program. It makes no network calls. Install the packages specified in the dependency file, then run `python code/ch01_counting.py`.

The sampling checks compare discrepancies with six times the analytical standard error of the sample mean and a large-sample standard-error estimate for the sample variance. They are broad implementation diagnostics, not a confidence statement that validates the physical model. Different pseudorandom-library versions can change the sampled column; exact probabilities and the mathematical relations are the reproducible targets. Tail probabilities in the table are computed from the laws, rather than estimated by hoping to sample enough rare events.

A counter measures a transformed law

To model a fixed detection efficiency, let each of the X incident events be recorded independently with probability $\eta \in [0, 1]$. Call the detected count Y. Conditional on X, the detected count is binomial: $\mathbb{E}[Y \mid X] = \eta X$ and $\operatorname{Var}(Y \mid X) = \eta(1-\eta)X$. Taking conditional expectations gives

$$\mathbb{E}[Y] = \eta\mu, \quad \operatorname{Var}(Y) = \eta^2\sigma^2 + \eta(1-\eta)\mu, \quad F_Y = 1 + \eta(F_X - 1) \quad (\eta\mu > 0). \tag{1.20}$$

For the variance identity, write $Y - \mathbb{E}[Y] = (Y - \mathbb{E}[Y \mid X]) + (\mathbb{E}[Y \mid X] - \mathbb{E}[Y])$. The cross term has zero expectation after conditioning on X; the two remaining squares are the mean conditional variance and the variance of the conditional mean. Substitution yields (1.20). This derivation states precisely why detection noise is added even when the incident count is fixed.

At efficiency 0.25, all four input models have detected mean one. Their detected Fano factors become 0.95, 1, 2, and 1.25, in table order. The observed departures from one shrink. Constant independent thinning cannot move a Fano factor across one; a varying efficiency, correlated detection, or detector recovery requires a different model. This short executable calculation implements (1.20):

```
def detected_moments(mean, variance, efficiency):
    if not 0 <= efficiency <= 1:
        raise ValueError("efficiency must lie in [0, 1]")
    detected_mean = efficiency*mean
    detected_variance = (
        efficiency**2*variance + efficiency*(1-efficiency)*mean
    )
    return detected_mean, detected_variance


for incident_variance in (3.2, 4.0, 20.0, 8.0):
    mean, variance = detected_moments(4.0, incident_variance, 0.25)
    print(f"mean={mean:.2f}, variance={variance:.2f}")
```

```
mean=1.00, variance=0.95
mean=1.00, variance=1.00
mean=1.00, variance=2.00
mean=1.00, variance=1.25
```

The snippet accepts the stated finite nonnegative input moments and a fixed efficiency, returns detected moments, and prints the four comparisons without writing a file. It is a moment calculation, not a simulator of detector timing. The full program additionally retains the count-law and sampling checks that support the table.

1.11 What a count law captures and misses

The derived laws answer a single-window question under different hypotheses: a fixed set of independent opportunities, a geometric waiting or occupation model, a rare-event limit, or a specified quantum state and measurement. Their generating functions make exact additions and transformations possible. Cumulants organize those operations when their moment-generating functions exist.

A histogram does not reveal the order of events or the correlations between windows. A Poisson marginal is insufficient to establish a Poisson process, and a geometric marginal does not identify a thermal mechanism. Those require additional measurements and a state description, developed in chapters 5–6. The gamma-mixture construction also shows that the negative-binomial shape alone is not a physical mode counter.

The photon statements concern a specified independent mode or an explicitly stated sum of modes. A complete field-coherence analysis requires observables beyond number probabilities. The detection calculation assumes fixed independent efficiency; it excludes dead time, afterpulsing, varying gain, and background correlations. Event and signal dynamics enter the later pulse and noise chapters. For a count-based alarm, the practical output here is a choice of full probability law together with its assumptions and tail probabilities, rather than a threshold inferred from the mean alone.

1.12 Exercises with worked solutions

Exercise 1.1 — Shared sources defeat independent-noise subtraction (conceptual)

Let Z be Poisson with mean four, and let two channels both report the same count: $X = Z$ and $Y = Z$. Each channel separately has a Poisson histogram. Find the mean and variance of their sum and difference. Explain why the independent-source formulas fail.

Worked solution

The channels share the same realization, so $X + Y = 2Z$ and $X - Y = 0$. Their sum has mean eight and variance $2^2 \operatorname{Var}(Z) = 16$; their difference is identically zero and has mean and variance zero. The covariance is $\operatorname{Cov}(X, Y) = \operatorname{Var}(Z) = 4$. In the variance formulas of Section 1.7, the covariance term contributes plus eight to the sum and minus eight to the difference. Two independent copies would instead have sum variance eight and difference variance eight. A pair of marginal Poisson laws does not determine the joint law or license independent-noise subtraction. Inspecting paired observations or cross-covariance is necessary.

Exercise 1.2 — Matching mean and variance does not identify the law (counterexample)

Add an independent Poisson count of mean two and a geometric count of mean two. Compare its mean, variance, and empty-window probability with a negative-binomial count of shape four and total mean four. Can fitting only the first two moments distinguish them?

Worked solution

The Poisson variance is two; the geometric variance is $2(1+2)=6$. Independence therefore gives total mean four and variance eight. The negative-binomial model has variance $4+4^2/4=8$ as well. The first two moments agree, so a fit using only them cannot distinguish the models.

For the independent sum to be zero, both nonnegative summands must be zero. Its probability is $e^{-2}/3=0.04511176$. For the negative binomial, $p=r/(r+\mu)=1/2$, so its zero-count mass is $p^r=1/16=0.06250000$. The difference is directly observable in the full distribution. Their generating functions are $e^{2(z-1)}/(3-2z)$ and $(2-z)^{-4}$, respectively, which are unequal. Thus an effective negative-binomial fit does not generally prove that a coherent-plus-thermal decomposition has that same law. The calculation assumes independent count addition; coherent superposition within the same mode is another physical construction.

Exercise 1.3 — Derive the entire thinning transformation (derive)

For the fixed-efficiency model of Section 1.10, derive the detected probability-generating function from the conditional binomial law. Show that Poisson input remains Poisson and that a sum of r equal-mean thermal components remains negative binomial with the same r. Recover (1.20) by differentiating.

Worked solution

Given incident count $X=n$, the conditional detected generating function is $(1-\eta+\eta z)^n$. Averaging over n gives $G_Y(z)=G_X(1-\eta+\eta z)$. For Poisson input this is $e^{\eta\mu(z-1)}$, proving that the entire detected law, not just its first two moments, is Poisson of mean $\eta\mu$.

For (1.16), substitution gives $G_Y(z)=[1+(\eta\mu/r)(1-z)]^{-r}$. It is the same family with mean $\eta\mu$ and unchanged shape. Differentiating the general composition yields $G_Y'(1)=\eta G_X'(1)$ and $G_Y''(1)=\eta^2 G_X''(1)$. Inserting these in (1.3) gives $\mathrm{Var}(Y)=\eta^2\sigma^2+\eta(1-\eta)\mu$. Independent thinning changes a distribution in a constrained way; a general nonlinear detector need not obey this composition.

Exercise 1.4 — A third cumulant for the negative binomial (derive)

Use (1.16) to derive the first three cumulants at fixed mean μ and shape r. Check the geometric case and the fixed-mean Poisson limit. Evaluate the third cumulant for $\mu=r=4$.

Worked solution

Put $a_r = \mu/r$, and define $h(u) = 1 - a_r(e^u - 1)$, positive in a neighborhood of zero. The cumulant-generating function is $\mathcal{K}_X(u) = -r \log h(u)$. Its first derivative is $r a_r e^u / h(u)$. At zero, $e^0 = h(0) = 1$, giving $\kappa_1 = r a_r = \mu$.

Differentiating again gives $r[a_r e^u/h + a_r^2 e^{2u}/h^2]$. A third derivative gives $r[a_r e^u/h + 3a_r^2 e^{2u}/h^2 + 2a_r^3 e^{3u}/h^3]$. Evaluating at zero proves $\kappa_2 = \mu + \mu^2/r$ and $\kappa_3 = \mu + 3\mu^2/r + 2\mu^3/r^2$. Setting $r = 1$ gives $\kappa_3 = \mu(1+\mu)(1+2\mu)$ for the geometric law. Sending $r \to \infty$ at fixed mean makes all three displayed cumulants approach μ, as required for Poisson. At $\mu = r = 4$, the third cumulant is 24, which the lab checks against the third centered moment of the exact masses. It contains information beyond the common mean four.

Exercise 1.5 — Choose a justified rare-event approximation (compute/design)

A model with mean four has a known binomial mechanism. Available choices for the number of opportunities are twenty, one hundred, one thousand, or ten thousand, with success probability adjusted to preserve the mean. Choose the smallest listed count for which total-variation error relative to Poisson is below 0.002. Verify that the conclusion is stable to expanding the mass grid. State what the bound does and does not guarantee for a threshold alarm.

Worked solution

Run the lab's exact mass comparison. The distances are 0.056000051, 0.010242342, 0.001004976, and 0.000100309 in the listed order, so one thousand is the smallest listed acceptable count. The grid ends at count four hundred, far beyond the substantial mass of either distribution, and the program verifies the Poisson tail is below 10^{-30}. For a larger opportunity count, the binomial tail can also be checked directly; expanding this grid leaves the quoted digits unchanged.

By the definition of total variation, every event probability, including the probability of exceeding a chosen threshold, has absolute error below 0.002 for the accepted comparison. This is an absolute probability error: it may be a poor relative error for a very rare event. It also says nothing about whether the binomial mechanism describes the physical source. A numerical approximation test and model validation answer different questions. Monte Carlo alone is a poor way to certify the small difference when exact mass evaluations and tail controls are available.

Exercise 1.6 — Unequal modes conceal their number (compute/design)

Two independent thermal modes have mean occupations three and one. Compare their sum with two equal-mean thermal modes of occupations two and two. Compute both variances and zero-count probabilities. If shape is estimated from the first two moments using $r_{\text{eff}} = \mu^2/(\sigma^2 - \mu)$, does it recover the physical number of modes? Check the unequal case by simulation and predict the Fano factor after independent detection of efficiency 0.25.

Worked solution

The unequal pair has total mean four and variance $3(1+3)+1(1+1)=14$. Its zero-count probability is $[1/(1+3)][1/(1+1)]=1/8=0.125$. The equal pair has variance $2[2(1+2)]=12$ and zero-count probability $(1/3)^2=1/9$. Equal total mean and equal physical mode count do not ensure equal fluctuations.

For arbitrary independent thermal occupations μ_j, the variance is $\mu+\sum_j \mu_j^2$, where $\mu=\sum_j \mu_j$. Consequently $r_{\text{eff}}=(\sum_j \mu_j)^2/\sum_j \mu_j^2$. By the Cauchy–Schwarz inequality this is no larger than the physical number of modes, with equality for equal occupations. Here it is $16/(9+1)=1.6$, not two. The formula is a moment-based effective shape; it does not establish an exact negative-binomial law for unequal modes.

Generate independent geometric variates with success probabilities $1/4$ and $1/2$, subtracting one from each library variate to count failures, then add them. Compare ensemble mean and variance with four and fourteen, and zero frequency with one eighth. Use fresh ensembles or analytical standard errors to assess sampling scatter. The incident Fano factor is $14/4=3.5$; (1.20) predicts detected mean one and Fano factor $1+0.25(3.5-1)=1.625$. The equal-occupation pair instead gives detected Fano factor 1.5. The lab retains this comparison as a separate seeded check.

1.13 Sources and attribution

- V. Balakrishnan, IIT Madras, Discrete probability distributions, lecture 1 and lecture 2, NPTEL 115106089. Section links identify the relevant discussion starts; added derivations and examples need not occur at those timestamps.
- Official instructor notes, printed/PDF pages 7–21: count laws and physical examples; cumulants begin on page 21. The notes' main exposition covers lectures 1–11. This chapter supplies its own exposition, derivations, figures, and six exercises.
- R. J. Glauber, Coherent and Incoherent States of the Radiation Field, *Physical Review* 131, 2766–2788 (1963), printed page 2769, equations (3.1)–(3.9): coherent-state definition, number-state expansion, and Poisson number probabilities. The paper's broader field-correlation theory is not reproduced here.
- NIST, Digital Library of Mathematical Functions, equation 10.35.1: modified-Bessel generating identity. Equation (1.14) is derived here directly from the probability convolution.

Scientific qualifications relative to the source discussion: an ideal radioactive finite sample has an exact binomial decay count; a fixed observation volume retains fluctuations in the thermodynamic limit; ordinary thermal equilibrium omits a photon-number constraint because emission and absorption do not conserve that number; an arbitrary coherent/thermal combination is not guaranteed to be negative binomial. The derivations specify the hypotheses used for each statement. No lecture images or raw transcripts are redistributed.

Densities, transformations, and characteristic functions

A gas has a Gaussian distribution of each velocity component. An instrument reports kinetic energy instead. Replacing velocity by energy inside the Gaussian exponent gives the right Boltzmann factor but the wrong probability density: in one dimension it misses an inverse square root of energy, and in three dimensions it misses a positive square root. Those factors change threshold probabilities. The task is to predict the energy readout from a velocity model without losing probability during the coordinate change.

> Lecture 3 supplies the sequence from cumulants through continuous laws, characteristic functions, Gaussian statistics, and squared variables. The energy calculation in general dimension, counterexamples, laboratory, and exercises are developed here. The full caption text and three timed stills were inspected; the latter give partial notation checks, not continuous audiovisual coverage. Fourier signs are declared explicitly below.

2.1 What survives when the variable becomes continuous L3 0:33

The cumulant argument in Section 1.9 did not depend on an integer support. Let X now be a real random variable with mean μ, and assume its moment-generating function $\mathcal{M}_X(u) = \mathbb{E}[e^{uX}]$ is finite for real u in an open interval containing zero. The parameter u has units inverse to X. Define $\mathcal{K}_X(u) = \log \mathcal{M}_X(u)$ and $\kappa_r = \mathcal{K}_X^{(r)}(0)$ for positive integer r. The logarithm is locally well defined because the real expectation is positive and equals one at zero.

The first two derivatives give $\kappa_1 = \mu$ and $\kappa_2 = \mathbb{E}[X^2] - \mu^2$. Translation and scaling follow directly from $\mathcal{K}_{aX+b}(u) = bu + \mathcal{K}_X(au)$ for real constants a, b. Thus $\kappa_r(aX + b) = a^r \kappa_r(X)$ for $r \geq 2$. If X and Y are independent and both functions exist near zero, their exponentials factor inside the expectation, so $\mathcal{K}_{X+Y} = \mathcal{K}_X + \mathcal{K}_Y$. The same algebra works for a count, a velocity, or a voltage.

A useful check is the Poisson difference from chapter 1. If its independent component means are λ_+, λ_-, then $\mathcal{K}(u) = \lambda_+(e^u - 1) + \lambda_-(e^{-u} - 1)$. Differentiation yields $\kappa_r = \lambda_+ + (-1)^r \lambda_-$. Even cumulants record summed noise; odd cumulants record the imbalance. Continuity of the measured coordinate does not create additivity. Independence creates it.

The existence assumption remains essential. The finite-four-moment example in chapter 1 has no moment-generating function on an open interval around zero. Treating every expectation as a convergent power series would silently exclude such laws. A transform with a bounded integrand avoids this difficulty and provides the next tool.

2.2 Density is probability per unit coordinate L3 18:54

Define the *cumulative distribution function* $F_X(x) = \Pr(X \leq x)$ for real x. It is nondecreasing and right continuous, with limiting values zero and one at the two infinities. If the law is absolutely continuous, there is a

nonnegative integrable function ρ_X such that $\Pr(a < X \le b) = \int_a^b \rho_X(x)\,dx$ for every finite interval. This function is a *probability density*. Its integral over the real line is one; its units are inverse to the coordinate units.

A density can exceed one. A velocity uniform on an interval of width 0.1 meters per second has density 10 seconds per meter on that interval. Integrating it over the interval gives one. A value at a single point has zero probability for this law, because the integral over a point is zero. A large density is therefore compatible with a small probability for a narrow measurement bin.

Not every real-valued law has an ordinary density. An *atom* at x_0 has probability $a_0 = \Pr(X = x_0) > 0$ and produces a jump of size a_0 in F_X. A mixture with a stalled instrument can combine such an atom with a continuous velocity distribution. Physicists often write the atom as $a_0\delta(x - x_0)$, where the Dirac delta is defined through integration against test functions: $\int f(x)\delta(x - x_0)\,dx = f(x_0)$. It is not an ordinary function taking an infinite value at one point. The cumulative distribution handles both parts without that shorthand.

There are also continuous cumulative distributions without an ordinary density, so “no atoms” is weaker than “has a density.” The branch calculations later in this chapter assume an absolutely continuous input. An instrument with quantization or a finite probability of zero response needs its discrete masses treated separately.

2.3 A transform that always exists L3 21:29

Definition 2.1 — Characteristic function. For real k, define $\phi_X(k) = \mathbb{E}[e^{ikX}]$, where $i^2 = -1$. If X is dimensional, k has inverse-coordinate units. For a density, $\phi_X(k) = \int_{-\infty}^{\infty} e^{ikx}\rho_X(x)\,dx$. This chapter uses the positive sign in the forward transform; the lecture uses the negative sign.

The expectation exists for every probability law because $|e^{ikX}| = 1$. The triangle inequality gives $|\phi_X(k)| \le 1$, normalization gives $\phi_X(0) = 1$, and complex conjugation gives $\phi_X(-k) = \overline{\phi_X(k)}$. Dominated convergence proves continuity in k. These are necessary checks when someone proposes a transform. For example, e^{k^2} fails the magnitude bound for every nonzero k and cannot describe a probability law.

Another necessary condition exposes why pointwise checks alone are weak. For any complex numbers $c_1, \dots, c_m$ and real $k_1, \dots, k_m$, expanding a nonnegative squared modulus gives

$$\sum_{j,l=1}^{m} c_j \overline{c_l}\,\phi_X(k_j - k_l) = \mathbb{E}\left[\left|\sum_{j=1}^{m} c_j e^{ik_jX}\right|^2\right] \ge 0. \tag{2.1}$$

This property is called positive definiteness. The identity proves its necessity directly; a general converse is not needed here. When ϕ_X is integrable, Fourier inversion gives the density $\rho_X(x) = (2\pi)^{-1}\int e^{-ikx}\phi_X(k)\,dk$ at its continuity points. Atomic laws need an inversion statement for measures instead. A formal inverse that changes sign cannot be a density.

For independent X, Y, factorization gives $\phi_{X+Y}(k) = \phi_X(k)\phi_Y(k)$. If both have densities, the corresponding expression in coordinate space is the convolution $\rho_{X+Y}(s) = \int \rho_X(x)\rho_Y(s-x)\,dx$. The condition $Y = s - X$ supplies the argument of the second density; the product uses independence. The transform is often preferable because multiplication replaces an integral over every possible partition of the sum.

Finite moments allow derivatives: if $\mathbb{E}[|X|^r] < \infty$, differentiation under the expectation gives $\phi_X^{(r)}(0) = i^r \mathbb{E}[X^r]$. The domination is by $|X|^r$. An existing characteristic function therefore does not imply existing derivatives or moments. Nor does it license substituting imaginary arguments into a moment-generating function that never existed near zero.

2.4 Gaussian normalization, widths, and transforms L3 28:34

A centered velocity component at thermal equilibrium motivates a two-parameter family. A *Gaussian* or *normal* variable $X \sim \mathcal{N}(\mu, \sigma^2)$ has real mean μ, positive standard deviation σ, and density

$$\rho_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]. \tag{2.2}$$

To check the constant, put $z = (x-\mu)/\sigma$. The square of $I = \int_{-\infty}^{\infty} e^{-z^2/2}\,dz$ is an integral over the plane. Polar coordinates with radius $r \geq 0$ and angle $0 \leq \theta < 2\pi$ give $I^2 = 2\pi \int_0^\infty r e^{-r^2/2}\,dr = 2\pi$. Positivity fixes $I = \sqrt{2\pi}$. Symmetry gives centered mean zero; the moment calculation below gives variance σ^2.

The density falls to half its peak when $(x-\mu)^2/(2\sigma^2) = \log 2$. The full width at half maximum is therefore $2\sqrt{2\log 2}\sigma = 2.354820045\sigma$, as evaluated in the lab. It is a density-width convention, not a probability interval. Integrating (2.2) after the same coordinate change gives

$$F_X(x) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{x-\mu}{\sqrt{2}\sigma}\right)\right], \qquad \operatorname{erf}(z) = \frac{2}{\sqrt{\pi}}\int_0^z e^{-u^2}\,du. \tag{2.3}$$

The error-function definition is DLMF 7.2.1. The upper integration limit depends on its argument; using infinity there would make it a constant. Symmetry immediately checks $F_X(\mu) = 1/2$. Threshold probabilities require the cumulative distribution or its complement, not the value of the density at the threshold.

Proposition 2.1 — Gaussian transform. The characteristic function and real cumulant-generating function of (2.2) are

$$\phi_X(k) = e^{ik\mu - \sigma^2 k^2/2}, \qquad \mathcal{K}_X(u) = \mu u + \tfrac{1}{2}\sigma^2 u^2. \tag{2.4}$$

All cumulants above order two vanish.

Derivation.

For real u, completing the square in the absolutely convergent integral gives $-(x-\mu)^2/(2\sigma^2) + ux = -(x-\mu-\sigma^2 u)^2/(2\sigma^2) + \mu u + \sigma^2 u^2/2$. The shifted Gaussian integrates to one, so its remaining exponential is $\mathcal{M}_X(u)$. Taking its logarithm proves the second expression.

A real-variable derivation also checks the Fourier sign without shifting a contour. Let $Z = X - \mu$ and $\psi(k) = \mathbb{E}[e^{ikZ}]$. Its density satisfies $\rho_Z'(z) = -z\rho_Z(z)/\sigma^2$. Integration by parts, whose boundary term vanishes with the Gaussian tail, gives $\psi'(k) = -\sigma^2 k\psi(k)$. With $\psi(0) = 1$, this differential equation has solution $e^{-\sigma^2k^2/2}$. Restoring the translation multiplies it by $e^{ik\mu}$. Differentiating the quadratic logarithm proves the cumulant claim. ∎

The width in k is inverse to the width in x: a larger σ makes the exponential decay more rapidly with $|k|$. A physical temperature changes velocity scatter; it changes the transform curvature at zero in exactly the corresponding way. This is a transform statement, independent of whether a particular gas actually satisfies the equilibrium assumptions.

2.5 Gaussian moments and the limits of four-number diagnostics L3 35:44

Symmetry forces every odd centered Gaussian moment to vanish. For even moments, let $M_{2r} = \mathbb{E}[Z^{2r}]$ with $Z = X - \mu$ and integer $r \geq 1$. The density derivative used above gives

$$M_{2r} = -\sigma^2 \int z^{2r-1}\rho_Z'(z)\,dz = (2r-1)\sigma^2 M_{2r-2}. \tag{2.5}$$

The boundary term is zero because a Gaussian decays faster than every fixed polynomial. Starting at $M_0 = 1$ proves $M_{2r} = (2r-1)!!\sigma^{2r}$, where the double factorial is $1 \cdot 3 \cdots (2r-1)$. Thus $M_2 = \sigma^2$, $M_4 = 3\sigma^4$, and $M_6 = 15\sigma^6$. The coefficient counts pairings of $2r$ labeled objects: choose one object's partner in $2r-1$ ways, then pair the remaining objects recursively. This elementary pairing argument anticipates Gaussian contraction formulas without assuming field theory.

For any variable with finite positive variance and finite fourth moment, define skewness $\gamma_1 = \mathbb{E}[(X-\mu)^3]/\sigma^3$ and excess kurtosis $\gamma_2 = \mathbb{E}[(X-\mu)^4]/\sigma^4 - 3$. Both vanish for a Gaussian. Their converse fails. A variable taking $-\sqrt{3}, 0, +\sqrt{3}$ with probabilities $1/6, 2/3, 1/6$ has mean zero, variance one, zero third moment, and fourth moment three. It has the same first four moments as the standard Gaussian, yet it is a three-point law. Its sixth moment is nine instead of fifteen.

Consequently, zero excess kurtosis cannot identify Gaussianity, and the sign of kurtosis alone does not order every tail probability or fix a peak's shape. It is a fourth-moment diagnostic. A finite list of moments cannot generally determine all instrument threshold probabilities. The stronger result about a cumulant-generating function terminating exactly at a finite degree is a different theorem; no truncation of empirical cumulants is used as a probability model here.

2.6 A squared coordinate needs both inverse branches L3 50:07

Let $Y = g(X)$ be a transformed reading, where X has density ρ_X and g is differentiable on its monotone branches. At a regular value y, list every inverse point x_j satisfying $g(x_j) = y$, with nonzero derivative. On one such branch, probability conservation gives $\rho_X(x_j)|dx_j|$ in an infinitesimal output interval. Since $|dx_j/dy| = 1/|g'(x_j)|$, adding the disjoint input branches yields

Proposition 2.2 — Transformation formula on regular branches.

$$\rho_Y(y) = \sum_{x_j : g(x_j) = y} \frac{\rho_X(x_j)}{|g'(x_j)|}. \tag{2.6}$$

The formula is local away from critical values and presumes the stated branch decomposition; atoms or flat branches are handled separately.

Derivation.

Integrate the proposed density over any output interval contained in the regular region. The change of variables on each branch converts its integral to the input probability on that branch. The inverse images are disjoint except for endpoints of zero input probability. Summing their probabilities gives exactly $\Pr(Y \text{ lies in the interval})$. Thus (2.6) is the density there. The absolute derivative preserves nonnegativity on decreasing branches. ∎

For $Y = X^2$ and a centered Gaussian of variance σ^2, every $y > 0$ has two inverse points $\pm\sqrt{y}$, both with derivative magnitude $2\sqrt{y}$. Their equal contributions give

$$\rho_Y(y) = \frac{e^{-y/(2\sigma^2)}}{\sqrt{2\pi\sigma^2 y}}, \quad y > 0; \qquad \rho_Y(y) = 0 \quad (y < 0). \tag{2.7}$$

The density diverges at the critical value zero but is integrable: near zero its integral is proportional to $\sqrt{y}$. There is no atom at zero, because $\Pr(X = 0) = 0$. Its cumulative distribution offers a second derivation: $\Pr(X^2 \le y) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$ for $y \ge 0$; differentiating gives the two terms in (2.6). Retaining only the positive root halves the total probability, a defect caught by the laboratory normalization check.

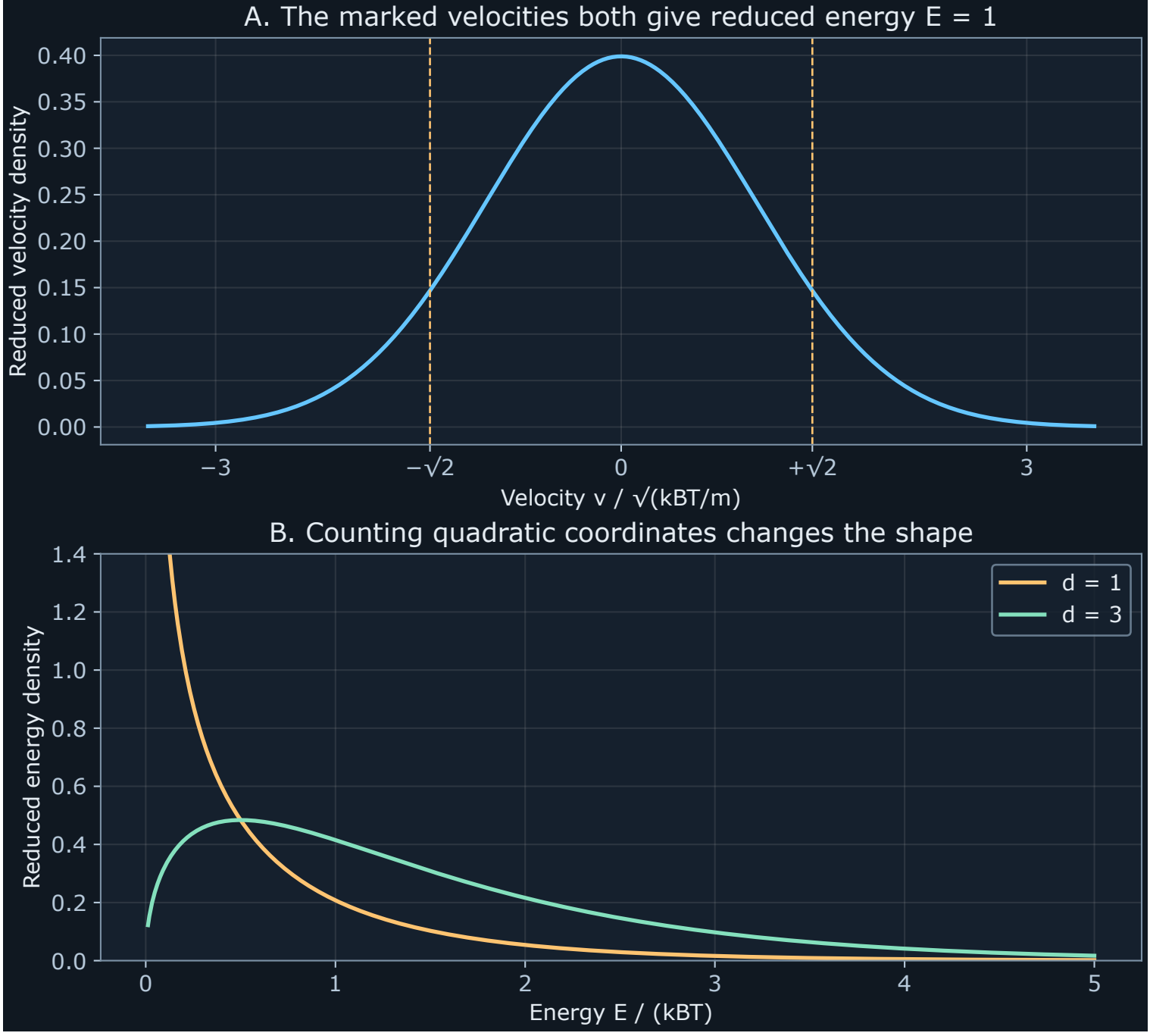


Figure 2.1. Two velocity preimages feed one energy bin. Panel A uses reduced velocity $v/\sqrt{k_BT/m}$; its two marked points both have $E/(k_BT) = 1$. Panel B shows gamma energy laws for one and three independent coordinates. The one-coordinate divergence exceeds the displayed vertical range close to zero; it is not an atom. Curves are exact theoretical densities, computed by the generator. Figure 2.1 explains why changing only the exponential loses the state-counting factor.

2.7 Counting quadratic coordinates gives a gamma energy law

Assume a classical nonrelativistic particle of mass $m > 0$ in equilibrium at absolute temperature $T > 0$, with d independent Cartesian velocity components $V_j \sim \mathcal{N}(0, k_BT/m)$. Here k_B is Boltzmann's constant and d is a positive integer. Write $\theta = k_BT$, with energy units, and define kinetic energy $E = \frac{1}{2}m\sum_{j=1}^{d} V_j^2$. The equilibrium Gaussian velocities are the physical input; the energy law is a consequence.

For one component, applying (2.7) and then the scaling $E = mY/2$ gives $\rho_E(e) = e^{-e/\theta}/\sqrt{\pi\theta e}$ for $e > 0$. Both velocity signs have already been counted. The moment-generating function of this energy is $(1 - \theta u)^{-1/2}$ for $u < 1/\theta$, obtained by integrating the Gaussian with a changed quadratic coefficient. Independence multiplies d copies.

Proposition 2.3 — Equilibrium kinetic energy.

$$\rho_E(e) = \frac{e^{d/2-1}\exp(-e/\theta)}{\Gamma(d/2)\theta^{d/2}}, \quad e > 0; \qquad \mathbb{E}[E] = \frac{d}{2}\theta, \quad \operatorname{Var}(E) = \frac{d}{2}\theta^2. \tag{2.8}$$

This is a gamma density of shape $d/2$ and scale θ.

Derivation.

The gamma function is $\Gamma(a) = \int_0^\infty s^{a-1}\exp(-s)\,ds$ for $a > 0$. Substitution $s = e/\theta$ normalizes the proposed density. Integrating it against $\exp(ue)$ replaces $1/\theta$ by $1/\theta - u$ and gives $(1-\theta u)^{-d/2}$, exactly the product derived from the independent velocities. Uniqueness of a moment-generating function finite around zero identifies the law; alternatively the radial calculation below obtains its density directly. Its logarithm is $-(d/2)\log(1-\theta u)$, whose first two derivatives at zero yield the stated mean and variance. ∎

The density-of-states factor can be seen without transforms. In d velocity dimensions, a shell of radius v has volume proportional to $v^{d-1}dv$. Since $e = mv^2/2$, the radial derivative gives $dv = de/(mv)$. Combining these powers gives $v^{d-2}de$, proportional to $e^{d/2-1}de$. Multiplication by the Boltzmann weight $\exp(-e/\theta)$ and normalization recovers (2.8). Thus the factor counts the amount of velocity space producing an energy bin. It is not a change in the temperature of the exponential.

For $d = 1$, the density diverges at zero; for $d = 2$, it is exponential; for $d = 3$, it vanishes at zero and peaks at $e = \theta/2$. The mean in three dimensions is $3\theta/2$, so the most probable and mean energies differ. At the threshold 3θ, their upper-tail probabilities are 0.01430588, 0.04978707, and 0.11161023. A detector that confuses a component energy with total three-dimensional energy therefore predicts a substantially different fraction above the same threshold. These are evaluations of the exact laws, not measured gas data.

The gamma integral and its parameter domains are independently recorded in DLMF 5.9.1. The instructor's notes, PDF page 31, give the three-dimensional Maxwell speed and energy laws. The derivation here extends the counting argument to arbitrary integer d while making the independence and quadratic-energy assumptions explicit.

2.8 Laboratory: catch a missing branch before fitting a temperature

Run `python code/ch02_energy.py` with the dependencies in requirements.txt. The program uses reduced units $m = k_B T = 1$ and generates 200,000 independent velocity vectors for each of $d = 1, 2, 3$, with seed 202611082. It squares and sums the coordinates, checks energy normalization by adaptive quadrature, and compares the empirical cumulative distribution with the exact gamma cumulative distribution. Outputs are written beside the program as JSON and text; no network or external data are used.

```
d=1: mass=1.000000000, mean=0.499668, variance=0.501427, KS=0.001491
d=2: mass=1.000000000, mean=1.003337, variance=1.005157, KS=0.001716
d=3: mass=1.000000000, mean=1.498797, variance=1.497460, KS=0.001876
Incorrect one-branch mass: 0.500000000
```

The reported Kolmogorov–Smirnov distance, abbreviated KS, is the maximum absolute gap between the empirical and exact cumulative distributions. It is a numerical diagnostic of this simulation. The sample means are checked against six analytical standard errors $\sqrt{(d/2)/200000}$; this broad check catches normalization or scale mistakes without treating one sample as a proof. The exact-law tail probabilities above are evaluated with a survival function, avoiding cancellation from subtracting a near-unit cumulative probability.

The following minimal calculation tests the transformation's most consequential failure. It takes no input, prints a normalization integral, and writes no files. The substitution $e = s^2$ removes the integrable singularity at zero:

```
import numpy as np
from scipy.integrate import quad

mass = quad(lambda s: 2*np.exp(-s*s)/np.sqrt(np.pi),
            0, np.inf)[0]
print(f"Both branches: {mass:.6f}")
print(f"One branch:   {mass/2:.6f}")
```

```
Both branches: 1.000000
One branch:   0.500000
```

Renormalizing a mistaken half-density would conceal this particular constant error in a symmetric example, but it would not repair a general transformation with unequal branches. The input law and its coordinate map must determine the normalization, rather than a fit absorbing a missing preimage into an arbitrary amplitude. The full lab also integrates Gaussian second, fourth, and sixth moments and checks its characteristic function by an independent cosine quadrature.

2.9 What the energy model captures and misses

The calculation translates a specified equilibrium velocity law into an energy distribution, with all inverse branches and coordinate volume included. It predicts both typical energy and threshold probabilities. It does not establish that a measured gas is equilibrated, that interacting particle velocities are independent, or that an instrument reports the full kinetic energy rather than one projection.

A nonzero bulk velocity gives noncentered Gaussian components and a different energy law. Quantum statistics, relativistic energy, constraints between coordinates, or an energy-dependent detector efficiency also change the model. The transformation formula still provides a method when its hypotheses hold, but the gamma law then needs a new derivation. Sums of independent fluctuations and the circumstances in which Gaussian behavior emerges belong to chapter 3; broad-tailed exceptions belong to chapter 4.

2.10 Exercises with worked solutions

Exercise 2.1 — A spike in the readout (conceptual)

An instrument reports exactly zero with probability $1/4$. Otherwise it reports a standard Gaussian velocity. Write its cumulative distribution and characteristic function. Does its continuous density integrate to one?

Worked solution

Let Φ be the standard Gaussian cumulative distribution. Conditioning on whether the instrument stalls gives $F(x) = \frac{1}{4}\mathbf{1}_{\{x \geq 0\}} + \frac{3}{4}\Phi(x)$, where the indicator is one on the specified set and zero elsewhere. Its jump at zero is $1/4$. Linearity of expectation gives $\phi(k) = 1/4 + (3/4)e^{-k^2/2}$. The continuous part has density $(3/4)(2\pi)^{-1/2}e^{-x^2/2}$, integrating to $3/4$; the atom supplies the remaining mass. Forcing the continuous part alone to normalize to one would erase the instrument's stall mechanism.

Exercise 2.2 — Gaussian diagnostics without a Gaussian law (conceptual)

Verify the first four moments of the three-point law in Section 2.5. Compare its probability of $|X| > 2$ with the standard Gaussian. What can zero skewness and excess kurtosis establish?

Worked solution

Symmetric outer points cancel every odd moment. The second moment is $2(1/6)3 = 1$, and the fourth is $2(1/6)9 = 3$. Thus both diagnostics vanish. Since its largest magnitude is $\sqrt{3} < 2$, its threshold probability is zero. The Gaussian assigns $2[1 - \Phi(2)] > 0$, because its density is positive beyond two. The numerical value is unnecessary for this strict counterexample. Vanishing diagnostics establish these specific moment relations, not a full density or any particular nontrivial tail probability.

Exercise 2.3 — Absolute speed and squared speed (derive)

For $X \sim \mathcal{N}(0, \sigma^2)$, derive the density of $R = |X|$, then transform R to $Y = R^2$. Explain precisely where the factor of two enters.

Worked solution

For $r > 0$, the two input intervals near r and $-r$ give $\rho_R(r) = \rho_X(r) + \rho_X(-r) = 2\rho_X(r)$. They account for both signs once. The second map $y = r^2$ is one-to-one on $r > 0$, so $\rho_Y(y) = \rho_R(\sqrt{y})/(2\sqrt{y})$. Substituting the Gaussian gives (2.7). Adding another factor of two at the second step would double-count signs already merged by R. Neither transformation creates an atom at zero, since the original Gaussian puts no mass on that point.

Exercise 2.4 — Energy cumulants in every dimension (derive)

Starting from the energy moment-generating function, derive the rth cumulant for integer $r \geq 1$. Find the relative standard deviation and explain its large-d behavior.

Worked solution

The logarithm is $\mathcal{K}_E(u) = -(d/2)\log(1-\theta u) = (d/2)\sum_{r\geq 1}(\theta u)^r/r$ for $|\theta u| < 1$. Comparing the coefficient of $u^r/r!$ gives $\kappa_r = (d/2)(r-1)!\theta^r$. The first two cases reproduce (2.8). Their ratio gives $\sqrt{\mathrm{Var}(E)}/\mathbb{E}[E] = \sqrt{2/d}$. The decrease follows because independent quadratic contributions add means and variances, while standard deviation is the square root of variance. Increasing temperature at fixed d changes both scales equally and does not reduce this relative scatter. Independence is crucial to the dimension argument.

Exercise 2.5 — Identify the measured coordinate (compute/design)

A simulated equilibrium detector at $k_B T = 1$ gives an energy mean near 1.5 and a fraction above energy three near 0.112. Compare one-, two-, and three-coordinate models using the lab. Can changing only the one-coordinate temperature fit both values?

Worked solution

The three-coordinate model has exact mean 1.5 and tail 0.11161023, consistent with the stated checks. One and two coordinates at the same temperature have means 0.5 and 1 and tails 0.01430588 and 0.04978707. To fit mean 1.5 with one coordinate requires scale $\theta = 3$, rather than one. Its probability above energy three is then $\mathrm{erfc}(1) = 0.15729921$, where $\mathrm{erfc}(x) = 1 - \mathrm{erf}(x)$. This differs from 0.11161023, so changing only the one-coordinate temperature cannot fit both the stated mean and tail.

The ratio $\mathrm{Var}(E)/\mathbb{E}[E]^2 = 2/d$ is two for one coordinate and $2/3$ for three coordinates, independent of temperature. Measuring variance supplies a second scale-free discrimination. The full distribution or a second threshold can also test the model; one mean alone cannot distinguish coordinate count from temperature.

Exercise 2.6 — A drifting gas (compute/design)

In one dimension let $V \sim \mathcal{N}(v_0, \sigma_V^2)$ and $E = mV^2/2$. Derive its energy density and mean. Design a numerical normalization check that does not assume equal inverse-branch weights.

Worked solution

For energy $e > 0$, put $r = \sqrt{2e/m}$. The two derivatives have magnitude mr, so $\rho_E(e) = [\rho_V(r) + \rho_V(-r)]/(mr)$. The two terms are unequal when $v_0 \neq 0$; replacing their sum by twice the positive term is wrong. From the original variable, $\mathbb{E}[E] = (m/2)(\sigma_V^2 + v_0^2)$. The second term is directed kinetic energy, so interpreting the total mean as thermal temperature overestimates it.

For a stable quadrature set $e = mr^2/2$ and integrate the proposed density times $de = mr\,dr$. The integrand becomes $\rho_V(r) + \rho_V(-r)$ on $r > 0$, whose integral must be one. This substitution removes the endpoint singularity and checks branch accounting separately from the energy mean. Compare that integral and the

energy-weighted integral with the normalization and mean above before fitting data. A sampled histogram alone can obscure a missing branch when one sign is rare.

2.11 Sources and attribution

- V. Balakrishnan, Continuous random variables, lecture 3, IIT Madras, NPTEL 115106089. Timestamp links navigate the corresponding topic starts.
- Official instructor notes, PDF pages 22–31: cumulants, densities, Gaussian moments, square transformation, and equilibrium energy. The error-function definition is corrected using the primary reference below. Characteristic functions here use the opposite, explicitly stated Fourier sign.
- NIST DLMF 7.2.1, error-function definition, and DLMF 5.9.1, gamma integral and domains. Both support the special-function normalizations used in the derivations.

The Gaussian characterization and transformation calculations are mathematical derivations under the stated models. Simulated outputs are synthetic checks. No raw captions, lecture images, or measured gas data are included in this reader.

The central limit theorem and the diffusion scale

A transport calculation replaces many independent microscopic steps by a Gaussian propagator. The replacement is attractive because one diffusion coefficient predicts the entire central profile. It can still be wrong by a factor of two at every reachable lattice point, assign probability outside a finite-step support, or fail completely when one step dominates the variance. The task is to identify the scaling and the kind of approximation that the Gaussian limit actually licenses.

> Lecture 4 develops Gaussian sums, uniform sums, and isotropic three-dimensional random flights in that order. The full caption track and two timed stills were inspected. The one-dimensional lattice comparison and numerical error audit are teaching extensions. The stated convergence theorem is qualified separately from density approximations and path-level limits.

3.1 Adding Gaussians is exact before any limit L4 0:49

Let $X_1, \dots, X_n$ be independent Gaussian variables with means μ_j and variances σ_j^2, and let a_j be real fixed coefficients. Define the measured sum $S_n = \sum_{j=1}^{n} a_j X_j$. Its mean is $m_n = \sum_j a_j \mu_j$, and independence gives variance $v_n = \sum_j a_j^2 \sigma_j^2$. The coefficients carry whatever units convert each input into the common output coordinate. Assume $v_n > 0$ when standardizing.

Proposition 3.1 — Exact Gaussian closure. Under these assumptions $S_n \sim \mathcal{N}(m_n, v_n)$ and $Z_n = (S_n - m_n)/\sqrt{v_n}$ is a standard Gaussian for every positive integer n.

> ***Derivation.***
>
> Use the positive-sign characteristic function from chapter 2. Independence factors it as $\phi_{S_n}(k) = \prod_j \exp(ika_j\mu_j - k^2 a_j^2 \sigma_j^2/2) = \exp(ikm_n - k^2 v_n/2)$. This is the characteristic function of the stated Gaussian. Translation and scaling give $\phi_{Z_n}(k) = \exp(-k^2/2)$. No large-$n$ approximation occurred. ∎

Independent inputs with variances one and four, combined as $X_1 - 2X_2$, give variance seventeen. Multiplying the second coefficient by minus two changes its mean contribution's sign but multiplies its variance by four. This checks the squared coefficients. If the inputs are correlated, the variance has additional terms $2\sum_{i<j} a_i a_j \operatorname{Cov}(X_i, X_j)$. A jointly Gaussian vector still gives Gaussian linear combinations, but Gaussian marginals alone do not imply a jointly Gaussian vector.

The exact theorem does not explain why non-Gaussian steps can produce an approximately Gaussian displacement. That requires a limit theorem. It also requires declaring what remains fixed as the number of terms grows.

3.2 A finite-variance central limit theorem L4 8:11

Assume now that $X_1, X_2, \ldots$ are independent and identically distributed, abbreviated iid, with common finite mean μ and variance $0 < \sigma^2 < \infty$. Their distribution may be discrete or continuous. Define $Y_j = (X_j - \mu)/\sigma$ and $Z_n = n^{-1/2} \sum_{j=1}^{n} Y_j$. The normalization keeps mean zero and variance one for every n. Without centering, the mean grows as n; without the square-root scaling, the variance grows as n.

Proposition 3.2 — The iid central limit theorem. Under these assumptions Z_n converges in distribution to $Z \sim \mathcal{N}(0,1)$. Here convergence in distribution means $\Pr(Z_n \le z) \to \Pr(Z \le z)$ for every real z, since the Gaussian cumulative distribution is continuous.

Derivation.

The finite second moment permits the small-argument expansion $\phi_Y(h) = 1 - h^2/2 + o(h^2)$. One justification avoids assuming a third moment. Write $e^{ix} = 1 + ix - x^2/2 + r(x)$, where $r(x)/x^2 \to 0$ as $x \to 0$ and $|r(x)| \le Cx^2$ for a fixed finite constant C. Splitting into $|x| \le 1$ and $|x| > 1$ supplies such a bound. Thus $r(hY)/h^2 \to 0$ pointwise and is dominated by CY^2, an integrable variable. Dominated convergence proves the expansion after taking expectations; $\mathbb{E}[Y] = 0$ and $\mathbb{E}[Y^2] = 1$ fix its coefficients.

Independence now gives $\phi_{Z_n}(k) = [\phi_Y(k/\sqrt{n})]^n = [1 - k^2/(2n) + o(1/n)]^n \to e^{-k^2/2}$ at every fixed real k. The final implication uses the *cited* Lévy continuity theorem: pointwise convergence of characteristic functions to the characteristic function of a probability law implies convergence in distribution to that law. The theorem is stated in Scott Sheffield's MIT 18.175 notes, PDF pages 41–43; pages 45–47 connect the finite-second-moment expansion to this central limit theorem. Applying it to the Gaussian transform completes the argument. ∎

The transform expansion is proved here; the continuity theorem is an invoked analytical result, rather than an unjustified exchange of a limit and a Fourier inversion integral. No smooth density for Z_n was assumed. For a lattice sum, such a density does not exist at finite n, yet convergence of cumulative probabilities still makes sense.

The iid assumptions keep no one term exceptional. Finiteness of each variance alone is insufficient for arbitrary nonidentical arrays: a single non-Gaussian summand can retain most of the total variance. Additional negligibility conditions are needed for those generalizations. Nor does convergence in distribution give a universal finite-sample error, a relative error for very small probabilities, or a guarantee that probabilities beyond a growing threshold are Gaussian.

3.3 A uniform distribution becomes triangular before becoming Gaussian L4 11:00

Take iid U_j uniform on $[0,1]$, so their density is one on that interval and zero elsewhere. Integration gives mean $1/2$ and variance $1/3 - 1/4 = 1/12$. For the sum $S_2 = U_1 + U_2$, the convolution is the length of the overlap of $[0,1]$ with $[s-1, s]$. Consequently,

$$\rho_{S_2}(s) = \begin{cases} s, & 0 \le s \le 1, \\ 2 - s, & 1 < s \le 2, \\ 0, & \text{otherwise.} \end{cases} \tag{3.1}$$

The overlap grows up to $s = 1$ and then shrinks. The triangular area is one, and the variance is $1/6$, agreeing with independent variance addition. A flat single-input law has become peaked without any input preferring the midpoint. More combinations of input values produce central sums than extreme sums.

For arbitrary n, standardize as $Z_n = (\sum_j U_j - n/2)/\sqrt{n/12}$. A centered uniform variable has characteristic function $\mathbb{E}[e^{ik(U-1/2)}] = \sin(k/2)/(k/2)$, obtained by integrating the exponential between $-1/2$ and $1/2$. Applying the scaling and multiplying n factors gives

$$\phi_{Z_n}(k) = \left[\frac{\sin(k\sqrt{3/n})}{k\sqrt{3/n}} \right]^n. \tag{3.2}$$

The ratio has value one at zero by continuity. Since $\sin x/x = 1 - x^2/6 + O(x^4)$, (3.2) tends to $e^{-k^2/2}$. For every finite n, however, Z_n lies in $[-\sqrt{3n}, \sqrt{3n}]$. The limiting law's unbounded support arises because these standardized bounds recede to infinity. It does not make an impossible finite-n sum possible.

Figure 3.1 compares the standardized densities for one, two, and eight uniforms. Standardizing matters: plotting unscaled sums would mix the changing shape with a drifting mean and a growing width. A finite-n cumulant check gives excess kurtosis $-6/(5n)$, derived in Exercise 3.3. Its decrease is one informative check of central Gaussian behavior, not a proof of all tail probabilities.

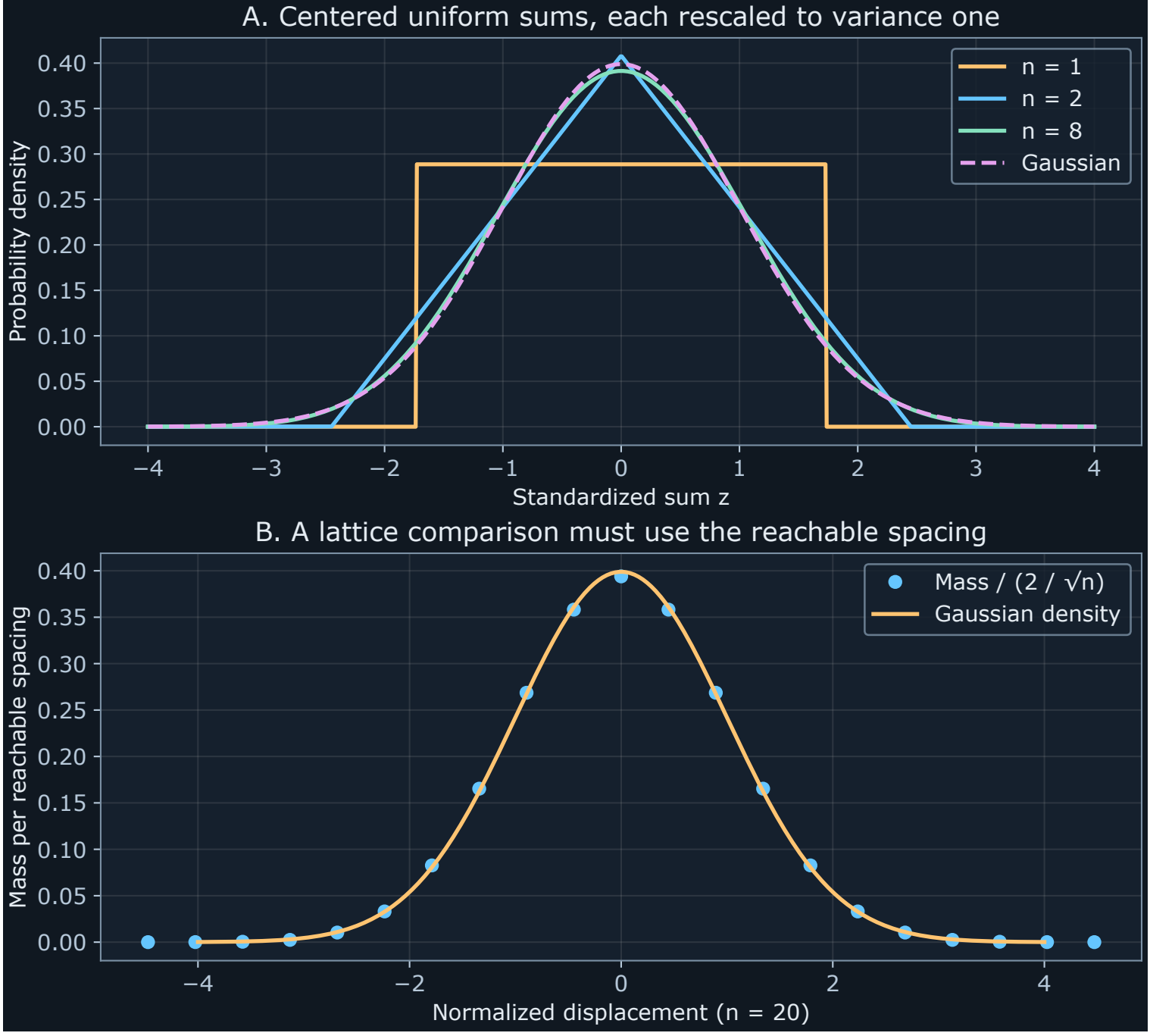


Figure 3.1. Exact laws compared at fixed standardized variance. Panel A shows why convolution and rescaling must be separated. Panel B divides lattice masses by $2/\sqrt{n}$, the spacing between reachable standardized positions, before comparing them with a continuous density. Using the underlying spacing $1/\sqrt{n}$ would double every plotted height. Curves and masses are theoretical, generated by the retained program. Figure 3.1 tests both shape convergence and a discretization failure.

3.4 Isotropic random flights and the geometry of one step L4 31:48

Consider independent three-dimensional step vectors $\mathbf{R}_j$, each of fixed length $\ell > 0$ and uniformly distributed in direction. Let $\mathbf{X}_n = \sum_{j=1}^{n} \mathbf{R}_j$ be the displacement after n steps, and let $\delta t > 0$ be the time between steps. Length has units meters and time seconds. Rotational symmetry gives $\mathbb{E}[\mathbf{R}_j] = \mathbf{0}$. Independence then removes every distinct-step cross term in the squared sum:

$$\mathbb{E}[|\mathbf{X}_n|^2] = \sum_j \mathbb{E}[|\mathbf{R}_j|^2] + 2\sum_{i<j} \mathbb{E}[\mathbf{R}_i] \cdot \mathbb{E}[\mathbf{R}_j] = n\ell^2. \tag{3.3}$$

This is the mean-square displacement, the sum of the three coordinate variances. It is not the variance of the distance $|\mathbf{X}_n|$, whose mean is generally positive. In an isotropic model each coordinate contributes $n\ell^2/3$. The root-mean-square displacement grows as $\sqrt{n}$, although the path has traveled total step length $n\ell$.

The one-step probability is concentrated on a sphere. In delta notation its volume density is $\rho_1(\mathbf{r}) = \delta(|\mathbf{r}| - \ell)/(4\pi\ell^2)$. Integrating with volume element $r^2 dr\, d\Omega$ gives one, and the delta supplies the extra inverse-length unit needed for an inverse-volume density. To calculate its characteristic function, orient the polar axis along the Fourier vector $\mathbf{k}$, and write $q = |\mathbf{k}|$. The direction cosine u is uniform on $[-1, 1]$, giving

$$\phi_1(\mathbf{k}) = \frac{1}{2}\int_{-1}^{1} e^{iq\ell u}\, du = \frac{\sin(q\ell)}{q\ell}, \qquad \phi_{\mathbf{X}_n}(\mathbf{k}) = \left[\frac{\sin(q\ell)}{q\ell}\right]^n. \tag{3.4}$$

The same sinc function appears as for a uniform scalar because the projection of an isotropic three-dimensional fixed-length step onto a fixed axis is uniform between $-\ell$ and ℓ. This is a statement about that projection; the three coordinates of one step are not independent, since their squared sum is fixed.

3.5 The diffusion limit fixes squared distance per unit time L4 53:36

Hold elapsed time $t = n\delta t > 0$ fixed while $n \to \infty$, and let $\ell \to 0$. Expansion of (3.4) gives $\log \phi_{\mathbf{X}_n}(\mathbf{k}) = -n\ell^2 q^2/6 + O(n\ell^4 q^4)$ for each fixed $\mathbf{k}$. A nontrivial limit therefore holds the positive diffusion coefficient $D = \ell^2/(6\delta t)$ fixed in three dimensions. Its units are square meters per second, and $n\ell^4 \to 0$ under this scaling. The transform tends to e^{-Dtq^2}.

Inverting that Gaussian transform is easiest in Cartesian coordinates, because the three factors separate. In d dimensions the isotropic limiting density has the form

$$\rho(\mathbf{x}, t) = \frac{\exp[-|\mathbf{x}|^2/(4Dt)]}{(4\pi Dt)^{d/2}}, \qquad \partial_t \rho = D\nabla^2 \rho, \qquad \mathbb{E}[|\mathbf{X}(t)|^2] = 2dDt. \tag{3.5}$$

Each coordinate is a centered Gaussian of variance $2Dt$, so the product density normalizes to one and proves the last identity. Direct differentiation verifies the diffusion equation: $\partial_t \rho/\rho = -d/(2t) + |\mathbf{x}|^2/(4Dt^2)$, while $\nabla^2\rho/\rho = -d/(2Dt) + |\mathbf{x}|^2/(4D^2t^2)$. The initial point mass is interpreted weakly: for bounded continuous f, $\int f(\mathbf{x})\rho(\mathbf{x}, t)d\mathbf{x} \to f(\mathbf{0})$ as $t \downarrow 0$. This follows by writing $\mathbf{x} = \sqrt{2Dt}\,\mathbf{z}$ and applying dominated convergence to the normalized standard Gaussian measure.

For an isotropic fixed-length flight in d dimensions, each coordinate step variance is ℓ^2/d, so matching coordinate variance to $2Dt$ gives $D = \ell^2/(2d\delta t)$. In one dimension, steps $\pm\ell$ therefore give $D = \ell^2/(2\delta t)$, rather than the three-dimensional factor six. Equation (3.5) uses the per-coordinate diffusion coefficient in every dimension.

The limit removes a microscopic speed scale. If steps are interpolated as straight flights of speed $\ell/\delta t$, that speed diverges as $\delta t^{-1/2}$ at fixed D. Holding a finite speed instead gives $\ell = v\delta t$, and then $D \to 0$ in this independent-step scaling. A finite-speed transport model requires an additional directional persistence mechanism, treated in chapter 19. The calculation here establishes fixed-time distributional convergence; a theorem about convergence of entire random paths is stronger.

3.6 A density cannot be compared directly with a point mass

A one-dimensional symmetric walk has $X_n = \ell \sum_{j=1}^{n} \epsilon_j$, where the independent signs ϵ_j are equally likely to be plus or minus one. If B_n counts positive steps, then $B_n \sim \mathbf{Binomial}(n, 1/2)$ and $X_n = \ell(2B_n - n)$. Its reachable locations differ by 2ℓ, not ℓ. At even n only even multiples of ℓ are reached; at odd n only odd multiples are reached. This is the parity constraint.

In a central region for large n, the local Gaussian comparison therefore has the form $\Pr(X_n = x) \approx 2\ell\rho(x, t)$ at reachable locations. At unreachable locations the exact probability remains zero. The factor 2ℓ converts density per length to probability mass in a reachable cell. It is not optional normalization. In standardized units $Z_n = X_n/(\ell\sqrt{n})$, the corresponding cell width is $2/\sqrt{n}$.

For $n = 50$, the exact central mass is 0.11227517 and the parity-aware approximation is 0.11283792. Dividing either mass by the elementary spacing $1/\sqrt{n}$ instead of the reachable spacing would produce twice the Gaussian density near the center. Yet cumulative probabilities would still converge: the central limit theorem averages over the increasingly fine jump structure. A local probability approximation and a cumulative convergence statement answer different questions.

3.7 Laboratory: separate central convergence, parity, and support

The laboratory evaluates exact binomial probabilities for standardized symmetric walks, and exact uniform-sum cumulative distributions. It writes JSON results and a numerical report beside the program. Run `python code/ch03_diffusion.py`. Its inputs are fixed synthetic parameters, its only effects are these output files, and it makes no network calls or Monte Carlo draws.

```
n=10: CDF distance=0.12304688, corrected grid error=0.00268616
n=50: CDF distance=0.05613759, corrected grid error=0.00054170
n=200: CDF distance=0.02817424, corrected grid error=0.00013603
n=1000: CDF distance=0.01261251, corrected grid error=0.00002717
```

The cumulative distribution distance checks the Gaussian against both sides of each exact lattice jump. A continuous curve cannot remove that jump. The smaller “corrected grid error” instead compares the exact cumulative probability at each reachable location with a Gaussian boundary halfway to the next reachable location. This continuity correction improves those selected event probabilities, but it is not the supremum distance between the two cumulative distribution functions. Reporting it under the same name would conceal the residual jump error.

A second check inverts (3.2) numerically for four uniforms at standardized value zero. Truncating the Fourier integral at $|k| = 100$ gives density 0.384900110; the independent exact convolution density is 0.384900179. Agreement tests the transform normalization and rescaling, while the finite cutoff leaves a controlled numerical target for refinement. It does not justify exchanging a limit and an integral in the theorem.

The following calculation exposes parity with a directly computable central mass. It accepts no input, prints the exact mass and local approximation, and does not write files:

```
import numpy as np
from scipy.stats import binom, norm

n = 50
exact = binom.pmf(n//2, n, 0.5)
reachable_spacing = 2/np.sqrt(n)
approximation = reachable_spacing*norm.pdf(0)
print(f"Exact: {exact:.8f}")
print(f"Local Gaussian: {approximation:.8f}")
```

```
Exact: 0.11227517
Local Gaussian: 0.11283792
```

The full program additionally checks normalization and variance exactly to numerical precision. These checks catch a wrong scaling constant before visual closeness becomes persuasive. Figure 3.1 is reproduced separately by `python figures/ch03_diffusion.py`; it uses theoretical probabilities and densities, not sampled histograms.

3.8 What the Gaussian limit captures and misses

The iid theorem determines the standardized central distribution at fixed thresholds. The diffusion scaling identifies a continuum propagator from a microscopic covariance per unit time. Together they explain a common Gaussian profile under independence and finite variance. They do not provide a universal number of steps needed for a chosen tolerance.

Finite support and lattice parity remain exact constraints at finite step number. Rare-event relative errors, persistent directions, state-dependent waiting times, correlated increments, and heavy-tailed jumps require additional analysis. In particular, an unbounded step distribution can still have finite variance, while a bounded distribution need not give an accurate Gaussian tail at a small finite sample. Tail boundedness and central-limit attraction are different properties. Chapter 4 examines stable alternatives when finite-variance scaling fails.

3.9 Exercises with worked solutions

Exercise 3.1 — Gaussian marginals without a Gaussian sum (conceptual)

Let G be standard Gaussian and let S be an independent sign equally likely to be plus or minus one. Define $X = G$ and $Y = SG$. Show that both marginals are Gaussian and uncorrelated, but $X + Y$ is not Gaussian.

Worked solution

Symmetry makes SG have the same law as G. Both means vanish and $\mathbb{E}[XY] = \mathbb{E}[S]\mathbb{E}[G^2] = 0$, so the covariance is zero. Conditional on $S = -1$, the sum is identically zero; conditional on $S = 1$, it is $2G$. Thus its law has an atom of mass $1/2$ at zero and a Gaussian continuous part of weight $1/2$. A nondegenerate Gaussian

has no atoms. The variance is two, matching variance addition for uncorrelated variables, yet the law is not Gaussian. Independence in Proposition 3.1 cannot be replaced by uncorrelated Gaussian marginals.

Exercise 3.2 — Finite variance in every row is insufficient (conceptual)

For each integer n, let a row contain a symmetric sign R and $n-1$ zeros. All variables in that row are independent and have finite variance. Center the row sum and divide by its standard deviation. Does it approach a Gaussian?

Worked solution

The sum is always R, with mean zero and variance one. Standardization changes nothing, and its two-point law remains fixed for all n. In particular, its cumulative probability at $1/2$ is $1/2$, while a standard Gaussian cumulative probability there exceeds $1/2$. There is no convergence to the Gaussian. The row is not identically distributed, and the lone sign carries all the variance. A general nonidentical-variable theorem needs a hypothesis that suppresses such dominant contributions. Counting summands alone supplies no averaging mechanism.

Exercise 3.3 — Excess kurtosis of a uniform sum (derive)

Derive the excess kurtosis $-6/(5n)$ of the standardized sum of n iid uniforms on $[0,1]$. Explain why its decay is consistent with, but does not prove, the central limit theorem.

Worked solution

For $A = U - 1/2$, direct integration gives $\mathbb{E}[A^2] = 1/12$ and $\mathbb{E}[A^4] = 1/80$. Its fourth cumulant is $1/80 - 3/144 = -1/120$. Adding n independent copies multiplies that cumulant by n. Dividing the sum by $\sqrt{n/12}$ multiplies the fourth cumulant by $(12/n)^2$, giving $-6/(5n)$. The standardized variance is one, so this fourth cumulant equals excess kurtosis. Its limit zero matches a Gaussian fourth-moment relation. The three-point example in chapter 2 shows why that one relation cannot establish convergence of the full law; the characteristic-function proof supplies the missing distributional statement.

Exercise 3.4 — Derive the diffusion coefficient from a covariance (derive)

Suppose iid zero-mean vector steps in d dimensions have covariance $2D\delta t\, I_d$, where I_d is the identity matrix and $D > 0$. Derive the mean-square displacement after n steps. Specialize to isotropic fixed-length steps and recover the dimension factor.

Worked solution

Covariances add under independence, so the sum has covariance $2Dn\delta t\, I_d$. Taking its trace gives $\mathbb{E}[|\mathbf{X}_n|^2] = 2dDt$ with $t = n\delta t$. For an isotropic step of fixed length ℓ, rotational symmetry makes its covariance cI_d, and its trace is $dc = \ell^2$. Thus $c = \ell^2/d$. Equating it to $2D\delta t$ gives $D = \ell^2/(2d\delta t)$. This second-

moment identity alone does not prove a Gaussian propagator: distributional convergence additionally needs conditions controlling higher or large-step contributions.

Exercise 3.5 — Choose a cumulative approximation honestly (compute/design)

Among $n = 10, 50, 200, 1000$, choose the smallest step count whose uncorrected cumulative distribution distance from the standard Gaussian is below 0.03. Can the much smaller continuity-corrected error be reported as that same distance?

Worked solution

The lab gives distances 0.12304688, 0.05613759, 0.02817424, and 0.01261251. Therefore 200 is the smallest listed acceptable count. At 50 steps the corrected grid error is already only 0.00054170, but that compares a selected lattice event with a shifted Gaussian boundary. It does not compare both cumulative distributions at the same real argument over the entire line. Since the exact central jump at 50 steps has size 0.11227517, any continuous approximation must differ by at least half that size on one side of the jump. That lower bound is 0.056137585, agreeing with the reported supremum distance and excluding the smaller figure as a substitute.

Exercise 3.6 — Refine the microscopic clock (compute/design)

A one-dimensional walk uses $\delta t = 0.01$ seconds and diffusion coefficient $D = 0.5$ square meters per second. Find its step length and one-second mean-square displacement. Refine the time step by a factor of four at the same D. What happens to step length, maximum reachable distance at one second, and flight speed?

Worked solution

The relation $\ell = \sqrt{2D\delta t}$ gives $\ell = 0.1$ meter. One second contains 100 steps, so $\mathbb{E}[X^2] = 100(0.1)^2 = 1$ square meter. The maximum reachable magnitude is ten meters, although reaching it requires every sign to agree. On changing δt to 0.0025 seconds, the step length becomes 0.05 meter and the step count becomes 400. The mean-square displacement remains one, while the maximum magnitude becomes twenty meters.

If interpolated as flights, the speeds are ten and twenty meters per second respectively. The speed grows as the step length shrinks, because the clock shrinks faster. This calculation explains how an unbounded-support continuum kernel can emerge from finite-support approximants. It also prevents interpreting the diffusion limit as a limit at fixed microscopic transport speed.

3.10 Sources and attribution

- V. Balakrishnan, Gaussian distribution and the central limit theorem, lecture 4, IIT Madras, NPTEL 115106089. Caption sequence determines the main topic order.
- Official instructor notes, PDF pages 32–39: Gaussian closure, iid theorem, uniform sums, three-dimensional flights, and diffusion scaling.

- Scott Sheffield, MIT 18.175, Characteristic functions and central limit theorem, PDF pages 41–43 for the continuity theorem and 45–47 for the second-moment expansion and iid limit. The continuity theorem is cited, not proved here.

Local lattice probabilities, approximation errors, counterexamples, and exercises are teaching constructions. The source's broader informal generalizations are restricted here to explicit hypotheses. No claim of complete path-level convergence or quantitative universal tail accuracy is made.

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Stable laws, heavy tails, and infinite divisibility

Repeating a noisy measurement is useful only if the chosen estimator concentrates. Gaussian sample means narrow as the inverse square root of the number of measurements. Cauchy sample means retain exactly the original distribution at every sample size. Both conclusions follow from independent measurements. The difference lies in the law being averaged, so independence alone cannot justify an error bar shrinking as $n^{-1/2}$.

> Lecture 5 supplies stable-law definitions, Gaussian/Cauchy/Lévy examples, transformations, and infinite divisibility. Its full caption text and two timed stills were inspected, with partial visual coverage. The mathematical parameter conventions below repair several informal source statements: a Cauchy location is not its expectation, a Gaussian-ratio scale follows the numerator variance, and the Lévy transform must match its density's scale. Numerical examples and exercises are independent teaching constructions.

4.1 Stability means closure after a specified rescaling L5 1:53

A probability law is *stable* if for iid copies $X_1, \dots, X_n$ and every integer $n \geq 2$, there are constants $a_n > 0$ and $b_n \in \mathbb{R}$ such that $(\sum_j X_j - b_n)/a_n$ has the same law as X_1. Equality here is equality of distributions, not equality of random realizations. If all b_n can be zero, the law is strictly stable. The definition asks for exact closure at every n, stronger than attraction to a limiting law as n grows.

The chapter's principal working family is the centered symmetric stable law with characteristic function $\phi_X(k) = \exp(-c|k|^\alpha)$, where $c > 0$ and $0 < \alpha \leq 2$. The exponent α is the stability index. Its units are fixed by requiring $c|k|^\alpha$ dimensionless: if X has length units, c has units length to the power α. Existence of this family for the stated index range is part of the stable-law classification given in the instructor notes, PDF pages 40–42; it is a cited result here. The scaling property itself is proved directly.

Proposition 4.1 — Symmetric stable scaling.

$$\sum_{j=1}^{n} X_j \overset{d}{=} n^{1/\alpha} X, \qquad \frac{1}{n}\sum_{j=1}^{n} X_j \overset{d}{=} n^{1/\alpha - 1} X. \tag{4.1}$$

The symbol $\overset{d}{=}$ denotes equality in distribution.

Derivation.

Independence gives $\phi_{\sum X_j}(k) = \exp(-nc|k|^\alpha)$. Scaling one copy gives $\phi_{n^{1/\alpha}X}(k) = \exp[-c|n^{1/\alpha}k|^\alpha]$, the same function. Characteristic functions identify probability laws. Dividing both random variables by n proves the sample-average identity. ∎

At $\alpha = 2$, the law is Gaussian with variance $2c$ and the sample mean narrows as $n^{-1/2}$. At $\alpha = 1$, the width is unchanged. When $\alpha < 1$, its quantile width grows with sample size under ordinary averaging. This last

statement concerns sample averages as defined random variables, not an expectation that might fail to exist.

The upper index limit can be checked without guessing a Fourier inverse. If $\exp(-c|k|^{\alpha})$ were a characteristic function for $\alpha > 2$, then $[1 - \Re\phi_X(k)]/k^2 \to 0$. For every fixed real x, $(1 - \cos kx)/k^2 \to x^2/2$, and the integrand is nonnegative. Fatou's lemma would imply $\mathbb{E}[X^2]/2 \le 0$. Thus $X = 0$ almost surely, contradicting a transform smaller than one away from zero. For $\alpha < 0$, continuity at zero fails; $\alpha = 0$ also fails to approach one for $c > 0$. These obstructions explain why this proposed transform cannot be extended beyond the stated range.

4.2 A Cauchy average does not become more precise L5 22:41

A *Cauchy distribution* with location μ and scale $s_C > 0$ has density $\rho(x) = s_C/[\pi((x-\mu)^2 + s_C^2)]$. The location is its symmetry center, median, and mode; it is not its mean. Integrating after the substitution $y = (x-\mu)/s_C$ gives cumulative distribution $F(x) = 1/2 + \pi^{-1}\arctan[(x-\mu)/s_C]$. Its central half of probability lies between $\mu - s_C$ and $\mu + s_C$, so its interquartile range is $2s_C$.

Fourier inversion of $e^{-s_C|k|}$ provides a short transform check: split the integral at zero and combine the two exponentials into a cosine. The elementary integral $\int_0^\infty e^{-s_C k}\cos(kx)\,dk = s_C/(s_C^2 + x^2)$ gives the centered density. Translation therefore yields $\phi_X(k) = \exp(ik\mu - s_C|k|)$ under the positive-sign convention.

Proposition 4.2 — Cauchy sample means. For iid Cauchy variables of location μ and scale s_C, every finite sample average has exactly that same Cauchy law.

Derivation.

The characteristic function of $\overline{X}_n = n^{-1}\sum_j X_j$ is $[\phi_X(k/n)]^n = \exp(ik\mu - s_C|k|)$. Its interquartile range therefore remains $2s_C$ for all n. By comparison a Gaussian average of input variance σ^2 has variance σ^2/n. ∎

The Cauchy tail has density proportional to $|x|^{-2}$. Consequently $\mathbb{E}[|X|]$ diverges logarithmically, and the positive and negative parts of the signed first moment do not define a finite expectation. A symmetrically truncated principal-value integral can equal μ, but that cancellation is a different operation. The second moment also diverges. Printing a finite sample variance does not repair either nonexistent population moment.

For the standard centered Cauchy, the probability $\Pr(|\overline{X}_n| > 10)$ is the same at one, one hundred, or any other positive sample size: $1 - (2/\pi)\arctan 10$. This gives an operational failure of averaging. It does not say that every estimator fails: the sample median uses quantiles, rather than a sum vulnerable to arbitrarily large observations. Its analysis is a different statistical problem.

4.3 A reciprocal square creates a one-sided heavy tail L5 28:23

Let Z be standard Gaussian, and set $L = c/Z^2$ for a positive scale c. The value $Z = 0$ has zero probability, so L is finite almost surely despite its unbounded support. For $x > 0$, the two inverse Gaussian values are $\pm\sqrt{c/x}$, each with absolute derivative $\sqrt{c}/(2x^{3/2})$. The branch formula from chapter 2 gives the *one-sided Lévy law*:

$$\rho_L(x) = \sqrt{\frac{c}{2\pi}}\, x^{-3/2} \exp\left(-\frac{c}{2x}\right), \qquad F_L(x) = \operatorname{erfc}\sqrt{\frac{c}{2x}}. \tag{4.2}$$

The cumulative formula follows by writing $L \le x$ as $|Z| \ge \sqrt{c/x}$. It tends to zero at $x \downarrow 0$ and to one at infinity, confirming normalization. The density decays faster than any inverse power near zero and as $x^{-3/2}$ at large x. The latter makes even its first moment infinite. The mode, obtained by differentiating the logarithm, is $c/3$; it is neither a mean nor a typical upper-tail cutoff.

The Laplace transform, with real $s \ge 0$ of inverse-x units, is $\mathbb{E}[e^{-sL}] = \exp[-\sqrt{2cs}]$. A useful integral verification uses $I(a,b) = \int_0^\infty \exp(-az^2 - b/z^2)\, dz$ for $a, b > 0$. Differentiating with respect to b and changing variables $z = \sqrt{b/a}/u$ gives $\partial_b I = -\sqrt{a/b}\, I$. Its value at $b = 0$ is $\sqrt{\pi}/(2\sqrt{a})$, so $I = \sqrt{\pi}\exp(-2\sqrt{ab})/(2\sqrt{a})$. Substituting $a = 1/2, b = sc$ proves the transform.

With the same density scale c, the positive-sign characteristic function is $\phi_L(k) = \exp[-\sqrt{c|k|}(1 - i\operatorname{sgn} k)]$. This is the boundary value of the Laplace expression on the principal square-root branch; bounded convergence from positive real damping gives that boundary. The factor multiplying $\sqrt{|k|}$ is $\sqrt{c}$, rather than c. This scale consistency matters when a density and transform are used in the same calculation.

Multiplying n Laplace transforms gives $\exp[-n\sqrt{2cs}] = \exp[-\sqrt{2(n^2c)s}]$. Thus the sum of n independent copies has the same law as $n^2 L$: it is strictly stable of index $1/2$, although it is not in the centered symmetric subclass of Section 4.1. Brownian first-passage times later supply a physical realization. No general reciprocal-power rule for arbitrary stable variables is inferred from this particular Gaussian transformation.

4.4 A ratio can destroy moments that both inputs possess L5 40:16

Let X_1, X_2 be independent centered Gaussian coordinates with standard deviations $\sigma_1, \sigma_2 > 0$. Define $R = X_1/X_2$, meaningful almost surely because $\Pr(X_2 = 0) = 0$. Use coordinates (r, y) with $x_1 = ry, x_2 = y$; the absolute Jacobian is $|y|$. Integrating the joint density gives

$$\rho_R(r) = \frac{1}{2\pi\sigma_1\sigma_2} \int_{-\infty}^{\infty} |y| \exp\left[-\frac{y^2}{2}\left(\frac{r^2}{\sigma_1^2} + \frac{1}{\sigma_2^2}\right)\right] dy = \frac{s_C}{\pi(r^2 + s_C^2)}, \quad s_C = \frac{\sigma_1}{\sigma_2}. \tag{4.3}$$

The integral uses $\int_0^\infty y e^{-Ay^2/2} dy = 1/A$. Near-zero denominators generate a heavy-tailed ratio even though both numerator and denominator have all Gaussian moments. If X_j are independent diffusing coordinates at a common time $t > 0$, then $\sigma_j^2 = 2D_j t$ and $s_C = \sqrt{D_1/D_2}$. Time cancels. Increasing numerator diffusion must broaden the ratio, an immediate check on which diffusion coefficient belongs on top.

4.5 Infinite divisibility asks for smaller component laws L5 49:56

A law is *infinitely divisible* if for every positive integer n it can be represented as the law of a sum $Y_{n,1} + \cdots + Y_{n,n}$ of iid variables. The component law may change with n and need not equal the final law. This is a property of distributions; it does not assert that one observed outcome has a unique microscopic decomposition.

Equivalently, for each n a characteristic function ψ_n must exist such that $\phi_X(k) = [\psi_n(k)]^n$. Taking an algebraic nth root of a complex function is insufficient: that root must be the transform of a nonnegative probability measure, with the positivity properties from chapter 2. In the symmetric stable family, $\psi_n(k) = \exp[-(c/n)|k|^\alpha]$ is valid because it is a scaled member of the same established family. The component is distributed as $n^{-1/\alpha}X$, rather than as an unscaled copy of X.

Proposition 4.3 — Poisson divisibility. A Poisson law of mean $\lambda \geq 0$ is infinitely divisible.

Derivation.

Choose iid components Poisson of mean λ/n. Their characteristic functions multiply to $[\exp((\lambda/n)(e^{ik} - 1))]^n = \exp(\lambda(e^{ik} - 1))$, the requested law. All roots here are actual Poisson laws, so the construction establishes more than formal algebra. ∎

A nondegenerate Poisson law is not stable in the real-variable rescaling sense. It has finite positive variance, so stability would require $a_n = \sqrt{n}$ by comparing variances. For $n = 2$, its rescaled lattice spacing becomes $1/\sqrt{2}$, incompatible with the original positive masses at every nonnegative integer after any translation. Thus infinite divisibility is strictly broader than stability.

The same Poisson construction proves that a Skellam law is infinitely divisible: divide each independent Poisson intensity by n, form n independent differences, then add them. A geometric law is also infinitely divisible. To see an actual root distribution, its generating function $p/(1 - qz)$ has nth root $p^{1/n}(1 - qz)^{-1/n}$, with $q = 1 - p$ and $0 < p \leq 1$. The binomial series has nonnegative coefficients $(1/n)(1/n + 1) \cdots (1/n + j - 1)q^j/j!$, and evaluating at $z = 1$ normalizes them. These roots are negative-binomial laws of positive shape $1/n$. Integer mode count was one construction of that family, not a restriction on all its valid parameters.

4.6 A bounded nondegenerate law cannot be infinitely divisible

The binomial count is a sum of a fixed number N of Bernoulli variables, so it is N-divisible. That does not imply divisibility into every number of iid real summands. A support argument supplemented by a variance bound gives a proof that does not assume the components are integer valued.

Suppose X has bounded essential support endpoints $a < b$, and $X = \sum_{j=1}^{n} Y_j$ in distribution with iid independent components. Each component must be bounded: if one can exceed arbitrarily large values while the others each lie above some fixed finite lower threshold with positive probability, independence makes the sum exceed b with positive probability. The lower-tail argument is the same. Let component essential endpoints be c, d. Independence and positive probabilities arbitrarily close to the endpoints imply $nc = a$ and $nd = b$.

Any variable in $[c, d]$ has variance at most $(d - c)^2/4$: its squared distance from the midpoint is at most that value, and variance is no larger than its mean squared distance from any fixed center. Therefore

$$\operatorname{Var}(X) = n \operatorname{Var}(Y_1) \le \frac{(b-a)^2}{4n}. \tag{4.4}$$

If such a decomposition exists for every n, the right side tends to zero. The only possibility is a point mass, contradicting nondegeneracy. A fixed-trial binomial with $0 < p < 1$ has bounded support and positive variance, hence is not infinitely divisible. Deterministic laws are the exception: a constant is the sum of n constants equal to its nth part.

4.7 Laboratory: measure quantile width instead of a nonexistent variance

The program constructs 20,000 independent sample averages at each listed sample size for standard Gaussian and Cauchy inputs, using seed 202611084. It reports the interquartile range, abbreviated IQR, rather than assigning a population variance to the Cauchy law. Its exact comparisons are $2\Phi^{-1}(0.75)/\sqrt{n}$ for Gaussian inputs and two for Cauchy inputs, where Φ is the standard Gaussian cumulative distribution. Run `python code/ch04_stable.py`; outputs are JSON and text, written beside the script without network access.

```
n=1: Gaussian=1.329102 (exact 1.348980), Cauchy=2.006741
n=4: Gaussian=0.672885 (exact 0.674490), Cauchy=1.997527
n=16: Gaussian=0.341904 (exact 0.337245), Cauchy=1.971913
n=64: Gaussian=0.166752 (exact 0.168622), Cauchy=2.023120
n=256: Gaussian=0.084087 (exact 0.084311), Cauchy=2.036893
```

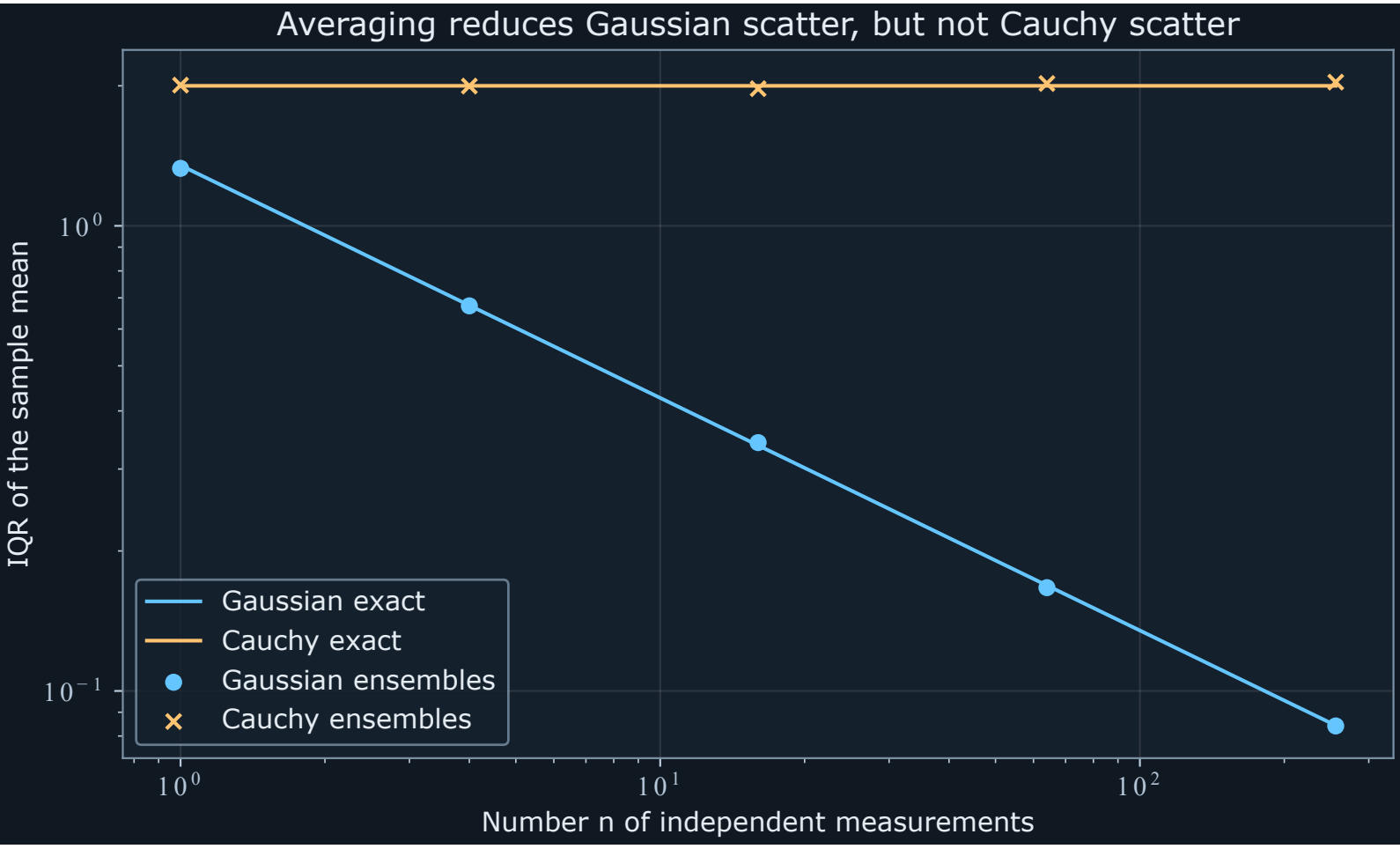


Figure 4.1. Quantile widths distinguish two exact stable scaling laws. Figure 4.1 overlays the finite-ensemble IQRs on the exact theoretical curves; the Gaussian width decreases while the Cauchy width remains fixed. The markers are synthetic samples, not measurements. The generator reads the retained laboratory results.

The full program also checks that a ratio with numerator standard deviation two and denominator standard deviation one has Cauchy scale two. Its cumulative-distribution discrepancy is 0.002944 in 200,000 draws. A separate reciprocal-square ensemble has discrepancy 0.002134 from the unit-scale Lévy law. These are implementation diagnostics. The independent Laplace quadratures at three positive arguments test the parameter matching in (4.2) without relying on an empirical moment that diverges.

This minimal calculation implements the exact quantile comparison and has no file effects:

```
import numpy as np
from scipy.stats import norm

for n in (1, 16, 256):
    gaussian_iqr = 2*norm.ppf(0.75)/np.sqrt(n)
    print(f"n={n}: Gaussian={gaussian_iqr:.6f}, Cauchy=2.000000")
```

```
n=1: Gaussian=1.348980, Cauchy=2.000000
n=16: Gaussian=0.337245, Cauchy=2.000000
n=256: Gaussian=0.084311, Cauchy=2.000000
```

4.8 Stable examples do not classify every heavy tail

The exact stable calculations require specified laws and independent copies. Having an infinite variance alone does not identify an index or prove attraction to a stable law. Nor does knowing that moments exist below a certain order suffice to specify a domain of attraction. Tail regularity, balance, and centering conditions matter. The chapter uses exact Gaussian, Cauchy, and Lévy constructions to establish the mechanisms without silently asserting such a general classification theorem.

The symmetric subclass, the one-sided Lévy law, and a general skew stable law have different phase and support structures. A Cauchy density is symmetric about its location; a skew index-one stable law should not be called that same location-scale Cauchy model. Infinite divisibility concerns distributional decompositions; a stochastic process with stationary independent increments needs consistent laws across time, addressed in later chapters. The repeated-measurement decision here is concrete: select a statistic whose defining population quantities exist, then check its actual scaling.

4.9 Exercises with worked solutions

Exercise 4.1 — Principal value is not expectation (conceptual)

For a centered unit Cauchy variable, integrate $x\rho(x)$ symmetrically between $-A$ and A. Then compute the positive contribution between zero and A. Explain the apparent disagreement.

Worked solution

The symmetric integral is zero for every A because the integrand is odd. The positive contribution is $\pi^{-1}\int_0^A x/(1+x^2)dx = (2\pi)^{-1}\log(1+A^2)$, which diverges. The negative part has the same divergent

magnitude. The signed expectation is not defined by subtracting those two infinities; choosing symmetric truncation defines a principal value instead. Thus the finite symmetric value cannot serve as the mean in a law of large numbers.

Exercise 4.2 — Same family versus same law (conceptual)

A Poisson variable of mean nine is split into three iid Poisson components. Find their means. Why does belonging to the same family not prove stability?

Worked solution

Each component has mean three; their sum has mean nine by independence and the Poisson generating function. Their distributions differ from the final mean-nine law. Infinite divisibility permits that change. Stability would instead require that a shifted and rescaled sum of copies of the mean-nine law recover the same mean-nine law. Finite variance fixes the rescaling to a square root, which conflicts with its lattice support as shown in Section 4.5. Family closure under parameter addition is weaker than real-variable stability.

Exercise 4.3 — Weighted Cauchy noise (derive)

For independent centered Cauchy variables of scales s_1, s_2, derive the law of $aX_1 + bX_2$ for arbitrary real a, b. Compare its scale with a Gaussian variance formula.

Worked solution

Its transform is $\exp[-s_1|ak| - s_2|bk|] = \exp[-(|a|s_1 + |b|s_2)|k|]$. Thus the scale is $|a|s_1 + |b|s_2$, with a deterministic zero result if both coefficients vanish. Negative weights do not subtract independent Cauchy scatter. For Gaussians, variances add as $a^2\sigma_1^2 + b^2\sigma_2^2$; for Cauchy variables there are no variances to add. Averaging n identical scales with weights $1/n$ keeps their scale unchanged.

Exercise 4.4 — Finite moment range for a power tail (derive)

Suppose a nonnegative density is bounded near zero and satisfies $\rho(x) \sim Cx^{-1-\alpha}$ as $x \to \infty$, with $C, \alpha > 0$. Determine which positive moments exist. If only this moment-existence cutoff were known, without the precise density asymptotic, would it prove stable-law attraction?

Worked solution

The near-zero integral of $x^q\rho(x)$ is finite for $q > 0$ by boundedness. For sufficiently large x, the asymptotic ratio bounds the density between positive constant multiples of $x^{-1-\alpha}$. The tail moment therefore converges exactly when $\int_M^\infty x^{q-1-\alpha}dx$ converges, namely $q < \alpha$. At equality it diverges logarithmically; above it diverges as a power.

This argument establishes the moment range. Knowing only that range does not prove a stable-domain-of-attraction statement; such a result needs tail information and a specified centering and normalization. The

precise power-density asymptotic assumed in this exercise supplies more information than the moment cutoff alone. The conclusion here concerns what that cutoff establishes, rather than claiming that attraction fails under the stronger density assumption.

Exercise 4.5 — Check the numerator scale (compute/design)

Two independent diffusing particles have diffusion coefficients four and one in common units. Predict the scale and IQR of $X_1(t)/X_2(t)$, and design a check that detects an inverted scale.

Worked solution

The standard-deviation ratio is $\sqrt{4/1} = 2$, so the Cauchy scale is two and IQR four at every $t > 0$. A wrong inverted scale would give IQR one, differing by a factor of four. The laboratory uses precisely this standard-deviation ratio and compares its cumulative distribution with Cauchy scale two; its discrepancy is 0.002944. Estimate quantiles rather than a variance, repeat at several times with each coordinate scaled by $\sqrt{t}$, and verify that the ratio distribution is time independent. Shared noise or a nonzero mean changes the joint construction and needs a new ratio analysis.

Exercise 4.6 — Test a proposed binomial decomposition (compute/design)

Let X be binomial with $N = 20$ and $p = 0.2$. Use the support-width bound to find an integer component count that is impossible for an iid real decomposition. Reconcile this with its twenty-Bernoulli construction.

Worked solution

The essential width is twenty and variance is $20(0.2)(0.8) = 3.2$. Bound (4.4) would require $3.2 \le 400/(4n) = 100/n$. Thus every integer $n \ge 32$ is impossible; for example, at 32 the upper bound is 3.125. This sufficient obstruction does not classify every smaller n. The known twenty-component Bernoulli construction satisfies the bound and proves twenty-divisibility. There is no contradiction: a decomposition at one component count does not imply all others. A numerical characteristic-function root must additionally be tested as an actual probability law, not accepted from its algebraic appearance.

4.10 Sources and attribution

- V. Balakrishnan, Stable distributions and infinite divisibility, lecture 5, IIT Madras, NPTEL115106089, and official instructor notes, PDF40–46. Stable-family existence is cited; scaling, transform checks, reciprocal square, ratio, and bounded-support obstruction are developed here.
- NIST DLMF7.2.1–2 provides error and complementary-error function conventions used in the Lévy cumulative distribution; these were verified in chapter2.

Corrections to the source's informal discussion: Cauchy location does not define a finite mean; the ratio scale is $\sqrt{D_1/D_2}$ for X_1/X_2; density scale c gives Lévy transform coefficient $\sqrt{c}$. No unrestricted reciprocal stable duality or moment-order-only generalized limit theorem is assumed. Raw source material is retained privately.

From random variables to Markov state descriptions

A fluctuating device spends half its observed times in each of two states. That histogram is compatible with independent switching, long persistent runs, or a deterministic alternation with a random starting phase. The same occupancy calibration can therefore predict very different waiting and relaxation behavior. The task is to decide which observations make the present state sufficient for predicting the future.

> Lecture 6 supplies the hierarchy of joint laws, Markov conditioning, stationarity, transition composition, and the opening master-equation and eigenvalue discussion. Its full caption text and two sampled stills were inspected. This chapter distinguishes time homogeneity from stationarity explicitly; the finite-chain constructions and numerical experiments are added examples.

5.1 A process requires joint laws across times L6 0:31

A *stochastic process* is a collection of random variables $\{X_t : t \in \mathcal{T}\}$ on one probability space, indexed by times in a set $\mathcal{T}$. The time set can be the nonnegative integers or nonnegative real numbers. The state space specifies possible values of each X_t. A realization fixes one outcome and gives a path $t \mapsto X_t$. An ensemble consists of all possible realizations with their probabilities.

For ordered times $t_0 < \cdots < t_m$, the joint law assigns probabilities to the entire tuple $(X_{t_0}, \ldots, X_{t_m})$. These finite-dimensional laws encode correlations and higher-order dependence. A one-time histogram supplies only one marginal of that hierarchy. Consistency requires that summing over one coordinate recover the joint law of the remaining times.

The probability chain rule is exact before any Markov assumption. For discrete states and histories of positive probability,

$$\Pr(X_{t_0} = i_0, \ldots, X_{t_m} = i_m) = \Pr(X_{t_0} = i_0) \prod_{r=1}^{m} \Pr(X_{t_r} = i_r \mid X_{t_0} = i_0, \ldots, X_{t_{r-1}} = i_{r-1}). \tag{5.1}$$

Each factor conditions on the whole known past. The product follows by applying $\Pr(A \cap B) = \Pr(A \mid B)\Pr(B)$ repeatedly. Multiplication here does not assume independence: dependence remains inside the conditionals. Zero-probability histories need not be assigned arbitrary empirical conditional ratios.

Compare independent signs $X_n = \pm 1$ with equal probabilities at every integer n against a constant random sign $Y_n = Z$, where Z is selected once with the same two probabilities. Both have zero mean and unit variance at each time. At nonzero lag, $\mathbb{E}[X_n X_{n+k}] = 0$, whereas $\mathbb{E}[Y_n Y_{n+k}] = 1$. The one-time law therefore cannot predict even the simplest two-time correlation. Figure 5.1 gives a less extreme persistent example.

5.2 The Markov property is sufficiency of the present state L6 9:57

Definition 5.1 — Markov property. Given any ordered finite collection of observation times, the conditional law of a future state, conditioned on the observed states at all earlier times, depends only on the latest state and its time. In discrete time this reads $\Pr(X_{n+1} = i \mid X_n = j, X_{n-1}, \dots, X_0) = \Pr(X_{n+1} = i \mid X_n = j)$ on histories of positive probability.

The property removes earlier observations from each conditional in (5.1). An initial law and transition kernels —the conditional probability rules from a current state—then generate the full joint hierarchy by multiplication. It does not imply that successive states are independent. A chain that stays in its present state with probability 0.9 is strongly persistent, yet the current sign contains all information needed for its next transition.

State choice determines whether this simplification is valid. A deterministic particle obeying $\dot{x} = v$ cannot have its future position predicted from present position alone if its velocity is unknown. Two particles at the same position with opposite velocities separate immediately. The pair (x, v) is sufficient under a specified deterministic force law with unique solutions. The example motivates state augmentation without claiming that every physical memory kernel can be captured by finitely many extra variables.

A finite exact counterexample makes the same point without differential equations. Let a hidden phase advance deterministically around four states $A \to B \to C \to D \to A$, and observe only $0, 0, 1, 1$ respectively. Start with phase uniform on the four states. If the current observation is zero and the preceding observation was zero, the phase must be B, so the next observation is one. If the preceding observation was one, the current phase must be A, so the next observation is zero. Present observation alone is insufficient, although the hidden phase process is Markov. Recording the last two observations resolves the phase and restores a sufficient state in this example.

5.3 Constant marginals, homogeneity, stationarity, and mixing L6 14:54

Strict stationarity means that every finite-dimensional joint law is unchanged when all its time arguments undergo the same allowed shift. Constant one-time marginals are necessary but insufficient. To prove insufficiency, take independent fair signs Z, W and define $X_0 = X_1 = Z$, with $X_n = W$ for every $n \geq 2$. Every marginal is a fair sign, but $\Pr(X_0 = X_1) = 1$ and $\Pr(X_1 = X_2) = 1/2$. A one-step shift changes a two-time law.

Time homogeneity of a Markov transition rule means that the kernel depends only on elapsed time, rather than the absolute starting time. A fixed discrete transition matrix is homogeneous even when the initial distribution is concentrated in one state and its marginals evolve. Strict stationarity additionally needs an invariant initial law. Calling an autonomous rate rule “stationary” without specifying its initial ensemble conflates these two statements.

For a homogeneous finite-state chain, let $K_{ij} = \Pr(X_{n+1} = i \mid X_n = j)$, with starting state j as the column index. If $p(n)$ is the column of state probabilities at step n, total probability gives $p(n+1) = Kp(n)$. A

probability column π is a *stationary distribution* when $K\pi = \pi$. Starting the chain with π makes each later marginal equal to π. More strongly, the Markov product formula for a time-shifted tuple uses the same initial marginal and the same elapsed-time kernels, so all finite-dimensional laws are unchanged. This proves strict stationarity for that initialization.

Mixing toward a stationary law here means that transition probabilities lose their initial-state dependence: $(K^n)_{ij} \to \pi_i$ for each pair of states. A strictly stationary process need not mix. The constant random sign example is stationary, but its initial sign is remembered forever. A periodically alternating sign initialized equally is also stationary, yet its lag dependence oscillates forever. Equilibrium occupancy and dynamical loss of memory are separate claims.

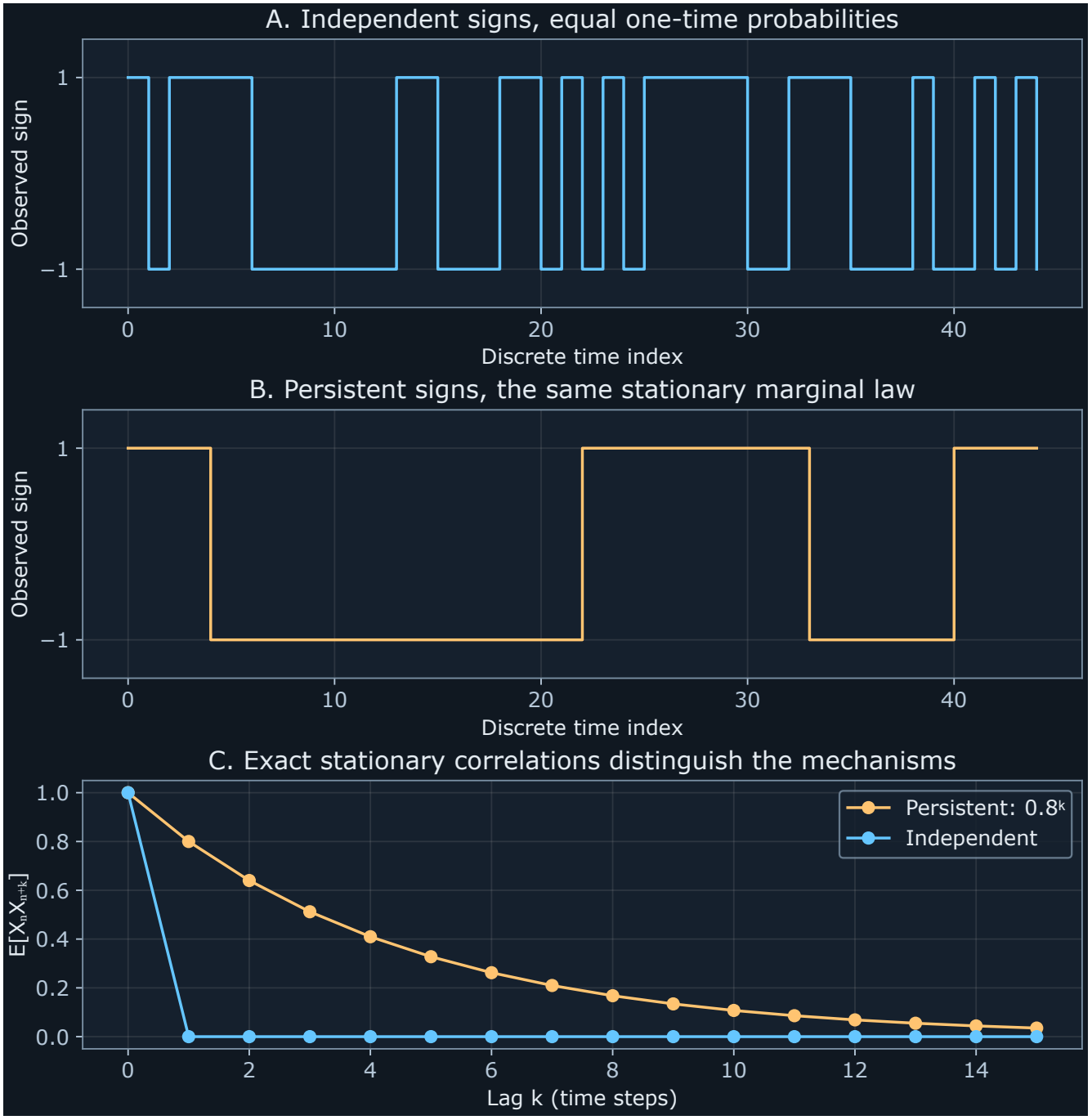


Figure 5.1. Identical stationary histograms conceal different dynamics. Panels A and B are illustrative synthetic realizations of independent and persistent signs; their finite paths need not have exactly equal observed occupancies. Panel C gives the exact stationary correlations, derived in Exercise 5.3. The persistent illustration starts at plus one, while its quoted stationary law requires an equal random initial sign; sign symmetry leaves the displayed correlation formula unchanged. Figure 5.1 is reproduced by the generator.

5.4 Composition sums over mutually exclusive intermediate states L6 25:32

For a time-homogeneous finite-state chain, define $K_{ij} = \Pr(X_{n+1} = i \mid X_n = j)$. The second index is the starting state; probability vectors are columns throughout this book. Each entry is nonnegative, and each column sums to one. Total probability gives $p_i(n+1) = \sum_j K_{ij} p_j(n)$, or $p(n+1) = Kp(n)$.

Proposition 5.1 — Chapman–Kolmogorov composition. A homogeneous Markov chain has m-step kernel K^m, and $K^{n+m} = K^n K^m$. For a continuous-time homogeneous transition family $P(t)$, the corresponding statement is $P(t+s) = P(t)P(s)$.

Derivation.

Condition on the state l at one fixed intermediate time. The events $X_m = l$ partition all paths into mutually exclusive classes. Markov conditioning gives
$\Pr(X_{m+n} = i \mid X_0 = j) = \sum_l \Pr(X_{m+n} = i \mid X_m = l) \Pr(X_m = l \mid X_0 = j)$. Homogeneity identifies the two factors as elapsed-time kernels. This is precisely matrix multiplication in the stated column convention. Iterating from the one-step kernel gives the powers. The same conditioning proves the continuous-time formula. ∎

The sum is over states at one chosen intermediate time, not over all possible intermediate times. Integrating the same path decomposition over every time would repeatedly count paths rather than form a partition. A renewal equation that integrates over a first event time is a different decomposition, because that first event time is unique for each path.

Composition of observed two-time kernels is a necessary consequence of a Markov model, but pairwise identities alone do not reconstruct every higher-order conditional. The hidden four-phase example fails even the elementary two-step check: its observed one-step matrix has every entry $\mathbf{1/2}$, whose square is unchanged. Its actual two-step observation always flips, giving matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. A pairwise fit to the one-step data therefore cannot supply a valid Markov prediction.

5.5 A stationary vector is not a convergence proof L6 19:56

A finite chain is *irreducible* if every state can reach every other with positive probability in some finite number of steps. The period of a state is the greatest common divisor of the positive return lengths; an irreducible chain is aperiodic when that period is one. A finite irreducible chain has a unique positive stationary law, and aperiodicity additionally guarantees convergence from every initial law. This general theorem is *cited* from the Cambridge Markov-chain notes, PDF page 18. A direct contraction proof for the strictly positive case explains its mechanism.

Proposition 5.2 — Contraction for a positive kernel. Let K be an r-state column-stochastic matrix with every entry at least $\varepsilon > 0$. For any two probability columns p, q, $\|Kp - Kq\|_1 \le (1 - r\varepsilon)\|p - q\|_1$, where $\|v\|_1 = \sum_i |v_i|$.

Derivation.

If $r\varepsilon = 1$, every column is the uniform distribution and all differences vanish. Otherwise subtract the constant matrix $\varepsilon \mathbf{1}\mathbf{1}^T$ from K and divide by $1 - r\varepsilon$ to form another column-stochastic matrix R. Because $\sum_i (p_i - q_i) = 0$, the constant part cancels and $K(p - q) = (1 - r\varepsilon)R(p - q)$. The triangle inequality gives $\|Rv\|_1 \le \sum_{i,j} R_{ij}|v_j| = \|v\|_1$. This proves the contraction. ∎

A stationary vector exists in finite state space: averages $a_N = N^{-1} \sum_{n=0}^{N-1} K^n p$ lie in the compact probability simplex and have a convergent subsequence. Since $Ka_N - a_N = (K^N p - p)/N \to 0$, its limit is invariant. For positive K, the contraction makes that invariant vector unique and drives every initial distribution to it geometrically. This proof supplies a mechanism stronger than merely locating eigenvalue one.

For the two-cycle $K = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, the stationary law is $(1/2, 1/2)^T$, but a point initial law alternates forever. The additional eigenvalue minus one preserves an oscillation. Adding a probability of remaining in place removes this periodic obstruction; disconnected closed classes would still prevent uniqueness. Infinite state spaces introduce further recurrence and normalization requirements, developed in later chapters.

5.6 From a short-time kernel to a rate equation L6 33:02

For a finite continuous-time state space, assume a homogeneous kernel has expansion $P(h) = I + hQ + o(h)$ as $h \downarrow 0$. Here I is the identity matrix, h has time units, and Q is the *generator*, with inverse-time units. Off-diagonal entries Q_{ij} are transition rates from j to i and must be nonnegative. Column normalization forces $Q_{jj} = -\sum_{i \ne j} Q_{ij}$. Rates are not finite-interval probabilities: a rate can exceed one in inverse-second units.

Insert the expansion into $P(t + h) = P(h)P(t)$, subtract $P(t)$, and divide by h. Taking the derivative gives $\dot{P}(t) = QP(t)$ and hence $\dot{p}(t) = Qp(t)$ for an arbitrary initial column. Component i gains $\sum_{j \ne i} Q_{ij} p_j$ and loses $p_i \sum_{j \ne i} Q_{ji}$. The homogeneous finite-dimensional solution is $p(t) = e^{tQ} p(0)$. Chapter 6 proves that this exponential remains a probability kernel and develops explicit models.

The generator's spectrum lies in the closed left half of the complex plane. A direct Gershgorin bound proves it. Apply the eigenvector equation to Q^T and choose an eigenvector component of largest magnitude. At its index j, division by that magnitude gives $|\lambda - Q_{jj}| \le \sum_{i \ne j} |Q_{ij}| = -Q_{jj}$. Thus every eigenvalue belongs to at least one disk centered at nonpositive real Q_{jj} with radius $-Q_{jj}$. Every disk lies in the closed left half-plane and touches the imaginary axis only at zero. The relevant set is the *union* of the disks. Conservation makes zero an eigenvalue.

This bound controls growth, but it does not make a stationary probability unique. A generator with two closed noncommunicating classes has at least two stationary laws. For a fixed finite generator, nonzero modes decay; defective eigenvalues can carry polynomial factors multiplying their decaying exponentials. The long-time limit must still be interpreted using the communicating classes and the initial distribution, rather than inferred from conservation alone.

5.7 Laboratory: test the state description before fitting its dynamics

The lab uses the column-stochastic matrix

$$K = \begin{pmatrix} 0.6 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.3 & 0.7 \end{pmatrix}, \qquad \pi = \frac{1}{34}\begin{pmatrix} 9 \\ 11 \\ 14 \end{pmatrix}. \tag{5.2}$$

It solves the stationary equations, enumerates every two-step path independently of matrix multiplication, and evolves a point initial distribution. It also computes the hidden four-phase observation's actual one- and two-step matrices and tests a two-state continuous-time semigroup. Run `python code/ch05_markov.py`; the only effects are writing JSON and text beside the script. There are no network calls or fitted measurements.

```
Three-state stationary law: 0.264705882, 0.323529412, 0.411764706
n=0: distance=0.735294118
n=1: distance=0.335294118
n=5: distance=0.033404706
n=20: distance=0.000003400
```

The distance is total variation, half the sum of absolute probability differences from π. Every entry of K is at least 0.1, so Proposition 5.2 supplies contraction factor 0.7 per step. The actual errors decay faster than that upper bound. A bound and a measured matrix error need not coincide; the bound protects all initial probability columns.

This small program makes the stationary-law check explicit, with no file writes:

```
import numpy as np

kernel = np.array([[0.6, 0.2, 0.1],
                   [0.3, 0.5, 0.2],
                   [0.1, 0.3, 0.7]])
stationary = np.array([9, 11, 14])/34
assert np.allclose(kernel.sum(axis=0), 1)
assert np.allclose(kernel @ stationary, stationary)
print("Normalization and stationarity pass.")
```

```
Normalization and stationarity pass.
```

The observed-cycle test has a different purpose: it checks whether the chosen state admits any valid first-order homogeneous Markov model with the fitted one-step kernel. Its failure cannot be repaired by more accurate matrix exponentiation. The state must first retain enough information.

5.8 What a Markov state captures and misses

The Markov model compresses a hierarchy of conditionals into an initial law and a transition rule because the present state is sufficient. Histograms alone do not justify that compression. A sampled trajectory can also fail

to reveal a hidden slow variable over the observation interval, so an empirically plausible Markov approximation still needs tests at multiple lags and relevant initial ensembles.

Time homogeneity describes the rule; strict stationarity describes the ensemble; an invariant vector is a candidate stationary initialization; mixing is a convergence property. Keeping these statements separate prevents equilibrium assumptions from entering a transient calculation unnoticed. Infinite state spaces, nonexponential residence times, and coarse observations require additional care. Chapter 22 separates a jump chain from its physical clock, and chapter 23 develops a non-Markov transport example.

5.9 Exercises with worked solutions

Exercise 5.1 — Homogeneous does not mean stationary (conceptual)

A two-state chain has $K = \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix}$ and starts in state one with probability one. Is it homogeneous? Is it strictly stationary? Find its one-step law.

Worked solution

The fixed kernel is time homogeneous. Its initial column is $(1, 0)^T$ and its next column is $(0.9, 0.1)^T$, so its one-time marginal changes and it is not strictly stationary. The stationary column is $(1/2, 1/2)^T$. Starting with that column would make the whole homogeneous Markov process stationary by the joint-law product argument, not merely make one histogram constant.

Exercise 5.2 — Stationarity does not remove hidden memory (conceptual)

For the uniformly initialized phase cycle with observations 0011, compute $\Pr(Y_{n+1} = 1 \mid Y_n = 0)$ and the same probability conditioned also on $Y_{n-1} = 0$. Explain why stationarity does not settle the Markov question.

Worked solution

Conditioning only on a zero observation leaves phases A, B equally likely. Their next observations are zero and one respectively, so the first probability is $1/2$. Conditioning on consecutive zeros forces phase B and makes the probability one. Uniform phase is invariant under deterministic cycling, so the observed joint laws are time-shift invariant. The failure comes from predictive information in the preceding observation, not from a changing ensemble. Augmenting the observed state to the last two bits distinguishes all four phases.

Exercise 5.3 — Correlation of persistent signs (derive)

Let a sign process stay unchanged with probability $1 - r$ and flip with probability r, independently of its earlier history given its present sign, where $0 < r < 1$. Derive its stationary covariance at lag k.

Worked solution

The symmetric stationary law assigns equal sign probabilities, with mean zero and variance one. Conditional expectation gives $\mathbb{E}[X_{n+1} \mid X_n] = (1 - 2r)X_n$. Iterating this conditional identity yields

$\mathbb{E}[X_{n+k} \mid X_n] = (1-2r)^k X_n$. Multiplying by X_n and averaging gives covariance $(1-2r)^k \mathbb{E}[X_n^2] = (1-2r)^k$. For $r = 0.1$ this is 0.8^k, as in Figure 5.1. At $r = 1/2$ signs become independent; as $r \uparrow 1$, the increasingly slow decay alternates in sign. At the excluded endpoint one, periodicity prevents decay altogether.

Exercise 5.4 — Two-step path enumeration (derive)

For (5.2), compute the probability of going from state one to state three in two steps by listing intermediate states. Check the column convention.

Worked solution

The three contributions are $K_{31}K_{11} = 0.1(0.6) = 0.06$, $K_{32}K_{21} = 0.3(0.3) = 0.09$, and $K_{33}K_{31} = 0.7(0.1) = 0.07$. Their sum is 0.22. This is entry $(3,1)$ of K^2: its column is the initial state and row the final state. The alternatives are mutually exclusive intermediate states. Multiplying the reverse-index entries would describe different paths and generally give a different answer.

Exercise 5.5 — Bound relaxation before computing it (compute/design)

Use Proposition 5.2 for the matrix (5.2) to bound the total-variation error at twenty steps from a point start. Compare with the lab. Does the bound certify an error below 0.001?

Worked solution

There are three states and minimum entry 0.1, so the contraction factor is 0.7. Total variation contracts by the same factor as the one-norm. Starting in state one gives distance $1 - 9/34 = 25/34$. Thus the bound is $(25/34)(0.7)^{20} < (0.7)^{20} < 0.001$, since the last power is below one thousandth. The lab's actual value is 0.000003400, smaller still. The bound certifies the requested tolerance without depending on this exact initial eigenmode decomposition; it does not estimate the actual error sharply.

Exercise 5.6 — A generator's conserved mode (compute/design)

For $Q = \left(\begin{smallmatrix} -2 & 1 \\ 2 & -1 \end{smallmatrix}\right)$ in inverse seconds, find its stationary law and eigenvalues. Starting in state one, derive its law at time t. State a check that catches a transposed generator.

Worked solution

Column sums vanish and $Q\pi = 0$ gives $2\pi_1 = \pi_2$, so $\pi = (1/3, 2/3)^T$. The trace is minus three and determinant zero, yielding eigenvalues zero and minus three. Conservation makes $p_2 = 1 - p_1$, so $\dot{p}_1 = 1 - 3p_1$ and $p_1(t) = 1/3 + (2/3)e^{-3t}$, with $p_2(t) = 2/3 - (2/3)e^{-3t}$. Both are nonnegative and sum to one. Check the derivative at zero, $\dot{p}(0) = (-2, 2)^T$, against the declared outgoing rate from state one. Transposing without changing the column convention gives nonzero column sums and fails probability conservation.

5.10 Sources and attribution

- V. Balakrishnan, Stochastic processes and the master equation, lecture 6, IIT Madras, and official notes, PDF47–51. These supply the joint-law, conditioning, and rate-equation spine.
- John Sylvester, Nicolás Rivera, Luca Zanetti, and Thomas Sauerwald, Cambridge, Lecture 2: Markov Chains, Lent2019, PDF18: finite irreducible aperiodic convergence theorem. The reference uses rows; this chapter consistently transposes that convention to columns.

> The positive-kernel contraction, hidden-state example, and laboratory are developed here. Strict stationarity and homogeneity are separated more explicitly than in the source terminology. Gershgorin localization uses the union of disks and does not establish a unique stationary law.

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Generators, detailed balance, and exactly soluble jump processes

A device switches between an active and inactive state. Its residence times determine both its occupancy fraction and its noise correlation time, but these time scales are different: two exit rates of two and one per second give mean stays of half a second and one second, yet a correlation time of one third of a second. A counter driven by its switching events introduces a second issue: randomizing the number of steps changes displacement variance even when the mean number of steps is fixed. Transition rates provide one model from which these distinctions can be derived.

> Lectures 7–8 supply the sequence from generators and detailed balance through two-state noise, Poisson events, and biased lattice walks. Both caption tracks were read fully. The finite-generator positivity proof, stationary ring counterexample, explicit clock comparison, and numerical laboratory are added derivations and constructions. Sampled lecture stills provide partial notation coverage; independent audio review was not performed.

6.1 A conservative generator produces a probability semigroup L7 0:17

Use a finite set of r states and a probability column $p(t)$. A generator Q has off-diagonal rate $Q_{ij} \geq 0$ for a jump from j to i, with units inverse time, and diagonal $Q_{jj} = -\sum_{i \neq j} Q_{ij}$. Thus each column sums to zero. The master equation is $\dot{p}(t) = Qp(t)$, with specified initial probability column $p(0)$. The conservation row $\mathbf{1}^T Q = 0$ gives $d \sum_i p_i / dt = 0$.

Proposition 6.1 — Finite conservative evolution. For such a finite generator, $P(t) = e^{tQ}$ is entrywise nonnegative, each column sums to one, and $P(t+s) = P(t)P(s)$ for nonnegative times.

Derivation.

Choose a rate $\Lambda > 0$ at least as large as every exit rate $-Q_{jj}$. Then $R = I + Q/\Lambda$ is a nonnegative column-stochastic matrix. Because I commutes with R,

$$e^{tQ} = e^{-\Lambda t} \sum_{n=0}^{\infty} \frac{(\Lambda t)^n}{n!} R^n. \tag{6.1}$$

Each R^n is a probability kernel. The nonnegative coefficients are Poisson masses summing to one, so their mixture is also a probability kernel. Matrix exponentials of the same fixed matrix compose as $e^{(t+s)Q} = e^{tQ} e^{sQ}$. If $Q = 0$, the identity evolution is immediate. ∎

This construction is called uniformization. It embeds the state process in a Poisson clock of rate Λ, at each clock event applying R. Some events leave the state unchanged. Their presence is allowed because the physical generator is $\Lambda(R - I) = Q$. Increasing the auxiliary rate changes the number of artificial self-events but preserves the law of the observed state process.

A time-independent rate rule is homogeneous; starting away from an invariant law still gives a transient distribution. A finite generator with an irreducible communication graph has a unique stationary probability column and converges to it. One way to connect this to chapter 5 is to choose Λ strictly above the maximum exit rate. Then R has positive self-loops and the same irreducible graph. Its finite-chain convergence, mixed through (6.1), yields continuous-time convergence. An infinite generator needs additional existence and nonexplosion conditions; column sums alone are insufficient there.

6.2 Zero net inflow is weaker than zero pairwise current L7 16:40

A stationary probability π satisfies $Q\pi = 0$, with nonnegative entries summing to one. Define the stationary probability current from j to i as $J_{ij} = Q_{ij}\pi_j - Q_{ji}\pi_i$. It has units inverse time and satisfies $J_{ij} = -J_{ji}$. The stationary equation requires $\sum_{j \neq i} J_{ij} = 0$ at each state: total incoming and outgoing flow balance.

Detailed balance imposes the stronger pairwise condition $Q_{ij}\pi_j = Q_{ji}\pi_i$ for every distinct pair. Each current then vanishes, immediately implying stationarity. For a connected rate graph, rate ratios can determine the probability ratios along edges, but their products around closed cycles must be consistent. A path with a zero rate cannot be handled by dividing by that rate.

A three-state clockwise ring with unit rates $1 \to 2 \to 3 \to 1$ has generator $Q = \left(\begin{smallmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{smallmatrix}\right)$. Its stationary law is uniform because every state has one unit-rate input and output. Yet the current $J_{21} = 1/3$ per second is nonzero. Probability circulates while every occupancy stays constant. Thus stationarity alone does not establish equilibrium reversibility.

If a positive stationary law satisfies detailed balance, let Π be the diagonal matrix with entries π_i. The matrix $\Pi^{-1/2} Q \Pi^{1/2}$ is symmetric: its (i, j) entry is $Q_{ij}\sqrt{\pi_j/\pi_i}$, which detailed balance makes equal to its reversed entry. The generator therefore has real eigenvalues in this case. This supplies a spectral consequence of reversibility beyond solving the stationary equations. Thermal interpretations need the physical time-reversal assumptions, including how velocities or external fields transform; an arbitrary rate network need not be a thermal bath model.

6.3 Two states can be solved without diagonalization L7 25:32

Let the process take values c_1, c_2 in states one and two, with exit rates $\lambda_1 > 0$ from one to two and $\lambda_2 > 0$ from two to one. Define $s = \lambda_1 + \lambda_2$, an inverse-time quantity. Its generator and stationary law are

$$Q = \begin{pmatrix} -\lambda_1 & \lambda_2 \\ \lambda_1 & -\lambda_2 \end{pmatrix}, \qquad \pi = \frac{1}{s}\begin{pmatrix} \lambda_2 \\ \lambda_1 \end{pmatrix}. \tag{6.2}$$

There is only one edge pair, so stationarity forces detailed balance. A larger exit rate from a state decreases its stationary occupancy. In particular, the stationary probability of state one depends on the rate leaving state two in the numerator; this is consistent with the residence-time interpretation below.

Direct multiplication gives $Q^2 = -sQ$. Every higher power is therefore a scalar multiple of Q. Substitution into the exponential series gives

$$P(t) = I + \frac{1 - e^{-st}}{s} Q = \frac{1}{s} \begin{pmatrix} \lambda_2 + \lambda_1 e^{-st} & \lambda_2 (1 - e^{-st}) \\ \lambda_1 (1 - e^{-st}) & \lambda_1 + \lambda_2 e^{-st} \end{pmatrix}. \tag{6.3}$$

At zero this is the identity, its derivative is Q, and its columns sum to one. At long time both columns tend to π, so the limiting state loses its initial-state dependence. For the running rates $\lambda_1 = 2$, $\lambda_2 = 1$ per second, state one occupancy tends to $1/3$, and $P_{11}(0.7) = 0.414970952$.

Let T_j be the residence time after entering state j. The Markov property and a constant exit rate give survival $S_j(t + h) = S_j(t)[1 - \lambda_j h + o(h)]$. Thus $\dot{S}_j = -\lambda_j S_j$ with $S_j(0) = 1$, yielding $S_j(t) = e^{-\lambda_j t}$ and mean residence $\tau_j = \int_0^\infty S_j(t)\,dt = 1/\lambda_j$. Alternating visits then spend mean fractions $\tau_1/(\tau_1 + \tau_2)$ and $\tau_2/(\tau_1 + \tau_2)$ in the two states, agreeing with (6.2). Staying in a state throughout an interval differs from leaving it and returning: $S_1(t)$ is generally not $P_{11}(t)$.

6.4 The covariance loses memory at the sum of exit rates L8 2:39

Initialize the two-state process with π and define its stationary mean $\mu = c_1 \pi_1 + c_2 \pi_2$. Its variance is $V = \pi_1 \pi_2 (c_1 - c_2)^2$. Define the centered autocovariance $C(\tau) = \mathbb{E}[(X(t) - \mu)(X(t + \tau) - \mu)]$, depending only on lag τ by stationarity. For a real scalar stationary process it is even in lag.

Proposition 6.2 — Two-state covariance.

$$C(\tau) = \frac{\lambda_1 \lambda_2 (c_1 - c_2)^2}{s^2} e^{-s|\tau|}. \tag{6.4}$$

For distinct state values the integral correlation time is $\tau_{\mathrm{corr}} = \int_0^\infty C(t)/C(0)\,dt = 1/s$.

Derivation.

Equation (6.3) gives $\mathbb{E}[X(t + \tau) \mid X(t) = x] = \mu + (x - \mu) e^{-s\tau}$ for nonnegative lag and either state value x. Multiply its centered version by $X(t) - \mu$ and average. The result is $C(\tau) = V e^{-s\tau}$. Stationary scalar symmetry extends it to negative lag, and integration gives $1/s$. ∎

In residence-time variables, $\tau_{\mathrm{corr}} = \tau_1 \tau_2/(\tau_1 + \tau_2)$. This is half the usual harmonic mean $2\tau_1 \tau_2/(\tau_1 + \tau_2)$, not that harmonic mean itself. When both exit rates equal ν and the values are $\pm a$, the mean vanishes and $C(\tau) = a^2 e^{-2\nu|\tau|}$. The mean residence is $1/\nu$, but the correlation time is $1/(2\nu)$. The factor two comes from losing the original sign through parity of the number of flips.

A covariance is only one contraction of the transition kernel with the observable values. It need not determine an arbitrary Markov chain, because many kernels can give the same covariance for one observable. Here a specified two-state model makes the single exponential diagnostic useful. More complex networks can have several decay modes, including damped oscillations when detailed balance fails.

6.5 Poisson arrivals use stationary independent increments L8 9:17

Let $N(t)$ count arrivals after time zero, with $N(0) = 0$. Assume independent counts on disjoint intervals, increments whose laws depend only on interval length, and a finite rate $\lambda \geq 0$ satisfying $\Pr(\Delta N = 1) = \lambda h + o(h)$ and $\Pr(\Delta N \geq 2) = o(h)$ over a short interval of length h. Multiple events are not impossible in a finite interval; their contribution is negligible only at first order as $h \downarrow 0$.

Writing $p_n(t) = \Pr(N(t) = n)$ gives $\dot{p}_0 = -\lambda p_0$ and $\dot{p}_n = \lambda p_{n-1} - \lambda p_n$ for $n \geq 1$. The boundary equation matters because no negative count can feed zero. Define $G(z,t) = \sum_{n \geq 0} p_n(t) z^n$, convergent at least for $|z| \leq 1$. Multiply by z^n and sum, including the zero equation. The incoming index shift contributes zG, so $\partial_t G = \lambda(z-1)G$ with $G(z,0) = 1$.

Proposition 6.3 — Homogeneous Poisson counts.

$$G(z,t) = \exp[\lambda t(z-1)], \quad p_n(t) = e^{-\lambda t}\frac{(\lambda t)^n}{n!}, \quad \mathbb{E}[N(t)] = \operatorname{Var}(N(t)) = \lambda t. \tag{6.5}$$

These are count laws under the stated increment assumptions.

Derivation.

The scalar differential equation solves to the displayed generating function. Expanding its factor $e^{\lambda t z}$ gives the coefficient formula; the exponential series normalizes it. Differentiating at $z = 1$ gives mean λt and factorial second moment $(\lambda t)^2$, hence variance λt. Independence of disjoint increments belongs to the model hypotheses, rather than being deduced from the marginal mass formula alone. ∎

For $\lambda > 0$, the count process itself is not stationary: its mean grows with t. Its increments are stationary. For $0 \leq s \leq t$, write $N(t) = N(s) + [N(t) - N(s)]$ with independent terms. This gives $\operatorname{Cov}(N(s), N(t)) = \lambda s$, another direct sign that the two-time count law is not a stationary covariance function of lag alone at positive intensity. At $\lambda = 0$, the count is identically zero and is a stationary degenerate process.

The waiting time to the next event has survival $\Pr(T > t) = p_0(t) = e^{-\lambda t}$, so its density is $\lambda e^{-\lambda t}$ and mean $1/\lambda$ when $\lambda > 0$. For a symmetric two-state fluctuator the flip times form such a Poisson process of rate ν. For unequal exit rates, inter-flip waits alternate their distributions and are not iid exponential with one common rate.

A finite radioactive sample requires a separate rate distinction. If each of M surviving nuclei has individual decay constant γ, then the total instantaneous event rate is $M\gamma$, decreasing as nuclei decay. The exact count in a fixed observation interval is binomial for independent nuclei. A constant-rate Poisson model approximates a sufficiently large weakly depleted sample over a sufficiently short interval; γ itself is not the sample's count rate. The same qualification separates a mathematical bus-arrival model from an actual timetable or correlated service.

6.6 An arbitrary clock start does not split a typical gap in half L8 24:08

Consider a homogeneous Poisson point process of positive intensity $\lambda > 0$ on the entire time axis, so an arbitrary observation time is not a privileged first event. Let A be the elapsed time since the preceding event and R the remaining time to the next. For $a, r \geq 0$, the event $A > a, R > r$ means there were no arrivals in the interval of length $a + r$ around the observation time. Its probability is $e^{-\lambda(a+r)} = e^{-\lambda a} e^{-\lambda r}$. Thus A and R are independent exponentials, each with mean $1/\lambda$.

The containing gap therefore has mean $2/\lambda$, although a gap sampled by starting at an event has mean $1/\lambda$. There is no contradiction: a uniformly chosen time is more likely to fall in a long interval than a short interval. The two experiments weight gaps differently. Memorylessness makes the remaining wait at an arbitrary time equal in law to a fresh exponential wait, not half that wait. A general renewal process need not share this property.

6.7 Randomizing the jump clock changes the displacement law L8 27:48

First fix the number n of jumps on a linear lattice. Let each sign be positive with probability p and negative with probability $q = 1 - p$, independently. The site index is $J_n = 2B_n - n$ for $B_n \sim \mathbf{Binomial}(n, p)$. Hence

$$\Pr(J_n = j) = \binom{n}{(n+j)/2} p^{(n+j)/2} q^{(n-j)/2}, \quad |j| \leq n, \quad n + j \text{ even}. \tag{6.6}$$

The mass is zero otherwise. Its Laurent generating function is $(pz + qz^{-1})^n$, its mean $n(p - q)$, and its variance $4npq$. The parity restriction follows because every left step lowers the all-right displacement by two. This fixed-step law recovers the lattice comparison in chapter 3.

Now put the jumps on a Poisson clock of rate λ, independent of their signs. Averaging the fixed-step generating function over $N(t)$ gives $\mathbb{E}[(pz + qz^{-1})^{N(t)}] = \exp[\lambda t(pz + qz^{-1} - 1)]$. Set $r_+ = \lambda p$ and $r_- = \lambda q$, the rates of positive and negative jumps. The two marked event streams are independent Poisson processes. One check uses their joint generating function: $\exp[\lambda t(pu + qv - 1)] = \exp[r_+ t(u - 1)] \exp[r_- t(v - 1)]$.

With lattice spacing ℓ, the physical displacement is $X(t) = \ell[N_+(t) - N_-(t)]$. Its generator equation and generating function are

$$\dot{p}_j = r_+ p_{j-1} + r_- p_{j+1} - (r_+ + r_-) p_j, \qquad G(z, t) = \exp\{t[r_+(z - 1) + r_-(z^{-1} - 1)]\}. \tag{6.7}$$

The initial law is $p_j(0) = \mathbf{1}_{\{j=0\}}$. This is exactly the Skellam construction from chapter 1 with Poisson means $r_+ t, r_- t$. Its mass at every integer is obtained by that earlier convolution or modified-Bessel formula. A zero jump rate is handled by the one-sided Poisson law rather than dividing by it in a Bessel prefactor.

Cumulant differentiation of $\log \mathbb{E}[e^{uX(t)}] = t[r_+(e^{u\ell} - 1) + r_-(e^{-u\ell} - 1)]$ gives

$$\mathbb{E}[X(t)] = \ell(r_+ - r_-)t, \qquad \mathbf{Var}(X(t)) = \ell^2(r_+ + r_-)t. \tag{6.8}$$

The variance differs from $4npq\ell^2$ with $n = \lambda t$. Conditional variance gives the reason: $\mathbb{E}[\mathbf{Var}(J \mid N)] = 4pq\lambda t$, while $\mathbf{Var}(\mathbb{E}[J \mid N]) = (p - q)^2 \lambda t$. Their sum is $\lambda t[4pq + (p - q)^2] = \lambda t$. Randomness of the clock adds noise

to a biased walk. At $p = 1$, fixed-count displacement is deterministic, while Poisson-clock displacement still fluctuates.

For small spacing, finite continuum drift v and diffusion coefficient D require $\ell(r_+ - r_-) \to v$ and $\ell^2(r_+ + r_-)/2 \to D$. At nonzero fixed fractional bias, holding D fixed would make drift diverge as $1/\ell$. A drift-diffusion limit must scale the bias too. The state-dependent versions of this transport generator follow in chapter 7.

6.8 Laboratory: exponential evolution and event-driven sampling

Run `python code/ch06_jumps.py`. The program compares the exact two-state formula with a matrix exponential and a truncated uniformization mixture, then generates 60,000 event-driven trajectories. Each residence time is exponential with the exit rate of the present state; the clock advances directly to the next jump. This avoids a finite time-step approximation. Seed202611086 fixes the synthetic experiment. Outputs are JSON and text beside the script; no network is used.

```
Two-state P11(0.7): exact=0.414970952, sample=0.413316667
Residence times: 0.5, 1; correlation time=0.333333333
Walk mean: exact=2.000000, sample=1.998558
Walk variance: exact=2.000000, sample=1.993127
Stationary ring current: 0.333333333
```

The walk uses $r_+ = 3$, $r_- = 1$ per second, $\ell = 0.5$ meter, and $t = 2$ seconds. Exact Skellam grid sums independently check mean and variance two in their respective units. The event-driven occupancy discrepancy is compared with six binomial standard errors, while the synthetic walk mean uses its known analytical standard error. These checks diagnose implementation; no synthetic ensemble proves that a physical switch follows the assumed rates.

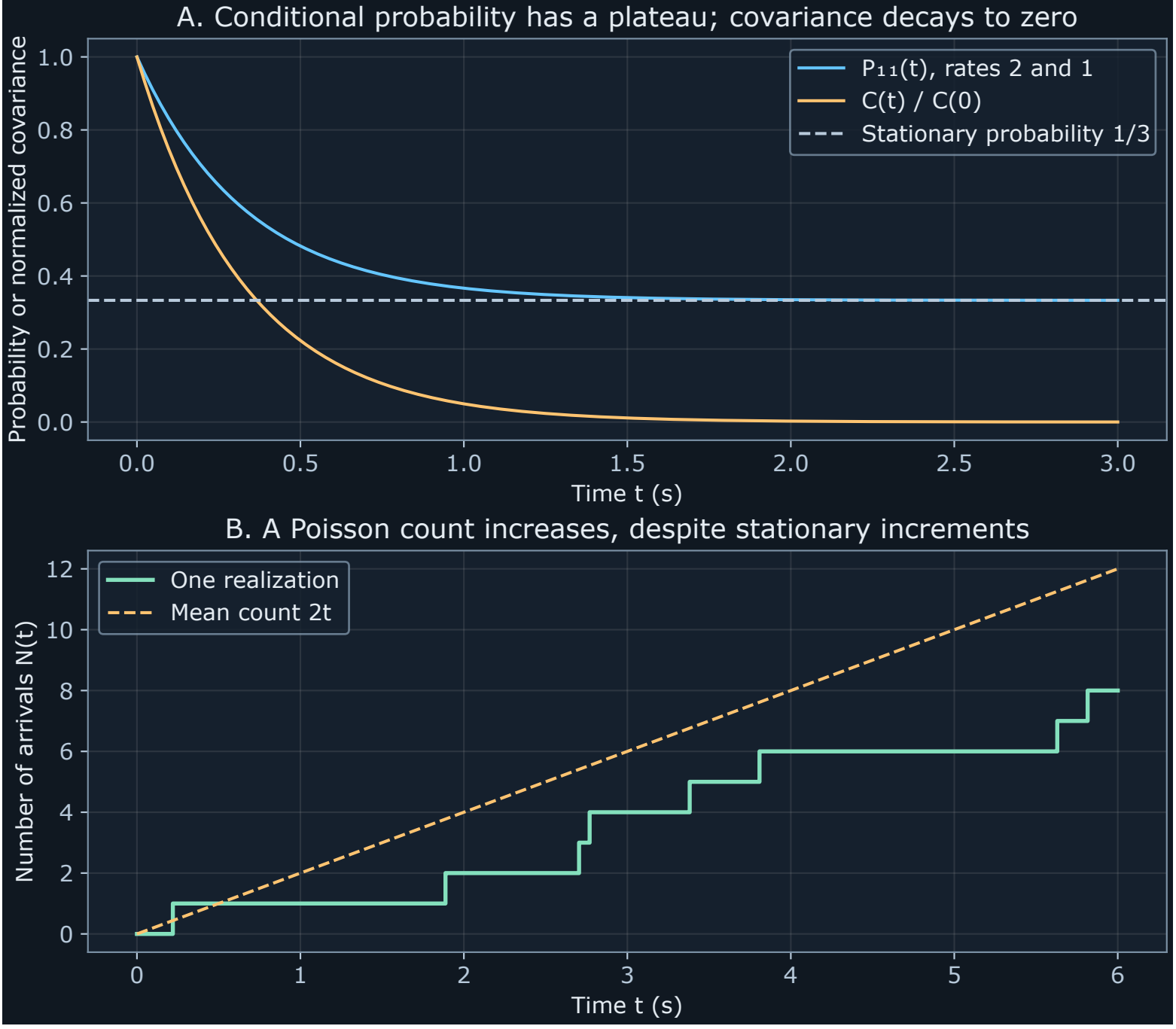


Figure 6.1. Different observables retain different long-time parts. Panel A uses exit rates two and one per second: the transition probability has stationary plateau $1/3$, while its centered covariance has none. Panel B's synthetic count has rate two per second and stationary increments, although its mean grows. Figure 6.1 is generated by the retained program; the staircase is one realization.

This direct exact-kernel implementation accepts positive exit rates and nonnegative time, returns the two-by-two transition matrix, and writes no files:

```
import numpy as np


def two_state_kernel(rate12, rate21, time):
    if min(rate12, rate21) <= 0 or time < 0:
        raise ValueError("positive rates and nonnegative time required")
    generator = np.array([[-rate12, rate21], [rate12, -rate21]])
    total = rate12+rate21
    factor = -np.expm1(-total*time)/total
    return np.eye(2)+factor*generator


print(f"P11 = {two_state_kernel(2, 1, 0.7)[0, 0]:.9f}")
```

```
P11 = 0.414970952
```

The function uses `expm1` to evaluate $e^x - 1$ accurately at a small argument. Its rate checks deliberately exclude absorbing endpoints; those are valid models but need their limits interpreted separately. The full program compares the exact two-state kernel with a matrix exponential and a truncated uniformization sum, checks the event-driven occupancy against its exact value, and verifies stationarity of the clockwise ring while reporting its nonzero current.

6.9 What the exact jump models capture and miss

A finite conservative generator supplies a probability-preserving evolution, detailed balance identifies a reversible subclass, and two-state rates give exact occupancy and covariance. A Poisson clock supplies independent stationary increments; independent marking then produces the biased lattice displacement. These are distinct model inputs, even though generating functions connect their solutions.

Neither a one-window Poisson histogram nor a single exponential fit establishes all temporal assumptions. Unequal switching rates do not produce a homogeneous Poisson flip clock. A finite radioactive population depletes, a general waiting law has an age-dependent hazard, and an infinite state-dependent generator may explode. The next chapters retain the gain-loss framework while relaxing constant event rates and testing stationary normalization.

6.10 Exercises with worked solutions

Exercise 6.1 — Stationarity with circulating flow (conceptual)

For the clockwise three-state unit-rate ring, verify uniform stationarity and compute all nonzero forward currents. Does a constant occupancy imply detailed balance?

Worked solution

Multiplying Q by $(1/3, 1/3, 1/3)^T$ gives zero because each row has one plus one and one minus one. The currents J_{21}, J_{32}, J_{13} each equal $1/3$ per second, and the reverse currents equal minus that value. At each state the net sum is zero, so the occupancy remains fixed. Every individual edge pair violates detailed balance because its reverse rate is zero. A histogram does not reveal the circulating probability current.

Exercise 6.2 — Survival is not return probability (conceptual)

For the running two-state rates, compare the probability of no exit from state one by time t with $P_{11}(t)$. Explain their long-time limits.

Worked solution

No exit has probability e^{-2t}. Equation (6.3) gives $P_{11}(t) = 1/3 + (2/3)e^{-3t}$. The latter also counts trajectories that leave state one and return one or more times. Its long-time limit is $1/3$, while uninterrupted survival tends

to zero. Both equal one at zero and have derivative minus two there, so a first-order short-time check cannot distinguish them. A finite-time or long-time check can.

Exercise 6.3 — Derive the symmetric covariance from flip parity (derive)

For a stationary sign process $X(t) = \pm a$ flipping at common rate ν, derive its covariance using the number of flips rather than the two-state kernel.

Worked solution

Over a lag $t \geq 0$, let $N(t)$ count flips. It is Poisson with mean νt. Each flip changes the sign, so $X(t) = X(0)(-1)^{N(t)}$. Thus $\mathbb{E}[X(0)X(t)] = a^2\mathbb{E}[(-1)^{N(t)}] = a^2 G(-1, t) = a^2 e^{-2\nu t}$. Equal stationary signs give mean zero, making this the covariance. Stationarity extends the expression to $|t|$. The correlation time is $1/(2\nu)$ even though each individual waiting time has mean $1/\nu$.

Exercise 6.4 — Add self-events without changing motion (derive)

At Poisson clock events of rate λ, jump right, left, or stay with probabilities p, q, r, where $p + q + r = 1$. Derive the displacement generating function and identify the observable jump rate.

Worked solution

A clock mark contributes generating factor $pz + qz^{-1} + r$. Averaging its product over the Poisson event count gives $G = \exp\{\lambda t[pz + qz^{-1} + r - 1]\} = \exp\{\lambda t[p(z-1) + q(z^{-1} - 1)]\}$. Thus observable right and left rates are $\lambda p, \lambda q$, and the total rate of actual state changes is $\lambda(p + q)$. The master equation loses $(p + q)\lambda p_j$ rather than λp_j. Artificial self-events cancel from its generator, exactly as in uniformization. If $p + q = 0$, the displacement is identically zero despite an arbitrarily busy event clock.

Exercise 6.5 — Fixed and random clocks at equal mean jump count (compute/design)

Take jump probabilities $p = 3/4, q = 1/4$, spacing $\ell = 1/2$, and compare eight fixed jumps with a Poisson number of mean eight. Compute physical mean and variance in each case. Explain a diagnostic that distinguishes the clocks.

Worked solution

The fixed-clock mean is $8(p - q)\ell = 2$ and variance $4(8)pq\ell^2 = 1.5$. The Poisson-clock mean is also two, but variance $8\ell^2 = 2$. The extra 0.5 is clock noise $(p - q)^2 \operatorname{Var}(N)\ell^2$. The lab's continuous-time walk has exactly these parameters and obtains sample variance1.993127. Count the number of actual jumps as well as displacement: a fixed count has no count variance and fixed parity, whereas a Poisson clock has count variance eight and mixes both parities. Measuring displacement mean alone cannot identify the clock.

Exercise 6.6 — Choose a finite bias in the continuum limit (compute/design)

Construct nonnegative right and left rates at spacing ℓ that give a chosen diffusion coefficient $D > 0$ and drift $v \in \mathbb{R}$ exactly. State when they remain valid and what happens as $\ell \downarrow 0$.

Worked solution

Solve $\ell(r_+ - r_-) = v$ and $\ell^2(r_+ + r_-) = 2D$. The solution is $r_\pm = D/\ell^2 \pm v/(2\ell)$. Both are nonnegative when $|v|\ell \le 2D$. At sufficiently small spacing this holds for any fixed finite v, D. The total event rate is $2D/\ell^2$, while the right-minus-left probability bias is $(r_+ - r_-)/(r_+ + r_-) = v\ell/(2D)$, tending to zero. Thus a finite drift arises from a vanishing per-jump bias combined with an increasingly rapid clock. Holding that bias fixed instead would produce divergent drift under the same diffusion scaling.

6.11 Sources and attribution

- V. Balakrishnan, lecture7 and lecture8, IIT Madras, NPTEL115106089, and official instructor notes, PDF52–63: master-equation solution, balance, two-state model, Poisson process, and lattice walks.
- NIST DLMF10.35.1 gives the modified-Bessel generating identity behind the Skellam mass formula already derived in chapter1. This chapter obtains its process meaning from independent Poisson marking.

Uniformization, explicit counterexamples, and numerical code are original exposition. Source terminology is qualified: Poisson increments are stationary while its count is not; correlation time is half the usual residence-time harmonic mean; an individual nucleus's decay constant differs from a whole sample's count rate. No raw lecture material is redistributed.

Lattice transport and state-dependent birth and death

A count accelerates as time passes. One explanation is a programmed event rate that rises at a known clock time. Another is reproduction: the more particles already present, the faster new particles arrive. These mechanisms can give similar increasing means but different fluctuations and conditional predictions. The task is to locate the dependence in the generator: on absolute time, on the present state, or on both.

> Lecture9 continues lattice cumulants and spatial spreading, then develops general one-step birth and death, deterministic time-dependent Poisson intensity, and closed low-order moments for linear rates. The full caption track was read. Its final moment identities are derived explicitly here, with boundary and existence hypotheses retained; synthetic examples test time dependence against state dependence.

7.1 Odd cumulants retain drift; even cumulants retain traffic L9 4:09

Let $N_+(t), N_-(t)$ be independent homogeneous Poisson processes with rates $r_+, r_- \geq 0$, and let $\ell > 0$ be the lattice spacing. Start at the origin and define $X(t) = \ell[N_+(t) - N_-(t)]$. Its cumulant-generating function in inverse-length variable u is $\mathcal{K}(u,t) = t[r_+(e^{u\ell} - 1) + r_-(e^{-u\ell} - 1)]$, finite for every real u.

Proposition 7.1 — Transport cumulants. For integer $k \geq 1$,

$$\kappa_k(X(t)) = \ell^k[r_+ + (-1)^k r_-]t. \tag{7.1}$$

In particular the drift velocity is $v = \ell(r_+ - r_-)$ and the variance growth rate is $2D = \ell^2(r_+ + r_-)$.

Derivation.

Differentiate the two exponential terms k times at $u = 0$. Each derivative gives ℓ on the positive stream and $-\ell$ on the negative stream. The constant terms disappear. This proves (7.1), with units length to the power k. The first two cumulants identify mean and variance as in chapter6. ∎

The mean-square displacement is $\mathbb{E}[X(t)^2] = 2Dt + v^2t^2$. Calling this entirely diffusive would conflate fluctuations about the drift with the systematic motion. Odd cumulants reverse sign on interchanging the rates, while even cumulants are unchanged. At fixed total traffic $r_+ + r_-$, bias changes the mean but not the variance for this Poisson-clock model.

For $r_+ = 3, r_- = 1$ per second, $\ell = 0.5$ meter, and $t = 2$ seconds, the first four cumulants are $2, 2, 0.5, 0.5$ in their respective powers of meters. The fourth central moment is $\kappa_4 + 3\kappa_2^2 = 12.5$ in fourth-power units, so the fourth cumulant must not be mistaken for the fourth moment. The lab independently checks these values by summing exact Skellam masses.

7.2 A vanishing fixed-site probability need not lose total mass L9 8:43

Assume both rates are positive and write $p_j(t) = \Pr(X(t) = \ell j)$. The exact mass is $e^{-(r_+ + r_-)t}(r_+/r_-)^{j/2} I_{|j|}(2t\sqrt{r_+ r_-})$. For fixed order j and positive argument $z \to \infty$, the cited asymptotic $I_j(z) \sim e^z/\sqrt{2\pi z}$ is DLMF10.40.1. Substitution yields

$$p_j(t) \sim \frac{(r_+/r_-)^{j/2} \exp[-t(\sqrt{r_+} - \sqrt{r_-})^2]}{\sqrt{4\pi t \sqrt{r_+ r_-}}}. \tag{7.2}$$

At a fixed site, nonzero bias produces exponential decay multiplied by $t^{-1/2}$; equal rates leave the $t^{-1/2}$ decay alone. This is a fixed-site long-time statement, not a uniform approximation around a mean that moves linearly with t. Interchanging these regimes would turn an asymptotic formula into an unjustified far-tail estimate.

The total probability remains one at every finite time. The mass spreads over an increasing collection of sites, so taking the limit at each site and then summing exchanges two operations that are not justified here. The limiting zero sequence is not a stationary probability. In fact no normalized stationary law exists for this homogeneous walk on the entire integer line. Suppose such a law π existed. Translation invariance would give $\pi_j = \sum_{i \in \mathbb{Z}} \pi_i p_{j-i}(t)$ for each fixed site j and every t. For each fixed i, the transition mass $p_{j-i}(t)$ tends to zero by (7.2), and it is bounded by one. Dominated convergence with the summable weights π_i therefore makes the right side tend to zero. Thus every π_j would be zero, contradicting normalization.

Unbounded state space by itself does not prohibit stationarity. State-dependent rates can confine a process even on all integers. The argument here concerns this homogeneous translating walk. A restoring-rate counterexample is developed in Exercise7.2, preventing the support geometry from being mistaken for a general nonexistence theorem.

7.3 Independent marked clocks produce a coordinate product L9 13:17

On a d-dimensional rectangular lattice, assign independent positive and negative Poisson streams of rates $r_{a,+}, r_{a,-}$ to coordinate a, for $a = 1, \dots, d$. Let spacing in that coordinate be ℓ_a. Each coordinate displacement is $X_a(t) = \ell_a[N_{a,+}(t) - N_{a,-}(t)]$. Independence gives the joint characteristic function

$$\phi(\mathbf{k}, t) = \prod_{a=1}^{d} \exp\{t[r_{a,+}(e^{ik_a\ell_a} - 1) + r_{a,-}(e^{-ik_a\ell_a} - 1)]\}. \tag{7.3}$$

The coordinate laws therefore factorize and their cross-covariances vanish. This is justified by independent event marking, not by the existence of Cartesian axes. At a fixed total number of jumps, counts in different coordinate directions are multinomial and constrained by that total; conditioning destroys this simple independent-coordinate construction.

For equal rates $\lambda/(2d)$ in each of $2d$ directions, (7.2) gives fixed-site decay proportional to $t^{-d/2}$, since there are d independent inverse-square-root factors. This anticipates the role of dimension in recurrence. It does not yet determine first-return probabilities, because occupancy at a time and first return at that time are distinct events.

7.4 One-step population equations require the correct boundary L9 22:10

Let $N(t)$ now denote a nonnegative integer population, unrelated to a homogeneous Poisson count unless its rates specify that case. From state n, a birth increases the population to $n+1$ at rate $\lambda_n \geq 0$, and a death decreases it to $n-1$ at rate $\delta_n \geq 0$. Set $\delta_0 = 0$ to prevent negative populations. Rates are functions of population, with inverse-time units; they need not sum to one.

Proposition 7.2 — One-step master equation. For a nonexplosive process with these rates and initial law $p_n(0)$,

$$\dot{p}_n = \lambda_{n-1}p_{n-1} + \delta_{n+1}p_{n+1} - (\lambda_n + \delta_n)p_n, \quad n \geq 1, \qquad \dot{p}_0 = \delta_1 p_1 - \lambda_0 p_0. \tag{7.4}$$

Derivation.

Over a short interval h, probability enters state n from $n-1$ with first-order coefficient $\lambda_{n-1}p_{n-1}$ and from $n+1$ with coefficient $\delta_{n+1}p_{n+1}$. It leaves through both exits with coefficient $(\lambda_n + \delta_n)p_n$. Subtract $p_n(t)$ and divide by h. At zero only the first surviving incoming neighbor and the birth exit remain, giving the separate boundary equation. Summing through a finite cutoff M cancels every interior flux and gives $d\sum_{n=0}^{M} p_n/dt = \delta_{M+1}p_{M+1} - \lambda_M p_M$. When the boundary flux at infinity vanishes for the nonexplosive law, total mass is conserved. ∎

The nonexplosion condition means finitely many jumps occur in each finite time interval almost surely. It cannot be replaced by formal cancellation of infinitely many terms. A pure birth model whose rates grow too quickly may make infinitely many jumps in a finite random time; its minimal finite-state probability can then lose mass to infinity. A finite numerical truncation that reflects at its top boundary silently defines another process.

Constant births with zero deaths recover a Poisson count. Linear births $\lambda_n = bn$ with $b > 0$, zero deaths, and initial population one model independent reproduction. Its mean grows exponentially rather than linearly; the rate increases because the population changed, without any explicit clock dependence. This distinction is central to the running inference problem.

7.5 A prescribed time-dependent intensity still gives Poisson increments L9 35:07

Let $\lambda(t) \geq 0$ be a deterministic locally integrable rate, and define cumulative intensity $\Lambda(t) = \int_0^t \lambda(s)ds$. A nonhomogeneous Poisson process has independent increments with count over $[s,t]$ Poisson of mean $\Lambda(t) - \Lambda(s)$. Unlike a population-dependent birth process, its conditional future rate is specified by the clock, independent of how many events have already occurred.

Proposition 7.3 — Deterministic time change. Starting with zero count, the generating function is $G(z,t) = \exp[(z-1)\Lambda(t)]$, giving count mean and variance both $\Lambda(t)$.

Derivation.

The pure-birth equations use the same rate $\lambda(t)$ in every state. Their generating-function equation is $\partial_t G = \lambda(t)(z-1)G$ with $G(z,0)=1$. Integrating its scalar coefficient gives the stated expression, and expansion gives Poisson masses with mean $\Lambda(t)$. A constructive version sets $N(t) = M(\Lambda(t))$, where M is a unit-rate homogeneous Poisson process. Deterministic disjoint clock intervals map to disjoint cumulative-intensity intervals, so increment independence is preserved. ∎

For rate two per second before one second and six per second afterward, $\Lambda(1)=2$ and $\Lambda(2)=8$. The slope changes, while count variance continues to equal the mean. Replacing a random intensity by its expectation is not the same operation. If an integrated intensity $A \geq 0$ has finite first and second moments and the count conditional on it is Poisson, then $\mathbb{E}[N \mid A] = \mathbf{Var}(N \mid A) = A$. Total variance therefore gives $\mathbf{Var}(N) = \mathbb{E}[A] + \mathbf{Var}(A)$. Fluctuating environments introduce excess variance and can correlate increments.

7.6 A generator can close moments before the full law is solved L9 43:06

For a function f of population, the backward generator acts as $(Lf)(n) = \lambda_n[f(n+1)-f(n)] + \delta_n[f(n-1)-f(n)]$, with the death term omitted at zero. Multiplying (7.4) by $f(n)$, summing, and shifting the incoming indices gives $d\mathbb{E}[f(N)]/dt = \mathbb{E}[(Lf)(N)]$ whenever moments, boundary terms, and the interchange are justified.

For $f(n)=n$ and $f(n)=n^2$, the increments are respectively ± 1 and $2n+1, -2n+1$. Hence

$$\frac{d\mathbb{E}[N]}{dt} = \mathbb{E}[\lambda_N - \delta_N], \qquad \frac{d\mathbb{E}[N^2]}{dt} = \mathbb{E}[(2N+1)\lambda_N + (-2N+1)\delta_N]. \tag{7.5}$$

These identities do not assume linear rates. Affine rates make the first equation depend only on the mean and the second on at most the first two moments. A quadratic rate generally introduces the next moment and breaks that closure. Linearity here refers to dependence on population, since the probability master equation is already linear in the probabilities.

For the pure reproduction model $\lambda_n = bn$, initial $N(0)=1$, equation (7.5) gives $m' = bm$ and $s_2' = 2bs_2 + bm$, where $m = \mathbb{E}[N]$ and $s_2 = \mathbb{E}[N^2]$. Solving yields $m = e^{bt}$ and $\mathbf{Var}(N) = e^{bt}(e^{bt}-1)$. The variance grows faster than the mean; a programmed deterministic Poisson rate chosen to match $m-1$ would not reproduce those fluctuations. The full reproduction law is derived in Exercise7.3.

Nonexplosion in this linear-birth example can be checked before using infinite-state moments. Stop the population when it first reaches a finite level M. Its finite-state mean satisfies $m_M' \leq bm_M$, so $m_M(t) \leq e^{bt}$. The probability of having reached M by time t is at most $m_M(t)/M \leq e^{bt}/M$, because the stopped population then remains at M. Letting $M \to \infty$ makes finite-time explosion probability zero. The normalized exact law in Exercise7.3 then verifies the finite moments used above.

7.7 Laboratory: distinguish clock acceleration from population feedback

Run `python code/ch07_populations.py`. The program generates 40,000 event-driven pure-birth populations with $b = 0.8$ per second at $t = 1.5$ seconds, starting at one. It samples each waiting time with the

current rate bn, changes the population, and repeats until the observation time. A separate unit-exponential clock simulates a nonhomogeneous Poisson process up to cumulative intensity eight. Outputs are JSON and text beside the program, with seed202611087 and no network access.

```
Yule mean: exact=3.320116923, sample=3.320250000
Yule variance: exact=7.703059458, sample=7.713489938
NHPP at t=2: exact mean=variance=8, sample mean=7.976950, variance=7.914869
Walk cumulants: 2.000000, 2.000000, 0.500000, 0.500000
```

The reproduction process is often called the Yule pure-birth process. Its exact geometric masses independently check the event-driven implementation and moments. The nonhomogeneous Poisson check operates in cumulative-intensity time, where independent unit exponential waits become valid; transforming those times back supplies physical event times. The finite simulation discrepancies are sampling error, while the exact moment identities are model predictions. Assertions compare sample means with six analytical standard errors and check exact mass moments separately.

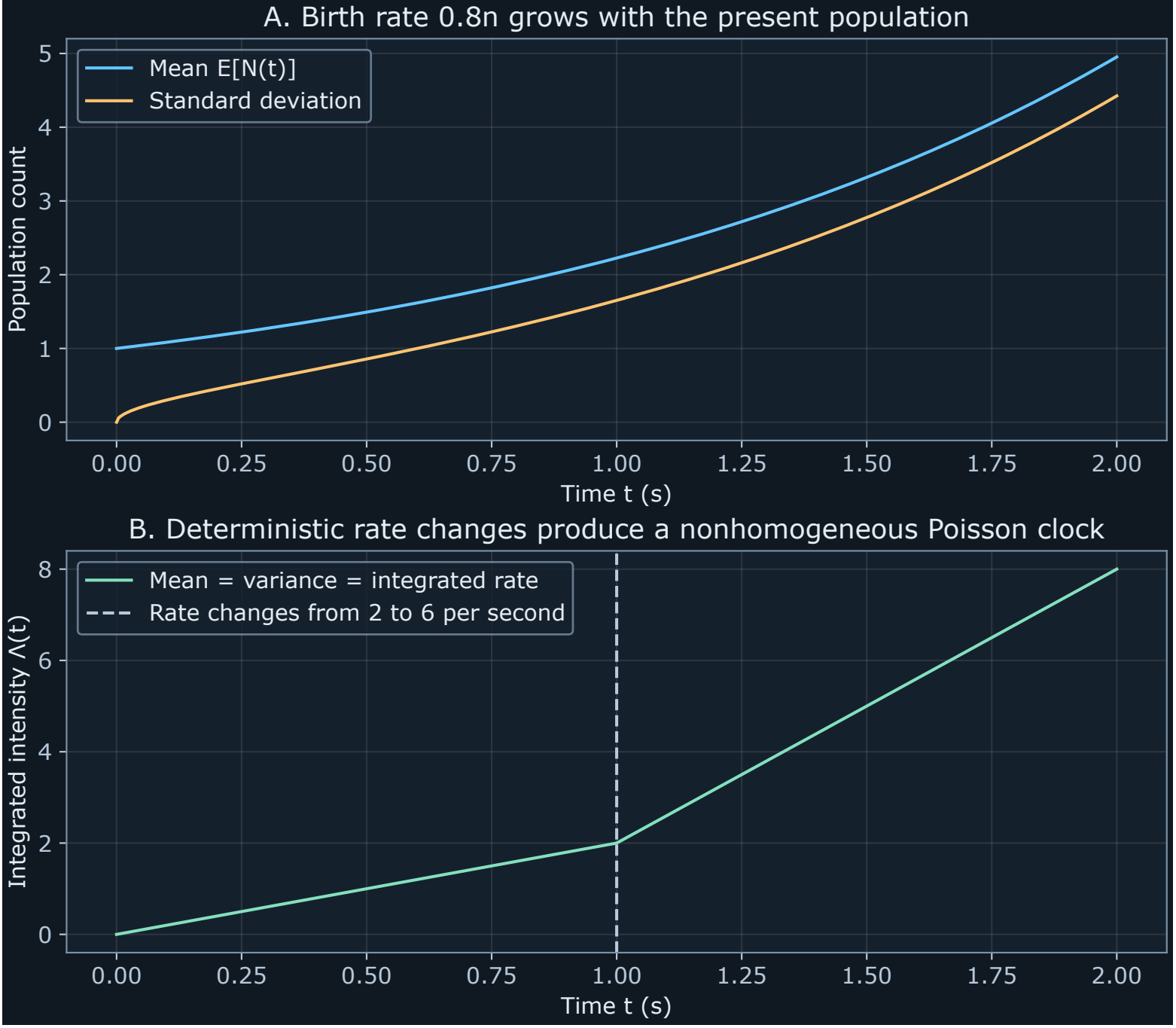


Figure 7.1. State dependence and explicit time dependence change different parts of the generator. Panel A uses pure-birth rate $0.8n$ per second, initial population one; its standard deviation approaches its mean from below as the population grows. Panel B uses a prescribed rate change with Poisson variance equal to its cumulative intensity. These are theoretical curves, reproduced by the generator. Figure7.1 contrasts mechanisms rather than fitting a common data set.

This exact moment calculation accepts a nonnegative time and positive birth-rate coefficient and writes no files:

```
import numpy as np


def reproduction_moments(birth_rate, time):
    if birth_rate <= 0 or time < 0:
        raise ValueError("positive rate and nonnegative time required")
    mean = np.exp(birth_rate*time)
    return mean, mean*np.expm1(birth_rate*time)


mean, variance = reproduction_moments(0.8, 1.5)
print(f"mean={mean:.9f}, variance={variance:.9f}")
```

```
mean=3.320116923, variance=7.703059458
```

The formula presumes one initial ancestor and independent per-capita reproduction. It does not model a carrying capacity or resource depletion. Large bt can overflow finite floating-point arithmetic even though the finite-time mathematical moments exist. An implementation intended for that regime should retain logarithmic values or use arbitrary precision.

7.8 The boundary and clock are part of the model

The transport cumulants distinguish drift from traffic, while the population generator identifies which present-state transitions are possible. A nonhomogeneous Poisson process changes a deterministic clock without changing its conditional count mechanism. Linear population feedback instead couples the event rate to the realized state and can produce overdispersion.

Nearest-neighbor structure alone does not imply independence of successive increments. A state-dependent process may be Markov while its increments are dependent. A two-sided unbounded lattice can support a stationary law with restoring rates, and a half-line can fail to support one if outward births dominate. Normalization, boundary flux, and nonexplosion must be checked for the particular rates. Chapter8 performs those checks and solves equilibrium population examples.

7.9 Exercises with worked solutions

Exercise7.1 — Equal means do not identify the clock (conceptual)

A pure reproduction population starts at one. A second model is one plus a nonhomogeneous Poisson count with cumulative intensity $e^{bt} - 1$. Compare their means, variances, and conditional future rates.

Worked solution

Both means are e^{bt}. Reproduction variance is $e^{bt}(e^{bt} - 1)$, while shifted Poisson variance is $e^{bt} - 1$. In the first model the next-event rate is $bN(t)$, depending on the realized population. In the second it is the deterministic derivative be^{bt}, independent of the realized count. Their ensemble-average rates agree, but their conditional predictions and fluctuations differ. A mean trajectory alone discards the information needed to distinguish feedback from clock programming.

Exercise7.2 — Stationarity on the full integer line (conceptual)

On all integers j, take upward rate $\lambda_j = e^{-j-1/2}$ and downward rate $\delta_j = e^{j-1/2}$ in common inverse-time units. Show that weights $\pi_j \propto e^{-j^2}$ satisfy detailed balance and normalize. What additional process issue must still be checked?

Worked solution

The neighbor ratio is $\lambda_j/\delta_{j+1} = e^{-2j-1} = e^{-[(j+1)^2-j^2]}$, exactly π_{j+1}/π_j. Thus every pairwise current vanishes. The sum of e^{-j^2} over all integers is finite, by comparison with a decaying geometric tail outside a finite interval, so the weights normalize. This disproves any blanket claim that an unbounded two-sided state space forbids stationary weights. One must additionally establish a nonexplosive process for which these balance equations define an invariant probability; the formal weight calculation alone is not that existence proof. Rates point strongly toward the origin, but that observation must be made into an appropriate existence argument before claiming a full stationary process.

Exercise7.3 — Derive the reproduction law (derive)

Verify that $p_n(t) = a(1-a)^{n-1}$ for $n \geq 1$, with $a = e^{-bt}$, solves the pure-birth equations $\lambda_n = bn$, starting at one. Recover its first two moments.

Worked solution

At $n = 1$, $\dot{p}_1 = -ba = -bp_1$. For $n \geq 2$, differentiation with $\dot{a} = -ba$ gives $\dot{p}_n = -ba(1-a)^{n-1} + ba^2(n-1)(1-a)^{n-2}$. Factoring $ba(1-a)^{n-2}$ leaves $(n-1)a - (1-a) = na - 1$. The gain-loss expression $b(n-1)p_{n-1} - bnp_n$ has the same factor and bracket. At zero time all mass is at one; the geometric series normalizes for positive time, with its endpoint interpreted by continuity.

This is the geometric law on positive integers with success parameter a. Its mean is $1/a = e^{bt}$ and variance $(1-a)/a^2 = e^{bt}(e^{bt}-1)$. These moments also solve (7.5), providing an independent check. The harmonic waiting-time sum for rates bn does not accumulate infinitely many jumps at finite time; in contrast, faster-growing birth rates can require a separate explosion analysis.

Exercise7.4 — Mixed intensities add noise (derive)

Choose a random rate R once per experiment, equal to one or three per second with equal probability. Given R, arrivals form a homogeneous Poisson process. Find the unconditional mean and variance of the count over two seconds, and covariance of counts in two disjoint one-second intervals.

Worked solution

The conditional two-second mean and variance are both $2R$. Since $\mathbb{E}[R] = 2$ and $\mathbf{Var}(R) = 1$, total expectation gives mean four, while total variance gives $\mathbb{E}[2R] + \mathbf{Var}(2R) = 4 + 4 = 8$. For two disjoint one-second intervals, conditional independence makes conditional covariance zero. Their conditional means are both R, so unconditional covariance is $\mathbf{Var}(R) = 1$. Replacing the random rate by its mean would falsely predict count variance four and zero cross-interval covariance.

Exercise7.5 — Invert the programmed clock (compute/design)

For rate two before time one and six afterward, give the inverse cumulative-intensity map for cumulative time $u \in [0, 8]$. Explain how to generate physical arrival times without using a fixed time grid.

Worked solution

For $0 \le u \le 2$, physical time is $t = u/2$. For $2 < u \le 8$, it is $t = 1 + (u-2)/6$. Generate successive independent unit exponential waits in u-time, accumulate them, retain sums no greater than eight, and apply this piecewise inverse to each sum. The resulting physical times have the intended nonhomogeneous Poisson law. Counts in the first and second one-second intervals have independent means two and six. A simulation using a single exponential rate of four per second would match the total mean eight but distribute events incorrectly between intervals.

Exercise7.6 — Close a moment equation and state its limit (compute/design)

For immigration-death rates $\lambda_n = a$ and $\delta_n = bn$, with $a, b > 0$, derive the mean equation. If the birth rate instead becomes $a + cn^2$, identify the new quantity needed and explain why replacing it by the squared mean is an approximation.

Worked solution

Equation(7.5) gives $m' = a - bm$ in the affine model. With mean initial value m_0, the solution is $m(t) = a/b + (m_0 - a/b)e^{-bt}$, tending to a/b. For quadratic birth it becomes $m' = a + c\mathbb{E}[N^2] - bm$. Since $\mathbb{E}[N^2] = m^2 + \mathrm{Var}(N)$, replacing it by m^2 discards $c\,\mathrm{Var}(N)$. No identity makes that variance vanish in a fluctuating population. One needs either another moment equation, a justified closure regime, or the full distribution. Rapidly growing birth rates can additionally raise explosion questions, so formal moment closure does not guarantee process existence.

7.10 Sources and attribution

- V. Balakrishnan, lecture9, IIT Madras, NPTEL115106089, and official instructor notes, PDF64–67: lattice spreading and one-step population framework. Nonhomogeneous intensity and moment equations follow the recording's discussion and are derived here.
- NIST DLMF10.40.1: modified-Bessel asymptotic at fixed order and large positive argument, used with its stated regime in (7.2).

> The Yule and programmed-clock comparison, explicit moment derivations, code, and exercises are teaching additions. No stationary-law nonexistence is inferred from unbounded support alone. A Markov state dependence does not imply independent increments. Raw lecture sources remain private.

Stationary populations, fluctuations, and physical balance

A population model produces positive stationary weights, but their sum diverges. Dividing by a finite numerical cutoff makes a plausible histogram whose mean changes whenever the cutoff changes. The task is to decide whether a rate model has an actual equilibrium probability, and then distinguish that equilibrium from a transient that merely has the same mean. Sedimentation, reservoir chemistry, and a thermal oscillator make the normalization test physically concrete.

> Lecture10 develops reflecting boundaries, stationary products, sedimentation, chemical and radiation examples, affine-rate moments, and an introduction to continuous-state master equations. The full caption track was read. The explicit immigration-death transient law, normalization diagnostics, and branching qualifications below extend the recorded calculations.

8.1 A stationary product is a probability only after normalization L10 9:04

Consider a birth-death process on the nonnegative integers with rates λ_n from n to $n+1$ and δ_n from n to $n-1$, with $\delta_0 = 0$. Assume all neighbor rates are positive where the corresponding edge exists, and assume a nonexplosive process. For a stationary candidate π_n, define the upward probability current $J_n = \lambda_n \pi_n - \delta_{n+1}\pi_{n+1}$. At an interior state, the stationary equation says $J_{n-1} - J_n = 0$.

The reflecting lower boundary gives $0 = \delta_1 \pi_1 - \lambda_0 \pi_0 = -J_0$. Therefore every J_n is zero. Unlike a ring, a reflecting line has no route for a stationary circulation; stationarity here forces neighbor detailed balance. Successive probability ratios are fixed by $\pi_{n+1}/\pi_n = \lambda_n/\delta_{n+1}$.

Proposition8.1 — Stationary product formula. Put $w_0 = 1$ and $w_n = \prod_{j=0}^{n-1} \lambda_j/\delta_{j+1}$ for $n \geq 1$. If $Z = \sum_{n=0}^{\infty} w_n < \infty$, the zero-current probability is

$$\pi_n = \frac{w_n}{Z}. \tag{8.1}$$

For a nonexplosive birth-death process with the stated boundaries this gives its invariant probability. If $Z = \infty$, no normalized zero-current law of this form exists.

Derivation.

Zero current fixes π_1 from π_0, then π_2 from π_1, and inductively $\pi_n = w_n \pi_0$. Positivity follows from the rate ratios. The sum can be one exactly when finite Z permits $\pi_0 = 1/Z$. Conversely these values satisfy every neighbor detailed-balance identity and the lower-boundary equation.

To establish invariance, consider a trajectory with finitely many jumps during a fixed time interval, weighted by the initial probability of its starting state. Its path density is that initial weight times the product of transition

rates along its jumps and the exponential survival factors during every holding interval, including the final interval. Reverse the order of its states, jumps, and holding intervals. Neighbor detailed balance makes the initial-weight-times-rate-product identical in the two directions: each adjacent probability ratio cancels the next one. The holding factors are unchanged because the same durations are spent in the same states. Integration over jump times and summation over finite paths therefore gives the transition-kernel detailed-balance identity.

Nonexplosion makes these finite-jump paths exhaust the transition probability. For a fixed terminal state, sum the kernel identity over all starting states. The reversed transition probabilities sum to one by conservativeness, leaving exactly the stationary weight of that terminal state. Thus the normalized weights are invariant. All path contributions are nonnegative, so these sums require no unjustified cancellation of infinite signed series. Both normalization and the process hypothesis are essential. ∎

On a finite range $0, \dots, M$, reflecting at the upper endpoint requires $\lambda_M = 0$, and the normalizer ends at M. This finite model always has a finite sum of positive weights. It is not evidence that the corresponding infinite model normalizes. Figure8.1 displays the difference between convergence of partial sums and an arbitrary truncation.

8.2 A reflecting floor turns downward bias into an exponential profile L10 15:09

Let height be $z_n = n\ell$ with spacing $\ell > 0$, and take constant upward rate $u > 0$ and downward rate $d > u$ away from a reflecting floor. The ratio is $q = u/d \in (0, 1)$, so (8.1) becomes $\pi_n = (1 - q)q^n$. Its mean height is $\ell q/(1 - q)$ and its most probable height is zero. A mode at the floor does not put all probability there; the full distribution balances random kicks with downward bias.

At equal rates $q = 1$, every weight is one and the infinite normalizer diverges. For $q > 1$ it diverges faster. In a finite reflecting column either case has a stationary law, but that law depends on the upper boundary. Removing the ceiling changes the normalization problem.

For thermal equilibrium under a constant downward force of magnitude $F > 0$, a sufficient neighbor balance condition is $u/d = \exp[-F\ell/(k_B T)]$. Here T is temperature and k_B Boltzmann's constant, so $F\ell$ and $k_B T$ are energies. Let $H = k_B T/F$, a length. Taking $\ell \downarrow 0$ gives

$$\frac{\pi_n}{\ell} \longrightarrow \rho(z) = \frac{1}{H} e^{-z/H}, \qquad z \geq 0. \tag{8.2}$$

The limit follows from $q = e^{-\ell/H}$ and $(1 - q)/\ell \to 1/H$. The limiting density has mean H and mode zero. This is a scale height, not a finite outer edge of an atmosphere. Applying it to a gas assumes isothermal equilibrium, approximately constant gravity, and a suitable independent-particle description over the considered heights.

The same profile arises from constant mobility M and diffusion coefficient D if the downward drift speed is MF and the fluctuation-dissipation relation is $D = Mk_B T$. Their ratio gives $D/(MF) = H$; viscous mobility cancels from equilibrium while governing relaxation speed. For a slowly sedimenting isolated sphere in a viscous fluid, Stokes drag is one possible mobility model, but its force must include buoyancy and its low-

Reynolds-number assumptions. The equilibrium ratio itself does not require importing that hydrodynamic approximation.

8.3 A maintained reservoir produces a Poisson population L10 29:03

Consider conversion $A \rightleftarrows B$ while a large maintained reservoir holds the available number of A approximately fixed. Let N count product molecules B. Constant input gives $\lambda_n = a > 0$, and independent first-order removal of each B gives $\delta_n = bn$ with $b > 0$. The parameter a has count-per-time units in the physical description and a numerical inverse-time rate on count states; b is the per-molecule inverse-time rate.

The stationary weights are $w_n = (a/b)^n/n!$. Summing the exponential series gives $Z = e^{a/b}$, so $\pi_n = e^{-a/b}(a/b)^n/n!$. The mean and variance both equal $\mu = a/b$. Reservoir maintenance is the physical idealization that makes creation independent of the present product count.

A closed system with a fixed total of M molecules is different. Forward creation then scales with the remaining A count: $\lambda_n = k(M - n)$, while reverse loss is $\delta_n = k'n$. Substitution into (8.1) gives binomial weights and a finite binomial equilibrium with success parameter $k/(k + k')$. A Poisson reservoir limit requires appropriate large-reservoir scaling; it is not the exact equilibrium of every reversible chemical conversion.

8.4 Stimulated transitions give geometric oscillator occupation L10 32:51

For a quantum oscillator with equally spaced excitation energies $E_n = n\varepsilon$, $\varepsilon > 0$, use upward rates $\lambda_n = A(n + 1)$ and downward rates $\delta_n = Bn$, with $A \geq 0$ and $B > A$. These are an assumed effective weak-coupling occupation model; deriving their quantum transition matrix elements is outside the present scalar jump calculation. The factor $n + 1$ permits excitation out of the ground state, while the death rate vanishes at zero.

The ratio $\lambda_n/\delta_{n+1} = A/B = q$ cancels the neighboring occupation factor. Thus $\pi_n = (1 - q)q^n$, with mean $q/(1 - q) = A/(B - A)$. For thermal equilibrium, neighbor detailed balance requires $q = \exp[-\varepsilon/(k_B T)]$, giving mean $1/[\exp(\varepsilon/(k_B T)) - 1]$ and variance $\mu(1 + \mu)$. This reproduces the single-mode equilibrium law from chapter1 through rates instead of direct canonical weighting.

The exponential must have the negative energy sign: a positive sign would make $q > 1$ and destroy normalization at positive temperature. Nor does a geometric stationary distribution alone prove that a bath is thermal; a nonthermal effective rate model with the same ratio would give the same occupation histogram. Observables beyond one-time occupation may be needed to distinguish mechanisms.

8.5 Immigration-death transients separate surviving ancestors from new arrivals L10 5:03

Return to $\lambda_n = a$, $\delta_n = bn$, starting from a deterministic population n_0. Put $q_t = e^{-bt}$ and $\mu = a/b$. Each initial molecule independently survives to time t with probability q_t, so surviving ancestors have a binomial law with parameters n_0, q_t. An immigrant arriving at time s survives with probability $e^{-b(t-s)}$. Independent

thinning of the Poisson immigration stream leaves a Poisson number of surviving immigrants with mean $a\int_0^t e^{-b(t-s)}ds = \mu(1-q_t)$.

The two contributions are independent because their underlying initial lifetimes and immigration events are independent. Consequently the exact population is a binomial-plus-Poisson sum. Its probability-generating function is

$$G(z,t) = [1 - q_t + q_t z]^{n_0} \exp[\mu(1-q_t)(z-1)]. \tag{8.3}$$

This constructs the full transient law and checks normalization at $z = 1$. It also establishes nonexplosion: up to any finite time there are finitely many Poisson immigrants, and deaths cannot exceed initial population plus total arrivals.

Proposition8.2 — Immigration-death moments. With deterministic initial population n_0,

$$m(t) = n_0 q_t + \mu(1-q_t), \qquad V(t) = n_0 q_t(1-q_t) + \mu(1-q_t). \tag{8.4}$$

Both approach μ at long times.

Derivation.

Add the independent binomial and Poisson means and variances from (8.3). Equivalently, the moment identities of chapter7 give $m' = a - bm$ and $V' = a + bm - 2bV$, with $m(0) = n_0, V(0) = 0$. Substitution of (8.4) verifies both equations and initial conditions. ∎

Even if $n_0 = \mu$ is an integer, the deterministic initial population is not the stationary Poisson ensemble. The mean then stays constant, but $V(t) = \mu(1 - q_t^2)$ rises from zero. This is a direct example where a constant measured mean hides a nonstationary fluctuation law. At $n_0 = 12, a = 3, b = 1, t = 2$, exact mean and variance are 4.218017549 and 3.998229882.

8.6 Linear births and deaths expose separate growth and noise rates L10 36:52

A branching population instead has $\lambda_n = an$ and $\delta_n = bn$, with $a, b \geq 0$. Zero is absorbing: without immigration an extinct population cannot restart. For a deterministic positive initial population n_0, define growth rate $r = a - b$. The first two moment equations give $m' = rm$ and $V' = 2rV + (a+b)m$. Solving gives

$$m(t) = n_0 e^{rt}, \qquad V(t) = \begin{cases} n_0 \dfrac{a+b}{r} e^{rt}(e^{rt} - 1), & r \neq 0, \\ 2a n_0 t, & a = b. \end{cases} \tag{8.5}$$

For $r \neq 0$, multiply the variance equation by e^{-2rt} and integrate its forcing $(a+b)n_0 e^{-rt}$; this gives (8.5). The critical result is its continuous limit as $r \to 0$, with $a + b \to 2a$. The mean measures the difference of birth and death rates, while noise growth measures their sum. At $a = 0.7, b = 0.4, n_0 = 2, t = 2$, mean and variance are 3.644237601 and 10.985319564.

Exponential growth of the mean is not finite-time explosion, and it does not preclude extinction. Linear rates are nonexplosive by domination of births by a pure-birth process with per-capita rate a and by the finite number of possible deaths before any fixed birth count. At $a = b > 0$, the mean remains n_0 while variance grows linearly. Extinction and survival-conditioned statistics explain why an ensemble mean can remain finite even as many individual trajectories vanish. The absorbing point mass at zero is always a stationary law; the irreducible positive-rate assumptions of Proposition8.1 do not apply across that absorbing boundary.

8.7 A continuous state can still make discrete jumps L10 45:22

A state coordinate can vary continuously even when a path changes by jumps. Let $\rho(x, t)$ be a probability density and $w(y \mid x)dy$ the transition rate from present x into a neighborhood of y. Its units are inverse time per output-coordinate unit. With finite total exit rates and the required integrability, summing gain and loss over all source states gives

$$\partial_t \rho(x,t) = \int [w(x \mid y)\rho(y,t) - w(y \mid x)\rho(x,t)]dy. \tag{8.6}$$

The double integral cancels under an interchange justified by integrability, preserving total probability. The conditional initial state x_0 is a delta measure, not an ordinary initial density. A continuous state space therefore does not automatically imply continuous sample paths or a second-order diffusion equation. Chapter9 derives the local diffusion approximation and its boundary conditions from this gain-loss structure.

8.8 Laboratory: test normalization before simulating equilibrium

The program computes stationary weights by neighbor ratios, compares them with a Poisson mass function, and evaluates the exact transient convolution (8.3). It separately generates 40,000 event-driven immigration-death trajectories with seed202611088. At each event, the total rate is $a + bn$; an independent mark chooses birth with probability $a/(a + bn)$, otherwise death. The zero-state death probability is zero automatically.

```
Immigration-death mean: exact=4.218017549, sample=4.219325000
Immigration-death variance: exact=3.998229882, sample=3.997671544
Stationary mean=variance=3.000000
Linear branching mean=3.644237601, variance=10.985319564
```

Run `python code/ch08_equilibrium.py` with the shared dependencies. It writes JSON and text beside the script and makes no network calls. Exact-grid normalization and moments test the convolution; the sample-mean check uses six analytical standard errors. The branching values are exact-formula evaluations, rather than outcomes of the immigration-death simulation.

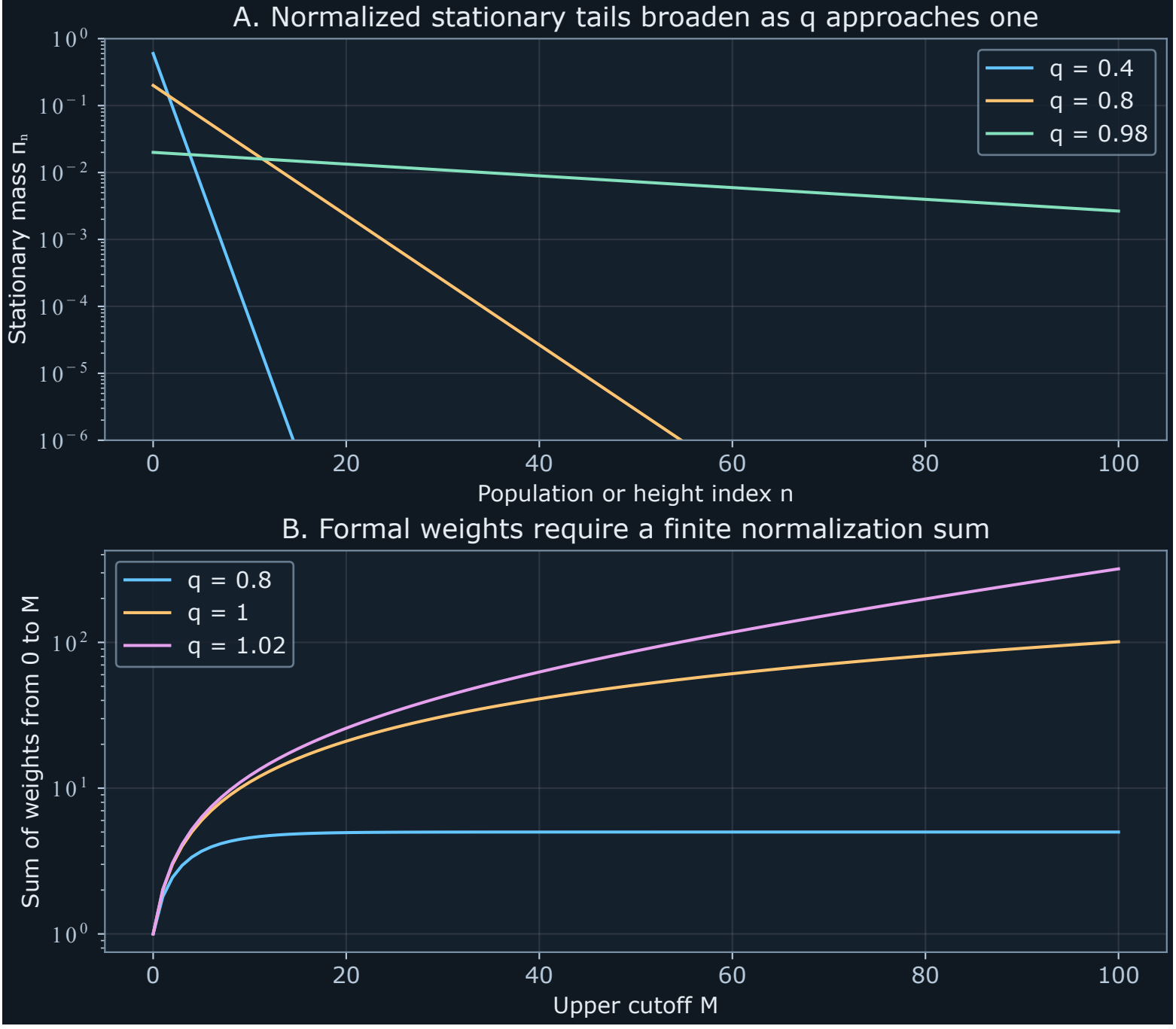


Figure 8.1. Formal stationary weights need a finite normalizer. Panel A displays actual normalized laws with ratios below one, without renormalizing the shown range. Panel B shows partial weight sums: convergence at ratio0.8 contrasts with divergence at1 and1.02. Figure8.1 is reproduced by the retained generator.

This short check implements the stationary ratio test in a parameter range where the geometric series is known exactly. It accepts a ratio, returns a mean when normalized, and refuses the divergent cases rather than silently truncating them:

```
def stationary_geometric_mean(ratio):
    if not 0 <= ratio < 1:
        raise ValueError("infinite geometric weights do not normalize")
    return ratio/(1-ratio)


for ratio in (0.4, 0.8, 0.98):
    print(f"q={ratio}: mean={stationary_geometric_mean(ratio):.6f}")
```

```
q=0.4: mean=0.666667
q=0.8: mean=4.000000
```

```
q=0.98: mean=49.000000
```

A positive finite sum at a numerical cutoff proves nothing about the infinite normalizer without a tail argument. For general rate products, ratios or logarithmic weights can guide a convergence test, but a borderline limit of one needs more information. Simulating a finite reflecting truncation also cannot by itself establish equilibrium of the infinite chain.

8.9 Equilibrium existence, relaxation, and extinction are separate questions

The product formula gives a stationary probability under explicit normalizability and process assumptions. Thermal rate ratios explain Boltzmann weights; reservoir assumptions explain Poisson populations; stimulated oscillator rates explain geometric occupation. The same probability law can arise from several mechanisms, so the physical interpretation requires the rate model as well as the histogram.

Time to approach that law depends on rate magnitudes, which cancel from many equilibrium ratios. Absorbing boundaries can support stationary extinction instead of a positive fluctuating population. A closed chemical system, a maintained reservoir, a killed process, and a reflecting truncation have different state spaces and boundary fluxes. Declaring the boundary is part of declaring the process.

8.10 Exercises with worked solutions

Exercise8.1 — A normalized truncation can mislead (conceptual)

At equal upward and downward rates on a reflecting finite chain $0, \dots, M$, find the stationary law and mean. What happens as the ceiling is removed?

Worked solution

Every neighbor ratio is one, so all weights equal one and $\pi_n = 1/(M+1)$. Symmetry or summation gives mean $M/2$. At each fixed state the mass tends to zero as $M \to \infty$, and the mean diverges. The limiting zero sequence is not a probability distribution. Finite reflecting normalization therefore does not create an infinite half-line equilibrium at ratio one. The upper boundary supplies confinement that disappears in the limit.

Exercise8.2 — A constant mean can conceal equilibration (conceptual)

For immigration-death with integer $\mu = a/b$, start at exactly $N(0) = \mu$. Compare its mean and variance with a process initialized from the stationary Poisson law.

Worked solution

The deterministic initialization gives mean μ at every time and variance $\mu(1 - e^{-2bt})$ from (8.4). The stationary initialization has both mean and variance μ at every time. The former's zero initial variance is enough to disprove stationarity despite its constant mean. For a random initial variance V_0, independent thinning adds a

term $q_t^2 V_0$ to the deterministic-initial variance formula with the same mean. Taking $V_0 = \mu$ cancels the transient deficit exactly.

Exercise8.3 — Thermal ratios and degeneracy (derive)

Suppose each energy level E_n has one state and neighbor rate ratio is $\exp[-(E_{n+1} - E_n)/(k_B T)]$. Derive its stationary weights. How would a specified degeneracy g_n modify a level probability?

Worked solution

Multiplying neighbor ratios telescopes the energy differences: $w_n = \exp[-(E_n - E_0)/(k_B T)]$. Normalization gives weights proportional to $\exp[-E_n/(k_B T)]$ when the partition sum is finite. If a level contains g_n distinct equally weighted microstates, its aggregate probability is proportional to $g_n \exp[-E_n/(k_B T)]$. A coarse level-rate model must then have ratio $\lambda_n/\delta_{n+1} = (g_{n+1}/g_n)\exp[-(E_{n+1} - E_n)/(k_B T)]$ to reproduce that level law. Degeneracy is a state-counting factor; it cannot be added to a transition model without checking the resulting ratios.

Exercise8.4 — Critical branching variance (derive)

For per-capita birth and death rates both $a > 0$, derive mean and variance from the generator. Explain why a constant mean does not describe deterministic population conservation.

Worked solution

The drift $\lambda_n - \delta_n$ vanishes, so $m' = 0$ and $m = n_0$. The second moment equation becomes $d\mathbb{E}[N^2]/dt = 2a\mathbb{E}[N] = 2an_0$, since the quadratic drift terms cancel but birth and death noise add. Subtracting the constant squared mean gives $V(t) = 2an_0 t$. Individual jumps change the population; only the ensemble drift cancels. This is not a fixed-total conservation law, which would force variance zero for a deterministic initial total.

Exercise8.5 — Infer two rates from drift and noise (compute/design)

A branching model has initial deterministic population ten, initial mean derivative three per second, and initial variance derivative eleven per second. Infer its per-capita birth and death rates. State the model-dependent assumption behind the inference.

Worked solution

At zero time, $m' = (a - b)10 = 3$ and $V' = (a + b)10 = 11$, because $V(0) = 0$. Thus $a - b = 0.3$ and $a + b = 1.1$, yielding $a = 0.7, b = 0.4$ per second. Mean evolution alone would give only their difference. The inference assumes independent one-at-a-time births and deaths with rates proportional to the current count, a

deterministic initial population, and accurately measured derivatives. Immigration, batch births, correlated events, or initial variance would change the equations and can invalidate this identification.

Exercise8.6 — Choose the correct chemical equilibrium (compute/design)

A closed conversion system has total population twenty and equal forward and reverse per-molecule rates. Compare its stationary product-count mean and variance with an immigration-death reservoir model fitted to the same mean.

Worked solution

The closed system has binomial equilibrium with $M = 20, p = 1/2$, so mean ten and variance five. A reservoir model with $a/b = 10$ has Poisson equilibrium with mean and variance ten. Their means agree but the closed system's fixed total suppresses variance and caps the count at twenty. Checking the Fano factor gives one half versus one; checking support supplies another distinction. The rates, boundary at twenty, and reservoir maintenance must therefore be specified before an equilibrium histogram is interpreted.

8.11 Sources and attribution

- V. Balakrishnan, lecture10, IIT Madras, NPTEL115106089; official instructor notes, PDF65–68: stationary products, sedimentation, chemical reservoir, oscillator, and continuous-state opening. The recording supplies the branching moment discussion.
- The generating-function, thinning, and independent-sum results used in (8.3) are derived in chapter1 and chapter6; this chapter applies them to an explicitly constructed transient.

Scale height is distinguished from total atmospheric extent. The oscillator equilibrium ratio uses a negative energy exponent, correcting the positive sign printed in the official PDF66. Exact-formula evaluations and seeded simulation are separate evidence classes. No experimental rate measurements are claimed.

From master equations to Fokker–Planck and Smoluchowski dynamics

A microscopic transport model jumps between lattice sites. A macroscopic calculation evolves a smooth probability density. Shrinking the jumps while increasing their rate can connect these descriptions, but a Taylor expansion alone does not ensure a valid probability equation. Keeping a fourth derivative can even make the proposed characteristic function exceed one. The task is to derive the diffusion approximation while preserving normalization, nonnegativity, and the reflecting boundary that creates sedimentation equilibrium.

> Lecture11 supplies the continuous master equation, Kramers–Moyal expansion, diffusion scaling, boundary currents, Smoluchowski dynamics, and a stochastic differential equation interpretation. Its full caption text was read. The truncation condition is checked against Pawula's original paper, printed page187, and stated with its even-order hypothesis. The Itô convention, noise normalization, and distinction between homogeneous evolution and stationary initialization are explicit below.

9.1 Continuous state space does not eliminate jumps L11 1:39

Let $X(t)$ be a real-valued Markov jump process, and write $w(y \mid x)dy$ for the rate of jumping from x to a neighborhood of y. It has units inverse time per unit output coordinate. For a finite-rate jump model the total exit rate is $q(x) = \int w(y \mid x)dy < \infty$. Assume the process is nonexplosive and the required probability integrals exist. Its density obeys

$$\partial_t \rho(x,t) = \int [w(x \mid y)\rho(y,t) - w(y \mid x)\rho(x,t)]dy. \tag{9.1}$$

The first term adds arrivals from every source; the second removes departures to every destination. A conditional initial value x_0 means initial law $\delta(x - x_0)$. There can remain an atom representing no jump, so (9.1) may need interpretation for measures rather than ordinary smooth densities. Assuming continuous values is not the same as assuming continuous paths.

Multiply (9.1) by a smooth compactly supported test function $f(x)$ and integrate over x. Assume differentiation under that integral is justified and the gain and loss integrals are absolutely integrable; a sufficient condition for these bounded test functions is $\int q(x)\rho(x,t)dx < \infty$. Fubini's theorem then permits exchanging the integrations. Interchanging the dummy variables x, y in the gain term writes it as $\int \rho(x,t) \int f(y)w(y \mid x)dy\,dx$. Subtracting the loss term $\int \rho(x,t)f(x)q(x)dx$ gives the weak evolution identity and its jump generator:

$$\frac{d}{dt}\mathbb{E}[f(X(t))] = \mathbb{E}[(Lf)(X(t))], \qquad (Lf)(x) = \int [f(y) - f(x)]w(y \mid x)dy. \tag{9.2}$$

The difference vanishes for constant f, expressing probability conservation. This weak form is also the cleanest route to the local approximation because derivatives first act on the known test function, not on an unknown

density.

9.2 The moment expansion is formal until its remainder is controlled L11 3:15

Let the jump increment be $\xi = y - x$, and set $W(\xi \mid x) = w(x + \xi \mid x)$. Define rate moments $A_n(x) = \int \xi^n W(\xi \mid x) d\xi$ when the corresponding absolute moments are finite. Their units are coordinate to the power n per time. Taylor expansion of the known function in (9.2) gives, formally, $Lf = \sum_{n \geq 1} A_n f^{(n)}/n!$. The sum starts at one because the zero-order terms cancel between gain and loss.

Integrating each derivative by parts transfers it to the coefficient-density product and introduces $(-1)^n$. With vanishing boundary terms and justified interchanges, the Kramers–Moyal expression is

$$\partial_t \rho = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \partial_x^n [A_n(x) \rho(x,t)]. \tag{9.3}$$

Neither the factorial nor the missing zero term is a matter of notation when A_n is defined as the raw rate moment above. Some references absorb $1/n!$ into their coefficient; the convention must be checked before comparing formulas. Equation(9.3) can fail as a convergent pointwise series for a general jump law. It is not an automatic exact replacement for (9.1).

A controlled small-jump approximation instead uses a finite Taylor formula with remainder. For f with bounded third derivative, the error after the quadratic term is bounded by $\|f'''\|_\infty \int |\xi|^3 W(\xi \mid x) d\xi / 6$. If that absolute third rate moment tends to zero in a scaling family, while A_1, A_2 tend to finite coefficients, the test-function generator approaches a second-order operator. This exposes the assumption needed for the approximation without requiring an infinite Taylor series.

9.3 Positivity restricts finite truncations L11 7:10

For a general Markov transition law, define infinitesimal moments $A_n(x) = \lim_{h \downarrow 0} h^{-1} \mathbb{E}[(X(t+h) - x)^n \mid X(t) = x]$, when finite. These limits need not arise from a regular finite-rate kernel. Pawula's truncation lemma states that, under existence of all the relevant finite moment limits, vanishing of an even moment coefficient of order at least four forces all coefficients of order three and above to vanish. Consequently an exact finite Kramers–Moyal termination beyond second order is incompatible with a probability transition law. This is the result in Pawula's original paper, printed page187, SectionIII.

The even-order condition matters. A symmetric compound Poisson process with jumps $\pm \ell$ equally likely and total rate λ has $A_3 = 0$ but $A_4 = \lambda \ell^4 > 0$. Symmetry cancels an odd moment without removing jumps. The lemma also does not certify that a second-order approximation is accurate for a particular experiment; it constrains exact positive transition laws.

The simplest moment inequality explains part of the mechanism. For a short-time increment ΔX, Cauchy–Schwarz gives $[\mathbb{E}(\Delta X)^3]^2 \leq \mathbb{E}(\Delta X)^2 \, \mathbb{E}(\Delta X)^4$. Dividing by h^2 and taking finite limits gives $A_3^2 \leq A_2 A_4$. Thus

$A_4 = 0$ forces $A_3 = 0$; the general lemma uses related inequalities to propagate this restriction through the higher coefficients.

A fourth-order truncation of the symmetric jump generator has Fourier growth exponent $-A_2k^2/2 + A_4k^4/24$. At sufficiently large $|k|$ it is positive, so the proposed characteristic function after positive time has magnitude greater than one. A characteristic function must have magnitude at most one. This failure is not a subtle numerical oscillation: the truncated operator cannot describe a probability law at all frequencies.

9.4 A biased lattice model gives drift and diffusion with different scalings L11 15:32

Take spacing $\ell > 0$ and constant right/left rates r_+, r_-. For $x = \ell j$, a smooth-density approximation uses $p_j(t) \approx \ell\rho(x,t)$ because a point mass and density have different units. Taylor expansion of $r_+\rho(x-\ell) + r_-\rho(x+\ell) - (r_+ + r_-)\rho(x)$ gives drift $a = \ell(r_+ - r_-)$ and second moment rate $A_2 = \ell^2(r_+ + r_-)$.

Proposition9.1 — Constant drift-diffusion scaling. Choose

$$r_\pm = \frac{D}{\ell^2} \pm \frac{v}{2\ell}, \qquad D > 0, \qquad |v|\ell \le 2D. \tag{9.4}$$

As $\ell \downarrow 0$, the fixed-time characteristic functions converge to $\exp(ikvt - Dk^2t)$, corresponding to a Gaussian of mean vt and variance $2Dt$.

Derivation.

The exact exponent is $t[r_+(e^{ik\ell} - 1) + r_-(e^{-ik\ell} - 1)]$. Its linear term is $ikvt$ and its quadratic term is $-Dk^2t$. The remaining terms vanish at fixed k, t because the third signed and fourth rate moments are $v\ell^2$ and $2D\ell^2$, and the full exponential Taylor remainder is controlled for the two vanishing jump sizes. The Gaussian transform follows, and the continuity theorem cited in chapter3 supplies fixed-time distributional convergence. Rate nonnegativity is precisely the stated inequality. ∎

The forward equation of that Gaussian is $\partial_t\rho = -v\partial_x\rho + D\partial_x^2\rho$. Under the raw-moment convention, $A_1 = v$ and $A_2 = 2D$, not D. The distinction prevents a factor-of-two error in the noise amplitude. At finite spacing, Poisson clocks mix all step parities, so the reachable spacing is ℓ, unlike the fixed-step parity spacing 2ℓ of chapter3.

9.5 An Itô equation determines the forward diffusion operator L11 40:12

A standard Wiener process W_t starts at zero, has continuous paths, and has independent Gaussian increments of mean zero and variance equal to elapsed time. Its increments have square-root-time units. Consider the *Itô stochastic differential equation*

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t. \tag{9.5}$$

The drift a has coordinate-per-time units and noise amplitude b has coordinate-per-square-root-time units. The Itô convention evaluates coefficients at the left endpoint of each approximating increment, using information available at that time. With suitable conditions such as globally Lipschitz coefficients and linear growth, a nonexplosive solution is well posed. More singular coefficients or constrained boundaries require separate existence and boundary analysis.

For a twice continuously differentiable test function with compact support, the Itô change-of-variable formula gives $df(X_t) = f'(X_t)a\,dt + f'(X_t)b\,dW_t + \frac{1}{2}f''(X_t)b^2dt$. The stochastic integral has mean zero under integrability assumptions. Here the change-of-variable formula is used as a quoted theorem: see Gamarnik, MIT15.070J Lecture17, Theorem1, printed p.2. Its quadratic-variation mechanism is developed later in this course. It gives generator $Lf = af' + \frac{1}{2}b^2f''$. Taking its adjoint by integration by parts gives

$$\partial_t\rho = -\partial_x(a\rho) + \frac{1}{2}\partial_x^2(b^2\rho). \tag{9.6}$$

Thus $A_1 = a$ and $A_2 = b^2$. This is the Fokker–Planck or forward Kolmogorov equation under the Itô convention. It is a statement about the probability law, while (9.5) constructs random trajectories. Coefficients and boundary behavior must be specified together to identify the same process. The existence of a formal partial differential equation alone does not prove uniqueness of the corresponding process.

White noise notation $\dot{X} = a + b\eta(t)$ abbreviates (9.5), with η a generalized derivative of W. It is not a finite-variance ordinary random variable at each time: its formal covariance is a Dirac delta. Integrals against it are defined, and their convention matters when b depends on X. A Stratonovich equation with the same written symbols generally has a different Itô drift. For this conversion, assume spatial differentiability of b and sufficient regularity, for example bounded smooth time-independent coefficients with bounded derivatives through second order. The scalar conversion adds $\frac{1}{2}b\partial_x b$ to the Stratonovich drift to obtain the Itô drift. No such correction occurs for constant b. See Baudoin, Lecture30, the conversion following the existence theorem, for this sufficient regularity setting.

9.6 A stationary current is constant; a reflecting boundary makes it zero L11 31:41

Write (9.6) as a continuity equation $\partial_t\rho = -\partial_x J$, with probability current $J = a\rho - \frac{1}{2}\partial_x(b^2\rho)$. Integrating over an interval $[L, R]$ gives $d\Pr(L \le X_t \le R)/dt = J(L,t) - J(R,t)$. The current has probability-per-time units in one dimension. Its sign is positive toward increasing x.

A reflecting endpoint requires zero current there at every time. An absorbing endpoint for a nondegenerate killed diffusion removes probability through the boundary; its interior transition density has decreasing total mass and commonly satisfies a zero-density boundary condition. These conditions describe different processes. Merely specifying an initial density is insufficient to select between them.

At stationarity $\partial_x J = 0$, so the current is spatially constant. If one endpoint is reflecting, that constant is zero everywhere. This does not force transient interior current to vanish: probability can redistribute within a reflecting domain. For a zero-current stationary density with $b^2(x) > 0$, integration of $\partial_x(b^2\rho) = 2a\rho$ gives

$$\rho_{\mathrm{st}}(x) = \frac{C}{b^2(x)} \exp\left[\int^x \frac{2a(y)}{b^2(y)}\,dy\right]. \tag{9.7}$$

The constant C must normalize the density over the specified domain. An arbitrary integration reference point is absorbed into C. Normalizability, endpoint behavior, and existence of a conservative diffusion remain requirements, rather than consequences of writing the formula. The prefactor $1/b^2$ matters for variable noise amplitude.

9.7 Thermal drift and diffusion agree on the Boltzmann density L11 38:44

Let $U(x)$ be a differentiable potential energy, so force is $-U'(x)$. For constant positive mobility M, the overdamped drift is $a(x) = -MU'(x)$, with coordinate-per-time units. Take constant diffusion coefficient $D > 0$ and $b = \sqrt{2D}$. The Smoluchowski equation is

$$\partial_t \rho = M\partial_x(U'\rho) + D\partial_x^2 \rho, \qquad J = -MU'\rho - D\partial_x \rho. \tag{9.8}$$

Zero current gives $\rho \propto \exp[-MU/D]$. If a thermal bath imposes $D = Mk_BT$, it becomes $\rho \propto \exp[-U/(k_BT)]$, provided that density normalizes. The physical fluctuation-dissipation relation is a model input; the equilibrium density is derived from it and the zero-current condition. A force alone has the wrong units to serve as a drift without mobility.

For a half-line $x \geq 0$ with constant downward drift $-v_d$, $v_d > 0$, a reflecting floor gives $J = -v_d\rho - D\rho' = 0$. Thus $\rho = (v_d/D)e^{-v_d x/D}$. The scale height D/v_d matches chapter8's lattice limit. Figure9.1 shows equal and opposite drift and diffusion currents in reduced units, explaining how a nonuniform equilibrium can have zero total flux.

For $U(x) = kx^2/2$ on the full line, with stiffness $k > 0$, (9.7) gives a Gaussian of variance k_BT/k. The drift is restoring and linear, producing the Ornstein–Uhlenbeck process treated dynamically in chapter10. Its stationary Gaussian initialization differs from its conditional evolution from a prescribed point. No uniqueness statement about all Gaussian Markov processes is needed for this example.

9.8 Free diffusion is homogeneous and Gaussian but not stationary L11 47:42

With zero drift and constant amplitude $\sqrt{2D}$, (9.5) integrates to $X_t = x_0 + \sqrt{2D}W_t$. Its density at positive time is Gaussian of mean x_0 and variance $2Dt$, so its marginal law changes. It is Markov with a homogeneous transition rule and stationary independent increments, but it is not a strictly stationary position process.

For $s \leq t$, independent increments give $\mathrm{Cov}(X_s, X_t) = 2Ds$ from a deterministic initial position, or $2D\min(s,t)$ for arbitrary nonnegative times. This depends on both times, not just their lag. The Gaussian density tends to zero at every fixed point as it spreads while retaining integral one. On the entire line, an invariant zero-drift density would satisfy $\rho'' = 0$ and hence be affine; no nonnegative normalized affine density exists there. On a finite reflecting interval, zero current instead selects a normalized constant density.

Integrating white noise produces continuous Wiener paths, but continuity does not imply differentiability. The stochastic equation uses increments rather than an ordinary finite instantaneous velocity. The path regularity and quadratic variation needed for this distinction are developed in chapter12; the current chapter establishes the law and its generator.

9.9 Laboratory: test the approximation and a positivity failure separately

Run `python code/ch09_diffusion_limit.py`. The program fixes $v = 0.7$ meter per second, $D = 0.4$ square meter per second, and time one second, then shrinks the lattice spacing using (9.4). It compares exact Skellam masses with Gaussian probabilities integrated over cells of width ℓ around each site. That comparison puts two probability masses on the same footing; it does not compare a mass directly with a density.

```
ell=0.5000: cell-mass TV=0.037884309, A4=0.200000000
ell=0.2500: cell-mass TV=0.009462083, A4=0.050000000
ell=0.1250: cell-mass TV=0.002385749, A4=0.012500000
ell=0.0625: cell-mass TV=0.000597397, A4=0.003125000
```

Total variation, abbreviated TV, is half the sum of absolute cell-mass differences. The grid extends more than twelve continuum standard deviations, and the program verifies exact lattice normalization and its first two moments. The observed error decreases approximately by four when spacing halves for these parameters. This is a numerical convergence check, not a universal rate theorem.

A separate symmetric-jump example uses total rate two per second and jump magnitude0.5 meter, giving $A_3 = 0$ and $A_4 = 0.125$ in appropriate units. At Fourier variable ten inverse meters and time one second, the exact characteristic-function exponent is -1.432675629, while its fourth-order truncation gives27.083333333. The latter would produce a magnitude greater than one. Thus the program tests both a successful second-order scaling limit and an invalid attempt to improve it by a finite higher-order truncation.

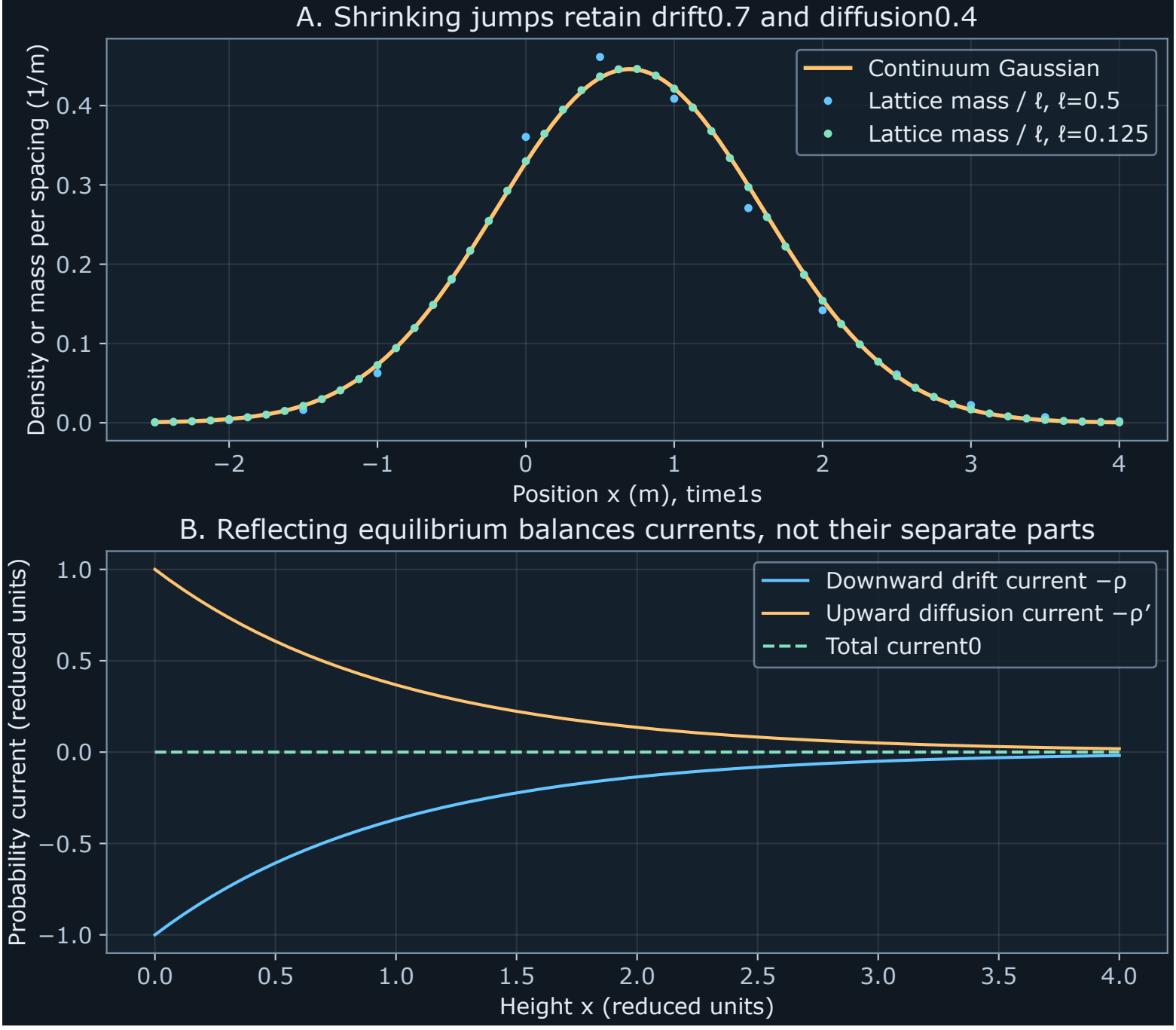


Figure 9.1. Diffusion requires both a scaling and a boundary current. Panel A uses the same parameters as the cell-mass error calculation; points show mass divided by spacing for a density-scale visual comparison. Panel B uses reduced coefficients $v_d = D = 1$ and density e^{-x}, so its two current contributions cancel. Figure9.1 is reproduced by the retained generator.

The program writes JSON and text beside the script, with no network calls or pseudorandom draws. Its final independent check normalizes a harmonic equilibrium density and verifies cancellation of its drift and diffusion currents. The following minimal calculation returns nonnegative lattice rates for admissible inputs and refuses parameters that cannot define jump probabilities:

```
def lattice_rates(spacing, drift, diffusion):
    if spacing <= 0 or diffusion <= 0:
        raise ValueError("positive spacing and diffusion required")
    if abs(drift)*spacing > 2*diffusion:
        raise ValueError("spacing is too coarse for this drift")
    right = diffusion/spacing**2+drift/(2*spacing)
    left = diffusion/spacing**2-drift/(2*spacing)
    return right, left
```

```
print(lattice_rates(0.5, 0.7, 0.4))
```

```
(2.3, 0.9000000000000001)
```

9.10 What a diffusion approximation preserves and discards

A controlled small-jump limit preserves finite drift and second-moment growth while discarding higher increment rates. A finite-rate jump process and a diffusion have different microscopic path structures even when their central laws agree closely. For a regular nonnegative finite-rate jump kernel, a vanishing fourth rate moment would force all nonzero jumps to disappear; a nontrivial diffusion is obtained through singular small-jump/high-rate scaling rather than by that substitution inside a fixed regular kernel.

Boundary layers, large jumps, rare tails, multiplicative noise conventions, and times comparable to a neglected bath correlation time require extra analysis. Pawula's condition does not supply an approximation error bound. A zero-current formal density does not establish normalizability or uniquely specify a boundary process. These limits identify where the microscopic model, the local generator, and the observed time scale must be checked together.

9.11 Exercises with worked solutions

Exercise9.1 — An odd moment can vanish without diffusion (conceptual)

A compound Poisson process jumps by plus or minus one with equal probability at total rate three per second. Compute its first four infinitesimal moment rates. Which vanishing coefficient would legitimately trigger the cited truncation lemma?

Worked solution

Over time h, one jump occurs with probability $3h + o(h)$, and multiple jumps contribute higher-order probability. Therefore $A_n = 3\mathbb{E}[\xi^n]$: the first four values are0,3,0,3 in coordinate-to-the-order per second. The zero third coefficient comes from symmetry. It does not eliminate the positive fourth coefficient or the jumps. Under the stated finite-moment hypotheses, a zero even coefficient at order four or higher would trigger Pawula's conclusion; this example satisfies no such condition.

Exercise9.2 — A reflecting condition is a flux condition (conceptual)

For drift $-v_d$ and constant diffusion D on $x \geq 0$, compare imposing $\rho'(0,t) = 0$ with imposing reflection. Are they equivalent?

Worked solution

The current is $J = -v_d\rho - D\rho'$. Reflection requires $J(0,t) = 0$, hence $\rho'(0,t) = -(v_d/D)\rho(0,t)$. A zero derivative alone would leave current $-v_d\rho(0,t)$, generally nonzero, allowing probability to cross the boundary.

The Neumann zero-derivative condition agrees with reflection only when the drift contribution vanishes there, such as zero drift. The stationary exponential satisfies the correct nonzero-slope flux condition.

Exercise9.3 — Derive the zero-current density with variable noise (derive)

On a finite reflecting interval, take zero Itô drift and strictly positive smooth $b^2(x)$. Find the stationary density. Is it generally uniform?

Worked solution

Zero current requires $\partial_x(b^2\rho) = 0$, so $b^2(x)\rho(x) = C$. Thus $\rho = C/b^2$ with $C^{-1} = \int dx/b^2$ over the interval, assuming that integral is finite. It is uniform only when b^2 is constant. The position-dependent noise amplitude changes residence statistics under the Itô convention. Choosing a Stratonovich model with the same displayed zero drift would define a different Itô drift and can change its stationary law.

Exercise9.4 — Recover moment equations from the forward operator (derive)

For (9.6), derive evolution of mean and second moment assuming all integration-by-parts boundary terms vanish. Apply it to free diffusion.

Worked solution

Multiplication by x and integration gives $d\mathbb{E}[X]/dt = \int a\rho dx$, because one integration by parts removes the drift derivative and two on the diffusion term leave the second derivative of x, zero. For x^2, the first derivative gives $2x$ and the second gives two, yielding $d\mathbb{E}[X^2]/dt = \mathbb{E}[2Xa + b^2]$. With $a = 0, b^2 = 2D$, the mean is constant and the second moment grows at rate $2D$. For a deterministic initial position, variance is therefore $2Dt$, consistent with the Gaussian law.

Exercise9.5 — Choose a microscopic spacing for a probability tolerance (compute/design)

For the lab's fixed drift, diffusion, and time, choose the largest listed spacing whose cell-mass TV error is below0.003. Does that certify an accurate Brownian path model at arbitrarily short times?

Worked solution

The listed errors are$0.037884309, 0.009462083, 0.002385749, 0.000597397$ as spacing decreases. Thus0.125meter is the largest listed acceptable spacing. The comparison certifies this numerical one-time cell-probability tolerance for the tested parameter set and grid. It does not control joint path distributions, very short times before many jumps occur, or relative errors in rare cells. At finite spacing the path still has discrete jumps and exponential waiting times, which a continuous Brownian trajectory does not.

Exercise9.6 — An equilibrium density can hide the relaxation coefficient (compute/design)

In harmonic potential $U = kx^2/2$, compare two thermal overdamped models with mobilities M and $2M$ at the same temperature. Find their stationary variance and mean relaxation rates. What must be observed to infer mobility?

Worked solution

With $D = Mk_BT$, both stationary densities are proportional to $\exp[-kx^2/(2k_BT)]$, so their variance is k_BT/k. The drift is $-Mkx$, giving mean equation $m' = -Mkm$ from Exercise9.4. Doubling mobility doubles both the drift relaxation rate to $2Mk$ and the diffusion coefficient, preserving equilibrium variance. An equilibrium histogram alone therefore cannot determine mobility. Measure a transient mean after displacement or the stationary time covariance, with a declared time calibration, to identify the dynamical rate.

9.12 Sources and attribution

- V. Balakrishnan, lecture11, IIT Madras, and official instructor notes, PDF68–73: continuous master equation, diffusion operator, currents, and OU preview. Factorials and raw-moment coefficients are kept consistent here.
- R.F.Pawula, Approximation of the Linear Boltzmann Equation by the Fokker–Planck Equation, *Physical Review*162,186–188(1967), printed187, SectionIII; readable paper copy. The finite-moment even-order truncation condition is cited; the symmetric-jump failure calculation is developed here.

Corrections and limits: an odd coefficient can vanish by symmetry; $A_2 = 2D$ under the declared convention; Itô and Stratonovich coefficients are distinguished; white noise is generalized; a homogeneous kernel need not have a stationary ensemble. No continuous audiovisual transcription or empirical transport measurements are claimed.

Velocity relaxation and the emergence of spatial diffusion

A particle-tracking instrument measures displacements at successively shorter delays. A diffusion fit assumes that their variance is proportional to the delay. At a delay equal to the velocity relaxation time, that fit overestimates the actual thermal-start mean-square displacement by a factor of about 2.72. At shorter delays, the particle retains its velocity and the displacement variance is quadratic in time. The task is to decide which temporal resolution permits a diffusion description, while retaining a model that remains consistent with thermal equilibrium.

> The topic sequence follows NPTEL lectures 12 and 13: thermal Langevin balance, velocity relaxation, covariance, spatial diffusion, and joint phase space. The conditional covariance formulas, exact joint-update laboratory, finite-resolution comparison, and exercises are developed here. Caption reading and sampled-frame coverage are recorded separately; neither establishes continuous audiovisual review.

10.1 A random force needs a dissipative partner L12 4:03

Let $m > 0$ be a particle's mass, X_t its position on a line, and V_t its velocity. Time t is nonnegative. Their units are kilograms, meters, meters per second, and seconds. The particle sits in a stationary fluid bath at absolute temperature $T > 0$. Write $\theta = k_B T/m$ for the thermal velocity variance, with units of squared velocity; k_B is Boltzmann's constant. The bath is assumed large enough that observing the particle does not change its temperature.

A standard Wiener process W_t starts at zero, has continuous paths with probability one, and has independent Gaussian increments with mean zero and variance equal to the elapsed time. Thus dW has square-root-time units. A force consisting only of uncorrelated kicks would give $dV = b\,dW$, for a constant noise amplitude b with units velocity per square root of time. Its solution is $V_t = v_0 + bW_t$ for a specified initial velocity v_0. Consequently $\mathbb{E}[V_t^2] = v_0^2 + b^2 t$. The kinetic energy grows without bound. Changing white kicks to a finite positive-correlation force alone does not supply the energy removal required by equilibrium.

At low particle velocity relative to the bath, introduce a linear drag force $-m\gamma V$, with relaxation rate $\gamma > 0$ in inverse seconds. The dimensional drag coefficient is $m\gamma$, not γ. Our inertial Langevin model is

$$dX_t = V_t\,dt, \qquad dV_t = -\gamma V_t\,dt + b\,dW_t. \tag{10.1}$$

The two deterministic equations say that position advances at the present velocity and that velocity relaxes toward zero. Noise acts in the acceleration equation. Position is therefore differentiable with derivative V_t, even though the idealized velocity is not differentiable. This distinction will determine which threshold-crossing formulas can be used in chapter 14.

Proposition 10.1 — Thermal noise strength. Requiring the Maxwell velocity variance θ to remain invariant under (10.1) fixes $b^2 = 2\gamma\theta$.

Derivation.

For a short interval h, conditional on current velocity v, the stochastic increment has $\mathbb{E}[\Delta V \mid v] = -\gamma v h + o(h)$ and $\mathbb{E}[(\Delta V)^2 \mid v] = b^2 h + o(h)$. The latter term survives because a Wiener increment has variance h, although its mean is zero. Expanding $(v + \Delta V)^2 - v^2 = 2v\Delta V + (\Delta V)^2$, dividing its conditional expectation by h, and taking the limit gives the quadratic moment rate $-2\gamma v^2 + b^2$. Averaging over the finite-second-moment velocity law yields $d\mathbb{E}[V_t^2]/dt = -2\gamma\mathbb{E}[V_t^2] + b^2$. A stationary second moment equal to θ therefore requires $b^2 = 2\gamma\theta$. This is the quadratic instance of the Itô rule; the noise variance term would be lost by ordinary differentiable calculus. ∎

The condition is an equilibrium consistency requirement within the white-noise, constant-friction model. It does not prove that every fluid has memoryless drag. Once imposed, it ties the fluctuation strength and damping to the same temperature. Increasing γ at fixed T strengthens both the velocity damping and the kicks per unit time, keeping the stationary velocity variance unchanged. It reduces the persistence of motion and therefore reduces spatial diffusion.

10.2 The complete velocity transition law L12 16:02

Specify $V_0 = v_0$ deterministically and assume future Wiener increments are independent of that initial value. Multiplying the velocity equation by $e^{\gamma t}$ cancels its drift. Integration gives

$$V_t = v_0 e^{-\gamma t} + \sqrt{2\gamma\theta}\int_0^t e^{-\gamma(t-s)}\,dW_s. \tag{10.2}$$

The stochastic integral is a limit of linear combinations of independent Gaussian increments. Its deterministic integrand makes that limit Gaussian. The mean is zero, and the Itô isometry says its variance is the integral of the squared integrand. This supplies the Gaussian property that a moment calculation alone could not establish.

Proposition 10.2 — OU transition density. For $t > 0$, the conditional law of V_t is Gaussian with mean $v_0 e^{-\gamma t}$ and variance $q_v(t) = \theta(1 - e^{-2\gamma t})$. Its density is

$$\rho(v, t \mid v_0) = \frac{\exp[-(v - v_0 e^{-\gamma t})^2/(2q_v(t))]}{\sqrt{2\pi q_v(t)}}. \tag{10.3}$$

Derivation.

Apply the preceding Gaussian-integral argument to (10.2). The integral's variance is $2\gamma\theta\int_0^t e^{-2\gamma(t-s)}\,ds = \theta(1 - e^{-2\gamma t})$. Adding a deterministic shift supplies the mean and leaves the variance

unchanged. Normalization is the Gaussian integral. This proves the entire law, rather than merely matching its first two moments. ∎

At $t \downarrow 0$, its variance approaches zero and its mean approaches v_0. The density tends to a Dirac mass in the distributional sense: its integral against every bounded continuous test function tends to that function evaluated at v_0. Substituting zero into the positive-time density formula would divide by zero. At long times the law approaches the Maxwell Gaussian of variance θ, independently of v_0.

The corresponding forward equation is $\partial_t \rho = \gamma \partial_v (v\rho) + \gamma\theta \partial_v^2 \rho$. It has probability current $J_v = -\gamma v\rho - \gamma\theta \partial_v \rho$, so $\partial_t \rho = -\partial_v J_v$. On the whole velocity line, impose vanishing probability flux at infinity. A stationary density then has zero constant current, giving $\partial_v \log \rho_{\mathrm{eq}} = -v/\theta$ and hence $\rho_{\mathrm{eq}} = e^{-v^2/(2\theta)}/\sqrt{2\pi\theta}$. Normalizability and the selected boundary conditions matter; a stationary differential expression alone is not a boundary-value problem.

A non-Gaussian initial velocity distribution generally remains non-Gaussian at finite times. Equation (10.2) adds an independent Gaussian to a contracted initial variable; it does not instantly erase all initial cumulants. Gaussian transition kernels therefore coexist with non-Gaussian evolving marginals. Full stationarity requires the invariant initial law, not merely the correct noise amplitude.

10.3 Which ensemble makes velocity stationary? L13 4:51

Now draw V_0 from the zero-mean Maxwell Gaussian with variance θ, independently of future noise. For $t \geq s$, split the solution at time s: $V_t = e^{-\gamma(t-s)} V_s + \sqrt{2\gamma\theta} \int_s^t e^{-\gamma(t-r)} dW_r$. The second term is independent of V_s and has zero mean. It follows that the stationary covariance is

$$C_v(\tau) = \mathbb{E}[V_{s+\tau} V_s] = \theta e^{-\gamma|\tau|}. \tag{10.4}$$

The one-time variance stays equal to θ by Proposition 10.1. All finite-dimensional vectors are Gaussian because they are linear functions of the Gaussian initial value and Wiener increments. Their zero means and time-shift-invariant covariance determine their laws, establishing strict stationarity. This argument requires a jointly Gaussian process, a stronger condition than Gaussian histograms at individual times.

For deterministic v_0, the centered two-time covariance is instead $\theta[e^{-\gamma|t-s|} - e^{-\gamma(t+s)}]$. This follows by integrating the product of the two noise kernels only up to $\min(s,t)$. The extra term remembers the zero initial variance. Adding the random thermal initial velocity contributes $\theta e^{-\gamma(t+s)}$ and cancels it. Conditional and stationary averages must not be interchanged.

A precise Gaussian–Markov qualification

A useful part of the classification mentioned in lecture 13 can be proved directly. Let Y_t be a real, centered, stationary Gaussian Markov process with positive variance c_0 and continuous covariance. Define its normalized covariance $r(t) = \mathbb{E}[Y_t Y_0]/c_0$. Gaussian regression gives $\mathbb{E}[Y_{s+t} \mid Y_s] = r(t) Y_s$. The Markov property permits conditioning on the past to be replaced by conditioning on Y_s. Multiplying by Y_0, taking expectations, and using stationarity gives $r(s+t) = r(s) r(t)$ for nonnegative s, t.

Continuity and $r(0) = 1$ imply $r(t) > 0$: if it vanished at some time, repeated halving would contradict continuity at zero; also $r(t) = r(t/2)^2 \geq 0$. The continuous additive function $\log r(t)$ is linear, so $r(t) = e^{-\gamma t}$. The covariance bound $|r(t)| \leq 1$ requires $\gamma \geq 0$. The case $\gamma = 0$ is a time-constant random variable; the nonconstant case has $\gamma > 0$ and the OU covariance. Joint Gaussianity then fixes its finite-dimensional law. This proves the stated restricted classification, including its degenerate exception.

Continuity, stationarity, or exponential correlation alone is insufficient. The two-state process of chapter 6 has exponential correlation but is non-Gaussian. A mixture of independent OU modes with different relaxation rates is stationary Gaussian, yet its summed covariance is not a single exponential and its scalar observation generally loses the Markov property. Correlation shape therefore tests a state model under specific distributional assumptions.

10.4 The displacement crosses from ballistic to diffusive L13 15:24

Take $X_0 = x_0$ fixed and the stationary thermal velocity ensemble. The displacement $\Delta X_t = X_t - x_0 = \int_0^t V_s ds$ has zero mean. Its mean-square displacement, abbreviated MSD, equals its variance. Squaring the integral and taking expectations gives an integral of the covariance over a square in the two time variables. The covariance is even, so the two triangular halves contribute equally.

Proposition 10.3 — Thermal-start crossover. With the specified initial ensemble,

$$\mathbb{E}[(\Delta X_t)^2] = \frac{2\theta}{\gamma^2}(\gamma t - 1 + e^{-\gamma t}) = 2D\left[t - \frac{1 - e^{-\gamma t}}{\gamma}\right], \qquad D = \frac{\theta}{\gamma}. \tag{10.5}$$

Derivation.

Integrate (10.4) over the lower triangle: $2\theta \int_0^t ds \int_0^s dr\, e^{-\gamma(s-r)}$. The inner integral is $(1 - e^{-\gamma s})/\gamma$. Integrating once more gives (10.5). All terms have squared-length units, and D has units length squared per time. ∎

For $\gamma t \ll 1$, expand the exponential through its quadratic term. Its constant and linear terms cancel, leaving MSD $\theta t^2 + O(\theta \gamma t^3)$. The particle travels almost at its initial thermal velocity: random velocities across the ensemble produce ballistic ensemble scatter. For $\gamma t \gg 1$, the bounded correction in brackets is small relative to t, leaving MSD asymptotic to $2Dt$. Successive sufficiently long displacement increments become approximately decorrelated, but finite-time increments remain correlated.

Define dimensionless delay $u = \gamma t$ and dimensionless MSD $M(u) = \gamma^2 \mathrm{MSD}/\theta = 2(u - 1 + e^{-u})$. Its local logarithmic slope is $d\log M / d\log u = u(1 - e^{-u})/(u - 1 + e^{-u})$. This exponent interpolates between two and one; it need not equal either at intermediate delays. At $u = 1$, $M = 0.73575888$ and the slope is 1.718282. A pure diffusion prediction would be $M = 2$, giving the opening discrepancy factor $e \approx 2.72$. At $u = 10$, the MSD is still about ten percent below $2u$.

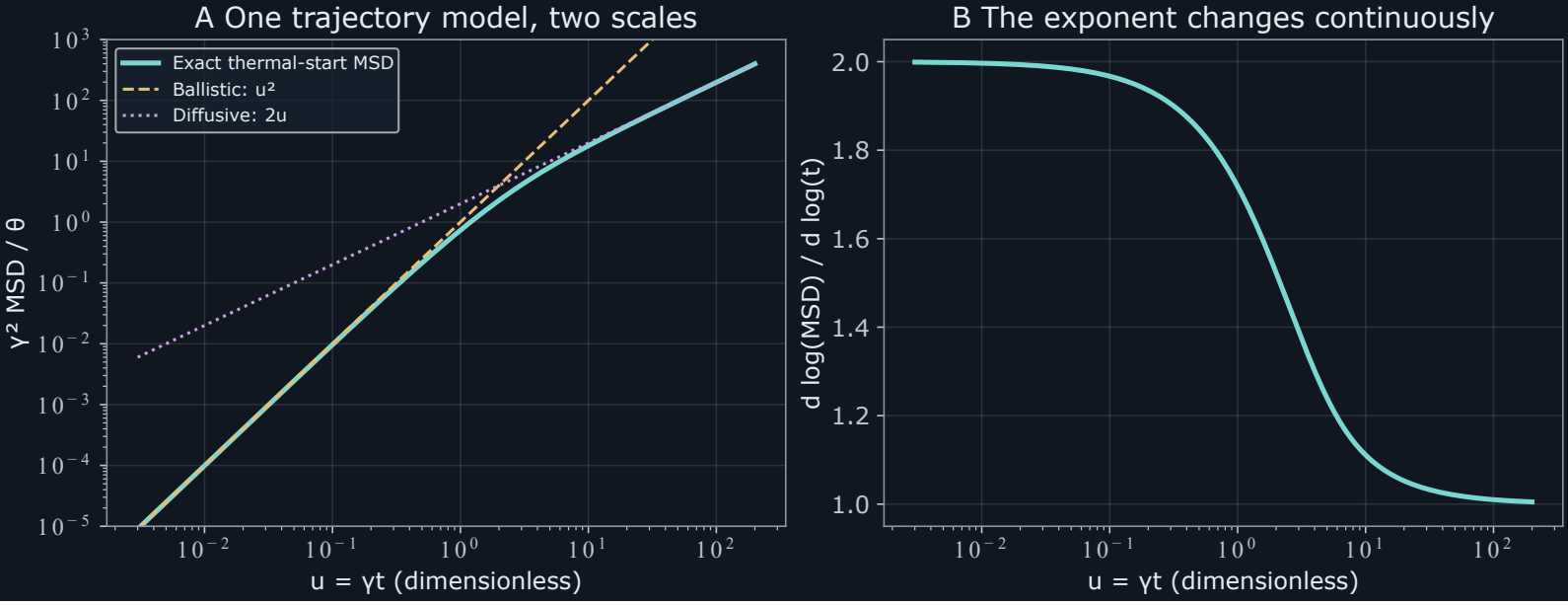


Figure 10.1. Inertial crossover for a thermal initial velocity. Panel A compares the exact MSD with its two asymptotes. Panel B makes the intermediate apparent exponent explicit. The axes use dimensionless quantities, so the curves apply to any positive relaxation rate and thermal variance in this model. Generator.

The Einstein diffusion coefficient $D = k_B T/(m\gamma)$ follows from the long-time limit. If a spherical particle in a continuum fluid satisfies no-slip Stokes drag, the additional hydrodynamic input is $m\gamma = 6\pi\eta a$, where η is dynamic viscosity and a is particle radius. Then $D = k_B T/(6\pi\eta a)$. Temperature changes may also change viscosity, so this does not establish a universally linear temperature dependence of D. The original Uhlenbeck–Ornstein paper, printed page 826, explicitly distinguishes the finite-inertia formula from the long-time Einstein approximation. Here the stochastic-integral derivation states the initial ensemble separately.

10.5 Diffusion from a general stationary velocity L13 25:49

The square-integral argument does not depend on Gaussianity, exponential decay, or a Markov state. Let V_t now be any real, zero-mean stationary velocity with finite variance and locally integrable covariance C_v. Stationarity and commutativity of scalar multiplication give $C_v(-s) = C_v(s)$; microscopic time-reversal symmetry is not needed for this scalar identity. Rearranging the covariance integral gives

$$\mathrm{Var}(\Delta X_t) = 2\int_0^t (t-s)C_v(s)\,ds. \tag{10.6}$$

For each lag s, the available earlier times occupy an interval of length $t - s$. That geometric weight explains the factor in the integral. The expression is nonnegative because it is a variance, even if some covariance lags are negative.

Proposition 10.4 — A sufficient Green–Kubo condition. If $\int_0^\infty |C_v(s)|ds < \infty$, then the limit $D = \lim_{t\to\infty} \mathrm{Var}(\Delta X_t)/(2t)$ exists and equals $\int_0^\infty C_v(s)ds$.

Derivation.

Divide (10.6) by $2t$ and extend the integrand by zero beyond t. For every fixed s, the multiplier $(1 - s/t)\mathbf{1}_{[0,t]}(s)$ tends to one and has absolute value at most one. The integrable function $|C_v(s)|$ dominates

the product. The dominated convergence theorem therefore passes the limit through the integral. Since each finite-time ratio is nonnegative, the limiting integral is nonnegative as well. ∎

The sufficient assumption is absolute integrability. A vaguely stated “finite correlation time” is not a substitute for it; conditionally convergent covariance integrals require a separate argument. A zero integral is possible and does not produce ordinary diffusion with positive coefficient. A divergent integral can produce anomalous growth, but divergence alone does not identify a universal power law. For the OU covariance, the integral is θ/γ, recovering (10.5).

In the tracking problem, measuring a covariance provides a second route to the diffusion coefficient. It also exposes where a finite observation window truncates the required integral. A long positive tail missed by the record biases the estimate downward; a negative tail can have the opposite effect. A diffusion fit and a covariance integral should be compared using consistent centering, initial ensembles, and time windows.

10.6 Position and velocity form the Markov state L13 35:16

Restore the OU model and fix both $X_0 = x_0$ and $V_0 = v_0$. Integrating (10.2) once and exchanging the deterministic time integral with the square-integrable stochastic integral gives

$$\Delta X_t = \frac{1-r}{\gamma} v_0 + \frac{\sqrt{2\gamma\theta}}{\gamma} \int_0^t (1 - e^{-\gamma(t-s)}) dW_s, \qquad r = e^{-\gamma t}. \tag{10.7}$$

The same noise drives position and velocity. Their joint conditional law is bivariate Gaussian, with mean column $(x_0 + (1-r)v_0/\gamma,\, r v_0)^{\mathsf{T}}$. Its covariance matrix $C_{\mathrm{cond}}(t)$, ordered as position then velocity, has entries

$$C_{\mathrm{cond},vv} = \theta(1-r^2), \quad C_{\mathrm{cond},xv} = C_{\mathrm{cond},vx} = \frac{\theta}{\gamma}(1-r)^2, \quad C_{\mathrm{cond},xx} = \frac{2\theta}{\gamma}\left[t - \frac{2(1-r)}{\gamma} + \frac{1-r^2}{2\gamma}\right]. \tag{10.8}$$

These formulas follow by integrating products of the noise kernels in (10.2) and (10.7). In particular the cross term is $2\theta \int_0^t (1 - e^{-\gamma s}) e^{-\gamma s} ds = \theta(1-r)^2/\gamma$. The matrix is positive definite for $t > 0$: a linear combination of its two unequal time kernels cannot vanish almost everywhere unless both coefficients vanish. Gaussian density normalization is therefore well defined. Its exponent is $-(y-\bar{y})^{\mathsf{T}} C_{\mathrm{cond}}^{-1} (y-\bar{y})/2$, with prefactor $1/(2\pi\sqrt{\det C_{\mathrm{cond}}})$, where $y = (x, v)^{\mathsf{T}}$.

Given a fixed initial velocity, displacement variance starts as $2\gamma\theta t^3/3$, rather than θt^2. The missing ballistic ensemble term is restored by adding the variance of $(1-r)V_0/\gamma$ when V_0 is thermal. This is a concrete use of the law of total variance. It explains why two correct formulas can disagree when their initial conditioning differs.

The joint forward equation is $\partial_t \rho = -v\partial_x \rho + \gamma \partial_v (v\rho) + \gamma\theta \partial_v^2 \rho$. The streaming term transports probability in position without adding a separate position-noise term. The pair (X_t, V_t) is Markov because its future is a function of the present pair and future independent Wiener increments. Position alone generally is not: its recent slope contains velocity information that a single position value omits. Nonstationarity of position and failure of its Markov property are distinct statements; a free Wiener position is nonstationary but Markov.

At long times, $C_{\mathrm{cond},xx} \sim 2Dt$, $C_{\mathrm{cond},vv} \to \theta$, and $C_{\mathrm{cond},xv} \to \theta/\gamma$. The correlation coefficient tends to zero because the position variance grows. A finite mean shift v_0/γ and bounded covariance corrections become negligible on the $\sqrt{t}$ displacement scale. This is the precise asymptotic sense in which the joint density approaches a thermal velocity factor and a diffusion-scale position factor. Exact finite-time factorization is false.

10.7 Laboratory: preserve the shared noise

The executable laboratory uses dimensionless units $\gamma = \theta = 1$. It evaluates (10.8), independently integrates its noise-kernel products by numerical quadrature, and checks positive eigenvalues. It adds the thermal initial component and verifies (10.5). Finally it generates 250,000 exact joint updates with seed 2026110810, using a Cholesky factor of $C_{\mathrm{cond}}(h)$: a lower-triangular matrix L satisfying $LL^{\mathsf{T}} = C_{\mathrm{cond}}(h)$. Multiplying a pair of independent unit Gaussians by L generates the required correlated noise. The reported sample covariance must agree with the exact value within the stated sampling tolerance.

Run `python code/ch10_ou.py` after installing the dependencies. The program makes no network calls and writes JSON and text output beside its source. Invalid nonpositive covariance parameters raise `ValueError`. Small-time cancellation limits direct evaluation of (10.8) at extremely tiny steps; this teaching implementation checks times down to 0.01 and is not an arbitrary-precision kernel.

```
t       stationary MSD  fixed-v0 variance  local slope
  0.01      0.00009967         0.00000066     1.996672
  0.10      0.00967484         0.00061892     1.967218
  1.00      0.73575888         0.33618248     1.718282
 10.00     18.00009080        17.00018160     1.111055
100.00    198.00000000       197.00000000     1.010101
```

An exact velocity update alone is $V_{n+1} = rV_n + \sqrt{\theta(1-r^2)}Z_n$ for independent standard normal Z_n. Euler instead uses $V_{n+1} = (1-\gamma h)V_n + \sqrt{2\gamma\theta h}Z_n$. Solving its stationary variance recurrence gives $\theta/(1-\gamma h/2)$ when $0 < \gamma h < 2$. At $\gamma h = 0.1, 0.5, 1$, the relative equilibrium variance biases are 0.052632, 0.333333, and 1. Stability is therefore weaker than accurate equilibrium sampling. These values are computed in the lab.

```
import numpy as np


def exact_velocity_step(v, h, gamma, theta, rng):
    if h < 0 or gamma <= 0 or theta <= 0:
        raise ValueError("invalid step, relaxation rate, or thermal variance")
    r = np.exp(-gamma*h)
    noise_variance = -theta*np.expm1(-2*gamma*h)
    return r*v + np.sqrt(noise_variance)*rng.normal(size=np.shape(v))
```

This pure update accepts a scalar or array of velocities and a caller-owned random generator; it returns the next velocities without writing files. The `expm1` form avoids subtracting nearly equal numbers for the velocity variance. Updating position by hV_n is still a separate discretization. For exact position–velocity propagation, use the joint noise of (10.8), not two independent draws with the correct marginal variances.

10.8 What the white-noise model omits

The model captures finite momentum relaxation, thermal equilibrium, and the crossover to diffusion. It neglects bath memory, hydrodynamic long-time tails, boundaries, interactions among tracked particles, and instrument blur. Its formal short-time ballistic asymptote is valid within the model; application below the bath-correlation time requires a microscopic or colored-noise description. The two scale separations are different: observation times can be long compared with molecular collisions while remaining comparable to momentum relaxation.

Sending γ to infinity at fixed θ makes the covariance area $2\theta/\gamma$ vanish, so it is not a nonzero white-noise limit. A nontrivial delta limit holds the area, equivalently D, fixed while the covariance narrows. The usual experimental diffusion approximation instead examines long delays at fixed parameters. In a trap, displacement variance eventually saturates; chapter 11 adds that restoring force.

10.9 Exercises with worked solutions

Exercise 10.1 — Equilibrium versus conditional relaxation (conceptual)

A thermal OU particle is selected at time zero only if its velocity is exactly v_0. Is its conditional velocity variance still θ? Explain without violating equilibrium.

Worked solution

Conditioning removes the initial ensemble scatter. Equation (10.3) gives conditional variance $\theta(1 - e^{-2\gamma t})$, initially zero. Equilibrium refers to the unconditioned ensemble. Averaging the conditional means $v_0 e^{-\gamma t}$ over a Maxwell initial law contributes another variance $\theta e^{-2\gamma t}$. Their sum is θ. The conditional relaxation and the stationary unconditioned variance answer different statistical questions.

Exercise 10.2 — Hidden velocity (conceptual)

Two particles occupy the same position but have sharply different observed recent slopes. Why is a position-only transition rule inadequate on momentum time scales?

Worked solution

For a short future interval h, the mean displacement conditional on current velocity is $V_t(1 - e^{-\gamma h})/\gamma \sim V_t h$. Recent differentiable position data reveal V_t, so the histories predict different future displacements even at equal present position. A Markov rule based only on that position would assign the same future conditional law to both histories and lose this predictive information. The phase-space pair retains it. This argument concerns inertial position; it does not prohibit a position-only Brownian limit at much longer time scales.

Exercise 10.3 — An arbitrary initial velocity variance (derive)

Let V_0 have mean μ_0 and variance s_0^2, independently of future noise. Derive the displacement mean and variance.

Worked solution

Equation (10.7) gives mean $\mu_0(1-r)/\gamma$. The initial term is independent of the zero-mean noise integral, so variance is $s_0^2(1-r)^2/\gamma^2 + C_{\mathrm{cond},xx}(t)$. Thus deterministic initial velocity gives $s_0^2 = 0$, while a thermal initial law gives $s_0^2 = \theta$ and (10.5). The mean-square displacement equals this variance plus $\mu_0^2(1-r)^2/\gamma^2$. Calling all three quantities "MSD" without specifying centering can conceal the initial drift.

Exercise 10.4 — Correlated displacement increments (derive)

For stationary thermal velocity, calculate the covariance of consecutive displacements $A = \int_0^h V_s ds$ and $B = \int_h^{2h} V_s ds$.

Worked solution

Every second integration time exceeds the first. Thus $\mathrm{Cov}(A,B) = \theta \int_0^h ds \int_h^{2h} dr\, e^{-\gamma(r-s)}$. The integrals factor into $(e^{\gamma h}-1)/\gamma$ and $(e^{-\gamma h}-e^{-2\gamma h})/\gamma$, leaving $\theta(1-e^{-\gamma h})^2/\gamma^2$. This is positive. For small h, both increments share nearly the same velocity, and their covariance is θh^2. At large h, their covariance stays bounded while each variance grows linearly, so their correlation tends to zero. Independent increments are an approximation at long scales, not an exact OU-position property.

Exercise 10.5 — A one-percent diffusion fit (compute/design)

Use relative error with respect to the exact MSD: $[2Dt - \mathrm{MSD}]/\mathrm{MSD}$. Give a simple sufficient delay for this error to be at most one percent.

Worked solution

In terms of $u = \gamma t$, the ratio is $(1-e^{-u})/(u-1+e^{-u})$. Its numerator is at most one and its denominator is at least $u-1$. Therefore $u \geq 101$ is sufficient for error at most 0.01. This is a conservative analytical condition; direct root finding can sharpen it slightly. It concerns only model approximation, not sampling uncertainty or localization error. The lab gives MSD 198 at $u = 100$, whose diffusive prediction 200 has relative error $2/198 \approx 0.010101$, so a hundred relaxation times narrowly misses the specified criterion.

Exercise 10.6 — Exact marginals, wrong joint law (compute/design)

At a step $h = 0.2$ with $\gamma = \theta = 1$, an implementation samples the two conditional noise marginals independently. Diagnose the error and design a check.

Worked solution

The conditional cross-covariance should be $(1 - e^{-0.2})^2$, strictly positive. Independent marginal draws set it to zero. With thermal initial velocity, the common deterministic initial contribution adds $r(1 - r)$ to the cross-covariance, but the required total is $1 - r$. The missing amount remains $(1 - r)^2$. A sample covariance check on the noise pair, or on final position and velocity with the initial contribution accounted for, reveals the error. Generate LZ with $LL^{\mathsf{T}} = C_{\text{cond}}(h)$ to preserve both marginals and their coupling. The laboratory compares the full sample matrix with its exact target, so this defect cannot hide behind separate variance checks.

10.10 Sources and attribution

- V. Balakrishnan, NPTEL 115106089, lecture 12 and lecture 13. Timestamp links identify topic starts, not verbatim reproduction.
- G. E. Uhlenbeck and L. S. Ornstein, On the Theory of the Brownian Motion, *Physical Review* 36, 823–841 (1930). Printed pages 823–826 were checked for the model assumptions, drag convention, and finite-inertia versus diffusion distinction; primary-paper scan. All displayed modern stochastic-integral derivations and numerical examples are developed here.

> Qualifications: stationarity requires the invariant initial law; the Gaussian–Markov classification above includes a constant-process exception and is proved in its stated restricted form; scalar stationary covariance evenness needs no time-reversal assumption; the Green–Kubo limit is proved under absolute integrability. The official course notes' main exposition stops before these lectures.

Confinement, phase space, and the overdamped limit

A trapped Brownian particle has the correct equilibrium position histogram in both an inertial model and its overdamped approximation. That agreement does not certify the dynamics. With natural frequency set to one and damping rate 0.5, the exact normalized position covariance at time 0.5 is 0.887137; the overdamped approximation predicts 0.367879. This chapter determines when eliminating velocity preserves the slow relaxation and when an equilibrium test conceals a large dynamical error.

> Based on NPTEL lecture 14, including its recap of free phase-space motion, harmonic confinement, high-friction approximation, general Kramers equation, and magnetic-field bridge. The covariance calculations, numerical comparisons, and exercises are original developments. The lecture's suggestions of scalar Markov behavior are qualified below: the full inertial state is the position–velocity pair.

11.1 Restoring force creates two relaxation modes L14 15:19

Retain the mass $m > 0$, temperature $T > 0$, thermal velocity variance $\theta = k_B T/m$, drag rate $\gamma > 0$, and standard Wiener process W_t of chapter 10. Let the potential energy be $U(x) = m\omega_0^2 x^2/2$, where $\omega_0 > 0$ is the undamped angular frequency in inverse seconds. Its force is $-U'(x) = -m\omega_0^2 x$. The model is

$$dX_t = V_t dt, \qquad dV_t = (-\gamma V_t - \omega_0^2 X_t)dt + \sqrt{2\gamma\theta}\, dW_t. \tag{11.1}$$

Noise and drag retain the thermal relation established in chapter 10. Confinement changes the position behavior: excursions now store potential energy and generate a restoring force. A finite position variance is possible at equilibrium, whereas the free particle has unbounded displacement variance.

Proposition 11.1 — Stable mean dynamics. For finite initial means, $\bar{x} = \mathbb{E}[X_t]$ satisfies $\ddot{\bar{x}} + \gamma\dot{\bar{x}} + \omega_0^2\bar{x} = 0$. Its characteristic roots are

$$r_\pm = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega_0^2}}{2}. \tag{11.2}$$

Both have negative real part for positive γ, ω_0.

Derivation.

The stochastic integral has zero mean. Taking expectations of (11.1) gives $\dot{\bar{x}} = \bar{v}$ and $\dot{\bar{v}} = -\gamma\bar{v} - \omega_0^2\bar{x}$. Eliminating $\bar{v}$ gives the second-order equation and quadratic roots. If the discriminant is negative, both real parts are $-\gamma/2$. If it is nonnegative, its square root is strictly smaller than γ, so both real roots are negative. ∎

For $\gamma < 2\omega_0$, the mean oscillates with damped frequency $\Omega = \sqrt{\omega_0^2 - \gamma^2/4}$ inside a decaying envelope. At $\gamma = 2\omega_0$, the repeated root yields $(a + bt)e^{-\omega_0 t}$, with constants set by initial position and velocity. For $\gamma > 2\omega_0$,

there are two decaying exponentials. “Overdamped” identifies the nonoscillatory root regime; the controlled elimination of velocity needs the stronger separation $\gamma \gg \omega_0$.

Eigenvalues describe mean relaxation. They do not alone prove thermal equilibrium, identify its variance, or license replacing every observed coordinate by a scalar OU process. The restoring force couples velocity to position, so knowing only velocity does not determine its future drift. The same issue holds for position alone at finite inertia.

11.2 The phase-space equation and its thermal density L14 39:18

Replace the harmonic energy by a differentiable potential $U(x)$. Assume its force is regular enough for a nonexplosive Langevin solution and for the forward equation and integrations below. The position–velocity probability density $\rho(x, v, t)$ obeys the Kramers equation

$$\partial_t \rho = -v\partial_x \rho + \frac{U'(x)}{m}\partial_v \rho + \gamma \partial_v (v\rho) + \gamma\theta\partial_v^2 \rho. \tag{11.3}$$

The first two terms stream probability along Hamiltonian trajectories; the last two describe damping and velocity diffusion. Writing $\partial_t \rho = -\partial_x J_x - \partial_v J_v$ identifies $J_x = v\rho$ and $J_v = -(U'/m + \gamma v)\rho - \gamma\theta\partial_v \rho$. Probability conservation follows by integrating this divergence when boundary fluxes vanish.

Proposition 11.2 — A stationary Gibbs density. If $Z = \int_{\mathbb{R}^2} \exp[-(mv^2/2 + U(x))/(k_B T)]\,dx\,dv$ is finite and the required boundary conditions hold, then

$$\rho_{\mathrm{eq}}(x, v) = Z^{-1} \exp\left[-\frac{mv^2/2 + U(x)}{k_B T}\right] \tag{11.4}$$

is stationary for (11.3).

Derivation.

The derivatives are $\partial_x \rho_{\mathrm{eq}} = -U'\rho_{\mathrm{eq}}/(k_B T)$ and $\partial_v \rho_{\mathrm{eq}} = -v\rho_{\mathrm{eq}}/\theta$. Therefore $-v\partial_x \rho_{\mathrm{eq}}$ cancels $(U'/m)\partial_v \rho_{\mathrm{eq}}$. Separately, $v\rho_{\mathrm{eq}} + \theta\partial_v \rho_{\mathrm{eq}} = 0$, so the two dissipative terms cancel. All four terms sum to zero. Finite Z supplies normalization. ∎

At equilibrium, the dissipative velocity current vanishes, but the reversible phase-space currents generally do not: $J_x = v\rho_{\mathrm{eq}}$ and the remaining velocity current is $-U'\rho_{\mathrm{eq}}/m$. Their divergences cancel. Requiring every current component to vanish would incorrectly remove Hamiltonian streaming. Equilibrium reversibility treats velocity as odd under time reversal; it is not the same statement as zero spatial and velocity current separately.

For harmonic U, (11.4) factors into independent Gaussians with $\mathbf{Var}(X) = \theta/\omega_0^2$, $\mathbf{Var}(V) = \theta$, and $\mathbf{Cov}(X, V) = 0$. Each quadratic energy contributes $k_B T/2$ to the mean. The free limit $\omega_0 \downarrow 0$ makes the position normalization diverge on the whole line, consistently recovering the absence of a free spatial equilibrium. A box would introduce a different domain and boundary problem.

Kramers's original paper, printed pages 287–289, constructs the phase-space diffusion equation, imposes stationarity of the Boltzmann distribution, and separates the large-viscosity approximation. Its notation sets mass and Boltzmann's constant to one. Restoring units and separating reversible from dissipative currents makes the assumptions in (11.3) explicit. Existence of a stationary candidate by substitution still does not establish convergence or a barrier escape rate.

11.3 Covariance proves relaxation in the linear model L14 27:09

Define the state column $Y_t = (X_t, V_t)^{\mathsf{T}}$, drift matrix $A = \begin{pmatrix} 0 & 1 \\ -\omega_0^2 & -\gamma \end{pmatrix}$, and noise column $B = (0, \sqrt{2\gamma\theta})^{\mathsf{T}}$. The matrix form is $dY = AY\,dt + B\,dW$. Its solution is $Y_t = e^{At}Y_0 + \int_0^t e^{A(t-s)}B\,dW_s$. For deterministic initial state the noise is Gaussian. Its covariance $C_{\mathrm{cond}}(t)$ is the integral of $e^{As}BB^{\mathsf{T}}e^{A^{\mathsf{T}}s}$ from zero to t.

Let $\Sigma = \mathbf{diag}(\theta/\omega_0^2, \theta)$, where $\mathbf{diag}$ denotes a diagonal matrix with the listed entries. Direct multiplication gives $A\Sigma + \Sigma A^{\mathsf{T}} + BB^{\mathsf{T}} = 0$, called the stationary covariance or Lyapunov equation. Differentiating $e^{At}\Sigma e^{A^{\mathsf{T}}t}$ and integrating proves

$$C_{\mathrm{cond}}(t) = \Sigma - e^{At}\Sigma e^{A^{\mathsf{T}}t}. \tag{11.5}$$

Stability established in Proposition 11.1 makes e^{At} tend to zero. The conditional mean tends to zero and $C_{\mathrm{cond}}(t)$ tends to Σ, so the Gaussian transition laws converge to the Gaussian equilibrium. This supplies the distributional relaxation argument missing from mean stability alone. For random initial data independent of future noise, the contracted initial contribution also vanishes in probability under ordinary finite-moment assumptions.

With equilibrium initial data, the lag covariance matrix is $e^{At}\Sigma$ for $t \geq 0$. Define normalized position covariance $f(t) = \mathbb{E}[X_t X_0]/\Sigma_{xx}$. It satisfies the damped mean equation with $f(0) = 1$ and $f'(0) = 0$. For unequal roots,

$$f(t) = \frac{-r_- e^{r_+ t} + r_+ e^{r_- t}}{r_+ - r_-}. \tag{11.6}$$

The numerator coefficients enforce both initial conditions. At critical damping, take the repeated-root limit or solve directly: $f(t) = (1 + \omega_0 t)e^{-\omega_0 t}$. For underdamping, an explicitly real form is $e^{-\gamma t/2}[\cos(\Omega t) + (\gamma/(2\Omega))\sin(\Omega t)]$. These formulas recover $f'(0) = 0$, consistent with differentiable position and zero equal-time cross-covariance. A scalar OU covariance instead has a cusp at zero. The inertial harmonic position is stationary Gaussian when initialized at equilibrium, but its covariance shows that it is generally not scalar Markov.

The equilibrium displacement MSD is $2\Sigma_{xx}[1 - f(t)]$. It approaches $2\theta/\omega_0^2$ for large lag, rather than growing indefinitely. The positional variance relative to a fixed origin approaches θ/ω_0^2. The factor of two arises because an equilibrium displacement subtracts two decorrelated random endpoints; it is not a disagreement about equipartition.

The same correlation area can conceal a different time law

The tracking case exposes a second possible false validation. Define the integrated normalized position covariance $\tau_x = \int_0^\infty f(t)dt$. Because the stable solution and its derivative decay exponentially, integrating $f'' + \gamma f' + \omega_0^2 f = 0$ gives $[f']_0^\infty + \gamma[f]_0^\infty + \omega_0^2\tau_x = 0$. The initial conditions make the first term zero and the second term $-\gamma$. Therefore $\tau_x = \gamma/\omega_0^2$ exactly, even when the motion is underdamped. The reduced exponential has the same integral.

Thus matching the equilibrium variance and this integrated covariance still does not establish the reduced process. At weak damping, the covariance oscillates slowly while positive and negative lobes cancel, producing a small signed area. A short signed correlation area is not necessarily rapid loss of all temporal dependence. The envelope lifetime is $2/\gamma$ in the oscillatory regime and becomes long when damping is weak. The high-friction position correlation time becomes large for a different reason: its slow real mode has rate approximately ω_0^2/γ.

The lab integrates the exact matrix-exponential covariance by independent quadrature. For $\omega_0 = 1$ and damping rates 0.5, 2, and 10, the areas are 0.5, 2, and 10, although their covariance shapes differ in Figure 11.1. An experiment that estimates only a covariance area would miss the discrepancy in the opening example. Retaining the lag dependence or measuring a frequency response tests more of the model.

Initial-state relaxation and equilibrium motion answer different questions

For deterministic initial state y_0, the conditional mean is $e^{At}y_0$ and conditional covariance is (11.5). For example, if the trap center is suddenly shifted relative to an initially displaced particle, the observed ensemble mean relaxes with both characteristic modes. Equilibrium time correlation instead averages over an initially random Gaussian state and follows the first column of $e^{At}\Sigma$. Its particular combination of modes is fixed by $f'(0) = 0$; a displaced deterministic state with arbitrary initial velocity can have a different combination. The roots belong to the dynamics, but amplitudes belong to the preparation and observable.

This distinction also explains the critical expression. The coefficient of $te^{-\omega_0 t}$ is set by a repeated root together with the initial derivative, rather than introduced as an ad hoc correction to an exponential fit. Matrix propagation handles that degeneracy continuously. An implementation based on separately evaluating the two nearly equal roots can lose numerical precision through subtraction even though the physical covariance is smooth in the damping parameter.

11.4 Eliminate velocity only after separating the time scales L14 30:21

Suppose $\varepsilon = \omega_0/\gamma \ll 1$. Expanding the square root in (11.2) gives the slow root $r_+ = -\omega_0^2/\gamma + O(\omega_0^4/\gamma^3)$ and fast root $r_- = -\gamma + \omega_0^2/\gamma + O(\omega_0^4/\gamma^3)$. Velocity adjusts over time $1/\gamma$ while position relaxes over time γ/ω_0^2. Their ratio is ε^2. The slow-root approximation is controlled by that small parameter, rather than merely by the absence of oscillations.

On slow time scales, the constant-friction reduction is the Smoluchowski equation associated with

$$dX_t = -\frac{U'(X_t)}{m\gamma}dt + \sqrt{\frac{2k_BT}{m\gamma}}dW_t. \tag{11.7}$$

For harmonic confinement it is a scalar OU process with decay rate ω_0^2/γ, stationary variance θ/ω_0^2, and normalized equilibrium covariance $e^{-\omega_0^2 t/\gamma}$. It preserves the Gibbs position density but loses the fast mode and differentiable microscopic position paths. It should not be used to predict sub-momentum-time slopes or crossing counts without a resolution qualification.

A more informative derivation than deleting an acceleration symbol uses the first two velocity moments of (11.3). Define position density $p(x,t) = \int \rho\, dv$, position probability flux $j(x,t) = \int v\rho\, dv$, and second moment density $M_2(x,t) = \int v^2\rho\, dv$. Integrating gives $\partial_t p = -\partial_x j$ and $\partial_t j = -\partial_x M_2 - (U'/m)p - \gamma j$. Local thermalization approximates $M_2 \approx \theta p$; neglecting the fast derivative of j yields $j \approx -[U'p/m + \theta\partial_x p]/\gamma$. Inserting this into continuity yields the forward equation of (11.7). The approximations are now visible: near-local Maxwellization, slow flux, and constant temperature and drag.

For rapidly varying forces, initial boundary layers, or position-dependent drag, those steps require further analysis. A Maxwell-shaped velocity marginal does not imply that the velocity distribution conditional on position is exactly local Maxwellian during transport. Kramers explicitly notes this distinction on printed page 289. Even in the linear trap, comparing (11.6) with the reduced covariance is a stronger dynamical check than matching histograms.

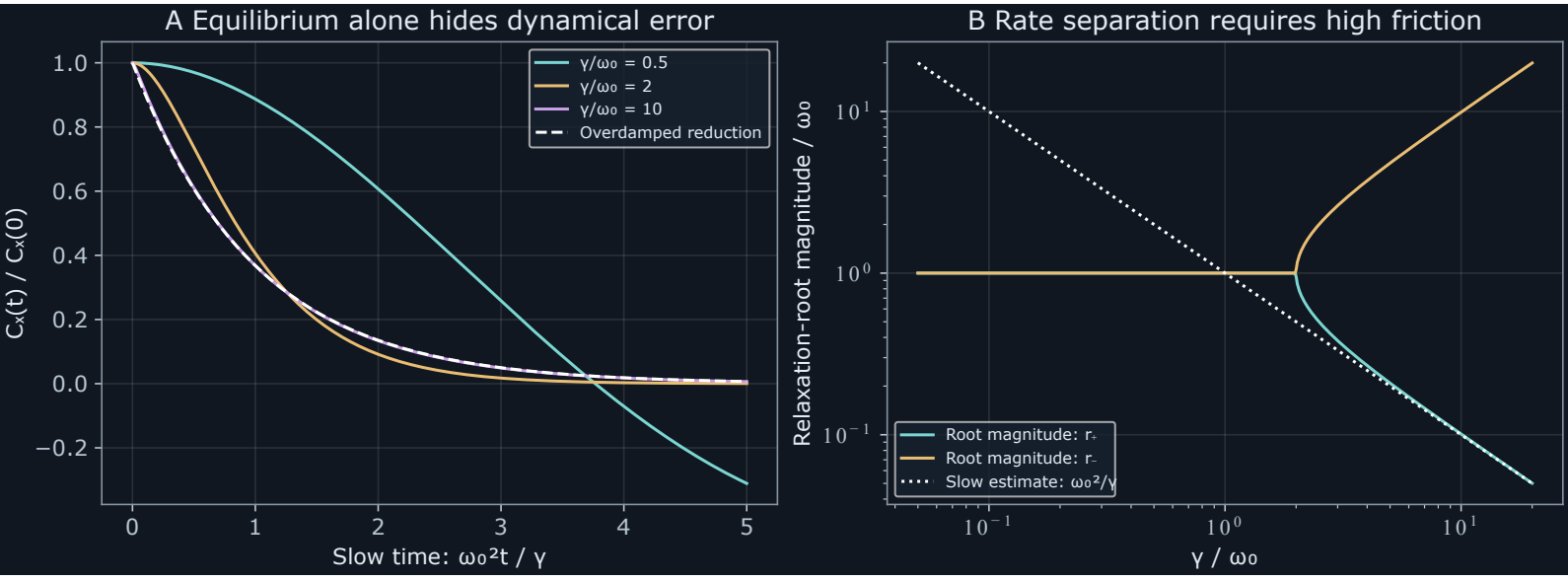


Figure 11.1. Equilibrium variance can agree while temporal correlations disagree. Panel A compares exact harmonic position covariance with the reduced slow exponential. Panel B plots characteristic-root magnitudes; in the oscillatory regime magnitude is not a decay rate, whose envelope is $\gamma/2$. High friction separates the real slow and fast roots. Generator.

11.5 Laboratory: test a reduction with correlations

The laboratory sets $\theta = \omega_0 = 1$ and compares damping rates 0.5, 2, and 10. It computes matrix exponentials rather than dividing by nearly coincident roots. An independent continuous Lyapunov solver must return the known equilibrium covariance. Numerical integration of the noise kernel must reproduce (11.5), and the critical case must reproduce $(1+t)e^{-t}$. A seeded exact Gaussian update also checks that equilibrium covariance is preserved in an ensemble of 200,000 states.

Run `python code/ch11_trap.py`. It makes no network calls and writes JSON results and a text report beside the script. The values below compare at one nominal slow relaxation time $t = \gamma/\omega_0^2$ for each model. Agreement at this single delay does not bound the entire covariance curve; Figure 11.1 supplies that broader comparison.

```
gamma  t       exact Cxx     overdamped Cxx
  0.5    0.5    0.88713672      0.36787944
  2.0    2.0    0.40600585      0.36787944
 10.0   10.0    0.36789873      0.36787944
```

```
import numpy as np
from scipy.linalg import expm


def position_correlation(t, gamma, omega0):
    if t < 0 or gamma <= 0 or omega0 <= 0:
        raise ValueError("invalid time, damping, or trap frequency")
    drift = np.array([[0., 1.], [-omega0**2, -gamma]])
    return expm(drift*t)[0, 0]
```

The function returns the dimensionless stationary normalized position covariance for positive model parameters. It neither simulates a path nor writes a file. Its matrix form handles underdamped, critical, and overdamped cases uniformly. The full lab independently tests its normalization and covariance consequences rather than treating agreement between two rearrangements of the same scalar formula as sufficient.

11.6 What confinement and reduction establish

Harmonic confinement admits an exactly soluble Gaussian phase-space model. It has finite equilibrium scatter and no long-range free diffusion. General confining potentials can have a Gibbs stationary density without Gaussian transients or a single relaxation time. Multiple wells introduce rare escape events whose scales are not obtained by replacing a well with one oscillator. The Kramers equation is the starting point for that analysis, not an automatic escape-rate formula.

The overdamped reduction preserves the equilibrium position distribution under its constant-friction thermal assumptions. It changes path regularity and short-time dynamics. Spatially varying noise or mobility requires a specified stochastic convention and a thermodynamically consistent drift; those issues cannot be resolved by substituting a variable $\gamma(x)$ into every constant-coefficient formula. Chapter 13 makes the noise-induced drift explicit. A magnetic Lorentz force is linear but velocity dependent and does no work; chapter 15 uses the same matrix machinery with a rotation term.

11.7 Exercises with worked solutions

Exercise 11.1 — Correct histogram, wrong model (conceptual)

Why does agreement with the harmonic Gibbs position histogram fail to validate a high-friction approximation?

Worked solution

Both the inertial model and its reduced scalar OU model have position variance θ/ω_0^2. Their one-time equilibrium laws coincide. Their two-time laws differ: the inertial normalized covariance has initial slope zero and may oscillate, while the reduced covariance has initial right slope $-\omega_0^2/\gamma$ and decays as one exponential. A histogram discards time ordering. Measuring correlations or response across momentum and trap scales tests the eliminated dynamics.

Exercise 11.2 — Is there an equilibrium current? (conceptual)

A student rejects (11.4) because $J_x = v\rho_{\mathrm{eq}}$ is not zero. Identify the missing distinction.

Worked solution

Stationarity requires the divergence of total probability current to vanish, not every component of current. Hamiltonian streaming has $J_x = v\rho_{\mathrm{eq}}$ and $J_v = -U'\rho_{\mathrm{eq}}/m$. Their divergences cancel by the derivatives used in Proposition 11.2. Drag and diffusion contribute equal and opposite dissipative currents. Time reversal reverses velocity, so reversible phase-space transport can coexist with equilibrium. Imposing zero current componentwise imports a purely overdamped criterion into an inertial state space.

Exercise 11.3 — Critical damping without a singular denominator (derive)

Find the critical normalized position covariance directly from its differential equation.

Worked solution

At $\gamma = 2\omega_0$, the equation has repeated root $-\omega_0$. Write $f(t) = (a + bt)e^{-\omega_0 t}$. The condition $f(0) = 1$ gives $a = 1$. Differentiating at zero gives $f'(0) = b - \omega_0 a = 0$, so $b = \omega_0$. Therefore $f = (1 + \omega_0 t)e^{-\omega_0 t}$. This is finite although the unequal-root expression appears to be zero divided by zero. For $\omega_0 = t = 1$, it gives $2/e = 0.73575888$, checked by the matrix exponential.

Exercise 11.4 — Derive the saturation of displacement scatter (derive)

For a stationary harmonic particle, derive the displacement MSD and its short- and long-time limits.

Worked solution

Expand $(X_t - X_0)^2$ and use equal stationary variances to obtain $2\Sigma_{xx} - 2\mathbb{E}[X_t X_0] = 2\theta[1 - f(t)]/\omega_0^2$. Since $f(0) = 1$, $f'(0) = 0$, and the differential equation gives $f''(0) = -\omega_0^2$, its short-time form is θt^2. It is ballistic, consistent with finite velocity variance. Stability makes $f(t) \to 0$, giving long-time limit $2\theta/\omega_0^2$. A zero-initial-position variance instead tends to one equilibrium variance, explaining the different factor.

Exercise 11.5 — Quantify slow-root error (compute/design)

For $\omega_0 = 1$ and $\gamma = 10$, estimate the slow root through the first correction and compare with the exact root.

Worked solution

Expanding $\sqrt{1 - 4/\gamma^2}$ through its quadratic term gives $r_+ = -1/\gamma - 1/\gamma^3 + O(\gamma^{-5})$, hence -0.101. The exact expression is $(-10 + \sqrt{96})/2$, approximately -0.1010205. The leading reduced rate is -0.1; its relative rate error is about one percent, consistent with the small parameter $\omega_0^2/\gamma^2 = 0.01$. The lab retains exact roots in JSON. This rate comparison does not uniformly bound covariance error at every time, because the exact covariance also includes a fast component and its amplitude.

Exercise 11.6 — Design a covariance-preserving simulator (compute/design)

Given step $h > 0$, equilibrium covariance Σ, and stable harmonic drift A, construct an exact Gaussian update and its strongest simple equilibrium check.

Worked solution

Compute $E = e^{Ah}$ and $C_{\mathrm{cond}} = \Sigma - E\Sigma E^{\mathsf{T}}$. Factor $C_{\mathrm{cond}} = LL^{\mathsf{T}}$ and set $Y_{n+1} = EY_n + LZ_n$, with independent two-component standard Gaussian columns Z_n. If Y_n has covariance Σ and is independent of Z_n, the next covariance is $E\Sigma E^{\mathsf{T}} + C_{\mathrm{cond}} = \Sigma$ exactly. Also check lag covariance $\mathbb{E}[Y_{n+1}Y_n^{\mathsf{T}}] = E\Sigma$: an incorrect independent resampling of equilibrium states would pass the equal-time test but fail this one. Use numerical quadrature of the noise kernel to verify C_{cond} independently and monitor roundoff at tiny steps.

11.8 Sources and attribution

- V. Balakrishnan, NPTEL lecture 14, course 115106089. The complete acquired caption text supplies the topic spine; selected stills support only partial visual checking.
- H. A. Kramers, Brownian motion in a field of force and the diffusion model of chemical reactions, *Physica* 7, 284–304 (1940), printed pages 287–289, especially equations (8)–(11); verified scan. This chapter uses its phase-space and large-viscosity distinctions, with independent modern derivations.

Wiener paths, roughness, and exceptional time statistics

A tracking algorithm estimates velocity by dividing a measured position increment by its sampling interval. For mathematical Brownian motion, refining that interval makes the estimated velocity variance diverge, even while the positions form a continuous path. Meanwhile the sum of squared increments settles toward a finite reproducible limit. The task is to understand these apparently incompatible observations and to distinguish properties of one path from distributions across an ensemble.

> The sequence follows NPTEL lecture 15: vector Langevin correlations, mathematical Brownian motion, covariance and path properties, occupation statistics, iterated logarithm, and the OU–Wiener relation. Proofs of finite-mesh relations and the OU time change are supplied here. Deeper path theorems are explicitly cited, not claimed proved by simulation.

12.1 From physical persistence to a mathematical path model L15 3:34

For an isotropic free thermal particle, each Cartesian velocity component follows the OU model of chapter 10. Write $\boldsymbol{V}_t = (V_{1,t}, V_{2,t}, V_{3,t})^{\mathsf{T}}$ and $\theta = k_B T/m$ for the equilibrium component variance. Independent component Wiener noises and an independent isotropic Maxwell initial velocity give $\mathbb{E}[V_{i,t} V_{j,0}] = \theta \delta_{ij} e^{-\gamma t}$ for $t \geq 0$, where the Kronecker symbol δ_{ij} is one for equal indices and zero otherwise. Summing diagonal entries gives $\mathbb{E}[\boldsymbol{V}_t \cdot \boldsymbol{V}_0] = 3\theta e^{-\gamma t}$.

A uniform magnetic field couples the velocity components through the Lorentz force. Independence of driving noises then does not imply independence of the outputs at different times. The drift matrix rotates velocity as it damps it, while leaving thermal speed variance unchanged. This opening bridge is developed fully in chapter 15. It illustrates why one should solve covariance dynamics rather than infer them from noise labels.

For time scales much longer than momentum relaxation, position is approximated by diffusion. Mathematical Brownian motion idealizes that approximation at every positive time scale. It is therefore a different path model from the time integral of continuous inertial velocity. Their long-time position laws can agree while their short-time differentiability differs. The finite-correlation physical velocity cannot be recovered by differentiating the ideal diffusion path.

12.2 Covariance encodes independent increments L15 18:07

Definition 12.1 — Standard Wiener process. A real process W_t, $t \geq 0$, has $W_0 = 0$, continuous paths with probability one, and independent increments such that $W_t - W_s$ is Gaussian with mean zero and variance $t - s$ for $t \geq s$. The standard normalization gives W square-root-time units.

The physical diffusion position is $\boldsymbol{X}_t = \boldsymbol{x}_0 + \sqrt{2D}\boldsymbol{W}_t$ for fixed initial position $\boldsymbol{x}_0$ and diffusion coefficient $D > 0$ of units length squared per time. Its increments have length units and variance $2D(t-s)$. The positive-

time transition density is the diffusion kernel $\rho(x, t \mid x_0) = (4\pi D t)^{-1/2} \exp[-(x - x_0)^2/(4Dt)]$. This kernel specifies transitions in addition to the path-continuity condition; a one-time Gaussian density alone does not define a process.

Proposition 12.1 — Nonstationary position, stationary increments. $\mathbb{E}[W_t W_s] = \min(t, s)$. For disjoint time intervals the increments are independent, and their distributions depend only on interval lengths.

Derivation.

For $t \geq s$, write $W_t = W_s + (W_t - W_s)$. The second term is independent of W_s with zero mean, so the covariance is $\mathbb{E}[W_s^2] = s$. Symmetry gives the minimum formula. The increment assertions are part of Definition 12.1, and Gaussian variance depends only on elapsed time. Conversely, for a centered jointly Gaussian process with this covariance, disjoint increments have zero covariance; joint Gaussianity makes them independent. A continuous version is still specified when discussing paths. ∎

The variance of W_t grows with t, so the process is not stationary. The increment $W_{t+h} - W_t$ has the same law for every starting time t, giving stationary increments. The two statements concern different random variables. In particular, $\mathbb{E}[(W_t - W_s)^2] = |t - s|$ is an increment variance, not the covariance of two displacements measured from time zero. That covariance is still $\min(t, s)$.

For the tracking algorithm, this distinction determines the variance of differences between estimates. Overlapping increments share noise and are correlated. If intervals of lengths h_1, h_2 overlap for duration a, their Wiener increments have covariance a. Divide by the interval lengths to obtain the covariance of estimated velocities. Treating overlapping increments as independent would understate uncertainty even if their individual histograms were correct.

12.3 Squared increments retain a finite accumulated effect

Fix an observation duration $t_* > 0$ and divide it into n equal cells of length $h = t_*/n$. Define increments $\Delta_j W = W_{jh} - W_{(j-1)h}$ and the quadratic-variation sum $Q_n = \sum_{j=1}^{n} (\Delta_j W)^2$. The phrase quadratic variation refers to the limit of these squared-increment sums as the mesh shrinks; it is not ordinary total variation, which uses absolute increments.

Proposition 12.2 — Mean-square quadratic variation. On the stated uniform partitions, $\mathbb{E}[Q_n] = t_*$ and $\operatorname{Var}(Q_n) = 2t_*^2/n$. Hence $Q_n \to t_*$ in mean square.

Derivation.

Each increment is an independent Gaussian of variance h. Its square has mean h and variance $3h^2 - h^2 = 2h^2$, using the Gaussian fourth moment. Independence adds the n square variances, giving $2nh^2 = 2t_*^2/n$, while the means add to $nh = t_*$. The expected squared error relative to t_* is exactly this variance and tends to zero. ∎

The finite-difference velocity $U_h = (W_{t+h} - W_t)/h$ has variance $1/h$. Its standard deviation grows like $h^{-1/2}$. Small increments have size of order $\sqrt{h}$, and dividing by h magnifies them. By contrast, squaring those increments gives order h, which accumulates to a finite value across order $1/h$ cells. This is the mechanism behind the noise-induced drift in Itô's rule: the quadratic term survives refinement.

For physical position $X = x_0 + \sqrt{2D}W$, quadratic variation is $2Dt_*$ and finite-difference velocity variance is $2D/h$. Thus an estimator can extract D from squared increments without assigning a finite instantaneous velocity to the same idealized path. Localization noise and camera integration change this estimator; its exact expectation here assumes direct observations of the diffusion position.

Mean-square convergence is an ensemble statement. It implies convergence in probability, but a statement about every sample path along every possible partition needs stronger hypotheses. Along the dyadic nested sequence $n = 2^k$, Chebyshev's bound gives $\Pr(|Q_{2^k} - t_*| > \epsilon) \le 2t_*^2/(2^k\epsilon^2)$. The sum over k is finite, so the Borel–Cantelli lemma gives almost-sure convergence along that sequence. Arbitrary path-dependent partitions are not covered by this argument.

12.4 Continuity does not supply a tangent L15 29:02

A function is locally Hölder continuous with exponent $\beta > 0$ if, near each point, its increment magnitude is bounded by a finite constant times the time separation raised to β. The constant and neighborhood may depend on the point and the path. A bound with smaller β is weaker at short separations. Differentiability with bounded derivative would imply an order-h increment bound, but failure of one sufficient Hölder criterion alone does not prove nondifferentiability.

Cited path results: Wiener paths are almost surely everywhere locally Hölder for every $\beta < 1/2$ and almost surely nowhere differentiable. Their zero set on $[0, 1]$ has Hausdorff dimension $1/2$. See Mörters–Peres, Corollary 1.20, Theorem 1.30, and Theorem 4.24, printed pages 18, 21, and 108. These theorems are not proved here.

The divergence of finite-difference variance is evidence of roughness across the ensemble; it is not by itself a proof that every time of almost every path lacks a tangent. Quantifiers matter. A theorem at each fixed deterministic time need not automatically hold simultaneously at all uncountably many times. The nowhere-differentiability theorem supplies that stronger statement.

Hausdorff dimension characterizes how the cost of coverings changes with scale: cover a set by intervals of small diameter, sum their diameters raised to an exponent, and locate the transition from divergent to vanishing covering cost. It is different from the fraction of ordinary time spent in a set. Indeed the zero set has zero Lebesgue measure, proved directly by $\mathbb{E}[\int_0^1 \mathbf{1}_{\{W_t=0\}} dt] = \int_0^1 \Pr(W_t = 0) dt = 0$, since each positive-time Gaussian has no atom at zero. The nonnegative random duration is therefore zero almost surely. A complicated uncountable set can occupy no ordinary time.

Continuity also makes each individual path bounded on every compact time interval. The Gaussian ensemble can attain arbitrarily large finite displacements with nonzero probability, but a realized continuous path does not become unbounded in every short interval. Confusing unbounded distribution support with a pathwise lack

of bounds produces a false physical picture. Refining Figure 12.1 reveals additional fluctuations within a finite vertical range.

12.5 Occupation fraction is a path functional L15 33:20

For a fixed duration $t_* > 0$, define $A = \int_0^{t_*} \mathbf{1}_{\{W_s>0\}} ds$ and the positive occupation fraction $F = A/t_*$. This adds all positive-time intervals, including separated excursions; it is not the waiting time until the first zero crossing. Scaling $W_{t_* u}/\sqrt{t_*}$ gives another standard Wiener process in u, because its mean, covariance, independent increments, and continuous paths satisfy Definition 12.1. Therefore F has a distribution independent of t_*.

Cited occupation theorem: F has density $[\pi\sqrt{f(1-f)}]^{-1}$ on $0 < f < 1$. This is the occupation arcsine law, Mörters–Peres, Theorem 5.28, printed page 138. Its proof is not supplied here.

The cumulative distribution follows by the substitution $f = \sin^2\vartheta$, with $0 \le \vartheta \le \pi/2$. Then $df = 2\sin\vartheta\cos\vartheta d\vartheta$ cancels the square-root denominator, giving $\Pr(F \le f) = 2\arcsin\sqrt{f}/\pi$. The density's endpoint singularities are integrable. They create large probability near zero and one without placing atoms at either endpoint.

Symmetry gives $\mathbb{E}[F] = 1/2$, but a path need not spend nearly half its time on each side. The probability $F \le 1/4$ is $2\arcsin(1/2)/\pi = 1/3$. By symmetry, $F \ge 3/4$ also has probability one third, leaving only one third between one quarter and three quarters. A sequence of independent fair signs would concentrate its occupation fraction around one half; Brownian signs have long excursion dependence. The mean alone conceals that dependence.

For the tracking algorithm, time samples from a single path are therefore not interchangeable with independent ensemble replicates. An unusually one-sided trace does not by itself establish a drift. A drift test must use the appropriate path law and observation window rather than treating all signs as independent coin flips.

12.6 The iterated logarithm describes an envelope, not confinement L15 38:26

At one fixed time, the Wiener standard deviation is $\sqrt{t}$. A statement about excursions along one path over arbitrarily long times asks a different question. The slowly changing logarithmic correction distinguishes those extremes from one-time standard deviations.

Cited iterated-logarithm theorem: in dimensionless standard Wiener units, almost surely

$$\limsup_{t\to\infty} \frac{W_t}{\sqrt{2t\log\log t}} = 1, \qquad \liminf_{t\to\infty} \frac{W_t}{\sqrt{2t\log\log t}} = -1. \tag{12.1}$$

See Mörters–Peres, Theorem 5.1 and Remark 5.2, printed page 119; the small-time counterpart uses $\log\log(1/t)$, Corollary 5.3, page 121.

The logarithms require a dimensionless argument and a domain where their iterated value is positive. For dimensional time, choose a reference time before applying the expression. The theorem does not say the path remains near the two envelope curves. Its upper statement means that, for each positive tolerance, the path eventually lies inside the slightly enlarged envelope and reaches arbitrarily close to its upper edge at arbitrarily late times; the lower statement is symmetric. Between those times it can pass through the interior, including zero. A finite simulated path cannot certify a limit over unbounded times.

There is no contradiction with the Gaussian tail at a fixed time. The theorem selects exceptional times depending on the realized path; it does not replace the fixed-time distribution by a larger variance. This separation of pathwise extrema from ensemble scatter is essential when interpreting a long drifting-looking trace without imposing a deterministic trend.

12.7 The OU process is a scaled Wiener clock L15 41:39

Let Y_t solve $dY_t = -\gamma Y_t dt + b\,dW_t$ with deterministic $Y_0 = 0$, where $\gamma, b > 0$. Its zero-mean covariance, from the common-noise integral, is $b^2[e^{-\gamma|t-s|} - e^{-\gamma(t+s)}]/(2\gamma)$. Define the increasing clock $q(t) = (e^{2\gamma t} - 1)/(2\gamma)$ and another standard Wiener process B_u.

Proposition 12.3 — Zero-initial OU time change. The processes Y_t and $be^{-\gamma t}B_{q(t)}$ have the same finite-dimensional distributions.

Derivation.

Both are centered Gaussian processes. For $t \geq s$, the second process has covariance $b^2 e^{-\gamma(t+s)} \min(q(t), q(s)) = b^2 e^{-\gamma(t+s)}(e^{2\gamma s} - 1)/(2\gamma)$. Expanding gives exactly the covariance displayed above. Every Gaussian finite-dimensional vector is determined by its mean and covariance, proving the assertion. ∎

The subtraction of one in the clock enforces $q(0) = 0$. Omitting it changes the initial law. To represent a stationary OU process with variance θ, use dimensionless Brownian clock $\widetilde{B}$ and $Y_t = \sqrt{\theta} e^{-\gamma t}\widetilde{B}(e^{2\gamma t})$ in law. At time zero, $\widetilde{B}(1)$ is already a unit Gaussian; the covariance is $\theta e^{-\gamma|t-s|}$. The two representations answer different initial-conditioning questions.

Equality in law does not identify two arbitrary independently simulated paths. A coupling can be constructed from $B_{q(t)} = \int_0^t e^{\gamma s} dW_s$, whose increments are independent Gaussian with variance increments of q. That constructs a Brownian process on the clock's range using the same underlying noise. Without such a construction, matching laws permits distributional comparisons, not pointwise subtraction of realizations.

Smooth nonvanishing rescaling and smooth monotone time change retain local roughness. Finite OU correlation time does not make its velocity path differentiable. Inertial position is smoother because it integrates that continuous velocity once more. Time correlation and differentiability therefore measure different forms of regularity.

12.8 Laboratory: refine the same paths

The laboratory generates 6,000 independent standard Wiener meshes on $[0, 1]$, each with 2,048 cells and seed 2026110812. Coarser increments are sums of the same finest increments. This nesting compares resolution while keeping the realized paths coupled; independently redrawing each mesh would add unnecessary ensemble variation to the visual comparison. It checks quadratic-variation mean and variance against Proposition 12.2 and compares velocity variance with inverse step length.

Run `python code/ch12_wiener.py`. The program performs no network calls and writes numerical JSON and text output beside its source. It also checks the OU covariance clock algebra at unequal times. The occupation-time comparison is a finite-mesh numerical illustration with sampling and discretization error; its tolerance does not certify the cited theorem.

```
cells   mean QV   variance QV   velocity variance
   16   1.005521   0.13266246           16.08832
   64   0.995301   0.03102905           63.69926
  256   0.997707   0.00797148          255.41292
 1024   0.999141   0.00192018         1023.12038
 2048   0.999465   0.00096505         2046.90449
```

The quadratic-variation scatter shrinks while apparent velocity scatter grows. These trends test different predictions from the same increment law. Figure 12.1 pairs a path-resolution view with ensemble quadratic-variation variance. Neither a straight-line interpolant nor a finite sampled mesh reproduces all properties of an ideal continuous Brownian path.

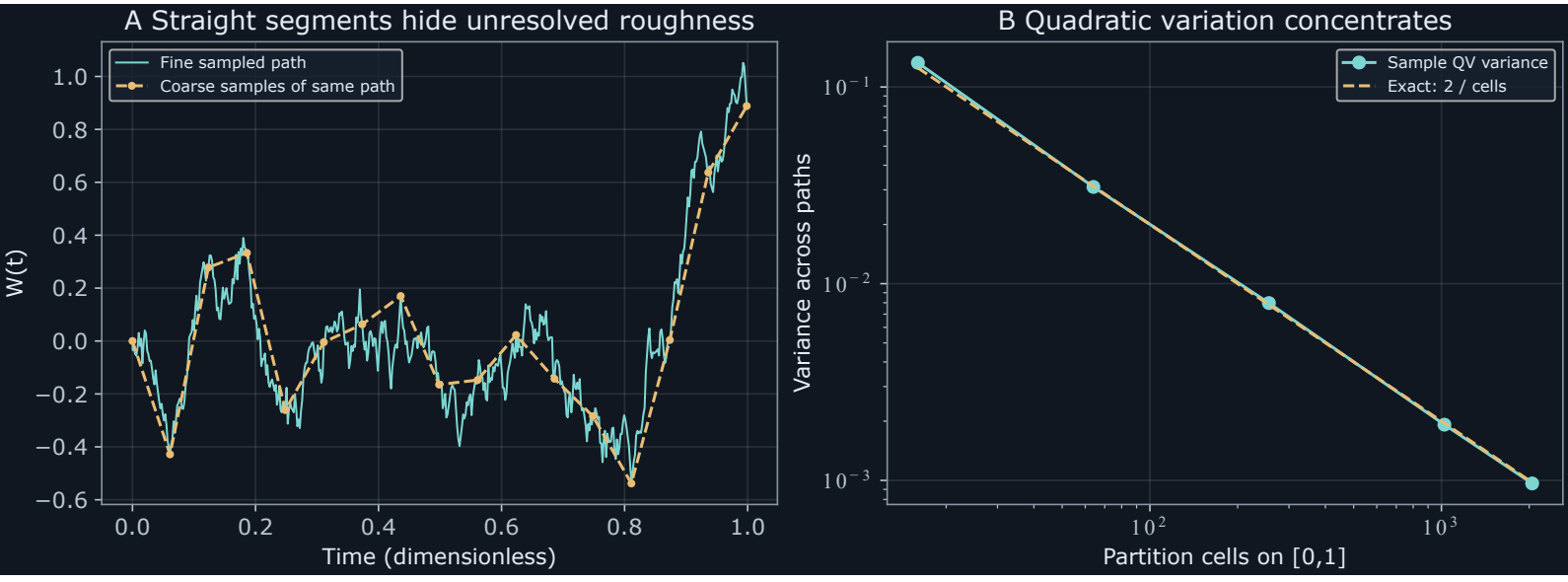


Figure 12.1. Refinement exposes roughness while stabilizing accumulated squared increments. Panel A draws two meshes of one synthetic realization. Panel B compares ensemble variance with the exact $2/n$ law on a unit interval. Lines joining observations are plotting interpolants. Generator.

```
import numpy as np


def coarsen_increments(fine, cells):
    paths, total = fine.shape
    if cells <= 0 or total % cells:
```

```
        raise ValueError("coarse cell count must divide fine cell count")
    return fine.reshape(paths, cells, total//cells).sum(axis=2)
```

The function accepts a two-dimensional array of increments, with one path per row, and an integer coarse cell count dividing the fine count. It returns sums preserving each path's endpoint. It draws no new randomness and writes no files. It is a coarsening operation, not a procedure for reconstructing unresolved fine increments from a coarse record.

12.9 Which conclusions survive physical resolution?

A measured trajectory has finite sampling rate and finite spatial precision. Its apparent derivatives, crossings, and variation statistics depend on those resolutions and on filtering. Mathematical Wiener motion is an ideal process whose increment law persists under arbitrary refinement. Inertial position behaves ballistically before momentum relaxation and has a continuous velocity; neither model describes infinitely fast real molecular dynamics.

The proved results concern Gaussian covariance, finite-mesh moments, and explicitly conditioned OU laws. The Hölder, nowhere-differentiability, zero-set dimension, arcsine, and iterated-logarithm statements are deeper cited theorems. A graph or histogram illustrates them but supplies no proof. Distinguishing pointwise support, compact-interval path boundedness, asymptotic envelopes, and occupation duration prevents superficially similar claims from being substituted for one another.

12.10 Exercises with worked solutions

Exercise 12.1 — Stationary increments do not imply stationarity (conceptual)

Compare W_1, W_4, $W_2 - W_1$, and $W_5 - W_4$. Identify equal and unequal laws.

Worked solution

The first two are centered Gaussians with variances one and four, so their laws differ. The last two are centered Gaussians with variance one and hence agree in law. They are also independent because their intervals are disjoint. The absolute process age changes its marginal variance, while the distribution of an increment depends only on interval length. Equal increment laws do not erase the distinguished initial condition $W_0 = 0$.

Exercise 12.2 — Continuous but unbounded? (conceptual)

A Gaussian increment has unbounded support. Does one continuous Brownian path attain arbitrarily large magnitude within $[0, 1]$?

Worked solution

No. A continuous real function on a compact interval attains a finite maximum and minimum. The bounds are random and need not have a deterministic uniform bound across the ensemble. For each fixed large threshold there are paths exceeding it with nonzero probability, which concerns the distribution across paths. The

quantifiers “for each threshold, some paths” and “one path, every threshold” cannot be exchanged. Nowhere differentiability concerns local slopes, not infinite position amplitude on a compact interval.

Exercise 12.3 — A partition with unequal cells (derive)

For deterministic partition lengths $h_1, \dots, h_n$ summing to t_*, derive the mean and variance of the squared-increment sum and a sufficient mesh condition for mean-square convergence.

Worked solution

Independent Gaussian increments have square means h_j and square variances $2h_j^2$. Thus the sum has mean t_* and variance $2\sum_j h_j^2$. Since $\sum_j h_j^2 \le (\max_j h_j)\sum_j h_j = t_* \max_j h_j$, a maximum cell length tending to zero makes the variance tend to zero. This proves mean-square convergence on deterministic shrinking meshes. It does not establish the same conclusion for arbitrary partitions selected using future path values, where increment independence can fail.

Exercise 12.4 — The occupation variance (derive)

Use the cited arcsine density to compute $\mathbb{E}[F^2]$ and $\mathrm{Var}(F)$.

Worked solution

With $f = \sin^2\vartheta$, the second moment is $(2/\pi)\int_0^{\pi/2}\sin^4\vartheta\, d\vartheta$. The identity $\sin^4\vartheta = (3 - 4\cos 2\vartheta + \cos 4\vartheta)/8$ gives the integral $3\pi/16$, hence $\mathbb{E}[F^2] = 3/8$. Symmetry gives mean one half, so variance is $3/8 - 1/4 = 1/8$. Its independence of observation duration is consistent with Brownian scaling. It is incompatible with an independent-sign sampling model whose fraction variance would shrink like the reciprocal sample count.

Exercise 12.5 — A required quadratic-variation precision (compute/design)

On a unit interval, how many equal cells make the exact root-mean-square error of one quadratic-variation sum at most 0.01? How does averaging independent paths change the answer?

Worked solution

Proposition 12.2 gives RMS error $\sqrt{2/n}$. Requiring it at most 0.01 gives $n \ge 20{,}000$. Averaging M independent path sums divides the variance by M, so the condition becomes $nM \ge 20{,}000$. Resolution and ensemble replication can both reduce this particular estimator's variance. They are not interchangeable for resolving individual-path features, and finite camera blur or localization error can add bias that this ideal calculation omits.

Exercise 12.6 — Detect a missing clock offset (compute/design)

Someone represents a zero-initial OU process by $be^{-\gamma t}B_{e^{2\gamma t}/(2\gamma)}$. Give two immediate diagnostics and the repair.

Worked solution

At $t = 0$, the proposed process has variance $b^2/(2\gamma)$ instead of zero. At general times its covariance is $b^2 e^{-\gamma|t-s|}/(2\gamma)$, which is stationary and lacks the zero-initial subtraction term. Both diagnostics identify a different initial ensemble rather than a small numerical error. Replace the clock by $(e^{2\gamma t} - 1)/(2\gamma)$. Match the resulting covariance at unequal times and check its zero-time value; testing only long-time variance would miss the defect.

12.11 Sources and attribution

- V. Balakrishnan, NPTEL lecture 15, course 115106089; source of topic ordering and physical bridges.
- P. Mörters and Y. Peres, Brownian Motion, author-hosted manuscript. Exact checked theorem and page locators appear beside the cited statements. The chapter supplies its own finite-mesh, covariance, clock, and numerical derivations.

Itô transformations, radial diffusion, and multiplicative noise

A microscope tracks three independent diffusing coordinates and reports their distance from the origin. There is no Cartesian force, yet the radius has an outward drift. Squaring a radial stochastic equation with the ordinary chain rule then predicts the wrong growth of mean squared distance. The task is to transform a noisy observable while preserving the probability law, including drift terms generated by unresolved fluctuations.

> Based on NPTEL lecture 16: Itô/Fokker–Planck correspondence, radial and squared-radial diffusion, powers, exponential transformations, and geometric Brownian motion. General-rule explanation, singular-boundary qualifications, numerical tests, and worked exercises extend the recording. The financial connection is a model distinction rather than a price forecast.

13.1 A second-order term survives mesh refinement L16 0:50

Let X_t solve the scalar Itô equation $dX_t = a(X_t, t)dt + b(X_t, t)dW_t$, where W is a standard Wiener process. The drift a has units of X per time and the noise amplitude b has units of X per square root of time. Assume a nonexplosive adapted continuous solution, locally regular coefficients, and the local integrability needed by the displayed integrals. Adapted means that the coefficient used at a given time depends only on information available by that time. The Itô stochastic integral is defined through left-endpoint, nonanticipating sums and their probability limits.

Take a coordinate function $g(x, t)$ with one continuous time derivative and two continuous space derivatives. Its Taylor expansion contains $g_x\Delta X + g_t\Delta t + g_{xx}(\Delta X)^2/2$. Wiener increments are order $\sqrt{\Delta t}$ in mean square. Their squares accumulate to elapsed time by quadratic variation. Mixed products of dt with dW and squares of dt vanish under refinement, while $(dW)^2$ contributes dt. These are rules for accumulated stochastic limits, not literal algebra on ordinary infinitesimals.

Proposition 13.1 — Itô transformation rule. Under the stated regularity and integrability hypotheses,

$$dg(X_t, t) = \left(g_t + ag_x + \frac{b^2}{2}g_{xx}\right)dt + bg_x dW_t. \tag{13.1}$$

Derivation.

The Taylor sums have linear-in-noise term converging to $\int bg_x dW$, finite-variation terms converging to $\int (g_t + ag_x)dt$, and quadratic-noise term converging to $\int b^2 g_{xx}dt/2$. On compact stopped ranges, continuity controls Taylor remainders by a vanishing modulus of continuity times bounded quadratic-variation sums. Localization removes that stopping under the assumed nonexplosion and integrability. This gives the mechanism and proof framework; the construction of the stochastic integral and weighted quadratic-variation

limit is the underlying theorem. A precise Brownian version and proof are given in Mörters–Peres, Theorems 7.12–7.14, printed pages 196–198. ∎

For $dX = \sqrt{2D}dW$ with $D > 0$ and $g(x) = x^2$, the rule gives $d(X^2) = 2X\sqrt{2D}dW + 2Ddt$. The second derivative term generates the positive mean-square growth. Discarding it would predict a zero mean square from zero initial position, despite the exact Gaussian variance $2Dt$.

The corresponding forward equation is $\partial_t \rho = -\partial_x(a\rho) + \frac{1}{2}\partial_x^2(b^2\rho)$. The entire product is differentiated when b varies with position. A Stratonovich equation uses midpoint rather than left-endpoint sums and admits the ordinary chain rule, but its drift must be changed to represent the same process: $a_{\text{Itô}} = a_{\text{Strat}} + b\partial_x b/2$ for one smooth scalar noise. Changing the convention while retaining the same drift generally changes the law.

13.2 Radius has an entropic outward drift L16 5:06

Let d be a positive integer and let the position vector $\boldsymbol{X}_t$ have independent coordinates $dX_{i,t} = \sqrt{2D}dW_{i,t}$ for $i = 1, \ldots, d$, with independent standard Wiener processes. Define radius $R_t = (\sum_i X_{i,t}^2)^{1/2}$. Away from $R = 0$, the derivatives of the radius are $\partial_i R = X_i/R$ and $\partial_i^2 R = 1/R - X_i^2/R^3$. Summing the second derivatives gives the Laplacian $\nabla^2 R = (d-1)/R$.

Proposition 13.2 — Radial equation away from the origin. Before encountering $R = 0$,

$$dR_t = \frac{D(d-1)}{R_t}dt + \sqrt{2D}dB_t. \tag{13.2}$$

Here B is the radial projected Wiener driver, not generally any one Cartesian driver.

Derivation.

The multidimensional Itô rule gives drift $D\nabla^2 R = D(d-1)/R$ and noise $\sqrt{2D}\sum_i (X_i/R)dW_i$. Define this projected integral through zero by taking the direction with components $H_i = X_i/R$ when R is positive and any fixed unit vector when $R = 0$. The bounded unit-length integrand makes it a continuous square-integrable martingale: an adapted process whose conditional expected future value, given its past information, equals its current value. It starts at zero and has quadratic variation $\int \sum_i H_i^2\, dt = t$ after this unit-vector completion. The quoted Lévy characterization says that a continuous martingale starting at zero with quadratic variation t is a standard Wiener process; see Bingham, Lecture23, printed p.3, Lévy theorem. It therefore identifies this driver as B. The radius equation itself is still asserted only away from zero. Equivalently, its generator acting on a smooth radial test function is $D[\partial_R^2 + (d-1)\partial_R/R]$, which gives the same local drift and diffusion. ∎

The radial drift reflects geometry, not a physical force doing extra work. More displacement directions increase radius than decrease it; angular degrees of freedom are discarded when radius alone is reported. For $D = 0.5$ and $d = 3$, the drift is $1/R$. It grows near the coordinate singularity, where a naive Euler step is especially unreliable. Figure 13.1 compares dimensions two and three.

For an origin-started vector, the Cartesian density is $(4\pi Dt)^{-d/2}e^{-R^2/(4Dt)}$ per unit d-dimensional volume. The radial density per unit radius is instead

$$\rho_R(r,t) = \frac{2\pi^{d/2}}{\Gamma(d/2)} r^{d-1}(4\pi Dt)^{-d/2}e^{-r^2/(4Dt)}, \qquad r > 0. \tag{13.3}$$

The prefactor before r^{d-1} is the surface area of the unit sphere in d dimensions; Γ is Euler's gamma function. This integrates angular volume and exposes the extra outward multiplicity. Substitution $z = r^2/(4Dt)$ reduces normalization to the gamma integral. In two dimensions (13.3) is the Rayleigh density $re^{-r^2/(4Dt)}/(2Dt)$. It vanishes at the origin although the Cartesian density is maximal there. The density's reference measure matters.

The radius equation is singular at zero. The coordinate construction defines its physical nonnegative version without arbitrary boundary choices. In $d = 1$, $R = |X|$ reflects at zero; simply deleting the $(d-1)/R$ drift does not give an unconstrained Brownian radius. Tanaka's formula adds local time: $d|X| = \operatorname{sign}(X)\sqrt{2D}dW + dL^0_X$, in the convention $|X_t| = |X_0| + \int_0^t \operatorname{sign}(X_s)dX_s + L^0_X(t)$. The general nonsmooth rule is cited in Theorem 7.33, printed page 210. Local time is the boundary accumulation term, not ordinary duration at zero.

13.3 Squared radius removes the coordinate denominator L16 18:20

Define $Z_t = R_t^2 = \sum_i X_{i,t}^2$. This smooth Cartesian observable can be handled even when radius itself is singular. Applying the square rule to each coordinate and adding gives deterministic drift $2ddDdt$ and noise $2\sqrt{2D}\sum_i X_i dW_i$. The latter has quadratic variation $8DZ_t dt$. Thus the nonnegative squared-radius process satisfies

$$dZ_t = 2ddDdt + \sqrt{8DZ_t}\, dB_t. \tag{13.4}$$

A projected Wiener driver can be completed through zero using the Cartesian construction. Its sign or orientation need not be the same driver as the original coordinates. For $d \geq 2$ and away from zero, applying Itô directly to R^2 gives $2R\,dR + (dR)^2$. The first term contributes $2D(d-1)dt$ and the second contributes $2Ddt$. Their sum is $2ddDdt$. Ordinary-chain-rule multiplication omits precisely the final $2D$ term.

Taking expectations of the Cartesian squares avoids integrability ambiguities in the singular radius equation: for deterministic origin start, $\mathbb{E}[Z_t] = 2ddDt$. Since independent Gaussian squares each have variance $2(2Dt)^2$, $\mathbf{Var}(Z_t) = 8dD^2t^2$. The law is $Z_t/(2Dt)$ equal in distribution to chi-square with d degrees of freedom, the sum of d independent standard Gaussian squares. At $D = 0.5, d = 3, t = 1$, mean squared radius is three and its variance is six, both checked numerically.

To state standard names without a scale ambiguity, $R_t/\sqrt{2D}$ is the dimension-d Bessel process built from standard coordinate Brownian motion, and $Z_t/(2D)$ is squared Bessel with drift d and noise amplitude $2\sqrt{Z/(2D)}$ in its own variable. Naming a process without its diffusion normalization can hide a factor of two. The forward equation for density of Z is $\partial_t \rho_Z = -2ddD\partial_z \rho_Z + 4D\partial_z^2(z\rho_Z)$, on a nonnegative domain with boundary behavior inherited from the coordinate process.

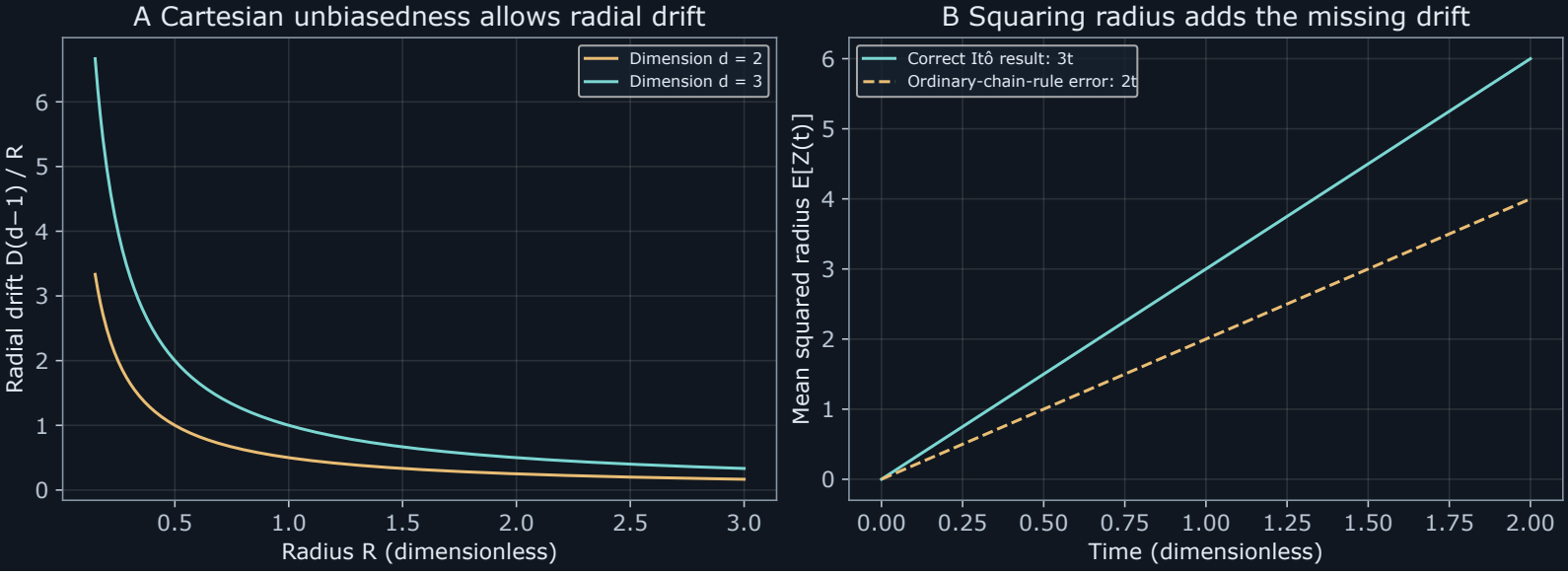


Figure 13.1. Geometry and quadratic variation both contribute drift. Panel A uses $D = 0.5$. Panel B uses dimension three and origin start, so correct mean squared radius is $3t$; squaring the radial equation with the ordinary chain rule would give $2t$. All inputs are synthetic model parameters. Generator.

13.4 Powers and exponentials test the rule L16 33:57

For $dX = \sqrt{2D}dW$ and a positive integer power n, (13.1) gives $d(X^n) = Dn(n-1)X^{n-2}dt + n\sqrt{2D}X^{n-1}dW$. At $n = 1$ the drift vanishes; at $n = 2$ it is $2D$. The drift couples the nth moment to the $(n-2)$th moment after averaging, provided the stochastic integral is integrable. Starting at zero, this recursion produces the familiar even Gaussian moments and vanishing odd moments. The stochastic equation for a transformed observable need not close solely in that observable when the transformation discards relevant information; inverse branches and symmetry must be examined.

Let $\ell > 0$ be a length scale and set $Y = e^{X/\ell}$ so its exponent is dimensionless. Its derivatives are $g_x = Y/\ell$ and $g_{xx} = Y/\ell^2$. Hence $dY = (D/\ell^2)Ydt + (\sqrt{2D}/\ell)YdW$. An unbiased Cartesian position produces a positively drifting exponential observable. This is a transformation effect: the exponential's convexity weights positive excursions more strongly than negative ones, not an added force on the particle.

The inverse transformation is single-valued for $Y > 0$. Starting from a Cartesian density ρ_X, its density is $\rho_Y(y) = \ell\rho_X(\ell \log y)/y$. The factor $1/y$ is the Jacobian. In contrast, transforming X to X^2 needs both preimages $\pm\sqrt{z}$, as already established in chapter 2. Itô calculus and density transformation give compatible laws when the support and branches are handled consistently.

13.5 Multiplicative motion remains positive L16 43:19

Let S_t be a positive quantity of specified units, with constant fractional drift a_S in inverse time and fractional noise amplitude b_S in inverse square root of time. Given $S_0 > 0$, consider $dS_t = a_S S_t dt + b_S S_t dW_t$. Applying (13.1) to the dimensionless logarithm $\log(S_t/S_0)$ gives drift $a_S - b_S^2/2$ and noise amplitude b_S. Integrating and exponentiating yields

Proposition 13.3 — Geometric Brownian solution.

$$S_t = S_0 \exp[(a_S - b_S^2/2)t + b_S W_t]. \tag{13.5}$$

It stays positive at every finite time almost surely and has mean $S_0 e^{a_S t}$.

Derivation.

The logarithmic derivation gives the candidate, and direct Itô differentiation of the exponential verifies its original drift and noise, including the cancellation of the half-variance correction. A continuous Wiener path is finite on compact intervals, so the exponential is strictly positive there. Finally W_t is Gaussian of variance t, so $\mathbb{E}[e^{b_S W_t}] = e^{b_S^2 t/2}$; multiplication by the deterministic factor gives the stated mean. ■

The median is $S_0 e^{(a_S - b_S^2/2)t}$, while variance is $S_0^2 e^{2a_S t}(e^{b_S^2 t} - 1)$. A growing mean can coexist with a falling median because rare large realizations dominate the average. At dimensionless $S_0 = t = 1$, $a_S = 0.1$, and $b_S = 1.2$, mean is 1.10517092 and median 0.53794444. Interpreting a single typical trajectory as the ensemble expectation fails even in this elementary model.

An exact logarithmic step is multiplication by $\exp[(a_S - b_S^2/2)h + b_S\sqrt{h}\,Z]$ with unit Gaussian Z. Euler multiplication by $1 + a_S h + b_S\sqrt{h}\,Z$ can be negative. For the same parameters and $h = 1$, that probability is 0.17965867, computed from the Gaussian lower tail. The numerical discretization has broken a structural property of the exact process, even though its one-step expected value is reasonable.

13.6 A price-process assumption is not an option-pricing theory L16 43:43

A positive asset-price model can use (13.5), but transforming a Wiener process supplies no rule for valuing a contingent payoff. Black and Scholes add a traded riskless asset, continuous hedging, and market assumptions to derive a pricing equation. Their original paper, printed pages 640–642, explicitly assumes constant interest and fractional variance rates, no dividends, frictionless trading, borrowing, and short selling. Itô's rule supplies the payoff's stochastic differential; the hedged-portfolio argument supplies the pricing constraint. A forward probability equation with a real-world drift is a different object from that pricing equation. This distinction limits what the mathematical transformation establishes and prevents treating a tractable price law as an empirically validated market model.

13.7 Laboratory: preserve transformed moments

The laboratory makes three independent checks. First it uses the exact finite-sum identity $X_n^2 - X_0^2 = 2\sum_j X_j \Delta X_j + \sum_j (\Delta X_j)^2$, so the missing chain-rule term is visible on each sampled path. Second it constructs radial motion directly from independent Gaussian coordinates and compares squared-radius mean and variance with their exact values. Third it checks geometric Brownian mean by Gaussian quadrature and Monte Carlo and measures the negative-value defect of one-step Euler.

Run `python code/ch13_transform.py`. It uses seed 2026110813 and 100,000 samples, makes no network calls, and writes JSON and text output beside the source. It intentionally avoids Euler simulation of the singular radial equation. Cartesian sampling preserves its nonnegative radius and correct origin behavior automatically.

```
Mean squared-increment correction: 0.99925612
dimension   sample mean R²   exact mean R²
        1       1.00103222      1.00000000
        2       2.00075036      2.00000000
        3       3.00118606      3.00000000
GBM exact mean=1.10517092, median=0.53794444
One-step Euler negative probability=0.17965867
```

```
import numpy as np


def geometric_step(s, h, drift, noise, normal):
    if np.any(np.asarray(s) <= 0) or h < 0:
        raise ValueError("positive s and nonnegative h required")
    return s*np.exp((drift-noise**2/2)*h + noise*np.sqrt(h)*normal)
```

The caller supplies standard-normal draws and positive initial values. The function returns the exact constant-coefficient step and writes no files. At extreme exponents, floating-point overflow or underflow remains possible; mathematical positivity does not guarantee representable machine values. Storing logarithms is appropriate when that range matters.

13.8 Transformations do not remove boundary and state questions

Itô's correction is determined by noise covariance and coordinate curvature. It does not create extra energy or select an interpretation without a specified integral convention. Radius, squared radius, and exponential coordinates inherit the original probability law only when inverse branches and boundary conditions are retained. Smooth invertible transformations preserve a Markov state; a many-to-one observation needs an additional closure argument. Radial Brownian symmetry supplies one here, while a general projection need not.

At zero radius the smooth formula is local and requires its coordinate construction or boundary extension. At zero multiplicative state the positive-logarithm transform is unavailable; the positive-initial exact solution never reaches zero at finite time. The same notation can therefore describe different boundary problems unless initial support and admissible extensions are stated.

13.9 Exercises with worked solutions

Exercise 13.1 — Outward drift without a force (conceptual)

Does positive radial drift imply positive mean Cartesian displacement or extra kinetic-energy injection?

Worked solution

No. Each Cartesian increment remains centered and its origin-started mean is zero. Radius discards direction and is nonlinear, so opposite Cartesian displacements do not cancel in radius. The $D(d-1)/R$ drift represents

angular multiplicity in this reduced observable. Its squared-radius growth $2dDt$ is exactly the sum of Cartesian variances, with no additional source. A physical potential would enter the Cartesian drift separately.

Exercise 13.2 — Radius in one dimension (conceptual)

Why is $dR = \sqrt{2D}dB$ an incomplete global equation for $R = |X|$?

Worked solution

Unconstrained Brownian motion crosses below zero, whereas a radius cannot. The derivative of $|x|$ is nonsmooth at zero, so the smooth Itô formula used away from zero does not cover returns to that point. Tanaka's local-time term supplies reflection in the coordinate construction. The squared radius avoids this derivative singularity and obeys (13.4) with $d = 1$, drift $2D$, and noise amplitude $\sqrt{8DZ}$. Dropping a singular term locally does not determine a global boundary behavior.

Exercise 13.3 — A fourth moment from Itô (derive)

For origin-started $dX = \sqrt{2D}dW$, derive $\mathbb{E}[X_t^4]$ without assuming its Gaussian fourth moment in advance.

Worked solution

At $n = 4$, the power rule gives $d(X^4) = 12DX^2dt + 4\sqrt{2D}X^3dW$. Its stochastic integral has zero mean under the finite Gaussian moments of this process. Since the square rule gives $\mathbb{E}[X_s^2] = 2Ds$, integrate $d\mathbb{E}[X_t^4]/dt = 24D^2t$ from zero to obtain $12D^2t^2$. This equals three times the squared variance $(2Dt)^2$, independently confirming the Gaussian fourth-moment relation through moment dynamics.

Exercise 13.4 — Recover squared-radius variance from its generator (derive)

Use (13.4) to derive the second moment of Z and its variance.

Worked solution

Apply the square rule again: $d(Z^2) = [4dDZ + 8DZ]dt + 2Z\sqrt{8DZ}dB$. Thus $d\mathbb{E}[Z^2]/dt = 4D(d+2)\mathbb{E}[Z] = 8d(d+2)D^2t$. Origin start gives $\mathbb{E}[Z^2] = 4d(d+2)D^2t^2$. Subtracting $(2dDt)^2$ leaves $8dD^2t^2$. The independent-coordinate calculation in the laboratory supplies a different check of both drift and noise normalization.

Exercise 13.5 — Diagnose the missing squared-increment term (compute/design)

For a sampled path with $D = 0.5$ on a unit interval, compare the ordinary chain sum with the actual squared endpoint.

Worked solution

Compute $2\sum X_j \Delta X_j$ at left endpoints and subtract it from $X_n^2 - X_0^2$. The difference is exactly $\sum(\Delta X_j)^2$, up to rounding. Its expected value is one at every deterministic mesh resolution, and its variance is $2/n$. The lab gives ensemble mean 0.99925612 for 128 cells. Refinement reduces its scatter but does not remove its mean. An implementation that hopes the omitted term vanishes under smaller steps converges to the wrong transformation.

Exercise 13.6 — Which growth does a typical path show? (compute/design)

For $a_S = 0.1, b_S = 1.2, S_0 = 1$, compare expected and median values after unit time and explain a suitable simulation check.

Worked solution

The log process has mean $0.1 - 1.44/2 = -0.62$ and variance 1.44. Exponentiating its median gives $e^{-0.62} = 0.53794444$, while averaging the lognormal law gives $e^{0.1} = 1.10517092$. A finite ensemble may estimate the mean noisily because of the upper tail; use its exact variance $e^{0.2}(e^{1.44} - 1)$ to assess the standard error. Compare log mean and variance as well as positivity, because checking the raw mean alone can hide discretization errors. One-step Euler's negative probability 0.17965867 is already a decisive structural failure at this large step.

13.10 Sources and attribution

- V. Balakrishnan, NPTEL lecture 16, course 115106089.
- P. Mörters and Y. Peres, Brownian Motion, author-hosted manuscript, Theorems 7.12–7.14, printed pages 196–198, and Theorem 7.33, page 210: stochastic-limit and nonsmooth-boundary foundations.
- F. Black and M. Scholes, The Pricing of Options and Corporate Liabilities, *Journal of Political Economy* 81, 637–654 (1973), printed pages 640–642, verified in the linked primary-paper scan in Section 13.6. Used only to distinguish price dynamics from pricing assumptions.

Threshold crossings and extremes of smooth random signals

A thermal oscillator drives an alarm whenever its position rises through a threshold. The continuous model predicts 0.159155 zero-level upcrossings per unit time. Sampling the same model once per unit time records about 0.139688, because excursions can begin and end between observations. A false-alarm calculation must specify whether it counts physical crossings, sampled sign changes, or windows containing at least one event. They are different observables.

> The chapter follows NPTEL lecture 17: threshold indicators, joint value–derivative laws, stationary Gaussian rates, the harmonic example, extrema, and a spectral bridge. Its finite-sampling laboratory and count-versus-probability counterexamples are independent teaching constructions. All crossing formulas are restricted to suitable path regularity.

14.1 Counting a level requires both value and slope L17 1:48

Let X_t be a real signal and let $V_t = \dot{X}_t$ denote its path derivative. Assume continuously differentiable paths, a regular joint density $p_t(x, v)$, and almost surely isolated non-tangent crossings of the chosen level u on the finite observation window. Here u has signal units and v has signal per time units. Further integrability and uniformity conditions are needed to interchange the short-time limit, integration, and expectation; assume finite crossing intensity. These hypotheses exclude Wiener motion and a bare OU coordinate.

An upcrossing at time t satisfies $X_t = u$ with positive slope. In a short interval of duration h, a nearly linear segment with slope $v > 0$ crosses when its starting value lies between $u - vh$ and u. The probability contributed by slopes in $[v, v + dv]$ is approximately $vh\, p_t(u, v)dv$. Divide by h and integrate over positive slopes.

Proposition 14.1 — Rice intensity. Under the stated crossing regularity,

$$\nu_+(u,t) = \int_0^\infty v p_t(u,v)dv, \qquad \nu_-(u,t) = \int_{-\infty}^0 (-v) p_t(u,v)dv. \tag{14.1}$$

Expected upcrossing count on $[t_1, t_2]$ is $\int_{t_1}^{t_2} \nu_+(u,t)dt$.

Derivation.

The swept-interval argument gives the local expected count to first order. Under finite-intensity domination, integrating in slope and passing to the short-interval limit yields (14.1); summing disjoint time cells and taking their limit yields the time integral. An equivalent path identity for simple crossings is $N_u = \int_{t_1}^{t_2} |\dot{X}_t| \delta(X_t - u)dt$: converting the delta function from value to time divides by the absolute slope, leaving unit weight per crossing. Restricting to positive slopes gives the upcrossings. The regularity assumptions ensure this formal delta identity corresponds to an integrable count. ∎

The indicator $I_t = \mathbf{1}_{\{X_t > u\}}$ jumps upward at upcrossings and downward at downcrossings. Hence $N_+ - N_- = I_{t_2} - I_{t_1}$ when endpoints avoid the threshold. In a stationary process, expectation of the endpoint difference vanishes, so mean up- and downcrossing rates agree. This equality does not require symmetry about zero. Without stationarity the rate can vary with time, and (14.1) still applies with p_t.

The original Rice analysis, Section 3.3, equations (3.3-5) and (3.3-6), derives the directional slope integrals and states continuity and uniform-convergence qualifications. The smoothness assumptions are mathematical content, not optional footnotes attached after using a formula.

14.2 The Gaussian rate separates amplitude and time scale L17 25:26

Assume now that X_t is a stationary, centered, differentiable Gaussian process satisfying the preceding regularity. Let $\sigma_X^2 = \mathbb{E}[X_t^2] > 0$ and $\sigma_V^2 = \mathbb{E}[V_t^2] > 0$ be finite. Stationarity gives $d\mathbb{E}[X_t^2]/dt = 2\mathbb{E}[X_t V_t] = 0$. The pair (X_t, V_t) is jointly Gaussian because differentiation is a limit of Gaussian finite differences under the assumed mean-square regularity. Thus zero covariance makes them independent.

Proposition 14.2 — Stationary Gaussian upcrossing rate.

$$\nu_+(u) = \frac{\sigma_V}{2\pi\sigma_X} \exp\left[-\frac{u^2}{2\sigma_X^2}\right], \qquad \nu_-(u) = \nu_+(u). \tag{14.2}$$

Derivation.

The independent Gaussian joint density at (u, v) is $\exp[-u^2/(2\sigma_X^2) - v^2/(2\sigma_V^2)]/(2\pi\sigma_X\sigma_V)$. Inserting it into (14.1) leaves the integral $\int_0^\infty v e^{-v^2/(2\sigma_V^2)} dv = \sigma_V^2$. One factor σ_V cancels the denominator. The negative-slope integral gives the same value. ∎

The exponential measures threshold height relative to typical amplitude. The prefactor has inverse-time units and measures how rapidly the process changes relative to its amplitude. Multiplying the signal by a positive constant changes both standard deviations by that factor, leaving zero-level crossing rate unchanged. Changing its time scale changes the slope variance and therefore the rate. A value histogram supplies σ_X but cannot supply σ_V.

For a nonzero stationary mean μ, replace u by $u - \mu$ after centering. For non-Gaussian signals, zero covariance does not generally imply independence, and the integral of the actual joint density is required. For a nonstationary Gaussian signal, value and slope can have nonzero covariance, changing the conditional slope distribution at the threshold. Equation (14.2) must not be transplanted without those conditions.

14.3 A thermal oscillator has a finite position-crossing rate L17 22:46

Use the equilibrium inertial oscillator of chapter 11, with mass m, positive drag rate γ, natural angular frequency ω_0, temperature T, and $\theta = k_B T/m$. Its position is continuously differentiable, with derivative equal

to the continuous velocity. Equilibrium gives $\sigma_X^2 = \theta/\omega_0^2$, $\sigma_V^2 = \theta$, and independent equal-time position and velocity. Therefore

$$\nu_+(u) = \frac{\omega_0}{2\pi} \exp\left[-\frac{\omega_0^2 u^2}{2\theta}\right]. \tag{14.3}$$

At zero threshold, total up-plus-down rate is ω_0/π, independent of damping. That does not mean the oscillator's trajectories or crossing intervals are independent of damping. The first moment uses only equilibrium position and velocity statistics; their temporal organization depends on γ. A heavily damped trace can cross in closely grouped excursions separated by long slow changes, retaining the same mean crossing rate.

For $\omega_0 = \theta = 1$, the upcrossing rates at levels zero, one, and two are 0.15915494, 0.09653235, and 0.02153928. At level two, the rate is reduced by e^{-2} relative to level zero. Temperature raises spatial variance and therefore increases the rate at a fixed nonzero physical threshold, while the zero-level prefactor stays fixed in this ideal thermal model.

An overdamped scalar OU approximation preserves σ_X but loses finite derivative variance. Applying (14.3) after deleting inertia is invalid: the effective diffusion path is nondifferentiable. This is a sharp example of a reduction that preserves equilibrium and slow correlations while changing a path-sensitive observable. The alarm must specify its detector bandwidth or retain sufficient inertial dynamics.

14.4 A mean crossing count is not an alarm probability

Let N be the upcrossing count in a window of duration Δ, so $\mathbb{E}[N] = \nu_+(u)\Delta$ in stationarity. The probability of at least one upcrossing obeys $\Pr(N \geq 1) \leq \mathbb{E}[N]$, since the indicator of a positive count is bounded by the count. Equality requires that at most one crossing occurs per window. A Poisson approximation would instead give $\Pr(N \geq 1) \approx 1 - e^{-\nu_+\Delta}$, but its independence and rarity conditions do not follow from Rice's first moment.

For a simple counterexample, one window model has $N = 1$ with probability 0.1 and zero otherwise. A second has $N = 10$ with probability 0.01 and zero otherwise. Both have expected count 0.1, but their event probabilities are 0.1 and 0.01. Clustering changes the operational alarm probability by a factor of ten at identical mean rate. These abstract counts isolate information missing from a first moment; they are not asserted to be the full crossing laws of the oscillator.

The event “signal exceeds the threshold at some time” also includes windows beginning above it, even if no subsequent upcrossing occurs. Initial exceedance probability and crossing events overlap and require joint treatment. Higher crossing moments involve two-time joint laws of values and slopes. A factorial second moment counts distinct pairs; a naive double integral for $\mathbb{E}[N^2]$ must also include the diagonal self-count $\mathbb{E}[N]$. This prevents losing a term when translating delta-function notation into count variance.

14.5 Extrema demand another derivative L17 33:16

For this section, strengthen the hypotheses to twice continuously differentiable regular paths with finite extrema intensity. Let $A_t = \ddot{X}_t$ be acceleration and let $p(x, v, a)$ be the stationary joint density. A nondegenerate local maximum has $V = 0$ and $A < 0$. Apply the directional crossing formula to velocity rather than position: the mean rate of maxima above level u is

$$\nu_{\max}(u) = \int_u^\infty dx \int_{-\infty}^0 (-a) p(x, 0, a) da. \tag{14.4}$$

The factor $-a$ counts downcrossings of zero velocity, while integrating $x \geq u$ selects the desired heights. Minima use $a > 0$ instead. Tangencies with vanishing curvature need separate regularity control. The original Brownian oscillator has continuous but nondifferentiable velocity, so its acceleration is a generalized-noise quantity; (14.4) cannot be applied to it without further smoothing.

For a centered stationary Gaussian signal with enough smoothness, define variances σ_X^2, σ_V^2, and σ_A^2. Differentiating the stationary equal-time identities yields $\mathbb{E}[XV] = 0$, $\mathbb{E}[VA] = 0$, and $\mathbb{E}[XA] = -\sigma_V^2$. Thus the covariance matrix of (X, V, A) is

$$C = \begin{pmatrix} \sigma_X^2 & 0 & -\sigma_V^2 \\ 0 & \sigma_V^2 & 0 \\ -\sigma_V^2 & 0 & \sigma_A^2 \end{pmatrix}. \tag{14.5}$$

Value and acceleration are generally correlated. A product of three independent Gaussian densities would therefore give incorrect height-selective extrema statistics. Positive definiteness requires $\sigma_X^2 \sigma_A^2 > \sigma_V^4$ for a nonsingular density; equality identifies a degenerate spectral case that needs its own treatment.

If all maximum heights are counted, integrate out X first. The remaining Gaussian pair (V, A) is independent, so the same one-sided integral gives total maxima rate $\sigma_A/(2\pi\sigma_V)$. For a smooth covariance $C_X(\tau) = \exp[-\tau^2/(2\ell_t^2)]$ with positive time scale ℓ_t, its derivatives give $\sigma_V^2 = 1/\ell_t^2$ and $\sigma_A^2 = 3/\ell_t^4$. The total maxima rate is then $\sqrt{3}/(2\pi\ell_t)$. This example has the extra smoothness that the inertial oscillator lacks.

14.6 High frequencies control derivative statistics L17 44:23

Define centered covariance $C_X(\tau) = \mathbb{E}[X_{t+\tau} X_t]$ and use the book's two-sided angular-frequency convention $S_X(\omega) = \int_{-\infty}^\infty C_X(\tau) e^{i\omega\tau} d\tau$. Its inversion has factor $1/(2\pi)$. This convention differs by a constant factor from the lecture's displayed normalized transform, so formulas must be converted consistently. Under differentiation and inversion integrability conditions,

$$\sigma_X^2 = \int \frac{d\omega}{2\pi} S_X(\omega), \quad \sigma_V^2 = \int \frac{d\omega}{2\pi} \omega^2 S_X(\omega), \quad \sigma_A^2 = \int \frac{d\omega}{2\pi} \omega^4 S_X(\omega). \tag{14.6}$$

Each derivative adds weight to high frequencies. An OU spectrum falls as ω^{-2}, which gives finite signal variance but divergent derivative variance. An inertial oscillator position spectrum falls as ω^{-4}: its first

derivative variance is finite, while its second derivative variance diverges. Smoothing can make these moments finite, but it also changes the crossing and extrema rates being measured.

Finite spectral moments are useful mean-square regularity conditions; they do not alone establish every pathwise hypothesis used in the crossing theorem. Chapter 16 derives the spectral normalization and filtering rule. In an alarm chain, specifying the transfer function and sampling interval is part of specifying the observable, not a cosmetic choice in plotting a fixed signal.

14.7 Laboratory: distinguish crossings from sampled signs

The laboratory generates 512 equilibrium oscillator paths for duration 200 each, with $\theta = \omega_0 = 1$, $\gamma = 0.8$, and fine step 0.01. Exact linear-Gaussian phase-space updates use a matrix exponential and correlated noise covariance, avoiding an Euler equilibrium bias. It counts sampled transitions from below to above each threshold, then repeats on coarser subsequences of the same paths. These counts can miss an even number of crossings between observations.

Run `python code/ch14_crossings.py`. It uses seed 2026110814 and writes JSON and text output beside the program, without a network call. In addition to continuous Rice targets, JSON records the exact sampled-endpoint rate obtained from the bivariate Gaussian law at lag h. That distinguishes finite-sampling bias from Monte Carlo scatter. The finite-step count is a lower-resolution observable, not a purported exact continuous crossing counter.

```
step  upcrossing rates at levels 0, 1, 2
0.01  0.15965820, 0.09607422, 0.02194336
0.05  0.15879883, 0.09561523, 0.02187500
0.20  0.15589844, 0.09371094, 0.02127930
1.00  0.13968750, 0.08168945, 0.01694336
Continuous Rice rates: 0.15915494, 0.09653235, 0.02153928
```

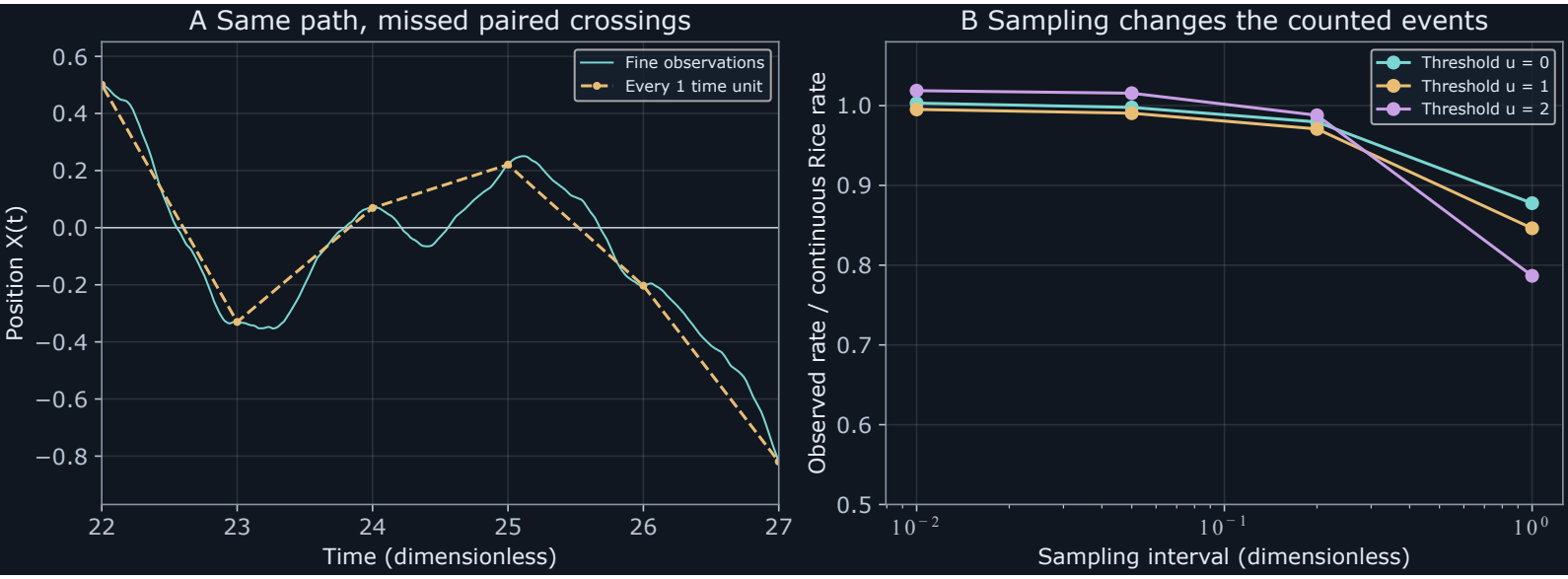


Figure 14.1. Sampling loses crossings while preserving the same underlying model. Panel A selects the first simulated path and coarse cell containing a pair of unresolved zero crossings, then shows its neighborhood. Straight segments are display interpolants. Panel B uses all simulated paths, without selection, to compare ensemble endpoint-sign rates with continuous Rice rates; deviations near unity include finite-ensemble scatter. Generator.

```
import numpy as np


def sampled_upcrossings(values, threshold):
    values = np.asarray(values)
    if values.ndim != 1:
        raise ValueError("one sampled path required")
    return np.count_nonzero(
        (values[:-1] < threshold) & (values[1:] >= threshold)
    )
```

The function returns an integer for a one-dimensional record and fixed threshold; it writes no files. An equality convention is explicit. A continuous Gaussian sample has zero probability of exactly equaling a fixed threshold, but quantized real measurements can have ties and need a deliberate event convention. Missing observations and hysteresis are outside this minimal counter.

14.8 What a crossing rate establishes

Rice's first moment converts a regular value–slope density into an expected event count. Gaussianity and stationarity simplify that density but do not make events Poisson. Nonstationary rates, clustered events, initial exceedance, sampling loss, and detector smoothing must be treated at the level of the actual alarm definition. Extrema add a derivative requirement and cannot be inferred from the mere continuity of a signal's first derivative.

14.9 Exercises with worked solutions

Exercise 14.1 — Equal rates without reflection symmetry (conceptual)

Why do stationary up- and downcrossing mean rates agree at any regular level, even for a skewed signal distribution?

Worked solution

The pathwise count difference is the difference of endpoint exceedance indicators. Stationarity makes their expectations equal, so the expected count difference is zero. Under finite-intensity stationarity, each expected count is its rate times window duration, giving equal rates. The argument uses no reflection of X about zero and does not require a Gaussian law. It does require the specified ordinary-crossing convention and excludes pathological infinite counts.

Exercise 14.2 — Why not count OU extrema? (conceptual)

A scalar OU trace is continuous and stationary. Is that sufficient for either (14.2) or the twice-differentiable extrema formula?

Worked solution

No. Its covariance has a cusp at zero and its high-frequency spectrum falls as ω^{-2}. Multiplication by ω^2 gives a nonintegrable derivative spectrum, so finite derivative variance fails. There is no ordinary derivative random variable with the density assumed in (14.1). The extrema formula needs still another derivative. Filtering can define a smoother observed signal, but its rates belong to that filtered process and depend on the filter.

Exercise 14.3 — Nonzero mean and amplitude rescaling (derive)

For $Y_t = cX_t + \mu$ with $c > 0$ and centered stationary Gaussian X, derive the upcrossing rate at level u.

Worked solution

The transformed level is $(u - \mu)/c$. Also $\sigma_Y = c\sigma_X$ and $\sigma_{\dot{Y}} = c\sigma_V$, so their ratio is unchanged. Substitution into (14.2) gives $\sigma_V \exp[-(u-\mu)^2/(2c^2\sigma_X^2)]/(2\pi\sigma_X)$. At its mean level $u = \mu$, amplitude scaling cancels entirely. At a fixed physical threshold it changes the exponential factor, because the threshold's normalized distance has changed.

Exercise 14.4 — The value–acceleration covariance (derive)

Derive the off-diagonal entry in (14.5) and explain its effect on maxima height.

Worked solution

Stationarity gives $\mathbb{E}[XV] = 0$. Differentiating in time yields $0 = \mathbb{E}[V^2 + XA]$, so $\mathbb{E}[XA] = -\sigma_V^2$. In a Gaussian model, the conditional acceleration mean given $X = x$ is therefore $-\sigma_V^2 x/\sigma_X^2$, after the independent zero-velocity condition is included. High positive values favor negative curvature. Replacing the joint density by three independent marginals would remove this dependence and mispredict how maxima heights are distributed, although an all-height marginal count can still be simpler.

Exercise 14.5 — A bound for one alarm window (compute/design)

For the normalized thermal oscillator at threshold two, bound the probability of at least one upcrossing in a window of duration five. What extra claim would a Poisson approximation introduce?

Worked solution

Equation (14.3) gives rate $e^{-2}/(2\pi) = 0.02153928$, so mean count in five units is about 0.1076964. The probability of at least one is at most that number. A Poisson approximation would replace the bound by $1 - e^{-5e^{-2}/(2\pi)}$, requiring a model for event dependence and rarity. Rice's expectation alone does not justify that replacement. If the alarm asks whether the signal is ever above two, initial exceedance must be included as a different component of the event.

Exercise 14.6 — Verify the sampled zero-level target (compute/design)

At sampling interval h, two standardized Gaussian endpoints have correlation $r = f(h)$. Show how their sign-change probability checks the sampled zero-level rate.

Worked solution

For a centered bivariate unit Gaussian pair, rotational symmetry of independent standard Gaussian coordinates gives the probability of opposite signs as $\arccos(r)/\pi$; the angular wedges between the two zero-projection lines have that total angular fraction. Half correspond to a negative-to-positive transition. Thus the expected sampled upcrossing rate is $\arccos(f(h))/(2\pi h)$. For differentiable oscillator position, $f(h) = 1 - \omega_0^2 h^2/2 + o(h^2)$, and $\arccos(1 - z) \sim \sqrt{2z}$ recovers $\omega_0/(2\pi)$ as $h \downarrow 0$. The bivariate Gaussian CDF in the lab evaluates the more general nonzero-threshold target. Compare the simulation with this finite-h target before attributing a discrepancy to the continuous formula.

14.10 Sources and attribution

- V. Balakrishnan, NPTEL lecture 17, course 115106089.
- S. O. Rice, Mathematical Analysis of Random Noise, *Bell System Technical Journal* 24 (1945), continuation of the 1944 paper. Primary-paper reprint scan, Section 3.3, equations (3.3-5), (3.3-6), (3.3-11), and (3.3-14), scan pages 58–61: directional crossing integrals, Gaussian zero/level crossing results, and regularity limitations. The chapter rederives its displayed formulas in the book's angular-frequency convention.

Charged Brownian motion and anisotropic diffusion

A magnetic field does no work on a charged particle, yet it can reduce transverse spatial diffusion by a factor of seventeen while leaving thermal velocity variance unchanged. The field redirects a persistent velocity before it carries the particle far. The task is to distinguish thermal scatter, signed rotational response, and the symmetric covariance growth that defines a diffusion tensor.

> The sequence follows NPTEL lecture 18: vector Langevin dynamics, rotation matrices, conditional velocity law, lag covariance, tensor Green–Kubo relation, and slow spatial reduction. All component signs below follow an explicitly stated SI convention; the lecture uses a different order for covariance indices.

15.1 The Lorentz drift is an antisymmetric rotation L18 1:19

Let a particle have signed charge q, mass $m > 0$, position column $\boldsymbol{X}$, and velocity column $\boldsymbol{V}$ in three Cartesian dimensions. A uniform static magnetic field is $\boldsymbol{B} = B\boldsymbol{n}$, with magnitude $B \geq 0$ and unit direction $\boldsymbol{n}$. In SI units the Lorentz force is $q\boldsymbol{V} \times \boldsymbol{B}$. Let $\gamma > 0$ be isotropic drag rate and $\theta = k_B T/m > 0$ the thermal variance per velocity component. Three independent standard Wiener drivers form the column $\boldsymbol{W}_t$. The equations are

$$d\boldsymbol{X}_t = \boldsymbol{V}_t dt, \qquad d\boldsymbol{V}_t = A\boldsymbol{V}_t dt + \sqrt{2\gamma\theta}\, d\boldsymbol{W}_t, \quad A = -\gamma I + \Omega J. \tag{15.1}$$

Here I is the identity matrix, $\Omega = qB/m$ is a signed cyclotron rate, and $J\boldsymbol{v} = \boldsymbol{v} \times \boldsymbol{n}$ defines the dimensionless matrix J. Its entries are $J_{ij} = \epsilon_{ijk} n_k$, with the totally antisymmetric Levi-Civita symbol ϵ_{ijk} and sum over k. The cyclotron magnitude is $\omega_c = |\Omega|$. For $\boldsymbol{n} = (0,0,1)^{\mathsf{T}}$, $J = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$: positive q gives $\dot{v}_x = \Omega v_y$, $\dot{v}_y = -\Omega v_x$ before drag and noise.

The identity $\boldsymbol{v} \cdot (\boldsymbol{v} \times \boldsymbol{n}) = 0$ proves that the magnetic term changes no instantaneous kinetic energy. Matrix antisymmetry $J^{\mathsf{T}} = -J$ expresses the same fact. Define parallel and transverse projectors $P_{\parallel} = \boldsymbol{n}\boldsymbol{n}^{\mathsf{T}}$ and $P_{\perp} = I - P_{\parallel}$. The vector triple-product identity gives $J^2 = -P_{\perp}$, so $J^3 = -J$. Summing the exponential series therefore yields

$$R(t) = e^{\Omega J t} = P_{\parallel} + P_{\perp}\cos(\Omega t) + J\sin(\Omega t), \qquad e^{At} = e^{-\gamma t} R(t). \tag{15.2}$$

The identity part commutes with J, permitting the factorization. Antisymmetry gives $R(t)R(t)^{\mathsf{T}} = I$, so this exponential is a rotation. In a deterministic rotating velocity frame $\boldsymbol{U}_t = R(-t)\boldsymbol{V}_t$, the magnetic drift disappears. The rotated isotropic noise is again Wiener noise because deterministic orthogonal transformations preserve its independent Gaussian increment covariance. This simplifies the velocity law; it does not simply remove the field from laboratory-frame spatial displacement.

15.2 Rotation preserves the thermal velocity covariance L18 23:38

Given deterministic initial velocity $\boldsymbol{v}_0$, the linear stochastic solution is $\boldsymbol{V}_t = e^{\boldsymbol{A}t}\boldsymbol{v}_0 + \sqrt{2\gamma\theta}\int_0^t e^{\boldsymbol{A}(t-s)}d\boldsymbol{W}_s$. Its stochastic integral is multivariate Gaussian. Its covariance is $2\gamma\theta\int_0^t e^{\boldsymbol{A}s}e^{\boldsymbol{A}^{\mathsf{T}}s}ds$. By (15.2), the product inside is $e^{-2\gamma s}\boldsymbol{I}$. Hence

$$\mathbb{E}[\boldsymbol{V}_t \mid \boldsymbol{v}_0] = e^{-\gamma t}\boldsymbol{R}(t)\boldsymbol{v}_0, \qquad C_{\mathrm{cond}}(t) = \theta(1-e^{-2\gamma t})\boldsymbol{I}. \tag{15.3}$$

The mean rotates and decays; the conditional covariance broadens isotropically just as in the zero-field OU model. The density is $\exp[-|\boldsymbol{v}-e^{-\gamma t}\boldsymbol{R}(t)\boldsymbol{v}_0|^2/(2s_v^2)]/(2\pi s_v^2)^{3/2}$, where $s_v^2 = \theta(1-e^{-2\gamma t})$. The squared norm contains the full vector difference. It is not a matrix power inside a scalar probability density.

At long times the mean vanishes and the density becomes the Maxwell Gaussian with covariance $\boldsymbol{\Sigma} = \theta\boldsymbol{I}$. Independently, the stationary covariance equation is $\boldsymbol{A}\boldsymbol{\Sigma} + \boldsymbol{\Sigma}\boldsymbol{A}^{\mathsf{T}} + 2\gamma\theta\boldsymbol{I} = \boldsymbol{0}$. Substituting $\boldsymbol{\Sigma} = \theta\boldsymbol{I}$ makes the antisymmetric magnetic terms cancel. Stability follows because the eigenvalues of $\boldsymbol{A}$ are $-\gamma$ and $-\gamma \pm i\Omega$; all real parts are negative. The transition covariances converge to the claimed equilibrium, establishing the full Gaussian limit.

The velocity forward equation is $\partial_t\rho = -\nabla_v\cdot(\boldsymbol{A}\boldsymbol{v}\rho) + \gamma\theta\nabla_v^2\rho$. In the magnetic term, $\nabla_v\cdot(\boldsymbol{J}\boldsymbol{v}) = \operatorname{tr}\boldsymbol{J} = 0$. For the Maxwell density, its radial gradient is perpendicular to $\boldsymbol{J}\boldsymbol{v}$, so rotation contributes zero to the stationary equation. The drag and diffusion terms cancel as in chapter 10.

The Czopnik–Garbaczewski primary analysis, Section 2, develops the explicit magnetic velocity transition law rather than inferring dynamics from a stationary histogram. Its Lorentz equation uses a Gaussian-unit factor of $1/c$; (15.1) uses SI units. The rotation and Gaussian-covariance derivations here make that convention conversion explicit.

15.3 Ordered lag covariance retains the rotation sign L18 27:09

Initialize velocity in its stationary Maxwell law, independently of future noise. Define $C_{ij}(\tau) = \mathbb{E}[V_i(t+\tau)V_j(t)]$, so the later-time index is first. This ordering is explicit because transposing it reverses every antisymmetric cross-correlation. For $\tau \geq 0$, write $\boldsymbol{V}_{t+\tau} = e^{\boldsymbol{A}\tau}\boldsymbol{V}_t +$ independent mean-zero future noise. Multiplication by $\boldsymbol{V}_t^{\mathsf{T}}$ and averaging gives

$$C(\tau) = \theta e^{-\gamma\tau}[P_{\parallel} + P_{\perp}\cos(\Omega\tau) + \boldsymbol{J}\sin(\Omega\tau)]. \tag{15.4}$$

With field along positive z, $C_{xx} = C_{yy} = \theta e^{-\gamma\tau}\cos\Omega\tau$, $C_{xy} = \theta e^{-\gamma\tau}\sin\Omega\tau$, $C_{yx} = -C_{xy}$, and $C_{zz} = \theta e^{-\gamma\tau}$. The sign check is immediate at very short lag: v_x responds positively to initial v_y for positive Ω, so $C'_{xy}(0+) = \theta\Omega$.

At equal times the covariance is symmetric and isotropic. At unequal times an ordered covariance matrix need not be symmetric. Stationarity implies $C(-\tau) = C(\tau)^{\mathsf{T}}$, rather than componentwise evenness. The diagonal entries are even in lag, and the transverse cross entries are odd. No change of magnetic field is required to write this stationarity identity. Physical time reversal additionally reverses the external magnetic field; that is a different transformation of the experiment.

Independent Cartesian driving noises coexist with unequal-time output cross-correlations because the drift mixes components. Reversing charge or field changes the sine-term sign while preserving cosine and equal-time variances. That supplies an experimentally meaningful test of handedness beyond checking the trace or speed distribution.

15.4 Only the symmetric integral grows displacement covariance L18 38:24

Take fixed initial position and stationary zero-mean velocity. The displacement column is $\Delta \boldsymbol{X}_t = \int_0^t \boldsymbol{V}_s ds$. Its covariance matrix is the double integral of the ordered velocity covariance. The two triangular time regions have opposite orderings, producing

$$\mathrm{Cov}(\Delta \boldsymbol{X}_t) = \int_0^t (t-s)[C(s) + C(s)^{\mathsf{T}}] ds. \tag{15.5}$$

The covariance is symmetric by definition, so an antisymmetric term cannot contribute to it. With absolute integrability of all matrix entries, the dominated-convergence argument of chapter 10 gives $D = \lim_{t\to\infty} \mathrm{Cov}(\Delta \boldsymbol{X}_t)/(2t) = \int_0^\infty [C(s) + C(s)^{\mathsf{T}}] ds/2$.

Proposition 15.1 — Magnetic diffusion tensor.

$$D = D_{\parallel} P_{\parallel} + D_{\perp} P_{\perp}, \quad D_{\parallel} = \frac{\theta}{\gamma}, \quad D_{\perp} = \frac{\theta\gamma}{\gamma^2 + \Omega^2}. \tag{15.6}$$

Derivation.

The sine term in (15.4) cancels under transposition. The parallel integral is $\int_0^\infty e^{-\gamma s} ds = 1/\gamma$. The transverse integral is the real part of $\int_0^\infty e^{-(\gamma - i\Omega)s} ds$, equal to $\gamma/(\gamma^2 + \Omega^2)$. Multiplication by θ proves the result. Positive coefficients make the tensor positive definite for finite positive γ, θ. ∎

At zero field the tensor is isotropic. At large ω_c/γ, transverse diffusion is suppressed relative to parallel by $1/[1 + (\omega_c/\gamma)^2]$. In the running dimensionless example $\gamma = \theta = 1$, $\omega_c = 4$, the ratio is $1/17 = 0.05882353$. Thermal speed variance is still one in each direction. Rapid directional rotation cancels contributions to long-range displacement without reducing instantaneous thermal scatter.

The finite-time transverse component variance is also explicit. With $a_c = \gamma - i\Omega$, integrate (15.5) to obtain $2\theta\,\mathrm{Re}[t/a_c - (1 - e^{-a_c t})/a_c^2]$. Its parallel counterpart is $2\theta[t - (1 - e^{-\gamma t})/\gamma]/\gamma$. Both begin as θt^2, because at sufficiently short times a stationary particle retains its initial velocity. Field-dependent diffusion appears after temporal integration, not in the initial ballistic coefficient.

The full integrated ordered covariance $M_C = \int_0^\infty C(s) ds = -\theta A^{-1}$ also contains $\theta\Omega J/(\gamma^2 + \Omega^2)$. This antisymmetric term describes a signed transport response, not a mean-square displacement. For constant coefficients its contraction with a symmetric Hessian is zero, so it also drops from the homogeneous bulk diffusion operator. Boundary currents or force response can still reveal it.

15.5 Mobility and the slow spatial noise give the same symmetric tensor L18 46:36

Define the damping–rotation matrix $G = \gamma I - \Omega J = -A$. It is invertible: its eigenvalues are γ and $\gamma \pm i\Omega$, none zero. Its inverse, obtained by separating the parallel direction and using $J^2 = -P_\perp$, is

$$G^{-1} = \frac{1}{\gamma} P_\parallel + \frac{\gamma P_\perp + \Omega J}{\gamma^2 + \Omega^2}. \tag{15.7}$$

A small constant applied force $\boldsymbol{F}$ produces stationary mean velocity $\bar{\boldsymbol{v}} = G^{-1}\boldsymbol{F}/m$. Thus force mobility is G^{-1}/m and has an antisymmetric Hall part. Its symmetric part multiplied by $k_B T$ is the displacement diffusion tensor, consistent with (15.6). The vector force must be coupled to the same mechanical coordinates for this equilibrium relation.

On coarse spatial time scales, the formal constant-coefficient reduction gives $d\boldsymbol{X} = \sqrt{2\gamma\theta}\, G^{-1} d\boldsymbol{W}$. Its noise covariance per unit time is $2\gamma\theta G^{-1}G^{-\mathsf{T}}$. The identity $G + G^{\mathsf{T}} = 2\gamma I$ implies

$$\gamma G^{-1}G^{-\mathsf{T}} = \frac{G^{-1} + G^{-\mathsf{T}}}{2}, \tag{15.8}$$

by multiplying on the left by G^{-1} and right by $G^{-\mathsf{T}}$. Half the noise covariance is therefore the same D as the symmetric Green–Kubo integral. A scalar square root applied entry by entry would not be a valid replacement for a matrix noise factor; covariance is the matrix product of the factor with its transpose.

This agreement identifies the asymptotic covariance and a diffusion approximation. It does not make the exact finite-inertia position process Markov. Its history retains velocity information. The primary analysis, Section 3 and its Smoluchowski discussion, similarly distinguishes phase-space propagation from reduced configuration-space descriptions. An asymptotic diffusion coefficient alone does not prove uniform accuracy for all temporal or boundary observables.

15.6 Laboratory: compare three tensor calculations

The laboratory checks (15.6) in three ways: quadrature of the ordered velocity matrix exponential, direct inverse-matrix integration, and the reduced noise covariance (15.8). It then builds the exact six-dimensional position–velocity covariance by integrating the linear noise kernel. A thermal initial velocity contribution is added explicitly. The displacement diagonal must agree with the independently integrated scalar cosine formulas.

Run `python code/ch15_magnetic.py`. With dimensionless $\gamma = \theta = 1$, signed $\Omega = 4$, and observation time 80, the program generates 200,000 exact Gaussian endpoints using seed 2026110815. It checks their displacement and velocity variances. It makes no network calls and writes JSON and text output beside the source. One-step Gaussian endpoints check moments efficiently; they are not a continuous-path animation.

```
D_parallel=1.00000000
D_perpendicular=0.05882353
Suppression ratio=0.05882353
At t=80: transverse MSD=9.51557093, parallel MSD=158.00000000
```

Dividing the finite-time transverse MSD by $2t$ has a bounded transient correction; it is not exactly the infinite-time coefficient at time 80. The final velocity variances remain near one despite the factor-seventeen transport anisotropy. That combination directly tests the opening physical distinction.

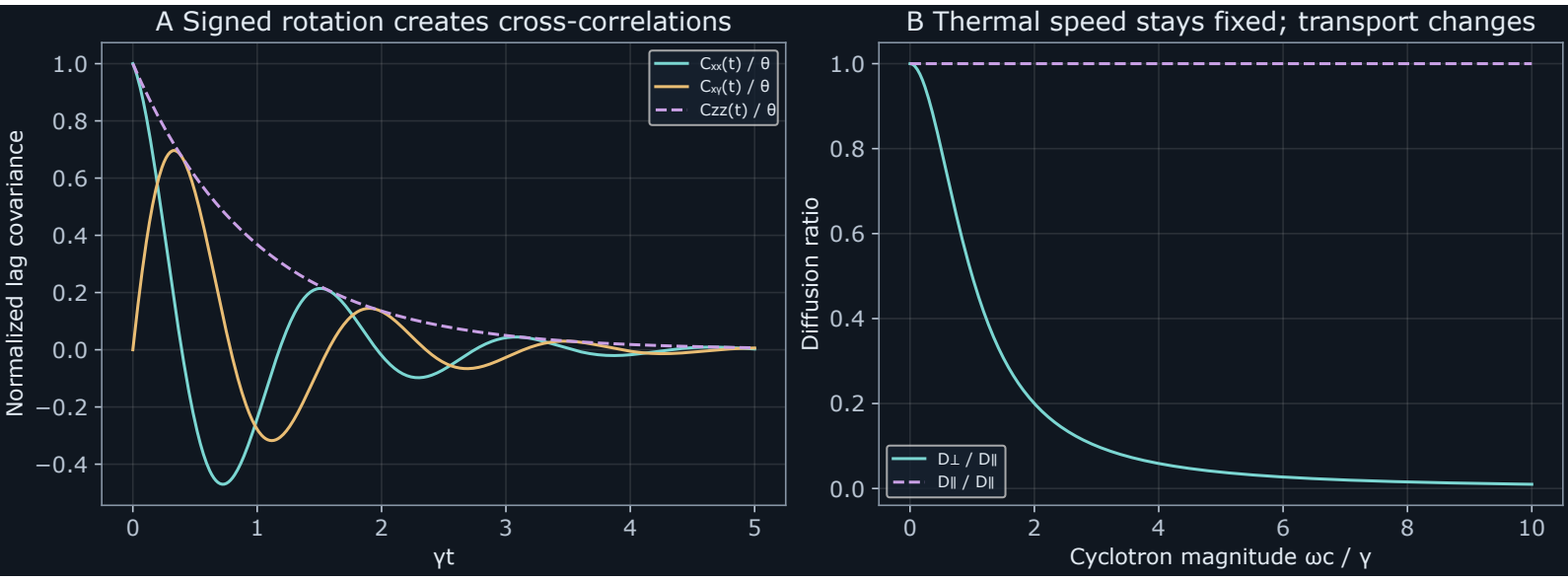


Figure 15.1. Thermal scatter and spatial transport respond differently to a field. Panel A uses signed $\Omega/\gamma = 4$ and the later-index-first covariance convention. Panel B shows the symmetric diffusion coefficients, which are even under field reversal. Generator.

```
import numpy as np


def diffusion_tensor(gamma, theta, signed_omega):
    if gamma <= 0 or theta <= 0:
        raise ValueError("positive damping and thermal variance required")
    transverse = theta*gamma/(gamma**2 + signed_omega**2)
    return np.diag([transverse, transverse, theta/gamma])
```

The function assumes the field direction is z, returns a symmetric diffusion matrix, and writes no files. It deliberately omits the antisymmetric mobility; conflating those outputs would defeat its definition. The full lab retains the ordered covariance integral to test the lost Hall part separately.

15.7 What the simple magnetic model omits

The model assumes a uniform imposed field, constant isotropic drag, classical equilibrium noise, and independent test particles. It omits collective plasma fields, collisions generating memory, spatially varying coefficients, confinement, and quantum orbital effects. The vanishing transverse diffusion limit is a statement at fixed positive drag and temperature as field grows; it does not describe every ordering of a collisionless or strong-field plasma limit. A constant magnetic field changes no kinetic energy through the Lorentz force, but drag and noise still exchange energy with the bath.

15.8 Exercises with worked solutions

Exercise 15.1 — No work, different diffusion (conceptual)

Explain why unchanged Maxwell variance does not force an unchanged diffusion coefficient.

Worked solution

Equal-time variance fixes the covariance at zero lag. Diffusion integrates covariance over positive lag. The field turns transverse covariance into a damped cosine whose positive and negative portions cancel. The initial thermal variance remains θ, but the integral decreases from θ/γ to $\theta\gamma/(\gamma^2+\Omega^2)$. A single-time speed histogram does not determine directional persistence.

Exercise 15.2 — Field reversal (conceptual)

Which of $C_{xy}(t)$, equal-time covariance, and transverse MSD changes sign when charge changes sign?

Worked solution

The signed rate Ω changes sign. The sine cross-covariance changes sign; cosine diagonal correlations and the equal-time Maxwell covariance do not. MSD integrates only the symmetric covariance and is even in Ω. Measuring only diagonal displacement variances therefore cannot determine charge sign in this model, whereas an ordered lag cross-correlation can.

Exercise 15.3 — Verify the inverse (derive)

Multiply (15.7) by G and show why no parallel singularity occurs.

Worked solution

On the parallel subspace $J=0$, so $G=\gamma I$ and inverse is I/γ. On the transverse subspace, $(\gamma I-\Omega J)(\gamma I+\Omega J)=\gamma^2 I-\Omega^2 J^2=(\gamma^2+\Omega^2)I$. Mixed projector products vanish. Adding the two inverse actions gives (15.7). The rotation generator itself has a zero eigenvalue parallel to the field, but G shifts it to positive γ, removing the singularity.

Exercise 15.4 — Derive tensor symmetrization (derive)

Why does integrating only $C_{ij}(s)$ fail to give a displacement covariance coefficient when $i\neq j$?

Worked solution

The displacement covariance double integral includes both time orderings. On the region $t_1>t_2$ its integrand is $C_{ij}(t_1-t_2)$; on the other triangle it is $C_{ji}(t_2-t_1)$. At lag s, each region has length weight $t-s$. Adding

them gives (15.5), and division by $2t$ yields the symmetric integral. An antisymmetric integrated matrix would give $D_{ij} = -D_{ji}$, impossible for covariance. It can instead represent an ordered response quantity.

Exercise 15.5 — Infer the field-to-drag ratio (compute/design)

A test-particle model predicts $D_\perp / D_\parallel = 1/17$. What magnitude ω_c/γ follows, and what can this measurement not determine?

Worked solution

Invert $r = 1/[1 + (\omega_c/\gamma)^2]$ to obtain $\omega_c/\gamma = \sqrt{1/r - 1} = 4$. The ratio does not determine charge sign or field direction's orientation, since it contains a square. It also does not separately determine γ, temperature, and mass. Parallel diffusion and independent thermal velocity variance would supply $\gamma = \theta/D_\parallel$ within this model. Model assumptions must be validated before interpreting a measured ratio that way.

Exercise 15.6 — A decisive simulator check (compute/design)

A simulation gets the Maxwell histogram correct but reports zero $C_{xy}(t)$ for every positive lag at nonzero field. Give two possible defects and a targeted check.

Worked solution

The drift may have omitted the off-diagonal Lorentz coupling, or the program may have averaged C_{xy} with C_{yx} before reporting an ordered correlation. Check the deterministic short-lag response $\partial_t \mathbb{E}[V_x(t) \mid \boldsymbol{v}_0]_{t=0} = -\gamma v_{0x} + \Omega v_{0y}$. Then compute the ordered sample covariance with later x and earlier y; its initial slope must be $\theta\Omega$. The symmetric cross combination should vanish in this field-aligned isotropic model, so labeling the ordering is essential to diagnose which quantity was actually measured.

15.9 Sources and attribution

- V. Balakrishnan, NPTEL lecture 18, course 115106089.
- R. Czopnik and P. Garbaczewski, Brownian motion in a magnetic field, *Physical Review E* 63, 021105 (2001); author preprint, Section 2 and Section 3's configuration-space discussion. Used for the explicit velocity law, unit-convention comparison, and the distinction between inertial phase space and reduced spatial descriptions. All displayed derivations and numerical cases are developed here.

Noise spectra, filters, and thermal fluctuations

A longer noise record makes Fourier bins narrower but does not make the unsmoothed power at one frequency reliable. In the OU laboratory below, increasing record duration eightfold leaves the relative periodogram variance near one. Averaging independent records reveals the Lorentzian underneath. The task is to infer a relaxation time and a physical noise variance while keeping transform conventions, estimator uncertainty, sampling aliases, and filtering distinct.

> The topic sequence follows NPTEL lecture 19: stationarity, finite-record Fourier power, covariance transforms, Langevin filtering, matrix spectra, and resistor thermal noise. The ensemble finite-window proof and Gaussian periodogram inconsistency argument make the limiting statements precise. The book uses a transform with no prefactor in the forward direction, differing from the lecture's spectrum by 2π.

16.1 Fix a measure before assigning spectral power L19 1:56

Let X_t be a real, zero-mean, second-order stationary process: its variance is finite and its covariance $C_X(\tau) = \mathbb{E}[X_{t+\tau}X_t]$ depends only on lag. For scalar classical variables, stationarity and commutativity give $C_X(-\tau) = C_X(\tau)$. Suppose C_X is absolutely integrable over the real line. Define the two-sided angular-frequency spectral density by

$$S_X(\omega) = \int_{-\infty}^{\infty} C_X(\tau) e^{i\omega\tau} d\tau, \qquad C_X(\tau) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} S_X(\omega) e^{-i\omega\tau}. \tag{16.1}$$

The inversion holds under the usual Fourier inversion hypotheses, in particular at continuity points when applicable. Angular frequency ω has units inverse seconds; S_X has units squared signal times seconds. Variance is the spectral area $C_X(0) = \int S_X d\omega/(2\pi)$, not simply $\int S_X d\omega$. A real scalar covariance gives a real even spectrum. Positive definiteness of covariance makes the spectrum nonnegative, as the finite-window argument below also shows.

For a nonzero mean μ, first center the signal. The uncentered correlation has an added constant μ^2, whose transform is $2\pi\mu^2\delta(\omega)$. This is a spectral line at zero frequency rather than a broad noise floor. Other stationary processes can have discrete spectral lines even when centered, so an integrable density is a restricted case of the more general spectral-measure description. A sample path need not itself be absolutely integrable over infinite time for a covariance spectrum to exist.

16.2 A finite-window identity supplies the spectral limit L19 5:14

For record duration $T_r > 0$, define Fourier amplitude $A_{T_r}(\omega) = \int_0^{T_r} X_t e^{i\omega t} dt$ and raw periodogram $I_{T_r}(\omega) = |A_{T_r}(\omega)|^2/T_r$. The subscript distinguishes record duration from thermodynamic temperature. A periodogram is a random estimate from one record; S_X is an ensemble property of a stationary law.

Proposition 16.1 — Ensemble Wiener–Khinchin limit. With the absolute-integrability assumptions above,

$$\mathbb{E}[I_{T_r}(\omega)] = \int_{-T_r}^{T_r} \left(1 - \frac{|\tau|}{T_r}\right) C_X(\tau) e^{i\omega\tau} d\tau \longrightarrow S_X(\omega). \tag{16.2}$$

Derivation.

Expand the modulus square into a double integral over $0 \le t, s \le T_r$. Taking expectation replaces $X_t X_s$ by $C_X(t-s)$. For fixed lag $\tau = t - s$, the allowed positions along the square have length $T_r - |\tau|$. Divide by record duration to obtain the triangular weight in (16.2). Its magnitude is at most one and it tends to one at each fixed lag. Absolute integrability of C_X permits dominated convergence. Since every expected periodogram is nonnegative, its limit S_X is nonnegative. ∎

This proof establishes an ensemble-mean limit without assuming ergodicity. Replacing covariance expectations by averages along one long trace asks a separate ergodic question. Interchanging that trace average with a frequency integral or a limit requires still more control. Ergodicity concerns equality of suitable long-time and ensemble averages; it does not by itself say a process literally visits every state or make a raw periodogram consistent.

The finite triangular lag weight is a useful bias diagnosis. Correlations at lags comparable with the record length are downweighted; longer lags are missing. Windowing a record changes the weight further. A longer record improves frequency resolution and changes this expectation bias, but its binwise randomness needs a different treatment.

16.3 A raw periodogram does not settle to one curve

The OU process supplies a precise counterexample to an overstrong pointwise limit. Its finite-record amplitude $A = A_{T_r}(\omega)$ is a centered complex Gaussian, because its real and imaginary parts are deterministic linear integrals of a jointly Gaussian process. Gaussian fourth-moment factorization gives $\mathbf{Var}(|A|^2) = (\mathbb{E}|A|^2)^2 + |\mathbb{E}[A^2]|^2$. Therefore the relative periodogram variance is at least one at every nontrivial record length, not a quantity forced to vanish by increasing duration.

At a fixed nonzero frequency, the nonconcentration can be stated more strongly. In $\mathbb{E}[A^2]$, changing to lag and sum-time coordinates leaves an oscillatory inner integral of $e^{2i\omega s}$. Its magnitude is bounded by $1/|\omega|$, independent of record length. Hence $|\mathbb{E}[A^2]|/T_r \le \|C_X\|_1/(T_r|\omega|) \to 0$. Equation (16.2) gives $\mathbb{E}|A|^2/T_r \to S_X(\omega)$.

Consequently the real and imaginary parts of $A/\sqrt{T_r}$ approach independent centered Gaussians of variance $S_X(\omega)/2$. Their squared sum has limit distribution $S_X(\omega)$ times a unit-mean exponential random variable. When $S_X(\omega) > 0$, it does not converge in probability to that mean. This Gaussian derivation directly separates convergence of expectation from convergence of the estimator.

Averaging M independent periodograms divides their variance by M while retaining the mean. Averaging neighboring frequency bins can also reduce scatter, with a bias-versus-resolution tradeoff and dependence conditions. In practice one often divides a long record into segments, applies a taper, and averages spectra. The effective number of independent estimates depends on overlap and correlations. A smooth-looking curve is not automatically a more accurate high-resolution estimate.

16.4 The OU spectrum is a normalized Lorentzian L19 24:10

For stationary thermal velocity V_t, write $\theta = k_B T/m$ and use covariance $C_V(\tau) = \theta e^{-\gamma|\tau|}$ with $\gamma > 0$. Evenness reduces its transform to $2\theta \int_0^\infty e^{-\gamma\tau}\cos\omega\tau\, d\tau$. The integral is the real part of $1/(\gamma - i\omega)$, giving

$$S_V(\omega) = \frac{2\theta\gamma}{\gamma^2 + \omega^2}. \tag{16.3}$$

The half-width at half maximum is γ, so the spectrum identifies a relaxation rate when the OU model is appropriate. Its area is $\int S_V d\omega/(2\pi) = \theta$, since $\int_{-\infty}^{\infty} d\omega/(\gamma^2+\omega^2) = \pi/\gamma$. A height fit without area normalization can confuse a change of variance with a change of correlation time.

The random force ζ_t in $m\dot V + m\gamma V = \zeta$ has covariance $2m\gamma k_B T\delta(\tau)$, hence flat spectrum $S_\zeta = 2m\gamma k_B T$. This white noise is a generalized process with no finite pointwise variance. The velocity response filters high frequencies and produces the integrable Lorentzian. The derivative transform has factor $-i\omega$ under (16.1), so the transfer function from force to velocity is $H(\omega) = 1/[m(\gamma - i\omega)]$. Multiplying S_ζ by $|H|^2$ reproduces (16.3).

The sampled OU law at spacing h is an autoregression with coefficient $r = e^{-\gamma h}$ and innovation variance $\theta(1-r^2)$. Its spectrum, normalized so that integration over $|\omega| \le \pi/h$ recovers variance, is $S_h(\omega) = h\theta(1-r^2)/(1 + r^2 - 2r\cos\omega h)$. This is the periodic alias sum of the continuous spectrum, not simply its restriction to the sampling band. At low frequency and fine sampling it approaches (16.3); near the sampling limit, the distinction is visible in Figure 16.1.

16.5 Filtering preserves spectral positivity L19 28:38

Let a stationary input column $\boldsymbol{X}_t$ drive a stable linear time-invariant filter with matrix impulse response $h(s)$. Its output is $\boldsymbol{Y}_t = \int h(s)\boldsymbol{X}_{t-s}\,ds$, assuming integrability sufficient for stationary output and exchanges of integrals. The transfer matrix is $H(\omega) = \int h(s)e^{i\omega s}\,ds$. Causality means $h(s) = 0$ for negative lag, but the spectral multiplication itself only needs a well-defined stable filter.

Use the later-index-first matrix covariance $C_X(\tau) = \mathbb{E}[\boldsymbol{X}_{t+\tau}\boldsymbol{X}_t^{\mathsf T}]$ and its transform S_X. Substitute the filter into the output covariance. The integrand becomes $h(s)C_X(\tau - s + r)h(r)^{\mathsf T}$. Transforming lag contributes $e^{i\omega(s-r)}$, which separates the two impulse-response integrals. Thus

$$S_Y(\omega) = H(\omega)S_X(\omega)H(\omega)^\dagger, \tag{16.4}$$

where dagger denotes complex conjugate transpose. The scalar rule is $S_Y = |H|^2 S_X$. The transfer function is the complex amplitude ratio H; its modulus squared is the power-gain ratio. Calling $|H|^2$ the full transfer function would discard phase and cross-channel information.

Stationarity gives $C_X(-\tau) = C_X(\tau)^{\mathsf{T}}$ for real signals, so $S_X = S_X^{\dagger}$ is Hermitian. For any complex column z, $z^{\dagger} S_X z \geq 0$, as follows from the corresponding expected squared Fourier projection. It is therefore positive semidefinite. Equation (16.4) preserves that property because $z^{\dagger} S_Y z = (H^{\dagger} z)^{\dagger} S_X (H^{\dagger} z)$. In particular $|S_{12}|^2 \leq S_{11} S_{22}$ for a two-channel spectral matrix. An arbitrary pair of cross-spectrum curves need not be physically admissible.

For the magnetic velocity model, imaginary cross-spectra retain the odd lag correlations whose symmetric contribution vanished from the displacement diffusion tensor. A Hermitian spectrum can have complex off-diagonal entries without being a real symmetric matrix. This is how a phase-sensitive frequency description preserves signed rotational information.

16.6 Johnson–Nyquist noise: convert conventions by preserving variance L19 47:28

Let $R_e > 0$ be a resistance in ohms at equilibrium temperature T. In the classical low-frequency regime, its open-circuit voltage-noise source has $S_e(\omega) = 2 k_B T R_e$ in the two-sided angular-frequency convention (16.1). To obtain the one-sided density per hertz, set $\omega = 2\pi f$. Evenness gives $\int_{-\infty}^{\infty} S_e d\omega/(2\pi) = \int_0^{\infty} 2 S_e(2\pi f) df$, so the one-sided density is $G_e(f) = 4 k_B T R_e$. The factor of two comes from folding negative frequencies; the 2π belongs to the change of integration variable.

Nyquist's original primary paper, printed page 112, equation (1), states the squared voltage in a positive cycles-per-second band as $4RkT\,d\nu$. Printed page 113, equations (5)–(6), carries it through a network transfer admittance. That is the physical source for the normalization; the modern convention conversion here preserves the same measured variance.

To check the coefficient, connect the resistor's source to a capacitor $C_e > 0$ through the series resistance and measure capacitor voltage. The low-pass transfer is $H_c(\omega) = 1/(1 - i\omega R_e C_e)$. Integration gives

$$\mathbb{E}[V_c^2] = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \frac{2 k_B T R_e}{1 + (\omega R_e C_e)^2} = \frac{k_B T}{C_e}. \tag{16.5}$$

The result follows by substituting $z = \omega R_e C_e$ and integrating $1/(1 + z^2)$ to π. It yields mean capacitor energy $C_e \mathbb{E}[V_c^2]/2 = k_B T/2$, precisely equipartition. An erroneous one-sided/two-sided factor would fail this physical check. Resistance changes bandwidth and relaxation time while leaving the integrated equilibrium capacitor variance unchanged.

The ideal white source is only a band approximation. At sufficiently high frequency or low temperature, quantum and frequency-dependent impedance effects matter; no finite real resistor emits a classical flat spectrum to arbitrarily high frequency. The chosen ideal RC integral is consistent when its dominant band lies in the regime where the classical source and ideal components are adequate. Thermal-current circuit analogies must also use the actual topology: $L\dot{I} + R_e I = e$ describes a series resistance–inductance loop.

16.7 Laboratory: resolution, averaging, and spectral area

The laboratory generates 1,024 exact stationary OU records with step 0.02 and 8,192 samples, using seed 2026110816. The discrete Fourier periodogram is $h|\sum_{j=0}^{n-1} x_j e^{i\omega jh}|^2/n$. At the discrete Fourier frequencies, Parseval's identity makes its two-sided bin sum equal to the record mean square. Real-input output stores positive frequencies, so interior bins are doubled for the area; zero and the sampling-limit endpoint are counted once.

Run `python code/ch16_spectra.py`. It writes JSON and text output beside the source, without network calls. The program compares periodogram means with the exact finite-record discrete triangular-covariance sum, measures relative variance at the same nonzero frequency for two record lengths, integrates the RC spectrum, and checks Hermitian positivity after filtering correlated two-channel inputs. These tests distinguish normalization, finite-length bias, and random estimation scatter.

```
duration   mean periodogram   finite-record mean   relative variance
   20.48         0.79962567           0.80602849          0.92260528
  163.84         0.79662209           0.79914090          1.01739133
RC filtered variance=0.33333333; kBT/C=0.33333333
```

The periodogram expectation changes only modestly at the chosen frequency, while its relative variance remains near one. Increasing duration eightfold has not supplied eight independent spectral estimates at that frequency. Figure 16.1 compares one raw record, the ensemble average, the continuous Lorentzian, and the exact sampled spectrum. Their high-frequency differences include aliasing, not just statistical scatter.

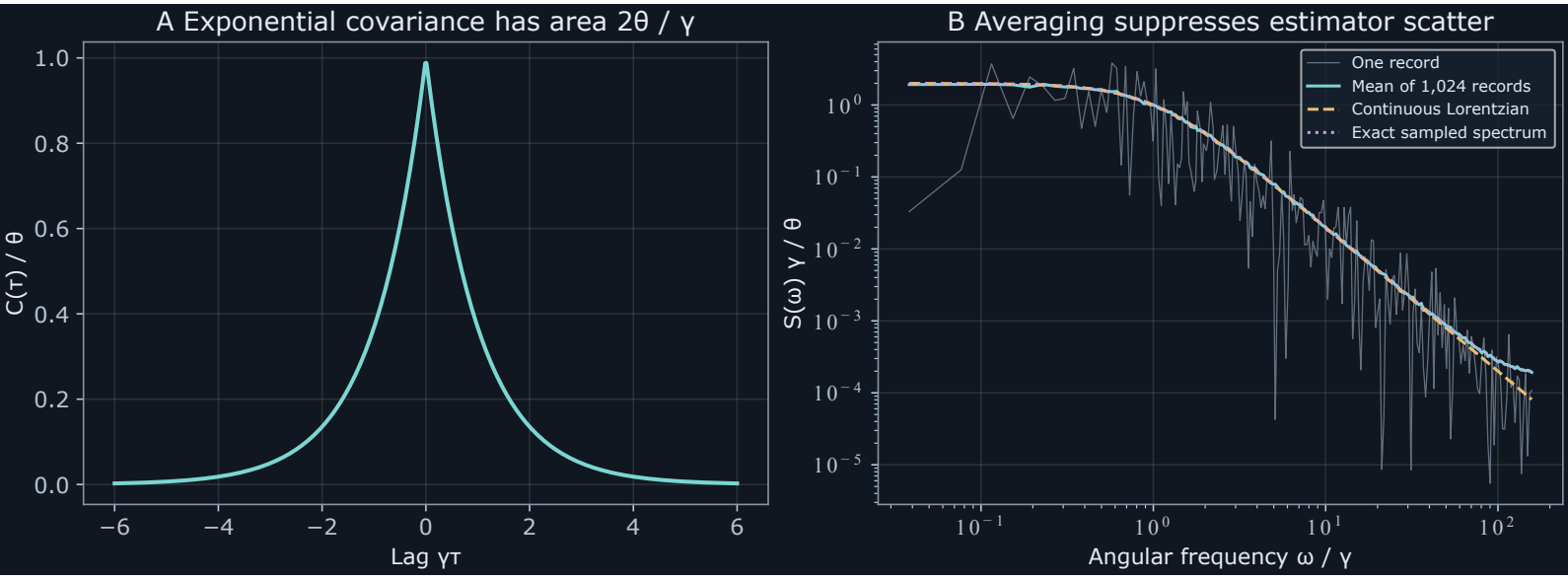


Figure 16.1. Time and frequency descriptions carry the same covariance information but different estimator noise. Panel A has normalized covariance. Panel B uses dimensionless $\gamma = \theta = 1$, finite sampling step 0.02, and 1,024 independent synthetic records. The raw periodogram remains irregular; averaging reduces scatter, while the sampled model differs from the continuous spectrum near the sampling limit. Generator.

```
import numpy as np


def ou_spectrum(omega, gamma, variance):
    if gamma <= 0 or variance < 0:
        raise ValueError("positive rate and nonnegative variance required")
```

```
    omega = np.asarray(omega)
    return 2*variance*gamma/(gamma**2 + omega**2)
```

The function returns a two-sided angular-frequency density, not power already integrated over a bandwidth. It accepts scalar or array frequencies and writes no files. The integration measure and any instrument filter must still be applied before comparing with a measured mean square.

16.8 What a spectrum identifies

A covariance spectrum describes second-order temporal structure. For a jointly Gaussian process, mean and covariance determine finite-dimensional laws. Without Gaussianity, identical spectra can coexist with different higher-order statistics and event structure. Stationarity is also a model condition, not guaranteed by drawing a finite window. Slow drift can contaminate low-frequency power, and ensemble stationarity does not remove finite-record uncertainty.

The filter rule gives the output spectrum for a specified linear stationary system. It does not make arbitrary observed noise thermal or establish fluctuation–dissipation. That stronger connection requires equilibrium and a conjugate perturbing force, developed in chapter 17.

16.9 Exercises with worked solutions

Exercise 16.1 — A factor-of-two diagnosis (conceptual)

A calculation integrates $4k_BTR_e$ over negative and positive hertz. What has gone wrong?

Worked solution

The coefficient $4k_BTR_e$ is a one-sided per-hertz voltage density. Integrating it over both signs counts the same real-signal variance twice. The two-sided per-hertz density is $2k_BTR_e$; equivalently use the angular density $2k_BTR_e$ with measure $d\omega/(2\pi)$. Preserving the integral, rather than memorizing isolated factors, resolves the convention. The RC equipartition check reveals the error directly.

Exercise 16.2 — Does a spectrum identify a mechanism? (conceptual)

A two-state stationary fluctuator and a centered Gaussian OU process have the same exponential covariance. Can their Lorentzian spectra distinguish them?

Worked solution

No. The spectrum is the transform of that shared covariance, so it is identical after matching variance and decay rate. Their marginal distributions and path structures differ: one jumps between two levels, the other is Gaussian and continuous. Higher moments, histograms, or resolved events can distinguish them. A Lorentzian fit establishes a second-order relaxation pattern under the assumed stationarity, not a unique microscopic model.

Exercise 16.3 — Derive Gaussian periodogram variance (derive)

Write $A = U + iV$ with centered jointly Gaussian real components. Prove the fourth-moment identity used in Section 16.3.

Worked solution

Let $a = \mathbb{E}[U^2]$, $b = \mathbb{E}[V^2]$, and $c = \mathbb{E}[UV]$. Gaussian moments give $\mathbb{E}[U^4] = 3a^2$, $\mathbb{E}[V^4] = 3b^2$, and $\mathbb{E}[U^2V^2] = ab + 2c^2$. Expanding $(U^2 + V^2)^2$ and subtracting $(a + b)^2$ gives variance $2a^2 + 2b^2 + 4c^2$. Meanwhile $(\mathbb{E}|A|^2)^2 + |\mathbb{E}[A^2]|^2 = (a + b)^2 + |(a - b) + 2ic|^2$ gives the same expression. Division by record duration squared proves the periodogram relation.

Exercise 16.4 — Derive the sampled OU spectrum (derive)

Transform the discrete covariance $\theta r^{|k|}$ at step h and recover S_h.

Worked solution

The density in physical angular-frequency units is $h\theta \sum_{k=-\infty}^{\infty} r^{|k|} e^{i\omega h k}$. Split off $k = 0$ and sum the two geometric series, giving $h\theta[1 + re^{i\omega h}/(1 - re^{i\omega h}) + re^{-i\omega h}/(1 - re^{-i\omega h})]$. Combining denominators yields $h\theta(1 - r^2)/(1 + r^2 - 2r\cos\omega h)$. Integration over one sampling band kills all nonzero Fourier modes, leaving variance θ. This verifies the required factor of h.

Exercise 16.5 — Choose independent averaging (compute/design)

If individual periodograms have relative standard deviation approximately one, how many independent records reduce it to approximately 0.1?

Worked solution

The average of M independent, identically distributed estimates has variance divided by M and unchanged mean, so relative standard deviation becomes $1/\sqrt{M}$. The target 0.1 requires approximately 100 records. Merely making one raw record a hundred times longer does not provide that same binwise averaging. Correlated overlapping segments have an effective independent count smaller than their nominal count and need an uncertainty analysis.

Exercise 16.6 — Test an impossible spectral matrix (compute/design)

A two-channel estimate proposes $S_{11} = S_{22} = 1$ and real $S_{12} = S_{21} = 1.2$ at a frequency. Is it admissible?

Worked solution

The matrix is Hermitian but has determinant $1 - 1.2^2 = -0.44$ and eigenvalues 2.2 and -0.2. It is not positive semidefinite. The difference channel would have density $S_{11} + S_{22} - 2S_{12} = -0.4$, impossible for an expected squared Fourier amplitude. The coherence bound $|S_{12}|^2 \le S_{11}S_{22}$ detects the same defect. A physically valid

estimation procedure should preserve the joint spectral covariance rather than fit each entry independently without constraints.

16.10 Sources and attribution

- V. Balakrishnan, NPTEL lecture 19, course 115106089. The book explicitly separates the ensemble spectral identity from raw-periodogram consistency and converts the lecture's transform normalization.
- H. Nyquist, Thermal Agitation of Electric Charge in Conductors, *Physical Review* 32, 110–113 (1928). Verified primary scan, printed pages 112–113, equations (1), (4)–(8), inspected directly. Used for thermal voltage normalization, physical filtering, and the classical-regime qualification.

Causality, susceptibility, and fluctuation–dissipation

A trapped particle can be characterized by driving it gently or by recording its spontaneous equilibrium fluctuations. Under precise equilibrium and coupling assumptions, the two measurements determine the same dissipative response. A stronger drive in a nonlinear trap can invalidate the comparison even when its weak response is predicted accurately. The task is to identify the observable, conjugate force, Fourier convention, and perturbation scale before converting a noise spectrum into a susceptibility.

> The sequence follows NPTEL lecture 20: driven Langevin mobility, the two fluctuation–dissipation relations, causal response, classical equilibrium correlation, analyticity, and LCR analogy. The canonical phase-space derivation, coordinate-versus-velocity distinction, finite-band dispersion check, and nonlinear control are developed here. Quantum response is identified as a separate extension.

17.1 Linear response is a causal convolution L20 18:04

Consider a stationary reference system with observable B and a prescribed small external field $F(t)$. Write $\delta\langle B_t\rangle$ for the change in its ensemble mean. Linearity to first order, time-translation invariance of the reference state, and causality give

$$\delta\langle B_t\rangle = \int_0^\infty R_{BA}(s)F(t-s)ds + O(F^2). \tag{17.1}$$

The retarded response kernel $R_{BA}(s)$ vanishes at negative lag; the label A will identify the observable conjugate to the field. A causal response depends on earlier fields, and stationarity makes the kernel depend only on elapsed time. “Retarded” expresses that causal ordering; it is not an additional synonym for equilibrium. Causality and linearity alone do not establish a fluctuation–dissipation theorem.

Use the book's positive-exponential transform: $\widetilde{F}(\omega) = \int F(t)e^{i\omega t}dt$. Substituting it into (17.1) and shifting integration time gives $\widetilde{\delta\langle B\rangle}(\omega) = \chi_{BA}(\omega)\widetilde{F}(\omega)$, with susceptibility

$$\chi_{BA}(\omega) = \int_0^\infty R_{BA}(s)e^{i\omega s}ds. \tag{17.2}$$

For a force $F_0\cos\omega t = \mathrm{Re}[F_0 e^{-i\omega t}]$, the steady linear response is $F_0[\mathrm{Re}\,\chi\cos\omega t + \mathrm{Im}\,\chi\sin\omega t]$. This fixes the phase convention and the sign of the dissipative component. The kernel has units observable divided by field and time; susceptibility has units observable divided by field. Initial transients must decay before this steady sinusoidal formula applies.

17.2 Free velocity response and thermal noise L20 1:21

For mass $m > 0$ and drag rate $\gamma > 0$, add a deterministic force to the free Langevin model: $m\,dV_t = (-m\gamma V_t + F(t))dt + \sqrt{2m\gamma k_B T}\,dW_t$. The white random force has zero mean and its amplitude

retains the equilibrium thermal relation. Averaging gives $m\dot{\bar{v}} + m\gamma\bar{v} = F$. Its impulse-response kernel is $R_{VX}(t) = e^{-\gamma t}/m$ for $t \geq 0$, since a unit force impulse changes momentum by one.

Transforming the exponential kernel gives dynamic mobility $\mu(\omega) = 1/[m(\gamma - i\omega)]$, a velocity-per-force susceptibility. The equilibrium velocity spectrum from chapter 16 is $S_V(\omega) = 2\gamma k_B T/[m(\gamma^2 + \omega^2)]$. Therefore

$$S_V(\omega) = 2k_B T \operatorname{Re}\mu(\omega). \tag{17.3}$$

This connects spontaneous fluctuations with the dissipative response to a force. The real part enters because mean force power is $\langle FV\rangle_{\text{cycle}} = F_0^2 \operatorname{Re}\mu/2$. For a passive equilibrium model this is nonnegative. The static mobility is $\mu(0) = 1/(m\gamma)$, so $D = k_B T\mu(0)$ recovers the Einstein relation.

The random-force spectrum itself is $S_\zeta = 2m\gamma k_B T$. This is often called the second fluctuation–dissipation relation, while (17.3) is called the first. Names vary across treatments; the objects remove ambiguity. One relates bath-force fluctuations to the friction appearing in a reduced equation, and the other relates an observed equilibrium spectrum to a driven response. The algebraic filter identity $S_V = |\mu|^2 S_\zeta$ is compatible with both but, without a thermal relation for S_ζ, does not imply either equilibrium statement.

The harmonic model is linear, so its deterministic mean equation responds linearly at every force amplitude within that model. “Small force” becomes essential when linear response approximates a more general nonlinear system or a bath whose properties change with driving. The laboratory includes an explicitly nonlinear trap to test this boundary rather than pretending a large-amplitude harmonic calculation must fail.

17.3 Equilibrium turns response into a correlation L20 28:20

To see where the thermal connection comes from, consider classical canonical phase space with coordinates q_j, p_j, Hamiltonian H_0, and normalized equilibrium density $\rho_0 = Z^{-1}e^{-\beta H_0}$, where $\beta = 1/(k_B T)$ and Z is the finite partition integral. Angular brackets now mean integration against this density. Define the Poisson bracket by $\{f, g\} = \sum_j (\partial_{q_j} f\, \partial_{p_j} g - \partial_{p_j} f\, \partial_{q_j} g)$. For an observable with no explicit time dependence, $\dot{A} = \{A, H_0\}$ under the reference dynamics.

Let the field perturb energy through $H(t) = H_0 - F(t)A$. This defines conjugacy: FA has energy units. A mechanical force couples to position, so $A = X$ for that example. Let B be the measured observable, which can differ from A. Assume smooth Hamiltonian flow, preserved phase-space measure, boundary decay permitting integration by parts, and integrability of the correlations below. A reduced Langevin model can inherit this equilibrium response relation; it is not a proof that its finite-dimensional stochastic dynamics are isolated Hamiltonian dynamics.

Proposition 17.1 — Classical equilibrium response. To first order in the field, for positive lag,

$$R_{BA}(t) = \beta\langle \dot{A}(0)B(t)\rangle_0. \tag{17.4}$$

Derivation.

The density obeys Liouville evolution $\partial_t \rho = \{H, \rho\}$. Write $\rho = \rho_0 + \delta\rho$ and retain terms linear in F. With $L_0 f = \{H_0, f\}$, this gives $\partial_t \delta\rho = L_0 \delta\rho - F\{A, \rho_0\}$. Differentiating the Gibbs density gives $\{A, \rho_0\} = -\beta\rho_0\{A, H_0\} = -\beta\rho_0\dot{A}$. Thus the perturbation source is $\beta F \rho_0 \dot{A}$.

Propagate that source from the earlier time to the observation time with the unperturbed flow. Integrating B against the propagated source is, by measure preservation, equivalent to propagating B forward against the initial source. The result is $\delta\langle B_t\rangle = \beta \int_{-\infty}^{t} F(s)\langle \dot{A}(0) B(t-s)\rangle_0 ds$, with suitable switch-on or convergence. Comparing with (17.1) proves (17.4). Constants in B do not contribute because $\langle \dot{A}\rangle_0 = 0$. ∎

The equilibrium density and energy coupling create the correlation formula. General causal filters do not supply those ingredients. Kubo's original analysis, printed pages 572–573, Section 2 and Theorem 1, constructs response to a perturbation $-AF$ with a classical Poisson-bracket expression and a separate quantum commutator expression. The derivation here explicitly performs the canonical Gibbs-density step in classical phase-space notation.

In a normalizable confining equilibrium, choosing $A = X$ and $B = V$ gives $\dot{A} = V$ and the mobility form $\mu(\omega) = \beta \int_0^\infty \langle V_0 V_t\rangle_0 e^{i\omega t} dt$. A free position on the whole line has no normalized spatial Gibbs factor, so the finite-Z phase-space proof above does not literally apply to that full free state. For the free Langevin velocity, Section 17.2 instead verifies the response and thermal covariance directly: $R_{VX}(t) = e^{-\gamma t}/m = \beta C_V(t)$. Transforming this identity reproduces $1/[m(\gamma - i\omega)]$. It is consistent with the Kubo form without assuming a nonexistent free-position equilibrium.

17.4 Coordinate response uses the imaginary susceptibility

Let the measured coordinate itself be $B = A = X$, with centered stationary covariance $C_X(t)$ and perturbing energy $-F(t)X$. Assume its covariance and derivative decay sufficiently for the following transforms and integrations by parts. Stationarity gives $\langle \dot{X}_0 X_t\rangle_0 = -C_X'(t)$: differentiating a common shift of the two observation times gives zero. Equation (17.4) therefore becomes $R_{XX}(t) = -\beta C_X'(t)$ for positive lag.

Proposition 17.2 — Classical coordinate FDT. In the book's two-sided angular-frequency convention,

$$S_X(\omega) = \frac{2k_B T}{\omega} \operatorname{Im} \chi_X(\omega), \qquad \omega \neq 0. \tag{17.5}$$

The zero-frequency value is the corresponding limit when it exists.

Derivation.

Integrate $\chi_X = -\beta \int_0^\infty C_X'(t) e^{i\omega t} dt$ by parts. The decaying upper boundary and initial covariance give $\chi_X = \beta C_X(0) + i\beta\omega \int_0^\infty C_X(t) e^{i\omega t} dt$. Its imaginary part is $\beta\omega \int_0^\infty C_X(t) \cos\omega t \, dt = \beta\omega S_X(\omega)/2$, using scalar evenness. Rearranging proves (17.5). ∎

At zero drive frequency, the same boundary calculation gives static susceptibility $\chi_X(0) = \beta \operatorname{Var}(X)$ when the coordinate relaxes. Positive-frequency dissipation is $\omega F_0^2 \operatorname{Im}\chi_X/2$: multiply the sinusoidal force by the

derivative of its displacement response and average over a cycle. Its positivity matches the nonnegative spectrum in (17.5). The factor ω distinguishes this displacement response from the velocity mobility formula (17.3).

For a harmonically trapped particle, the mean equation is $m\ddot{\bar{x}} + m\gamma\dot{\bar{x}} + m\omega_0^2\bar{x} = F$. It gives

$$\chi_X(\omega) = \frac{1}{m(\omega_0^2 - \omega^2 - i\gamma\omega)}, \quad S_X(\omega) = \frac{2\gamma k_B T/m}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}. \tag{17.6}$$

Direct force-noise filtering gives the same spectrum, and its integral returns $k_BT/(m\omega_0^2)$. At zero frequency the susceptibility is $1/(m\omega_0^2)$, also equal to β times that equilibrium variance. Three routes—static equilibrium, force response, and spontaneous spectrum—check the same spring stiffness.

Free position on an infinite line is not stationary, so the coordinate spectrum and finite static coordinate susceptibility required in (17.5) do not exist in this form. The free-particle velocity mobility relation remains well defined. Choosing the correct measured observable and conjugate force prevents an apparent divergence from being mistaken for a failed theorem.

17.5 Causality selects an analytic half-plane L20 48:38

For a real causal kernel $R \in L^1([0,\infty))$, extend frequency to $z = x + iy$ with $y > 0$. The transform $\chi(z) = \int_0^\infty R(t)e^{izt}dt$ converges absolutely because $|e^{izt}| = e^{-yt}$. On compact subsets of the upper half-plane, each differentiated integrand $t^nR(t)e^{-yt}$ is dominated by an integrable multiple of $|R(t)|$. Differentiation under the integral therefore proves analyticity there. On real frequencies, a real kernel gives $\chi(-\omega) = \chi(\omega)^*$; in the upper half-plane the reflection identity is $\chi(-z^*) = \chi(z)^*$.

For an unsubtracted dispersion relation, assume additionally that χ is continuous on the real axis, locally differentiable at the frequency in question, and $\chi(z) = O(1/|z|)$ uniformly on large upper semicircles. The Cauchy contour then has no contribution at infinity. Taking its boundary limit gives $\chi(\omega) = (i\pi)^{-1}\,\mathrm{PV}\int \chi(s)/(s-\omega)ds$; the small contour around the boundary point supplies the half-residue. Splitting real and imaginary parts yields $\operatorname{Re}\chi(\omega) = \pi^{-1}\,\mathrm{PV}\int_{-\infty}^{\infty} \operatorname{Im}\chi(s)/(s-\omega)ds$. Here PV is the Cauchy principal value, taking symmetric exclusions around the denominator zero. For a real kernel, oddness of the imaginary part gives the equivalent positive-frequency form

$$\operatorname{Re}\chi(\omega) = \frac{2}{\pi}\mathrm{PV}\int_0^\infty \frac{s\operatorname{Im}\chi(s)}{s^2 - \omega^2}ds. \tag{17.7}$$

These are unsubtracted Kramers–Kronig relations under the stated large-frequency decay. Instantaneous or nondecaying response components can require subtractions or distributional terms. Analyticity alone does not specify how a numerical finite-band integral should handle missing tails. The laboratory varies that band explicitly.

For the oscillator, poles solve $\omega_0^2 - z^2 - i\gamma z = 0$, giving $z_\pm = -i\gamma/2 \pm \sqrt{\omega_0^2 - \gamma^2/4}$. Positive damping keeps both poles in the lower half-plane. At critical damping they coincide; at stronger damping they separate along the negative imaginary axis. They are the frequency-plane form of the time-domain roots of chapter 11.

A causal susceptibility need not have a finite lower-half-plane singularity. For example, the integrable kernel $R(t) = 1$ on $0 \le t \le \tau_*$ and zero elsewhere has $\chi(z) = (e^{iz\tau_*} - 1)/(iz)$, with a removable value τ_* at zero. It is entire and grows in parts of the lower half-plane. Liouville's theorem would require boundedness throughout the entire complex plane, a hypothesis not supplied by causality. The oscillator's poles reflect its specific dynamics, not a universal requirement for every response.

17.6 A series LCR circuit separates charge response from admittance L20 55:38

Let $q_e(t)$ be circuit charge, $I = \dot{q}_e$ current, $L_e > 0$ inductance, $R_e > 0$ series resistance, and $C_e > 0$ capacitance. An applied voltage $e(t)$ drives $L_e\ddot{q}_e + R_e\dot{q}_e + q_e/C_e = e$. The circuit parameters corresponding to mechanical m, $m\gamma$, and $m\omega_0^2$ are respectively L_e, R_e, and $1/C_e$.

The charge-per-voltage susceptibility is $\chi_q(\omega) = 1/[C_e^{-1} - L_e\omega^2 - iR_e\omega]$. The electrical admittance is current divided by voltage, so $Y(\omega) = -i\omega\chi_q(\omega)$, with a numerator that must not be omitted. The two functions share the damped poles but have different units and zero-frequency limits: charge can acquire a static value while a series capacitor blocks steady current.

Thermal voltage noise with spectrum $2k_BTR_e$ gives current spectrum $S_I = 2k_BT\,\mathrm{Re}\,Y$ in the classical equilibrium model. The conjugate energy is $-eq_e$, and measuring current differentiates the charge response, just as velocity mobility differs from displacement susceptibility. This circuit check ties the response formula to an actual measured quantity rather than a name assigned to a denominator.

17.7 Laboratory: predict response without driving the sample

The executable ch17_response.py uses $m = \omega_0 = k_BT = 1$ and $\gamma = 0.6$. It compares independent routes to the oscillator susceptibility: an analytic force response, a fitted numerical driven solution, and an ensemble spectrum of an undriven equilibrium stochastic system. It also tests a truncated dispersion integral. NumPy and SciPy are required; run `python code/ch17_response.py` from the course folder. Results are written beside the script as JSON and plain text; the seed and parameters are recorded. There is no network dependency.

The stochastic part samples the exact two-dimensional position–velocity transition, not Euler steps: with $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & -0.6 \end{smallmatrix}\right)$ and equilibrium covariance I, its update matrix is $E = e^{Ah}$ and its independent innovation covariance is $I - EE^T$. An equilibrium Gaussian initial state avoids a warm-up ambiguity. The program advances 512 independent records with $h = 0.02$ and 32768 samples per record, computes their position periodograms, and averages the spectra. The comparison band is $0.1 < \omega < 5$. Exact transitions remove integration bias but do not remove finite record, sampling, or periodogram fluctuations.

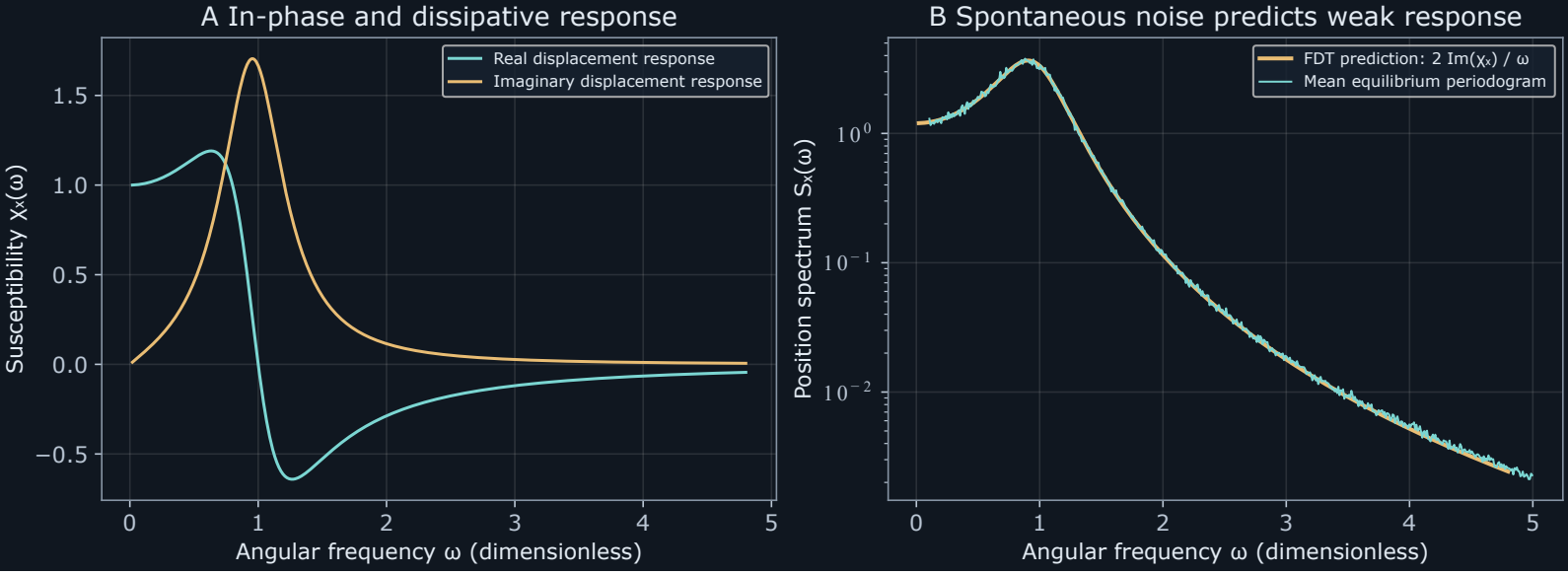


Figure 17.1. The oscillator's dispersive and dissipative responses have different shapes. The ensemble spectrum on the right predicts the imaginary displacement susceptibility through $S_X = 2\,\mathrm{Im}\,\chi_X/\omega$ for the stated unit thermal energy. The remaining jaggedness is statistical; averaging 512 records does not make a finite estimate exact.

The spectral integral independently gives position variance one, matching the canonical equilibrium covariance. Across the comparison band, the mean estimated spectrum has relative RMS discrepancy 0.05275627 from the continuous-time FDT prediction. This is a seeded numerical check, not a proof or an error bar for every frequency. The program's exact algebraic identity and spectral quadrature check the analytic formula separately from the random estimator.

```
python code/ch17_response.py
python figures/ch17_response.py
# Results: code/ch17-results.json and code/ch17-output.txt
# Figure: assets/ch17-response.svg
```

At drive frequency one, the deterministic mean equation is integrated to time 400, and only times after 200 enter a least-squares fit to cosine, sine, and a constant. With zero cubic stiffness, force amplitudes 0.05 and 1 both give response magnitude per force 1.6666667, agreeing with $|\chi_X(1)| = 1/0.6$. A second, explicitly deterministic control adds restoring force κx^3 with $\kappa = 0.5$. Its fitted fundamental response per force is 1.66664966 for amplitude 0.05, but 1.21065199 for amplitude one. The higher harmonics and changed fundamental are permitted by the nonlinear equation.

This control studies the deterministic small-amplitude limit about the stable origin. It is not a simulation of a thermally fluctuating Duffing oscillator: thermal excursions in that nonlinear potential would alter both its equilibrium covariance and its weak response. Applying the harmonic FDT curve to that different thermal model would be unjustified. Here the nonlinear control has the narrower purpose of showing why amplitude independence can be exact for a linear model and fail once nonlinear forces matter.

For the dispersion check, the principal-value integral is evaluated at $\omega = 0.7$, using finite upper cutoffs 3, 10, and 50. Its absolute error in the recovered real susceptibility falls from 0.005459119 to 0.00012896617 to 0.0000010191125. SciPy's Cauchy-weight quadrature handles the interior singularity; simply omitting one

frequency bin would mix discretization error with the tail error being measured. The result shows that a causal model obeying the full relation can still fail a finite-band implementation through omitted high frequencies.

17.8 What each assumption buys

Causality supports the retarded kernel and its analytic half-plane; equilibrium supports the canonical correlation formula; conjugate coupling identifies the correlation to use; and weak driving supports the truncation to first order. These are distinct assumptions. A stationary active process can have a spectrum and causal response while violating the equilibrium FDT. A harmonic thermal system can satisfy the same linear response at arbitrary deterministic drive amplitude because the modeled equation is exactly linear. Neither observation contradicts the restricted theorem.

The thermodynamic limit, slowly relaxing modes, conserved quantities, and unbounded coordinates can obstruct the decay conditions used in static limits and integration by parts. State the limiting procedure before interchanging infinite observation time, zero frequency, and system size. In particular, a free position diffuses rather than relaxing to a stationary coordinate distribution, whereas its velocity can have an integrable equilibrium covariance.

The derivation here is classical. Quantum mechanics replaces the classical phase-space response construction by operator response, with attention to operator order and the choice of fluctuation spectrum. One cannot obtain that result by inserting a quantum temperature factor into an unspecified classical correlation. Kubo's paper treats the classical and quantum formulations explicitly; the present chapter uses only the classical specialization stated above.

17.9 Six worked exercises

Exercise 17.1 — Which experiment tests equilibrium?

A stationary sample has a measured noise spectrum and a reproducible causal response to a weak periodic force. Does this establish an equilibrium fluctuation–dissipation theorem? Describe a falsifiable comparison and its required conventions.

Worked solution

Stationarity makes correlation depend on time difference, and weak probing supplies a linear response; neither determines the population weights of microscopic states. Equilibrium is additional information. First identify a coordinate X coupled through energy $-FX$, a temperature T, and the positive-exponential Fourier convention used here. Independently measure $S_X(\omega)$ and the response quadratures, extracting $\operatorname{Im}\chi_X$ from the sine component under a cosine force. The classical prediction is $\omega S_X/(2\operatorname{Im}\chi_X) = k_B T$ at frequencies where the denominator is nonzero, with controlled estimator and drive errors.

Failure of that comparison rejects at least one stated assumption or calibration. Agreement over a finite band supports the relation over that band but does not prove a canonical microscopic distribution. If the measured

observable is velocity rather than displacement, compare S_V to $2k_BT\,\mathrm{Re}\,\mu$ instead; using the wrong quadrature would manufacture a failure even for an equilibrium oscillator.

Exercise 17.2 — Can a causal response be entire?

For $R(t) = 1$ on $[0, \tau_*]$ and zero elsewhere, find $\chi(z)$, its value at zero, and its behavior at $z = -iy$ as $y \to \infty$. Explain why Liouville's theorem cannot force a finite pole.

Worked solution

Direct integration gives $\chi(z) = (e^{iz\tau_*} - 1)/(iz)$ for $z \neq 0$. The power series of the exponential makes the zero singularity removable, with value τ_*. Indeed, termwise integration on the finite time interval gives an everywhere-convergent power series, so the function is entire. For $z = -iy$, $y > 0$, it equals $(e^{y\tau_*} - 1)/y$, which grows exponentially.

Liouville's theorem applies to an entire function bounded throughout the complex plane. Causality bounds the transform in the upper half-plane for this integrable kernel; it does not bound the lower half-plane. The missing global boundedness prevents the proposed conclusion. Relaxation poles are powerful features of particular differential equations, but a finite-duration causal kernel provides a direct counterexample to their asserted necessity.

Exercise 17.3 — Recover mobility from unforced motion

Use $C_V(t) = (k_BT/m)e^{-\gamma|t|}$ and classical response to derive the complex mobility, its dissipative power, and the zero-frequency Einstein relation.

Worked solution

A force couples to position, so the formal conjugate pair is $A = X$, $B = V$, with $\dot{A} = V$. For the unconfined free Langevin model, use the velocity response directly established in Section 17.2 rather than assuming its whole phase space has a finite canonical partition integral. That calculation gives $R_{VX}(t) = e^{-\gamma t}/m = \beta C_V(t)$ for positive lag. Consequently $\mu(\omega) = m^{-1}\int_0^\infty e^{-(\gamma - i\omega)t}dt = 1/[m(\gamma - i\omega)]$. Its real part is $\gamma/[m(\gamma^2 + \omega^2)]$, which is positive.

A cosine force gives velocity $F_0[\mathrm{Re}\,\mu\cos\omega t + \mathrm{Im}\,\mu\sin\omega t]$. The mean product with force is $F_0^2\,\mathrm{Re}\,\mu/2$, since the mixed sine–cosine average vanishes. The same covariance yields $D = \int_0^\infty C_V(t)dt = k_BT/(m\gamma)$, while $\mu(0) = 1/(m\gamma)$; therefore $D = k_BT\mu(0)$. The factor of two in the two-sided spectrum follows from integrating both positive and negative lags, not from a different diffusion coefficient.

Exercise 17.4 — Charge susceptibility is not current admittance

Derive the series LCR current response and verify its equilibrium current-noise relation. What are its low-frequency limits?

Worked solution

With time convention $e^{-i\omega t}$, charge derivatives become $-i\omega q_e$ and $-\omega^2 q_e$. The circuit equation gives $\chi_q = 1/(a - iR_e\omega)$, where $a = C_e^{-1} - L_e\omega^2$. Current is $-i\omega q_e$, so $Y = -i\omega/(a - iR_e\omega)$. Multiplying numerator and denominator by $a + iR_e\omega$ gives $\operatorname{Re} Y = R_e\omega^2/(a^2 + R_e^2\omega^2) \geq 0$.

The resistor's two-sided voltage spectrum is $2k_BTR_e$. Filtering it to current multiplies by $|Y|^2 = \omega^2/(a^2 + R_e^2\omega^2)$, hence $S_I = 2k_BT \operatorname{Re} Y$. As $\omega \to 0$, $\chi_q \to C_e$ but $Y \sim -i\omega C_e \to 0$ and $\operatorname{Re} Y \sim R_e C_e^2 \omega^2$. A static voltage charges the capacitor while producing no sustained series current. Omitting the numerator would predict the wrong units, dissipation, and DC behavior.

Exercise 17.5 — Design a dispersion reconstruction

For the laboratory oscillator, estimate the real susceptibility at $\omega = 0.7$ from its imaginary part. Explain how the cutoff sequence distinguishes missing-tail error from a failed causal relation.

Worked solution

At the probe frequency, $a = 1 - 0.7^2 = 0.51$ and $b = 0.6(0.7) = 0.42$, so $\operatorname{Re}\chi = a/(a^2 + b^2) = 0.51/0.4365 \approx 1.16838488$. To reconstruct it, evaluate $(2/\pi)\,\mathrm{PV}\int_0^M s \operatorname{Im}\chi(s)/(s^2 - 0.7^2)ds$. Factor the denominator as $(s - 0.7)(s + 0.7)$ and supply the smooth factor $s \operatorname{Im}\chi(s)/(s + 0.7)$ to Cauchy-weight quadrature.

The program records absolute errors 0.005459119, 0.00012896617, and 0.0000010191125 for $M = 3, 10, 50$. At high frequency $\operatorname{Im}\chi(s) \sim 0.6/s^3$, so the integrand behaves as $0.6/s^4$ and the omitted tail decays as M^{-3}. The last two errors have approximately the predicted ratio $(50/10)^3 = 125$. This quantitative tail behavior supports a finite-band explanation. A single discrepant truncated integral would not justify rejecting causality; one must also control the pole treatment, calibration, Fourier sign, and missing bands.

Exercise 17.6 — Test the range of a linear model

At frequency one, compare the four fitted responses in the laboratory. Quantify the change produced by the cubic force at the stronger drive, and explain what can and cannot be concluded about thermal response.

Worked solution

For the harmonic equation, $\chi(1) = i/0.6$, giving magnitude 1.66666667. Both force amplitudes reproduce this value after dividing the fitted fundamental amplitude by force amplitude. The cubic control with $\kappa = 0.5$ gives 1.66664966 at force 0.05, a relative difference of about 1.02×10^{-5} from the harmonic value. At force one it gives 1.21065199, a reduction of about 27.36%. The dimensionless importance of the extra restoring force at displacement scale x_* is κx_*^2 relative to the unit linear stiffness, so its effect grows with amplitude.

The fit extracts the fundamental component, not the full waveform; nonlinear harmonics are allowed. The results demonstrate an amplitude-dependent deterministic response and the need to check a weak-drive range

when approximating nonlinear dynamics. They do not calculate the equilibrium FDT of a thermally populated nonlinear trap. That experiment would require its own thermal stochastic dynamics, equilibrium initialization, unforced correlation, and weak perturbation measurement. The harmonic spectrum cannot supply those missing data.

17.10 Sources and provenance

NPTEL lecture 20 supplies the chapter sequence. Its caption text was read sequentially; three timed lecture frames were inspected as bounded visual checks. Audio was not independently reviewed. The source map distinguishes those modalities and records the corrected circuit observable and analytic-function qualification.

R. Kubo, *Statistical-Mechanical Theory of Irreversible Processes. I. General Theory and Simple Applications to Magnetic and Conduction Problems*, Journal of the Physical Society of Japan 12, 570–586 (1957), doi:10.1143/JPSJ.12.570; primary-paper scan. Section 2, Theorem 1, printed pages 572–573, was inspected for the equilibrium response construction and its classical/quantum distinction. The canonical derivation, oscillator checks, and finite-band numerical experiment above are developed explicitly in the chapter.

The spectral normalization and Nyquist relation continue chapter 16; the exact oscillator covariance and roots continue chapter 11. Source coverage records sampled visuals rather than a full audiovisual audit.

← Chapter 16: noise spectra Course contents Chapter 18 →

Random pulses, counting cumulants, and shot noise

A detector's spectrum has regularly spaced minima. Do these reveal a preferred interval between events, or the duration of each detector pulse? Independent arrivals can produce spectral zeros without any preferred arrival spacing. The calculation below separates event timing, random pulse height, and deterministic pulse shape. It gives a concrete test: changing the pulse width moves the zeros even when the arrival process is unchanged.

> The topic sequence follows lecture 21: Poisson pulses, Campbell cumulants, spectral filtering, charge noise, magnetic pulses, and the bridge to persistent motion. The marked-process derivations, rectangular detector comparison, laboratory, and exercises are teaching extensions. The full English caption track was read; two timed frames were inspected, with limited legibility and partial occlusion. No continuous audiovisual review is claimed.

18.1 Distinguish an event train from its measured signal L21 1:08

Let $\{T_j\}$ be the event times of a homogeneous Poisson process on the entire real time axis. Its intensity $\lambda > 0$, in inverse seconds, means that the count in an interval of duration b is Poisson with mean λb, and counts in disjoint intervals are independent. Event times in their ordered sequence are not independent random variables: their ordered positions constrain one another. Independent counts on disjoint sets, or independent exponential successive gaps after a chosen event, are the useful statements.

Associate with each event an independent real amplitude A_j, all with the same distribution and independent of the event times. These amplitudes are called *marks*. Let the real deterministic function $h(u)$ be a pulse shape, taken dimensionless here; u is elapsed time since the event. A causal detector has $h(u) = 0$ for $u < 0$. The measured signal is

$$I(t) = \sum_j A_j h(t - T_j). \tag{18.1}$$

Its units are those of A_j. Overlap between pulses is allowed and does not create arrival correlations. The observation electronics superposes contributions linearly. Saturation, pulse suppression during recovery, or amplitudes depending on recent arrivals would change this model.

For a pulse of finite duration $w > 0$, only events in a finite interval affect any specified time, so the sum is almost surely finite. For pulses with infinite tails, require $\mathbb{E}|A| \int_{\mathbb{R}} |h(u)|\, du < \infty$ to obtain absolute convergence at each fixed time. This is a sufficient condition for the uncompensated sum used here. Square integrability alone can instead define a centered compensated random integral; it does not automatically make (18.1) an absolutely convergent ordinary sum.

Using arrivals on the whole line makes the signal stationary: translating every observation time is equivalent to translating the homogeneous event configuration. Every finite collection of measured values has the same joint law after that common translation. Starting arrivals at time zero produces a different, transient preparation. A

cumulative count from minus infinity is infinite for positive intensity; only counts in bounded intervals are ordinary finite random variables. The impulse train $J(t) = \sum_j \delta(t - T_j)$ is best interpreted as a random measure, with $\int_a^b J(t)\,dt$ equal to the interval count. A Dirac delta is a unit-integral distribution, not a finite-height random signal value.

18.2 Mean level: count first, then average L21 18:11

Write $m_r = \mathbb{E}[A^r]$ for an amplitude raw moment when it exists, and $m_I = \mathbb{E}[I(t)]$ for the signal mean. To establish the averaging rule, first restrict events to a bounded interval B of length b. Conditional on its Poisson count $N_B = n$, the unordered locations can be represented by n independent uniform samples from B. Therefore the expected sum of a deterministic integrable function g over these events is $\mathbb{E}[N_B]b^{-1}\int_B g(s)\,ds = \lambda \int_B g(s)\,ds$. Independent marks multiply this by their mean when g is a deterministic pulse contribution.

Proposition 18.1 — Stationary pulse mean. Under the absolute-integrability assumption in Section 18.1,

$$m_I = \lambda m_1 \int_{\mathbb{R}} h(u)\,du. \tag{18.2}$$

Derivation.

Apply the bounded-interval identity with $g(s) = h(t - s)$ and average over A. Increasing the interval to the whole line is justified for nonnegative contributions by monotone convergence; for signed contributions, apply it to their absolute values first and use the stated finite expectation to pass to the limit. The change of variable $u = t - s$ eliminates t, proving (18.2). ∎

This is the homogeneous marked version of Campbell's formula, Last–Penrose, Proposition 2.7, printed page 13. The mean identity itself needs an intensity measure, not Poisson independence. The covariance and higher cumulant identities below use that stronger assumption.

For the running detector, take $h(u) = 1$ on $0 \le u < w$ and zero elsewhere. Let amplitudes one and two signal units be equally likely. At intensity $\lambda = 3\ \mathrm{s}^{-1}$ and width $w = 0.4\ \mathrm{s}$, the mean is $\lambda w m_1 = 1.8$ signal units. The dimensionless product $\lambda w = 1.2$ is the mean number of simultaneously active pulses. A signal mean of 1.8 does not mean an event rate of 1.8 per second: detector width and amplitude intervene.

If arrivals start at zero and the detector is initially empty, the same calculation gives $\mathbb{E}[I(t)] = \lambda m_1 \int_0^t h(u)\,du$. For rectangular pulses this rises linearly until time w, then equals the stationary mean. For an exponential pulse it approaches the mean asymptotically. Stationary measurements require padding the arrival history or discarding the preparation interval; subtracting a measured mean cannot repair an incorrect transient covariance.

18.3 The same-event term survives centering L21 19:56

Define the centered covariance $C_I(\tau) = \mathbb{E}[(I(t) - m_I)(I(t+\tau) - m_I)]$, where τ is a real lag. Assume additionally that $m_2 < \infty$ and $\int h^2 < \infty$. For nonnegative rectangular pulses, a direct decomposition already identifies its origin: events affecting both observation times contribute to both signals; events affecting only one cannot create covariance after independent contributions are centered.

Proposition 18.2 — Covariance of marked Poisson pulses.

$$C_I(\tau) = \lambda m_2 \int_{\mathbb{R}} h(u)h(u+\tau)\,du. \tag{18.3}$$

Derivation.

Expand $I(t)I(t+\tau)$ as a double sum. Terms with the same event index have expected value $\lambda m_2 \int h(t-s)h(t+\tau-s)\,ds$, by the count identity. Terms with two distinct indices have expected value $\lambda^2 m_1^2 \iint h(t-s)h(t+\tau-r)\,ds\,dr = m_I^2$. To verify the second statement on a bounded interval, condition on N_B: there are $N_B(N_B-1)$ ordered distinct pairs, whose expectation is $(\lambda b)^2$ for a Poisson count. Their two uniform locations factorize. Integrability then allows the interval limit. Subtracting m_I^2 leaves the same-event term and proves (18.3). ∎

Amplitude variance would be the wrong coefficient. Even fixed identical amplitudes create fluctuations because the number of overlapping arrivals fluctuates. The coefficient is the *raw* second moment m_2, which includes m_1^2. The more general random-integral covariance, including its integrability assumptions, is recorded in Last–Penrose, Proposition 15.13, printed page 161.

The rectangle overlaps its translate for duration $(w - |\tau|)_+$, where $x_+ = \max(x, 0)$. Thus $C_I(\tau) = \lambda m_2 (w - |\tau|)_+$ is triangular, despite the absence of arrival-time correlations. In the detector example, $m_2 = 2.5$, so variance $C_I(0) = 3$ and the covariance at lag 0.2 seconds is 1.5. For lags at least 0.4 seconds, the two active-event intervals are disjoint and the signals are independent.

Higher cumulants retain the pulse non-Gaussianity

For bounded pulses of finite duration and bounded amplitudes, the moment-generating function exists near zero. Each event contribution has bounded absolute value, and only a Poisson number of events in a bounded interval can contribute. The Poisson generating function therefore bounds the exponential expectation. Let u have inverse-signal units and define $K_I(u) = \log \mathbb{E}[e^{uI(t)}]$. On a bounded event interval B, conditioning on its count gives a product of independent marked-location contributions. Summing the Poisson series yields

$$K_I(u) = \lambda \int_{\mathbb{R}} \left(\mathbb{E}[e^{uAh(s)}] - 1\right) ds, \qquad \kappa_r(I) = \lambda m_r \int_{\mathbb{R}} h(s)^r\,ds. \tag{18.4}$$

The first equality follows because $\mathbb{E}[z^{N_B}] = \exp[\lambda b(z-1)]$, with z the mean exponential contribution of one uniform marked event. Differentiating r times at zero proves the cumulant expression. For an unbounded-pulse, infinite-tail, or unbounded-mark model, require an integrable dominating exponential for the displayed

generating function and its derivatives; alternatively finite-order cumulants can be derived from characteristic-function derivatives under suitable absolute moment conditions. Existence of a few moments is insufficient for a moment-generating function near zero.

For the rectangular example every $\int h^r$ equals w, so $\kappa_3 = \lambda w m_3 = 5.4$ signal units cubed. Its positive third centered moment exposes skewness that a Gaussian approximation discards. Increasing intensity at fixed pulse shape makes standardized higher cumulants shrink: for $r > 2$, $\kappa_r / \kappa_2^{r/2}$ is proportional to $\lambda^{1-r/2}$ when the moments exist. This is a scaling observation about this family, not a proof that an arbitrary high-rate detector signal is Gaussian.

18.4 Pulse duration can create spectral zeros L21 22:53

Use angular frequency ω, in radians per second, and the two-sided centered convention from chapter 16: $S_I(\omega) = \int_{\mathbb{R}} C_I(\tau) e^{i\omega\tau}\, d\tau$, with inverse $C_I(\tau) = (2\pi)^{-1} \int S_I(\omega) e^{-i\omega\tau}\, d\omega$ when ordinary inversion applies. Define the pulse transform $H(\omega) = \int h(u) e^{i\omega u}\, du$, in seconds for our dimensionless h. This normalization differs from a transform that puts $1/(2\pi)$ in its forward integral; keeping the convention visible prevents a spurious factor in the measured noise level.

Proposition 18.3 — Centered pulse spectrum. With h absolutely and square integrable,

$$S_I(\omega) = \lambda m_2 |H(\omega)|^2. \tag{18.5}$$

The transform of the uncentered correlation contains the additional term $2\pi m_I^2 \delta(\omega)$.

Derivation.

Substitute (18.3) in the Fourier integral and set $v = u + \tau$. The exponential factors as $e^{i\omega v} e^{-i\omega u}$, so the double integral is the product $H(\omega) H(-\omega) = |H(\omega)|^2$ for real h. Absolute integrability justifies changing integration order. The uncentered correlation is $C_I(\tau) + m_I^2$; the Fourier transform of the constant term is the stated Dirac mass. ∎

The rectangle gives $H(\omega) = w e^{i\omega w/2} \operatorname{sinc}(\omega w/2)$, where $\operatorname{sinc}(x) = \sin x / x$ with value one at zero. This follows by integrating $e^{i\omega u}$ from zero to w and factoring out its midpoint phase. Hence $S_I(\omega) = \lambda m_2 w^2 \operatorname{sinc}^2(\omega w/2)$. The nonzero-frequency zeros occur at $\omega w = 2\pi n$ for nonzero integers n. At ordinary frequency $f = \omega/(2\pi)$, the first zero is $f = 1/w = 2.5$ Hz in our example. The limiting centered level at zero is $\lambda m_2 w^2 = 1.2$ signal units squared seconds; it is distinct from the mean-induced delta mass.

A finite record replaces a spectral distribution by bins with finite widths. Without subtracting the mean, a large near-zero bin reflects the windowed constant component. Calling that one bin a measured Dirac delta confuses an infinite-time definition with its estimator. Subtracting a sample mean also changes the very lowest-frequency estimate. The laboratory subtracts the known theoretical mean so that its centering convention is explicit.

18.5 Charge noise and magnetic pulses L21 27:48

For identical charge packets $e > 0$ arriving independently at rate λ, the ideal current measure is $i(t) = e \sum_j \delta(t - T_j)$. Its mean current is $\bar{i} = e\lambda$; its centered covariance is $e^2\lambda\delta(\tau)$ in the distributional sense. The two-sided angular spectrum is the constant $e^2\lambda = e\bar{i}$. Under the usual one-sided per-hertz convention, whose integral over $f \geq 0$ gives the variance, the level doubles to $2e\bar{i}$. These statements concern an ideal Poisson source; correlated electron transport, charge statistics, and the electronics can change the observed result. A real detector smooths the impulses, restoring the multiplicative $|H|^2$ response.

An exponentially decaying detector pulse $h(u) = e^{-u/\tau_p}$ for $u \geq 0$, with decay time $\tau_p > 0$, has $H(\omega) = \tau_p/(1 - i\omega\tau_p)$. Its centered spectrum is $\lambda m_2 \tau_p^2/(1 + \omega^2\tau_p^2)$, a Lorentzian. The covariance is $\lambda m_2 \tau_p e^{-|\tau|/\tau_p}/2$, obtained either by direct overlap integration or Fourier inversion. A Lorentzian alone therefore does not identify a thermal relaxation mechanism; random event pulses can produce one.

The lecture's magnetic-domain example motivates this exponential response. A changing magnetic flux produces a coil voltage proportional to its time derivative. One must specify whether the idealized jump in flux is included: differentiating $e^{-t/\tau_p}\mathbf{1}_{t\geq 0}$ produces both a delta term at onset and the subsequent exponential decay. If only the resolved decay after an unresolved onset is modeled, the exponential voltage pulse is a useful detector model. It does not establish independent domain events. Collective magnetic avalanches can violate the Poisson hypothesis; the algebra here predicts the statistics of the stated independent-pulse model.

18.6 An integral of colored noise remembers velocity L21 41:47

For comparison, let $V(t)$ take the values $+v_c$ and $-v_c$, switching direction at rate $\nu > 0$ in either state. Here v_c is a speed, not a pulse amplitude. In its symmetric stationary state, its mean is zero and its covariance is $v_c^2 e^{-2\nu|\tau|}$, as derived in chapter 6. Let position obey $X(t) = X(0) + \int_0^t V(s)\,ds$. The pair (X, V) is a Markov state; position alone generally omits the current direction.

This construction differs from adding independent pulses. Adjacent velocity intervals alternate sign, so treating their signs as independent shot-noise marks would change the process. Nevertheless the covariance-to-displacement calculation is immediate: $\mathbb{E}[(X(t) - X(0))^2] = 2\int_0^t (t - s) v_c^2 e^{-2\nu s}\,ds$ for the stationary symmetric initial velocity. Its short-time limit is ballistic, proportional to t^2; its long-time growth is diffusive. Chapter 19 develops the full state equations and the finite-speed probability fronts. A similar spectral shape need not imply a similar path law.

18.7 Computational laboratory: pad the stationary history

The laboratory constructs the rectangular detector exactly at sample times, by summing marks of arrivals in $(t - w, t]$. It draws a Poisson count over $[-w, T]$, then uniform ordered locations and independent heights one or two. Padding by one pulse width makes the sampled record stationary from its first observation. A prefix sum of sorted marks avoids approximating event times by grid cells. Inputs are the fixed parameters above, time step 0.01 seconds, duration 5,000 seconds, and seed 2026110821. Outputs are the retained numerical data and report; execution overwrites those two files and makes no network calls.

```
mean=1.800000; variance=3.000000; third cumulant=5.400000
stationary record mean=1.829676; variance=3.083746
integrated Welch variance=3.086729
first spectral zero=2.500000 Hz
```

The record is correlated, so its half-million samples are not independent moment estimates. A separate ensemble of 300,000 independent one-time counts tests the mean, variance, and third cumulant without that ambiguity. Independent thinning splits active arrivals into two Poisson counts of mean $\lambda w/2$; their weighted sum has exactly the required one-time law. The stationary record separately checks the triangular covariance at five lags. The Welch periodogram averages finite overlapping windows to reduce estimator scatter. For this even-length real-input Welch transform, only the interior positive-frequency bins are divided by two for comparison with (18.5) at $\omega = 2\pi f$; DC and Nyquist are undoubled endpoints and retain their returned values. Sampling aliases high-frequency power; the continuous-time envelope is a low-frequency comparison, not an exact sampled-spectrum identity.

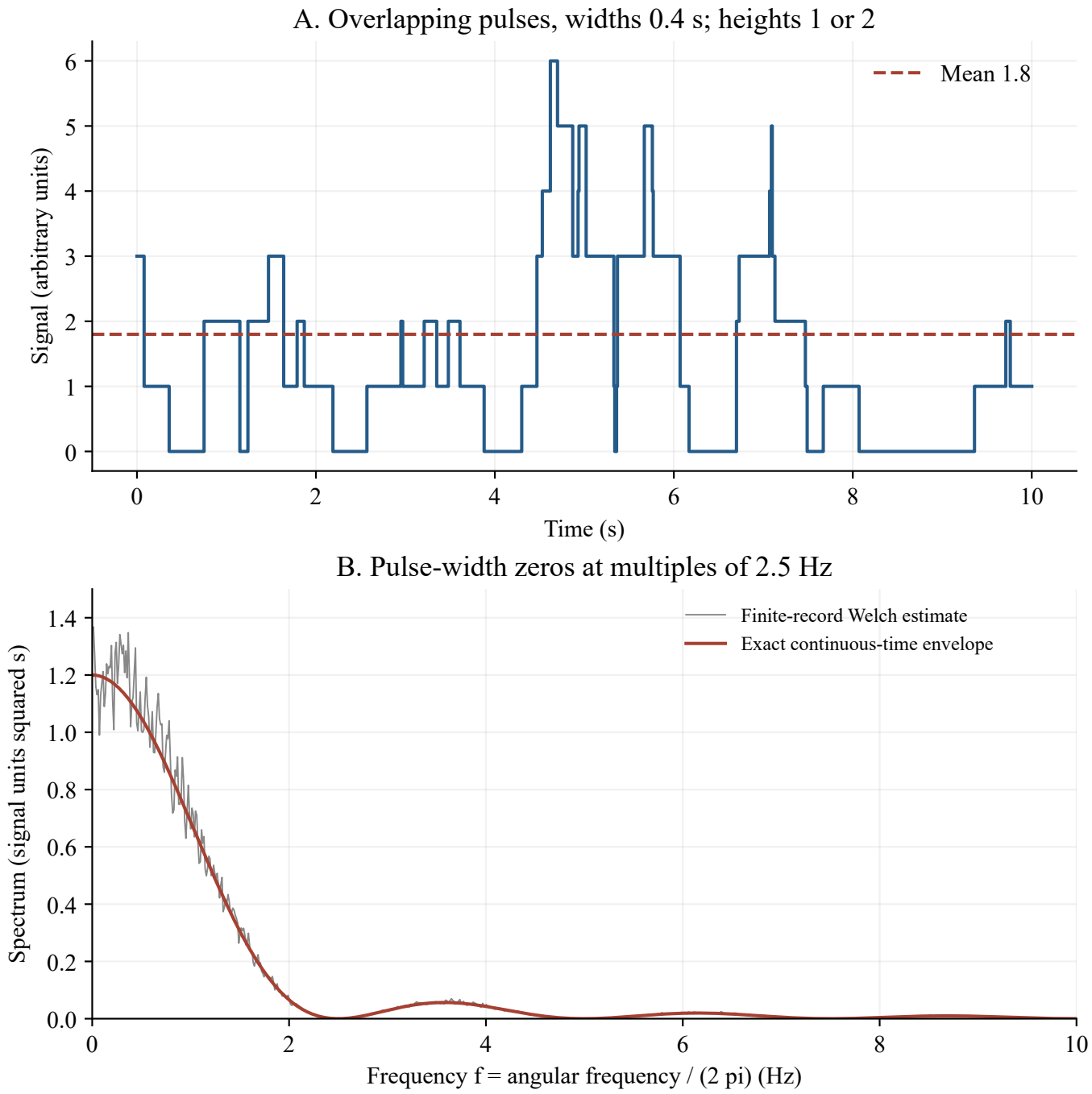


Figure 18.1. Independent arrivals produce a structured spectrum through pulse overlap. The gray curve is a finite sampled-record estimate; the red curve is the exact continuous-time centered spectrum. Moving the pulse width changes these zeros without changing the event-time law. The retained generator reproduces the figure from the laboratory data.

This compact calculation implements the exact moment predictions independently of the full simulation:

```
def rectangle_statistics(rate, width, amplitude_moments):
    if rate < 0 or width <= 0:
        raise ValueError("rate must be nonnegative; width positive")
    return [rate*width*moment for moment in amplitude_moments]


print(rectangle_statistics(3.0, 0.4, [1.5, 2.5, 4.5]))
```

The returned values are the first three cumulants, in their respective powers of signal units. They are not amplitude cumulants. Run `python code/ch18_shot_noise.py` followed by `python figures/ch18_shot_noise.py` using the declared dependencies to reproduce the complete comparison.

18.8 What can a spectrum identify?

A known pulse response and independent marks turn a spectrum into a test of arrival intensity and amplitude second moment. If pulse shape is unknown, its envelope and the event statistics are confounded. Mean and centered spectrum cannot distinguish all amplitude distributions, because they contain only m_1 and m_2; higher cumulants or resolved marks supply additional information. Correlated arrivals add distinct-event covariance terms that disappeared in Proposition 18.2. Random pulse widths require averaging the individual squared transforms, not squaring an averaged transform. Stationarity, linear response, independent marks, and sufficient integrability are model assumptions that must accompany any inferred event mechanism.

18.9 Exercises with worked solutions

Exercise 18.1 — Random counts with deterministic marks (conceptual)

Every pulse has height one, width w, and Poisson arrival intensity λ. A proposed covariance formula uses $\mathrm{Var}(A)$ and predicts no noise. Identify the missing randomness and compute the correct one-time distribution.

Worked solution

The signal at time t is the number of events in $(t-w, t]$. It is Poisson with mean and variance λw, although the amplitude variance vanishes. The random number of active pulses produces the noise. Its characteristic function is $\exp[\lambda w(e^{iu}-1)]$, so all one-time cumulants equal λw. The covariance uses $\mathbb{E}[A^2]=1$, giving $\lambda(w-|\tau|)_+$, not zero. The proposed formula incorrectly conditions on a fixed count and fails to restore count fluctuations.

Exercise 18.2 — A spectral minimum is not a quiet arrival process (conceptual)

The rectangular detector's width doubles while the arrival intensity and mark law stay fixed. Predict the mean, variance, low-frequency spectral level, and first zero relative to their old values. Explain why the zero does not demonstrate anticorrelated events.

Worked solution

The mean and variance are each proportional to w, so both double. The centered spectral level at zero is proportional to w^2, so it quadruples. The first positive zero $1/w$ halves. These conclusions follow from integrating the pulse and its overlap, with no change to the Poisson counts. The zero reflects cancellation of Fourier phases within one rectangular pulse: at $\omega w = 2\pi$, its transform is zero. It is not evidence of a forbidden arrival spacing. An independent measurement of pulse width, or a width-change experiment, resolves that ambiguity.

Exercise 18.3 — Derive an exponential detector covariance (derive)

For $h(u) = e^{-u/\tau_p}\mathbf{1}_{u\geq 0}$, derive the mean and covariance directly by integration. Check the variance by integrating its spectrum.

Worked solution

The pulse area is τ_p, giving $m_I = \lambda m_1 \tau_p$. For $\tau \geq 0$, the overlap integral is $e^{-\tau/\tau_p}\int_0^\infty e^{-2u/\tau_p}\,du = (\tau_p/2)e^{-\tau/\tau_p}$. Covariance is even for a real scalar stationary process, so $C_I(\tau) = \lambda m_2 \tau_p e^{-|\tau|/\tau_p}/2$. Integrating $\lambda m_2 \tau_p^2/(1+\omega^2\tau_p^2)$ over all angular frequencies and dividing by 2π gives $\lambda m_2 \tau_p/2$, since the integral of $(1+x^2)^{-1}$ on the real line is π. This agrees with $C_I(0)$ and checks both the width and Fourier normalization.

Exercise 18.4 — Random pulse durations (derive)

Each rectangular pulse now has an independent duration $W > 0$, independent of its amplitude. Assume $\mathbb{E}[W^2] < \infty$ and the required amplitude moments. Derive the mean, variance, covariance, and low-frequency spectral level.

Worked solution

Condition on the duration mark in the event-sum proof. Pulse area and squared-pulse area are both W, giving mean $\lambda m_1 \mathbb{E}[W]$ and variance $\lambda m_2 \mathbb{E}[W]$. The overlap for one marked pulse is $(W - |\tau|)_+$, so $C_I(\tau) = \lambda m_2 \mathbb{E}[(W-|\tau|)_+]$. Integrating over lag gives $S_I(0) = \lambda m_2 \mathbb{E}[W^2]$. More generally $S_I(\omega) = \lambda m_2 \mathbb{E}[W^2\,\mathrm{sinc}^2(\omega W/2)]$. Replacing W by its mean misses duration dispersion, even though it gives the first two one-time cumulants correctly here. If height and width are dependent, expectations must retain their joint products.

Exercise 18.5 — Design a rate comparison (compute/design)

Two otherwise identical detectors use widths 0.4 and 0.8 seconds with the mark law and intensity of the chapter. Compute the predicted first three cumulants and first spectral zeros. Give a simulation strategy that prevents a spurious preparation effect.

Worked solution

The raw amplitude moments are 1.5, 2.5, and 4.5. Multiplying them by λw gives cumulants $(1.8, 3, 5.4)$ and $(3.6, 6, 10.8)$, respectively. Their first zeros are 2.5 and 1.25 Hz. Generate one common Poisson event stream padded from at least minus 0.8 seconds, retain the same marks, and evaluate both moving active-event sums. Shared arrivals reduce the sampling noise of a paired comparison, but they make the two detector outputs correlated; their difference variance cannot be computed by adding variances. Starting at zero without padding understates early pulse overlap, especially for the wider detector. Compare ensembles or time-block uncertainty estimates rather than treating every sampled point as independent.

Exercise 18.6 — Equal spectra, unequal skewness (compute/design)

Replace the amplitudes one and two by a positive gamma-distributed amplitude with the same mean and variance. Does mean plus covariance distinguish the models? Derive the third-cumulant difference for the running rectangle.

Worked solution

The original amplitude mean is 1.5 and variance is $2.5 - 1.5^2 = 0.25$. A gamma amplitude of shape nine and scale $1/6$ has those moments: its mean is shape times scale and its variance is shape times scale squared. Its raw third moment is $9 \cdot 10 \cdot 11/6^3 = 55/12$, whereas the two-point amplitude has raw third moment $4.5 = 54/12$. Consequently both detector models have mean 1.8 and covariance $7.5(0.4 - |\tau|)_+$, but their third cumulants are $1.2(55/12) = 5.5$ and 5.4. A sample third moment needs substantially more data and uncertainty control than a mean estimate. Agreement of spectra cannot certify the full amplitude law. The gamma example's noninteger signal values also provide a direct resolved-pulse discriminator, if individual pulses can be measured.

18.10 Sources and attribution

- V. Balakrishnan, IIT Madras, lecture 21, NPTEL 115106089. Timestamp links follow the caption-verified topic changes; the notation and Fourier normalization here are stated independently.
- G. Last and M. Penrose, Lectures on the Poisson Process, author manuscript dated August 21, 2017: Proposition 2.7, p. 13; Theorem 3.9, pp. 23–24; Proposition 15.13, p. 161. These provide precise random-integral conditions and support the independently developed finite-interval derivations above.

The event-sum proofs, detector values, laboratory, original plot, and exercises are additions. The main scientific qualifications concern a stationary event history, raw amplitude moments, finite-intensity counts on bounded intervals, spectral normalization, and the idealized onset of a magnetic pulse. Raw captions and lecture frames are retained privately.

Finite-speed motion and the telegraph equation

A particle can travel at most one meter in one second. A Gaussian diffusion kernel assigns some probability to every distance, however short the observation time. The discrepancy is consequential for early arrivals at a distant target. A velocity that reverses randomly, while retaining a fixed speed between reversals, provides an exactly soluble model with both a finite propagation front and a diffusive long-time center.

> The sequence follows lecture 22: Poisson count covariance, a persistent lattice limit, coupled velocity states, second-order initial data, the mixed propagator, mean square displacement, diffusion scaling, and a crossing question. Its full caption track was read. One timed frame at 2995 seconds was inspected: front geometry is visible, while the detailed Bessel coefficients remain partly occluded. The derivation below checks those coefficients independently. No continuous audiovisual review is claimed.

19.1 Position retains an unobserved direction L22 0:46

Let $N(t)$ count events of a homogeneous Poisson process of rate $\nu > 0$, in inverse seconds, after time zero. Let the initial velocity $V(0)$ equal $+v_c$ or $-v_c$ with probability one half each, independently of N, where $v_c > 0$ has units of speed. Set $V(t) = V(0)(-1)^{N(t)}$ and $X(t) = \int_0^t V(s)\,ds$, starting at the origin. Each event reverses direction. The sample path of X is continuous and piecewise linear, with slopes restricted to these two values; its velocity jumps at reversal times.

Immediately $|X(t)| \le v_c t$ for every path. The same bound applies to displacement over any time interval. The state (X, V) is Markov: future Poisson increments are independent of the past and its present velocity determines the next displacement. Position alone generally fails this property. Two histories ending at the same position but moving in opposite directions predict different next short-time displacements. Retaining direction is a physical state enlargement, not a numerical correction to a Gaussian density.

Two related covariance identities should not be confused. Independent increments give $\operatorname{Cov}(N(s), N(t)) = \nu \min(s, t)$: for $s \le t$, split $N(t) = N(s) + [N(t) - N(s)]$ and use the Poisson variance of $N(s)$. The count is not stationary. By contrast, stationarity of the symmetric two-state velocity and the Poisson generating function give $\mathbb{E}[V(s)V(t)] = v_c^2 \mathbb{E}[(-1)^{N(|t-s|)}] = v_c^2 e^{-2\nu|t-s|}$. The mean time between reversals is $1/\nu$; the velocity correlation time is $1/(2\nu)$. They differ by a factor two because both an even and an odd number of switches affect the sign correlation.

19.2 The missing state becomes a second initial condition L22 19:14

Let $\rho_+(x, t)$ and $\rho_-(x, t)$ denote position probability densities in the positive and negative velocity states, understood as distributions when there are atoms. Their sum $\rho = \rho_+ + \rho_-$ is the position law. Their difference determines the probability current $j = v_c(\rho_+ - \rho_-)$, with units of inverse seconds in one spatial dimension.

Proposition 19.1 — Advection and switching. For the model above,

$$\partial_t \rho_+ + v_c \partial_x \rho_+ = \nu(\rho_- - \rho_+), \qquad \partial_t \rho_- - v_c \partial_x \rho_- = \nu(\rho_+ - \rho_-). \tag{19.1}$$

Consequently $\partial_t \rho = -\partial_x j$ and $\partial_t j = -v_c^2 \partial_x \rho - 2\nu j$, and

$$\partial_{tt} \rho + 2\nu \partial_t \rho = v_c^2 \partial_{xx} \rho. \tag{19.2}$$

Derivation.

During an interval dt, a positive-velocity path translates by $v_c dt$ and retains its state with probability $1 - \nu dt + o(dt)$. Paths from the negative state enter with probability $\nu dt + o(dt)$. Taylor expansion of the gain and retention terms gives the first equation; reversing the advection sign gives the second. Adding them cancels switching and gives probability conservation. Subtracting them and multiplying by v_c gives the current equation. Differentiate conservation with respect to time and substitute the spatial derivative of the current equation to obtain (19.2). ∎

A persistent lattice walk supplies the same scaling. Let a time step be h, spacing ℓ, and reversal probability β. The positive state at site q after a step receives probability $(1 - \beta)$ from the positive state at $q - 1$ and β from the negative state there. Take $\ell/h \to v_c$ and $\beta/h \to \nu$ as $h \to 0$. First-order Taylor terms produce (19.1). This is a finite-speed scaling; holding $\ell^2/(2h)$ fixed would instead be the different Brownian scaling of chapter 3.

The symmetric preparation has $\rho_\pm(x, 0) = \delta(x)/2$. The first-order equations then require $\partial_t \rho_+(x, 0) = -v_c \delta'(x)/2$ and $\partial_t \rho_-(x, 0) = +v_c \delta'(x)/2$, so $\rho(x, 0) = \delta(x)$ and $\partial_t \rho(x, 0) = 0$. Here δ' acts on a smooth test function g by $\int \delta'(x) g(x)\, dx = -g'(0)$. No square of a delta is involved. All three densities satisfy the same second-order differential equation, but their derivative data differ. Specifying only the initial position law loses this information.

If the initial positive-velocity probability is $b \in [0, 1]$, the same position delta is accompanied by $\partial_t \rho(x, 0) = -(2b - 1) v_c \delta'(x)$. The resulting mean displacement is $(2b - 1) v_c (1 - e^{-2\nu t})/(2\nu)$. The initial velocity bias decays, but its accumulated positional shift remains. A second-order scalar equation with arbitrary initial derivative also need not represent a positive probability law; the underlying nonnegative two-state initial data constrain which solutions have this interpretation.

19.3 A probability density alone omits the ballistic atoms L22 44:33

The probability of no reversal before t is $e^{-\nu t}$. Half of these paths are at $+v_c t$, and half at $-v_c t$. They form atoms: positive probability at individual points. Paths with at least one reversal supply an interior continuous law. Define $r = \sqrt{v_c^2 t^2 - x^2}$ for $|x| < v_c t$. For nonnegative argument z, the modified Bessel functions needed here can be defined by the convergent series $I_0(z) = \sum_{m \geq 0} (z/2)^{2m}/(m!)^2$ and $I_1(z) = \sum_{m \geq 0} (z/2)^{2m+1}/[m!(m+1)!]$. These are the integer-order cases of NIST DLMF 10.25.2.

Proposition 19.2 — Complete symmetric position law. At positive time the two atoms each have mass $e^{-\nu t}/2$, and the density on the open interior is

$$\rho_{\mathrm{int}}(x,t) = \frac{\nu e^{-\nu t}}{2v_c}\left[I_0\left(\frac{\nu r}{v_c}\right) + \frac{v_c t}{r} I_1\left(\frac{\nu r}{v_c}\right)\right]. \tag{19.3}$$

There is no probability outside $[-v_c t, v_c t]$.

Derivation.

Condition on $N(t) = n$. Its event times are the order statistics of n uniform samples on $(0,t)$, so the $n+1$ fractions of time between successive events are uniformly distributed on the simplex of nonnegative fractions summing to one. To see the resulting grouping law, let a of these intervals carry positive velocity and b carry negative velocity. The volume of the simplex slice where their positive sum is y is proportional to $y^{a-1}(1-y)^{b-1}$. Normalization gives the beta density $\Gamma(a+b)y^{a-1}(1-y)^{b-1}/[\Gamma(a)\Gamma(b)]$ on $(0,1)$.

The displacement is $x = v_c t(2y-1)$, with Jacobian $dy/dx = 1/(2v_c t)$. For $n = 2m+1$ reversals there are $m+1$ intervals of each sign, regardless of the initial sign. Multiply the beta density with parameters $(m+1, m+1)$ by the Poisson probability $e^{-\nu t}(\nu t)^{2m+1}/(2m+1)!$ and the Jacobian. Since $y(1-y) = r^2/(4v_c^2 t^2)$, summing these odd-switch contributions gives $\nu e^{-\nu t} I_0(\nu r/v_c)/(2v_c)$.

For $n = 2m$ with $m \geq 1$, the equally weighted initial signs give the mixture of beta densities with parameters $(m+1, m)$ and $(m, m+1)$. The factors proportional to y and $1-y$ add to one. Its density is $(2m)![y(1-y)]^{m-1}/[2m!(m-1)!]$. Multiplication by the even Poisson probability and the same Jacobian, followed by the substitution above, sums to the I_1 term in (19.3). The excluded $n = 0$ contribution gives exactly the two atoms. The construction conditions on mutually exclusive events with total probability one, so it establishes normalization and positivity as well as the formula. ∎

The apparent interior factor $I_1(\nu r/v_c)/r$ has a finite boundary limit, since $I_1(z) = z/2 + O(z^3)$. It does not represent the front atom. Integrating the finite interior density over a shrinking neighborhood of a front tends to zero; the separate atomic probability remains. At $\nu t = 1$, the continuous mass is $1 - e^{-1} = 0.63212056$, while each atom has mass 0.18393972. Renormalizing an interior-only histogram to unit area would therefore discard more than one third of the law and inflate its displayed density.

19.4 One covariance gives both transport scales L22 50:51

Let $M_2(t) = \mathbb{E}[X(t)^2]$ for the symmetric preparation. Because the mean is zero, it is also the centered variance. The double integral of velocity covariance is simpler than taking moments of Bessel functions. Stationarity gives $M_2(t) = \int_0^t \int_0^t v_c^2 e^{-2\nu|s-u|}\, ds\, du = 2v_c^2 \int_0^t (t-u)e^{-2\nu u}\, du$.

Proposition 19.3 — Ballistic to diffusive crossover.

$$M_2(t) = \frac{v_c^2}{\nu}\left[t - \frac{1 - e^{-2\nu t}}{2\nu}\right]. \tag{19.4}$$

It is $v_c^2 t^2 + O(t^3)$ as $t \downarrow 0$ and $2D_{\text{eff}} t + O(1)$ as $t \to \infty$, where $D_{\text{eff}} = v_c^2/(2\nu)$.

Derivation.

Differentiate the covariance integral: $M_2'(t) = v_c^2(1 - e^{-2\nu t})/\nu$. Integrating from zero proves (19.4). Expanding the exponential cancels the constant and linear terms inside the bracket and leaves $v_c^2 t^2$ at leading order. At long times the exponential vanishes, giving linear growth with slope v_c^2/ν. Alternatively multiplying (19.2) by x^2 and integrating distributionally gives $M_2'' + 2\nu M_2' = 2v_c^2$, with $M_2(0) = M_2'(0) = 0$, yielding the same result. ∎

The speed remains finite at every time even when the mean square displacement is linear. Long-time diffusion means cancellation of many alternating flight segments; it does not mean the microscopic trajectories become Brownian without a scaling limit. For $v_c = 1$ meter per second and $\nu = 1$ per second, the exact values at times 0.1, one, and five seconds are 0.00936538, 0.56766764, and 4.50002270 square meters. At one second neither the ballistic prediction one nor the long-time prediction one is accurate: the complete crossover is needed.

19.5 A diffusion limit suppresses the velocity relaxation scale L22 40:36

Define the characteristic function $\phi(k,t) = \mathbb{E}[e^{ikX(t)}]$, with k in inverse meters. Fourier transformation of the telegraph equation gives $\partial_{tt}\phi + 2\nu\partial_t\phi + v_c^2 k^2 \phi = 0$, with $\phi(k,0) = 1$ and $\partial_t\phi(k,0) = 0$. Its characteristic roots are $-\nu \pm \sqrt{\nu^2 - v_c^2 k^2}$. When $v_c, \nu \to \infty$ while $v_c^2/(2\nu) = D$ stays fixed and positive, the slow root tends to $-Dk^2$ and the fast root tends to minus infinity. The coefficient of the slow exponential tends to one. Thus at every fixed positive time and fixed k, $\phi(k,t) \to e^{-Dk^2 t}$.

Continuity of the limiting characteristic function establishes convergence in distribution at each fixed time to the Gaussian of variance $2Dt$. This argument does not by itself prove path-space convergence; that would require tightness and joint-time control. It does explain the disappearance of the atoms: their total mass $e^{-\nu t}$ vanishes while their front positions $\pm v_c t$ recede. At fixed parameters a separate long-time central scaling also leads to a Gaussian core, but a Gaussian approximation to that core still misses finite-speed support and rare no-switch flights. Deleting $\partial_{tt}\rho$ in (19.2) is a useful slow-scale approximation, not an identity valid near time zero.

19.6 Computational laboratory: preserve atoms and event times

The executable laboratory independently integrates the interior Bessel density, adds both front masses, and checks normalization and second moments at three times. It then draws 100,000 paths at each time with seed 2026110822. Given the Poisson switch count, it sorts uniform event times and sums actual signed flight durations. This simulation does not discretize switching or interpolate a Brownian path, and it does not sample the beta mixture used in the analytical proof. The speed is one meter per second and the switching rate one per second.

```
t=0.1: interior=0.09516258; atoms=0.90483742; MSD=0.00936538; sample=0.00937464
t=1: interior=0.63212056; atoms=0.36787944; MSD=0.56766764; sample=0.56775016
```

```
t=5: interior=0.99326205; atoms=0.00673795; MSD=4.50002270; sample=4.47136628
```

The tests compare the quadrature normalization and moment with exact values to numerical tolerance, and compare Monte Carlo values using broad sampling tolerances. They test an implementation of the model, not its validity for a particular material. The program writes data and output beside the source; it has no network effect. For much larger switching rates or times, unscaled Bessel evaluation can overflow, and arrays containing all event times can become expensive. Scaled Bessel functions and chunked event generation would be appropriate production extensions.

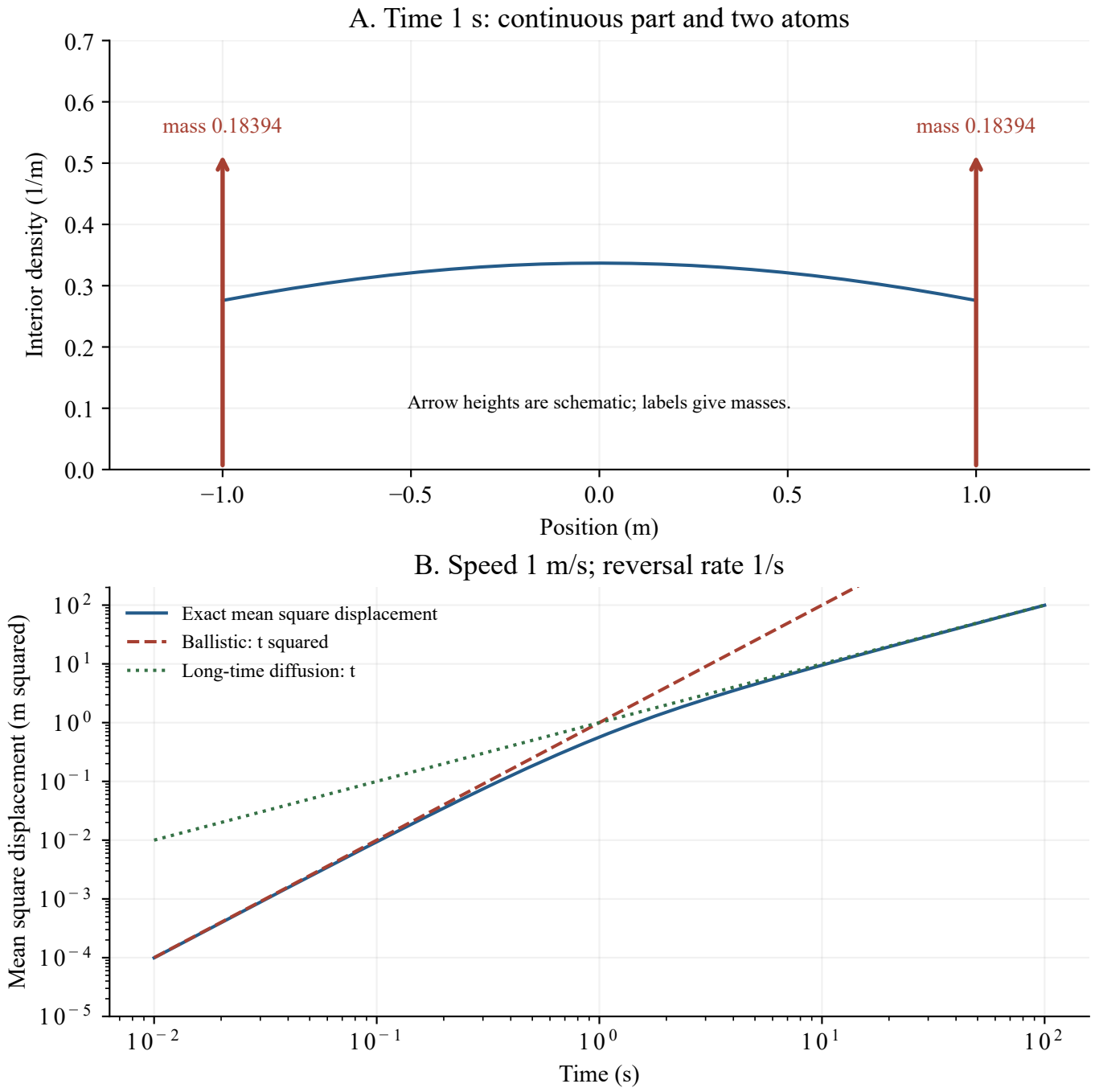


Figure 19.1. The probability law has an interior density and two atomic masses; arrow heights are schematic and their labels give probabilities. The lower panel compares the exact crossover with its two limiting powers. The generator uses the laboratory data. The diffuse core does not remove the finite-speed fronts.

For short times, evaluating $1 - e^{-2\nu t}$ by subtraction loses precision. The function `expm1` computes $e^{x} - 1$ accurately near zero. The following snippet returns (19.4) for ordinary positive times and a leading series for extremely small dimensionless times:

```
import math


def telegraph_msd(time, speed, rate):
    if time < 0 or speed <= 0 or rate <= 0:
        raise ValueError("require time >= 0 and positive speed, rate")
    if rate*time < 1e-5:
        return speed**2*time**2*(1-2*rate*time/3)
    return speed**2/rate*(time+math.expm1(-2*rate*time)/(2*rate))


print(f"{telegraph_msd(1.0, 1.0, 1.0):.8f}")
```

It prints 0.56766764. Its tiny-time branch is a truncated expansion with relative error of order $(\nu t)^2$; the full laboratory works at larger times and uses the exact exponential expression. Run the lab and figure generator in that order with the declared packages.

19.7 Boundaries of the finite-speed model

A constant speed with independent exponential reversal times idealizes a persistent moving object. It excludes speed variation, acceleration dynamics, unequal reversal rates, directional persistence with nonexponential flight durations, and interactions. Position is not a Markov variable merely because its density satisfies a local second-order equation. The full mixed law and its initial derivative condition carry the omitted velocity information.

For a positive target distance, no telegraph path can arrive before distance divided by v_c. Brownian first passage has a very small but nonzero density at every positive time. Thus an approximation that succeeds for the central propagator can fail for an early-arrival question. At a regular interior threshold the instantaneous mean crossing rate uses the state-resolved flux $v_c[\rho_+(a,t)+\rho_-(a,t)]$; a first arrival is different because later crossings must be excluded. Chapter 20 treats that stopping-time distinction and the role of absorbing boundaries.

19.8 Exercises with worked solutions

Exercise 19.1 — Same position, different future (conceptual)

Two ensembles have initial position zero, but one starts with velocity $+v_c$ and the other with $-v_c$. Why is the same initial position density insufficient? State the two initial time derivatives.

Worked solution

The first ensemble moves initially right except on an event of probability $O(dt)$; the second moves initially left. Their scalar densities both start at $\delta(x)$, while conservation gives $\partial_t\rho(x,0)=-\partial_x j(x,0)$. Initial currents are $+v_c\delta(x)$ and $-v_c\delta(x)$, respectively. Their derivatives are therefore $-v_c\delta'(x)$ and $+v_c\delta'(x)$. The second-order

equation needs both pieces of data. Observing the recent path reveals a direction that the present position alone omits.

Exercise 19.2 — A missing mass in a histogram (conceptual)

At $\nu t = 1$, an interior density is plotted and normalized to area one. What fraction of the actual probability has been omitted, and by what factor has the interior been exaggerated?

Worked solution

The omitted zero-switch event has probability $e^{-1} = 0.36787944$, split equally between the fronts. The correct interior area is $1 - e^{-1} = 0.63212056$. Renormalizing it multiplies every interior density by $1/(1 - e^{-1}) \approx 1.58197671$. A histogram may combine an atom and a neighboring density contribution in a boundary bin; changing that bin's width changes its height. Atomic masses should be stated separately, so a bin height is not mistaken for a continuous density value.

Exercise 19.3 — Derive the moment equation (derive)

Starting from the first-order state equations, derive a closed evolution for $M_2(t)$ without integrating the Bessel propagator. Work with the symmetric initial state.

Worked solution

The backward generator applied to a smooth observable $g(x, v)$ is $\mathcal{L}g = v\partial_x g + \nu[g(x, -v) - g(x, v)]$. For $g = x^2$, it gives $\mathcal{L}g = 2xv$, so $M_2' = 2\mathbb{E}[XV]$. For $g = xv$, the drift term is $v^2 = v_c^2$ and a reversal changes xv to $-xv$, giving $(\mathbb{E}[XV])' = v_c^2 - 2\nu\mathbb{E}[XV]$. Hence $M_2'' + 2\nu M_2' = 2v_c^2$, with both initial values zero. Integrating the first-order equation for M_2' gives $M_2' = v_c^2(1 - e^{-2\nu t})/\nu$, then (19.4). This avoids boundary errors from treating front atoms as a smooth density.

Exercise 19.4 — Rare reversals produce a uniform interior (derive)

Condition on exactly one reversal before t. Derive its position density and recover the leading interior term of (19.3) as $\nu t \downarrow 0$.

Worked solution

The single reversal time S is uniform on $(0, t)$. Starting right gives $X = v_c S - v_c(t - S) = v_c(2S - t)$, uniform on $(-v_c t, v_c t)$ with density $1/(2v_c t)$. Starting left reflects that same uniform law. Exactly one reversal has probability $\nu t e^{-\nu t}$, so its unconditional interior contribution is $\nu e^{-\nu t}/(2v_c)$. In (19.3), $I_0(z) = 1 + O(z^2)$ and the I_1 term is of higher switch order, recovering this term. Its integrated mass is $\nu t e^{-\nu t}$; the leading endpoint mass $e^{-\nu t}$ must still be retained.

Exercise 19.5 — Quantify a long-time approximation (compute/design)

For $v_c = \nu = 1$ in consistent SI units, find the smallest time at which the relative error of replacing exact $M_2(t)$ by t is below five percent. Define relative error with the approximation in the denominator. Give a conservative simple answer without root finding.

Worked solution

The specified relative error is $[t - M_2(t)]/t = (1 - e^{-2t})/(2t)$. It is strictly decreasing for positive t: its derivative has numerator $(2t + 1)e^{-2t} - 1 < 0$, since $e^{2t} > 1 + 2t$. The exact crossing solves $(1 - e^{-2t})/(2t) = 0.05$ and lies just below ten seconds. Thus ten seconds is a conservative simple choice: the error is $(1 - e^{-20})/20 < 0.05$. There is no smallest time with a strict inequality; the set is the open interval above that unique crossing. Changing the denominator to the exact value changes the threshold, illustrating why the error convention must accompany a quoted percentage.

Exercise 19.6 — Test a diffusion model for early arrivals (compute/design)

A target is two meters away, $v_c = 1$ meter per second, and $\nu = 1$ per second. Can matching the long-time diffusion coefficient justify predicted arrivals before one second? Design a numerical comparison that does not miss telegraph reversals.

Worked solution

The telegraph arrival probability before any $t < 2$ seconds is exactly zero by the pathwise speed bound. Its effective diffusion coefficient is $D = 0.5$ square meters per second, but a Brownian model with that coefficient has nonzero first-arrival probability at every positive time. Long-time variance matching cannot justify this early tail. Simulate exponential reversal gaps and integrate each linear segment until its endpoint or target crossing, recording a crossing inside a segment exactly. Compare with the Brownian image-method distribution developed in chapter 20 rather than relying on a time-grid crossing counter. This comparison isolates model error from the separate numerical error caused by missed between-sample crossings.

19.9 Sources and attribution

- V. Balakrishnan, IIT Madras, lecture 22, NPTEL 115106089: persistent lattice model, telegraph equations, mixed propagator, crossover, diffusion limit, and crossing motivation.
- NIST, DLMF 10.25.2: modified-Bessel series used to name the independently derived switch-count mixture.

The switch-count proof, exact-event laboratory, original figure, and worked solutions are additions. Density and atoms are always distinct measures; the diffusion claim proved here is fixed-time distributional convergence. A theorem about every possible correlated or finite-speed transport process is not inferred.

First passage, absorbing boundaries, and renewal

A diffusing searcher reaches a target with probability one, yet its mean arrival time can be infinite. A finite simulation will still print a finite average of its successful arrivals. To decide whether that average answers the intended question, distinguish eventual success, the distribution of the first success time, censored observations, and a mean conditioned on success. Absorbing boundaries make these distinctions mathematical rather than verbal.

> Based on lecture 23 and lecture 24. Both full caption tracks were read. One lecture-23 frame was sampled at 1355 seconds, with low-resolution notation; lecture-24 frames were not inspected. Added proofs check the half-line law, velocity-dependent absorbing condition, finite-interval backward equations, and simulation bias. Text, sampled visuals, and independent audio coverage are distinct; no full audiovisual review is claimed.

20.1 Repeated crossings and a first stopping event L23 0:22

A crossing counter records every passage through a threshold. A first-arrival clock stops at the earliest passage. Their expectations need different probability laws. For the finite-speed process of chapter 19, let $N_0(T)$ count crossings of the starting point during the open time interval $(0, T)$, excluding the initial presence at zero. The velocity magnitude is v_c, so the crossing flux through zero at positive time is $v_c \rho_{\mathrm{int}}(0, t)$. The front atoms are not at zero for positive t and create no later origin crossing. The expected count is therefore

$$\mathbb{E}[N_0(T)] = \frac{1}{2} \int_0^{\nu T} e^{-u} [I_0(u) + I_1(u)] \, du. \tag{20.1}$$

This follows by substituting $r = v_c t$ in the interior density (19.3), multiplying by v_c, and changing variable to $u = \nu t$. The same path may contribute many crossings. At long time the modified-Bessel asymptotic $e^{-u} I_j(u) \sim (2\pi u)^{-1/2}$ for fixed integer j yields $\mathbb{E}[N_0(T)] \sim \sqrt{2\nu T/\pi}$. This last step uses the positive-real large-argument expansion in DLMF 10.40.1; integrating a $u^{-1/2}$ tail gives the square-root growth. Finite velocity makes the crossing count locally finite, whereas a Brownian level can have infinitely many recrossings after a hit. The first-arrival problem remains meaningful for both.

For a process $X(t)$ and a target set A, define the first hitting time $\tau_A = \inf\{t \geq 0 : X(t) \in A\}$, with value infinity if the target is never reached. If the initial state lies in A, this definition gives zero; a *positive return* requires a different convention that excludes that initial visit. A stopping time means that deciding whether $\tau_A \leq t$ needs only information available up to t. The target geometry and path continuity determine whether the event is a hit, a crossing, or an overshoot.

20.2 Certain arrival with infinite mean time L23 17:11

Take one-dimensional Brownian diffusion $X(t) = \sqrt{2D} W(t)$, starting at zero, with $D > 0$ in square meters per second and standard Wiener process W of variance t. The target is $a > 0$ meters away and there is no left

boundary. Let τ_a be its first arrival. Define survival $S(t) = \Pr(\tau_a > t)$ and the density $f_a(t) = -S'(t)$ for positive time.

Proposition 20.1 — Half-line Brownian first arrival.

$$\Pr(\tau_a \le t) = \operatorname{erfc}\left(\frac{a}{\sqrt{4Dt}}\right), \qquad f_a(t) = \frac{a\, e^{-a^2/(4Dt)}}{\sqrt{4\pi D t^3}}, \quad t > 0. \tag{20.2}$$

The eventual arrival probability is one, but $\mathbb{E}[\tau_a] = \infty$.

Derivation.

Reflect a Brownian path about a after its first hit of a. The strong Markov property and symmetry of future increments preserve its law, pairing paths that hit and end below a with paths ending above a. Thus $\Pr(\tau_a \le t) = 2\Pr(X(t) \ge a)$. Integrating the centered Gaussian endpoint density gives the complementary error function, defined by $\operatorname{erfc}(z) = 2\pi^{-1/2} \int_z^\infty e^{-u^2}\, du$. Differentiation gives the displayed density. As $t \to \infty$, the argument tends to zero and $\operatorname{erfc}(0) = 1$, proving total mass one.

At large time $f_a(t) \sim a(4\pi D)^{-1/2} t^{-3/2}$. Multiplying by t leaves a nonintegrable $t^{-1/2}$ tail, so its first moment diverges. At zero the exponential suppresses every power singularity, so the divergence comes entirely from late arrivals. Equivalently, $S(t) = \operatorname{erf}(a/\sqrt{4Dt}) \sim a/\sqrt{\pi D t}$ and $\mathbb{E}[\tau_a] = \int_0^\infty S(t)\, dt = \infty$. ∎

The reflection step is developed in Pitman's Berkeley lecture 17, pages 17-2–17-3. The ordinary propagator $G(x,t) = e^{-x^2/(4Dt)}/\sqrt{4\pi D t}$ counts paths with any number of earlier target visits. The absorbing density on $x < a$ is instead $G(x,t) - G(2a - x, t)$; integrating it gives survival and differentiating its lost mass gives (20.2). This image subtraction works because reflected Brownian increments have the same law and the zero-density boundary enforces killing.

The mode of (20.2) is $a^2/(6D)$, found by setting the derivative of its logarithm to zero. It is finite even though the mean is not. A quantile is another finite measure of typical arrival. The distribution is a one-sided stable law of index one half, connecting this first-passage calculation with chapter 4. None of these typical scales makes a truncated sample average an estimator of a finite population mean.

20.3 An absorbing boundary must respect the velocity state L23 16:28

For the telegraph process, the target at $a > 0$ cannot be reached before a/v_c. A scalar zero-density image boundary cannot simply be imposed on the unbounded telegraph propagator. On the surviving region $x < a$, the forward condition is no injection from the target in the negative-velocity state, $\rho_-(a,t) = 0$. The positive-velocity flux $v_c \rho_+(a,t)$ leaves the domain and is the arrival measure. Setting both components to zero would incorrectly eliminate that outgoing flux. The boundary model must be derived before using images.

An exact transform illustrates the difference. Let $u_\pm(x,s) = \mathbb{E}_{x,\pm}[e^{-s\tau_a}; \tau_a < \infty]$, for $s > 0$ in inverse seconds and $x < a$. The backward generator gives $\pm v_c \partial_x u_\pm + \nu(u_\mp - u_\pm) = s u_\pm$. The solution decaying as $x \to -\infty$ has spatial factor $e^{k(x-a)}$, where $k = \sqrt{s(s+2\nu)}/v_c$. An approaching positive-velocity path reaches the target

immediately in the boundary limit, so $u_+(a,s)=1$. Its negative-velocity counterpart has the one-sided limit $u_-(a-,s)=\nu/[s+\nu+\sqrt{s(s+2\nu)}]$, obtained by substituting the exponential mode in the first backward equation. This limit concerns an approach from inside the surviving region, not the convention $\tau_a=0$ when initially at the target itself.

For equally likely initial velocities at zero the arrival transform is thus the independently derived expression

$$\widetilde{f}_{a,\mathrm{tel}}(s)=\frac{1}{2}\left[1+\frac{\nu}{s+\nu+\sqrt{s(s+2\nu)}}\right]\exp\left[-\frac{a}{v_c}\sqrt{s(s+2\nu)}\right]. \tag{20.3}$$

Its limit at $s\downarrow 0$ is one, establishing certain eventual arrival. Its leading correction is $1-[(a/v_c)+(1/(2\nu))]\sqrt{2\nu s}+O(s)$; its negative derivative diverges, proving infinite mean directly. A Gaussian central-limit approximation alone would not establish that tail property. At large s, the transform has leading term $\frac{1}{2}e^{-\nu a/v_c}e^{-sa/v_c}$, revealing an atomic arrival probability $e^{-\nu a/v_c}/2$ at the earliest allowed time. These are the initial-right paths with no reversal before arrival. The remainder of the arrival law is continuous at later times.

20.4 A bounded domain changes the question L23 26:09

Now let Brownian diffusion start at $x\in(0,L)$ and stop at either endpoint, where $L>0$ is an interval length. Write $T(x)=\mathbb{E}_x[\tau_{\{0,L\}}]$ for its mean exit time; T here is a function of position, not temperature. Let $b(x)=\mathrm{Pr}_x(X(\tau)=L)$ be the upper-exit probability. These functions depend on the starting state. The forward density equation instead differentiates the final position variable.

Proposition 20.2 — Mean exit and splitting probability.

$$DT''(x)=-1,\quad T(0)=T(L)=0,\qquad T(x)=\frac{x(L-x)}{2D}; \tag{20.4}$$

$$Db''(x)=0,\quad b(0)=0,\ b(L)=1,\qquad b(x)=\frac{x}{L}. \tag{20.5}$$

Derivation.

Over an initial short time h away from a boundary, the Markov property gives $T(x)=h+\mathbb{E}_x[T(X(h))]+o(h)$. Taylor expansion in the zero-mean increment with variance $2Dh$ yields $\mathbb{E}_x[T(X(h))]=T(x)+DhT''(x)+o(h)$, hence $DT''=-1$. The same expansion without elapsed-time cost gives $Db''=0$. At an absorbing endpoint no time remains and the exit side is already known, giving the boundary data. Integrating the differential equations determines the two displayed polynomials uniquely.

To identify the finite solution with the expectation rather than just a formal solution, apply the stopped-generator identity to the nonnegative bounded polynomial in (20.4) at $\tau\wedge t$. It gives $\mathbb{E}_x[\tau\wedge t]=T(x)-\mathbb{E}_x[T(X(\tau\wedge t))]\le T(x)$. Monotone convergence bounds $\mathbb{E}_x[\tau]$, so exit is almost sure and

the remaining bounded term tends to zero. This proves equality. The same stopped argument for the bounded affine function proves (20.5). ∎

At the center of a one-meter interval with $D = 1$ square meter per second, the mean exit time is 0.125 seconds. Removing the left absorbing endpoint returns the half-line problem with infinite mean. Making the left endpoint reflecting instead gives $T'(0) = 0$, $T(L) = 0$, and $T(x) = (L^2 - x^2)/(2D)$; the same starting point then gives 0.375 seconds. These three times answer different boundary-value problems.

Finite chains expose the backward transpose

For a discrete-time Markov chain, use the book's column convention $K_{ij} = \Pr(X_{n+1} = i \mid X_n = j)$. Let $K_{\mathcal{T}\mathcal{T}}$ retain only nonabsorbing states, and let $\mathbf{T}$ list their expected remaining step counts. First-step conditioning gives $T_j = 1 + \sum_{i \in \mathcal{T}} K_{ij} T_i$, hence $(I - K_{\mathcal{T}\mathcal{T}}^{\mathsf{T}})\mathbf{T} = \mathbf{1}$. An inverse exists when the absorbing set is reachable from every transient state of this finite chain. Finiteness of the graph alone is insufficient: a closed component disconnected from all targets can survive forever.

On the unbiased integer interval $-J, \dots, J$, each interior site jumps to either neighbor with probability one half. Its equation is $T_{j+1} - 2T_j + T_{j-1} = -2$, with endpoint values zero. The quadratic $T_j = J^2 - j^2$ satisfies both the recurrence and the boundaries. A physical time step h multiplies this answer by h; spacing ℓ with $D = \ell^2/(2h)$ recovers (20.4) after shifting the interval. Writing the adjoint explicitly prevents a common error on irregular graphs: the backward equation averages possible *destinations from the current state*.

For a diffusion with backward generator $\mathcal{L} = a(x)\partial_x + \frac{1}{2}b(x)^2\partial_{xx}$, finite exit moments $T_r(x) = \mathbb{E}_x[\tau^r]$ obey $\mathcal{L}T_r = -rT_{r-1}$, with $T_0 = 1$ and absorbing data $T_r = 0$ for $r \geq 1$. One derivation integrates $\mathbb{E}_x[\tau^r] = r\int_0^\infty t^{r-1}S_x(t)\,dt$ and the backward survival equation, checking the endpoint terms. An exponential survival tail in a regular bounded uniformly diffusive interval supplies the needed finiteness. Merely assuming all moments finite does not generally justify every pointwise boundary limit for an arbitrary density.

20.5 A first hit factors the remaining propagation L24 1:24

Let $p_{ij}(t)$ be a time-homogeneous Markov transition mass from state j to state i, and let $F_{aj}(dt)$ be the possibly defective first-arrival measure at target state a, starting from $j \neq a$. The word *defective* means its finite-time mass is below one because an atom at infinite arrival time remains. Decomposing paths by their first hit and applying the strong Markov property gives $p_{aj}(t) = \int_{[0,t]} p_{aa}(t-u)F_{aj}(du)$. The full state at arrival must be a fixed regenerating state. If the target is a set of states, sum or integrate over the state reached there; one scalar factor will generally not suffice.

Proposition 20.3 — Renewal transform. With $s > 0$ and Laplace transform $\widetilde{p}(s) = \int_0^\infty e^{-st}p(t)\,dt$,

$$\widetilde{F}_{aj}(s) = \frac{\widetilde{p}_{aj}(s)}{\widetilde{p}_{aa}(s)}. \tag{20.6}$$

The same formula holds for regular one-dimensional diffusion transition densities when the first-hit decomposition at the target applies.

Derivation.

The first hit occurs at one and only one time on every path ending at the target. Conditional on that hit, the remaining propagation begins afresh in state a, which gives the convolution. Multiplying by e^{-st} and integrating allows Tonelli's theorem because the kernels and arrival measure are nonnegative. With $v = t - u$, the integral becomes the product $\widetilde{F}_{aj}(s)\widetilde{p}_{aa}(s)$. Divide by the positive denominator. For a one-dimensional diffusion, the analogous transition-density identity follows from strong Markov conditioning and the existence of regular densities; it is not an assertion that the probability of a specified endpoint is positive. ∎

For Brownian diffusion, the free resolvent $\widetilde{G}(x, s)$ solves $(s - D\partial_{xx})\widetilde{G} = \delta(x)$ and decays at spatial infinity. Away from zero it is proportional to $e^{-|x|\sqrt{s/D}}$. Integrating across zero requires derivative jump $-D[\partial_x \widetilde{G}]_{0-}^{0+} = 1$, fixing its prefactor to $1/(2\sqrt{Ds})$. The ratio (20.6) gives $\widetilde{f}_a(s) = e^{-a\sqrt{s/D}}$. At zero its limit is one, while its derivative diverges like $-s^{-1/2}$, independently verifying both conclusions of Proposition 20.1.

For a nearest-neighbor discrete walk, let the right-step probability be $p \in (0, 1)$ and $q = 1 - p$. If $H(z)$ generates the step count to first hit $+1$ from zero, first-step decomposition yields $H(z) = z[p + qH(z)^2]$: a first left step requires two subsequent level advances to reach $+1$. The solution satisfying $H(0) = 0$ is $[1 - \sqrt{1 - 4pqz^2}]/(2qz)$. Reaching $+m$ requires m successive level advances, so its arrival generating function is $H(z)^m$. Sending $z \uparrow 1$ gives probability one when $p \geq q$ and $(p/q)^m$ when $p < q$. A drift away from the target therefore leaves a finite probability of never arriving. Conditional successful arrival statistics divide by this success probability; the unconditional hitting time is infinite with positive probability.

20.6 Two different dimensions enter transport L24 44:23

If a family of geometrically similar target domains of scale R has mean exit time proportional to R^{d_w}, the exponent d_w is a *walk dimension*. The quadratic lattice result gives $d_w = 2$. This is a dynamical exponent: it describes how time changes with distance under a specified walk and stopping rule. It is independent of the count of Euclidean coordinates for ordinary isotropic diffusion. An inverse relation $\mathbb{E}|X(t)|^2 \propto t^{2/d_w}$ is often compatible with a scaling theory, but one moment relation alone does not prove the other on every geometry or under every clock.

A geometric dimension answers a different question. The triadic Koch construction replaces each line segment by four segments of one third its length. At generation n, there are 4^n segments of scale 3^{-n}, and the polygonal length is $(4/3)^n$ times the initial span. The ratio $\log(4^n)/\log(3^n) = \log 4/\log 3$ gives its similarity dimension. For this separated self-similar construction it also gives the box dimension: the number of boxes of comparable scale needed to cover it differs from 4^n by bounded multiplicative factors, which disappear after dividing logarithms. Geometric covering, graph distance along a curve, and random-walk travel time are distinct measurements.

On graphs with loops and bottlenecks, distance can double while mean escape time grows by a factor other than four. The next chapter develops this distinction with return laws and finite graph calculations. A planar Brownian path having dimension two does not mean it hits every specified planar point. Neighborhood

recurrence and exact point hitting must be distinguished; chapter 22 treats the discrete-lattice and continuum difference.

20.7 Computational laboratory: a sampled path can hide a hit

The lab solves a tridiagonal backward equation for the absorbing interval on four grids. The quadratic exact solution has an exact second difference, so these particular grid errors are roundoff-level rather than evidence for a generic convergence order. A separate Monte Carlo experiment estimates hitting the target $a = 1$ by time one for $D = 1$. It compares an endpoint-only detector with a Brownian-bridge correction using 100,000 paths per time step and seed 2026110824.

Conditional on two below-target endpoints x, y separated by h, reflection gives crossing probability $\exp[-(a-x)(a-y)/(Dh)]$. This follows by dividing the image density $G(2a-y-x,h)$ by the free density $G(y-x,h)$ and simplifying their squared exponents. Independent conditional bridge segments make per-step Bernoulli draws a valid way to detect an unobserved crossing. The procedure corrects this one fixed target for the modeled Brownian paths; multiple competing boundaries require their joint bridge law.

```
Exact hit probability by t=1: 0.47950012
dt=0.100: endpoint=0.37440; bridge=0.47694
dt=0.020: endpoint=0.43318; bridge=0.48192
dt=0.005: endpoint=0.45707; bridge=0.48209
cutoff=10: partial first moment=1.328542
cutoff=100: partial first moment=5.155995
cutoff=1000: partial first moment=17.345701
```

The endpoint estimates improve as the grid is refined but remain visibly low. Bridge-corrected estimates fluctuate around the exact probability within the laboratory's sampling tolerance. The last three values integrate $\int_0^{t_c} t f_a(t)\,dt$, without renormalizing its finite-time mass. Their growth illustrates why a finite cutoff conceals an infinite mean. This partial moment differs from both $\mathbb{E}[\min(\tau_a, t_c)]$ and the mean conditional on arrival before t_c.

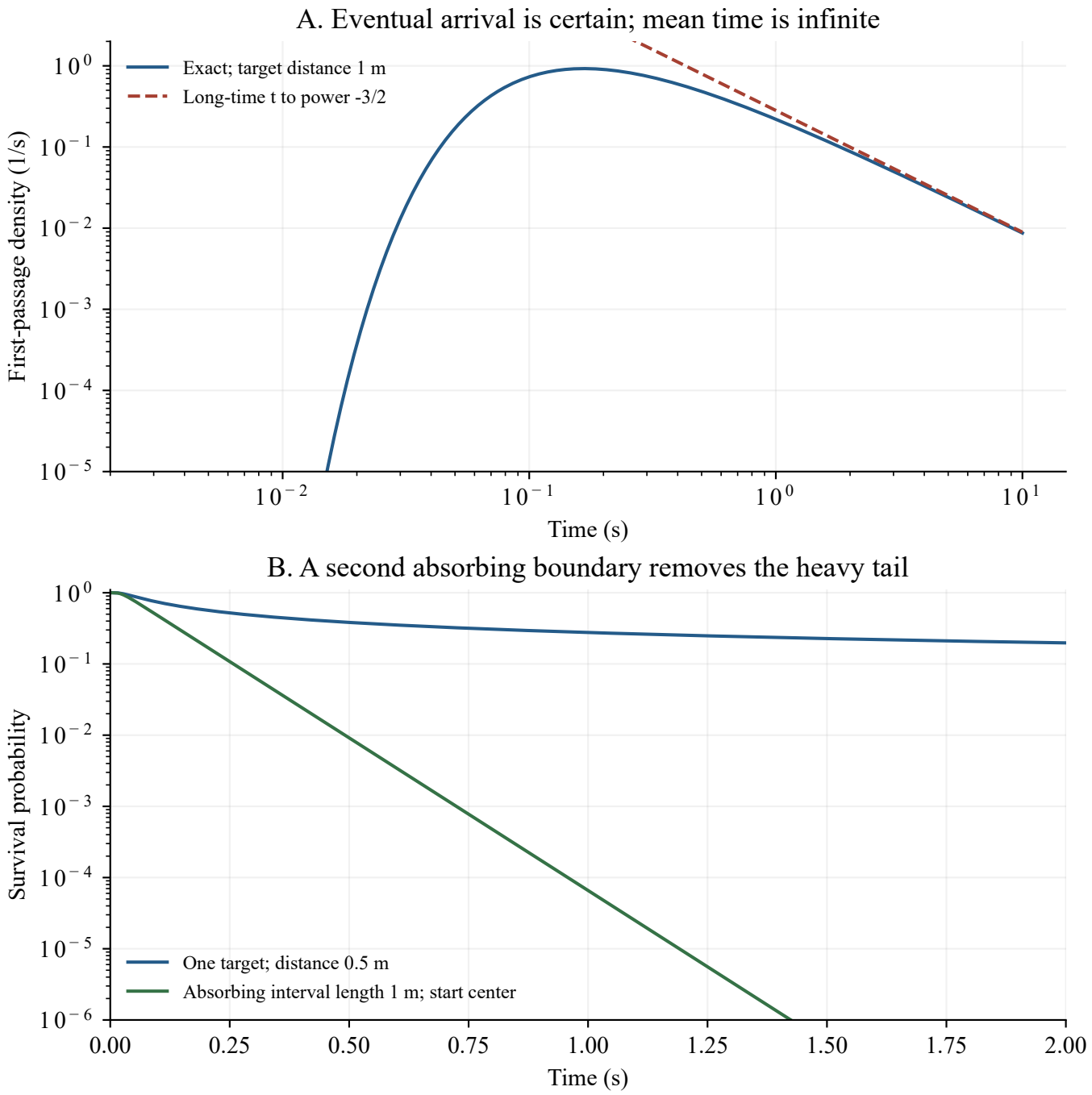


Figure 20.1. A certain half-line arrival retains a nonintegrable first-moment tail. A second absorbing boundary makes survival decay exponentially. The curves use $D = 1$ square meter per second and the stated target distances; the generator reads the retained laboratory data.

```
import math


def hit_probability(distance, diffusion, time):
    if distance <= 0 or diffusion <= 0 or time <= 0:
        raise ValueError("require positive distance, diffusion, time")
    return math.erfc(distance/math.sqrt(4*diffusion*time))


print(f"{hit_probability(1.0, 1.0, 1.0):.8f}")
```

This side-effect-free function returns the half-line cumulative arrival probability and prints 0.47950012. The full lab writes the report and numerical JSON, makes no network calls, and overwrites those files on rerun. Run it before the figure script with the declared dependencies. Synthetic Brownian paths and exact solvable boundary equations provide the checks; they are not measurements of a physical search process.

20.8 What an arrival statistic does and does not say

A survival integral must use the killed process, not the unrestricted propagator. Boundary conditions belong to the full Markov state, especially for velocity-dependent transport. Renewal at a target requires a regenerative state; it cannot be assumed for position alone when hidden velocity or waiting age remains relevant. Finite and infinite domains can have completely different mean-time behavior even when their local diffusion operator is identical. In data, distinguish completed arrivals from censored trials and report which mean or quantile is being estimated. In higher-dimensional continuous space, point targets require a separate hitting analysis; a lattice return formula cannot be transferred by replacing a mass with a density.

20.9 Exercises with worked solutions

Exercise 20.1 — Three finite-record averages (conceptual)

An experiment ends at cutoff t_c. Relate the partial first moment, the average among completed arrivals, and the mean of a clock capped at t_c. Which can approach a finite mean in the half-line Brownian problem?

Worked solution

Write $F(t_c) = \Pr(\tau \le t_c)$ and $M(t_c) = \int_0^{t_c} t f(t)\,dt$. Completed arrivals have conditional mean $M(t_c)/F(t_c)$. The capped clock has mean $M(t_c) + t_c[1 - F(t_c)] = \int_0^{t_c} S(t)\,dt$, because every unfinished trial contributes its full cutoff. Each is finite at fixed t_c. For the half-line law, $F(t_c) \to 1$ but both $M(t_c)$ and the capped mean diverge, with different leading coefficients. Reporting a finite observed value therefore does not establish a finite mathematical expectation. Discarding unfinished trials creates a different statistic whose conditioning must be stated.

Exercise 20.2 — A reflecting wall is not an absorber (conceptual)

Why does imposing $T(0) = 0$ instead of $T'(0) = 0$ at a reflecting left wall change the mean exit problem? Explain the corresponding forward probability condition.

Worked solution

Zero remaining time at the left wall declares that reaching it ends the process. A reflecting path instead continues and only the right endpoint ends its clock. For constant diffusion the backward reflecting condition is zero normal derivative, $T'(0) = 0$, while its forward density condition is zero probability current, $-D\partial_x \rho(0,t) = 0$. Absorption uses a killed density with zero boundary value and allows outgoing flux. The center-start means 0.375 and 0.125 seconds differ by a factor three for the chapter's parameters. They describe different experiments, not two approximations to one experiment.

Exercise 20.3 — Compute a bounded-domain second moment (derive)

For the interval $(0, L)$ absorbed at both ends, solve $DU'' = -2T$ with $U(0) = U(L) = 0$, where $U(x) = \mathbb{E}_x[\tau^2]$ and T is (20.4). Find the variance at the midpoint.

Worked solution

Substitute $T = x(L-x)/(2D)$ to get $U'' = -(Lx - x^2)/D^2$. Two integrations give $U(x) = [-Lx^3/6 + x^4/12]/D^2 + C_1 x + C_2$. The boundary at zero gives $C_2 = 0$; the one at L gives $C_1 = L^3/(12D^2)$. Thus $U = x(L^3 - 2Lx^2 + x^3)/(12D^2)$. At $x = L/2$, it is $5L^4/(192D^2)$, while $T^2 = L^4/(64D^2)$. Their difference is $L^4/(96D^2)$. These are finite because bounded-domain survival has an exponential tail. A half-line calculation with the same moment recursion would require a different boundary condition at infinity and has no finite nonnegative solution for the mean.

Exercise 20.4 — Derive the half-line transform without integration (derive)

Let $u(x, s) = \mathbb{E}_x[e^{-s\tau_a}]$ for $x < a$, with constant D and $s > 0$. Solve its backward equation and recover the arrival and mean conclusions.

Worked solution

First-step conditioning, including the discount e^{-sh}, gives $Du'' = su$. The bounded solution tending to zero as $x \to -\infty$ is $Ce^{x\sqrt{s/D}}$. Its target value is one, so $u(x, s) = e^{-(a-x)\sqrt{s/D}}$. At $x = 0$ this is the transform found by renewal. Letting $s \downarrow 0$ gives one. Differentiating for positive s gives $-ae^{-a\sqrt{s/D}}/(2\sqrt{Ds})$, which tends to minus infinity. Monotone convergence of $\mathbb{E}[\tau e^{-s\tau}]$ as $s \downarrow 0$ identifies the divergent expectation. The target and far-field conditions select the probability solution among the two formal exponential modes.

Exercise 20.5 — Choose a crossing estimator (compute/design)

Use the lab's three time steps to compare endpoint-only and bridge-corrected estimates with the exact target probability. Can the finest endpoint grid alone certify one-percent relative accuracy?

Worked solution

The finest endpoint value 0.45707 is below 0.47950012 by about 0.02243, or about 4.68 percent of the exact probability. Its Monte Carlo uncertainty is far smaller than that systematic discrepancy: for 100,000 independent Bernoulli outcomes the standard error is about $\sqrt{0.48(0.52)/100000} = 0.00158$. The bridge value 0.48209 is within roughly 1.64 such standard errors of the target. Thus merely using 200 time steps per second does not certify one-percent relative accuracy. Increase ensembles to quantify stochastic uncertainty and use a crossing-aware estimator to control discretization bias. A discretely observed physical detector may intentionally measure endpoint exceedances; that is then a different observation model.

Exercise 20.6 — A biased search with failed trajectories (compute/design)

An integer walk starts at zero, takes right steps with probability 0.4, and seeks $+2$. Find its eventual success probability. Explain how to report arrival-time data without treating failed trajectories as ordinary completed arrivals.

Worked solution

Here $p = 0.4$, $q = 0.6$, so the first-hit generating function has $H(1) = p/q = 2/3$. Success at $+2$ requires two successive level advances and has probability $H(1)^2 = 4/9$. The missing mass $5/9$ is an atom at infinite arrival time. If assigning infinity to never-arriving trajectories, the unconditional mean is infinite. A mean among eventual successes is instead obtained from the normalized transform $H(z)^2/(4/9)$, when its derivative is finite; it is not obtained by pretending $H(1)^2 = 1$. A finite cutoff adds censoring even among eventual successes. Report the estimated finite-horizon arrival fraction, cutoff, and conditional completed-time distribution separately, and compare them with the defective theoretical law.

20.10 Sources and attribution

- V. Balakrishnan, IIT Madras, NPTEL 115106089, lecture 23 and lecture 24: crossing counts, first-passage distributions, backward moments, renewal transforms, and walk dimension.
- James W. Pitman, Hitting Times and the Reflection Principle, Berkeley Stat205B, lecture 17 (2003), pages 17-2–17-3, scribed by Jonathan Weare: Brownian reflection and hitting distribution.
- NIST DLMF 10.40.1: fixed-order modified-Bessel positive-real asymptotics used in the crossing estimate.

The velocity-state absorbing calculation, boundary qualifications, explicit finite-interval solutions, numerical bias experiment, figure, and exercises are developed here. A finite graph must connect to its absorbing set; a finite mean needs an integrability argument; planar continuum point hitting is distinct from neighborhood recurrence.

Return laws, recurrence, and fractal transport

Doubling a travel distance need not multiply a diffusive search time by four. A hierarchy of loops can multiply it by five, even though every individual jump follows a Markov rule. Separately, an unbiased line walk returns to its starting site almost surely but has an infinite mean return time. Geometry, state sufficiency, eventual return, and mean return duration are four different properties; this chapter calculates them rather than inferring one from another.

> The order follows lecture 25: gasket geometry and exit scaling, renewal of Markov visits, recurrence criterion, biased line returns, and the unbiased heavy tail. The full caption track was read; one frame at 770 seconds confirms the gasket setup but partly obscures the algebra. Finite-graph computations, exact renewal checks, and exercises are additions. Independent audio was not reviewed.

21.1 A graph with loops at every scale L25 2:19

Start with a triangular graph of three vertices connected pairwise by three edges. This is generation zero. Generation $n+1$ consists of three copies of generation n, joined at their corner vertices to form a triangle of twice the side length, leaving the central upside-down triangle empty. Keep the shortest edge length equal to one. The resulting finite Sierpiński gasket graph has side length $R_n = 2^n$ and vertex count $(3^{n+1}+3)/2$. Each noncorner vertex has four neighbors; each of the three outer corners has two.

A discrete-time simple random walk chooses uniformly among the current vertex's neighbors at each step. Let one outer corner be the starting point and the other two be absorbing targets. The remaining outer corner has its ordinary degree-two transitions; it is not a trap. Let T_n be the expected number of steps to reach either target. This definition fixes both the starting state and the target geometry. A different target set, a fixed embedded-size graph with shrinking edges, or a variable-rate clock would require a different scaling statement.

On any finite graph with these reachable targets, the first-step equations are $T(x) = 1 + \sum_{y \notin A} K_{yx} T(y)$ for $x \notin A$, with $T = 0$ at the target set A. The column stochastic transition matrix K has entry $K_{yx} = 1/\deg(x)$ when y is adjacent to x, and zero otherwise. Consequently the vector of mean times solves $(I - K_{\mathcal{T}\mathcal{T}}^{\mathsf{T}})\mathbf{T} = \mathbf{1}$. The transpose is essential because the equation averages destinations from a specified starting vertex. These finite linear equations provide a direct numerical check before an asymptotic interpretation is attempted.

21.2 Eliminate the fine vertices before declaring an exponent L25 7:07

Generation zero has $T_0 = 1$: the first step reaches a target. For generation one, symmetry leaves three unknown mean times. Call the starting-corner value a, the two equivalent adjacent midpoint values u, and the midpoint on the opposite side v. The first-step equations are $a = 1 + u$, $u = 1 + (a + u + v)/4$, and $v = 1 + u/2$. The two target values vanish. Substitution gives $u = 4$, $a = 5$, and $v = 3$. A loop permits many unsuccessful excursions before an outer target is reached.

Proposition 21.1 — Gasket exit scaling. For the graph family and targets above,

$$T_n = 5^n, \qquad d_w = \frac{\log 5}{\log 2} = 2.32192809\ldots. \tag{21.1}$$

The exponent is defined by the exact relation $T_n = R_n^{d_w}$.

Derivation.

Observe a refined walk only when it reaches a different old-generation vertex. Inside a refined triangular cell, the calculation just made gives mean five fine steps from a retained corner to either of its two neighboring retained corners. Symmetry gives equal exit probabilities. At an old vertex incident to two cells, the same reflection symmetry among its four available old neighbors gives uniform coarse exit choices; the expected waiting time to a different old vertex is again five, by the identical local first-step system. Thus the retained-vertex trace has exactly the old simple-walk transition rule.

Apply the strong Markov property at successive retained-vertex hits. Conditional on each coarse starting vertex, its next coarse jump has mean fine-step cost five, whether that vertex has two or four neighbors. Summing expected costs up to absorption gives $T_{n+1} = 5T_n$; the expected number of coarse jumps is finite on these finite graphs. Induction from $T_0 = 1$ proves $T_n = 5^n$. Taking logarithms with $R_n = 2^n$ proves the exponent. ∎

The laboratory solves all vertex equations through generation six, independently of this elimination argument, obtaining mean times one, five, twenty-five, and their successive factors of five. The construction supplies a dynamical exponent beyond the Euclidean value two. The geometric similarity dimension is instead $d_f = \log 3/\log 2$, because the graph has three half-scale copies. One measures travel time; the other measures geometric proliferation. A general relation involving a spectral dimension requires a heat-kernel scaling theorem; it does not follow from merely taking these two logarithms.

The slower exit scaling is compatible with a displacement scale $R \propto t^{1/d_w}$ and mean-square scaling t^{2/d_w} in the corresponding infinite-fractal theory. The exact result proved here is the finite-family exit relation. The laboratory alone does not prove an infinite-graph displacement theorem, and a short finite-size fit on another graph would be evidence for an exponent rather than an exact renormalization law.

21.3 A first return starts after time zero L25 21:18

For a time-homogeneous discrete Markov chain started at state i, let $u_n = \mathbf{Pr}_i(X_n = i)$ with $u_0 = 1$. Define its positive first return time $\tau_i^+ = \inf\{n \geq 1 : X_n = i\}$ and $f_n = \mathbf{Pr}_i(\tau_i^+ = n)$. It may have value infinity. If self-loops are allowed, f_1 may be positive; for a nearest-neighbor line walk without holding, all odd-time returns are impossible. The sequence u_n includes repeated visits, while f_n excludes any earlier positive-time visit.

Proposition 21.2 — Return renewal and recurrence criterion. For $n \geq 1$,

$$u_n = \sum_{k=1}^{n} f_k u_{n-k}, \qquad U(z) = \frac{1}{1 - F(z)}, \tag{21.2}$$

where $U(z) = \sum_{n \geq 0} u_n z^n$ and $F(z) = \sum_{n \geq 1} f_n z^n$ for $|z| < 1$. The state is recurrent, meaning $\Pr_i(\tau_i^+ < \infty) = 1$, exactly when $\sum_{n \geq 0} u_n = \infty$.

Derivation.

Every path at i at time $n \geq 1$ has a unique first positive return $k \leq n$. Conditional on that event, the Markov property restarts its return probabilities from i for the remaining $n - k$ steps. Adding mutually exclusive first-return alternatives gives the convolution. Absolute convergence inside the unit disk allows multiplication of generating functions, yielding $U = 1 + FU$ and then (21.2). For real $z \uparrow 1$, monotone convergence gives $F(z) \uparrow r = \sum f_n$, the eventual return probability. If $r < 1$, then $U(1) = 1/(1 - r)$ is finite. If $r = 1$, $U(z)$ diverges. This proves the criterion. ∎

The sum of the u_n is the expected total number of visits including the initial one, not the probability of any visit. Visits at different times overlap as events and cannot be added to form a union probability. First-return alternatives are mutually exclusive and can be added. This distinction is the mathematical reason renewal transforms are so useful. A recurrent state may be *positive recurrent*, with finite mean positive return time, or *null recurrent*, with infinite mean. A transient state has $r < 1$.

21.4 Bias suppresses returns without forbidding them L25 35:06

Let the integer line walk take independent right steps with probability $p \in (0, 1)$ and left steps with $q = 1 - p$. Returning to zero after $2m$ steps requires m of each, giving $u_{2m} = \binom{2m}{m}(pq)^m$; odd-time values vanish. The binomial expansion of $(1 - x)^{-1/2}$ has coefficient $\binom{2m}{m}/4^m$, as follows by successively differentiating at zero. Therefore $U(z) = (1 - 4pqz^2)^{-1/2}$. The initial term one is included; omitting it changes the renewal algebra.

Proposition 21.3 — Line first-return law.

$$F(z) = 1 - \sqrt{1 - 4pqz^2}, \qquad f_{2m} = \frac{\binom{2m}{m}(pq)^m}{2m - 1}, \quad m \geq 1, \tag{21.3}$$

with $f_{2m+1} = 0$. Its total finite return probability is

$$r = 1 - |p - q| = 2\min(p, q). \tag{21.4}$$

Derivation.

Substitute the explicit U into $F = 1 - 1/U$. The binomial coefficient of z^{2m} in $1 - \sqrt{1 - 4pqz^2}$ is $\binom{2m}{m}(pq)^m/(2m - 1)$; it can also be verified by differentiating the generating function and matching its coefficients. Monotone convergence as $z \uparrow 1$ gives $r = 1 - \sqrt{1 - 4pq}$. Since $p + q = 1$, $1 - 4pq = (p - q)^2$, proving (21.4). ∎

At $p = 0.6$, return probability is 0.8, not zero. A walk can initially step against its drift or execute a finite excursion before drifting away permanently. At $p = q = 1/2$, return probability reaches one. At the unidirectional endpoints $p = 0$ or one, the limiting return probability is zero. The probability of hitting a specified point to the right is a different quantity: chapter 20 gave one with rightward drift and $(p/q)^m$ against leftward drift. Returning requires leaving and undoing that first displacement, so the two probabilities cannot be interchanged.

21.5 Null recurrence is not equilibrium sampling L25 53:06

For the unbiased law, Stirling's factorial asymptotic gives $\binom{2m}{m} \sim 4^m/\sqrt{\pi m}$. Applying it to (21.3) yields $f_{2m} \sim 1/(2\sqrt{\pi}\, m^{3/2})$. The probabilities sum to one, but $\sum_{m\geq 1} 2m f_{2m}$ has a divergent $m^{-1/2}$ tail. The same fact follows by differentiating $F(z) = 1 - \sqrt{1 - z^2}$: its derivative $z/\sqrt{1 - z^2}$ tends to infinity at one. Monotone convergence identifies this derivative limit with the infinite expected return step count.

A recurrent return can be almost surely finite while its expectation diverges. This is null recurrence. It supplies no normalized uniform equilibrium distribution on the entire integer line. To check directly, a stationary mass sequence would obey $\pi_j = (\pi_{j-1} + \pi_{j+1})/2$, so its second difference vanishes. The general sequence is affine in j; nonnegativity on both tails forces it to be constant, and no positive constant sequence is summable. Recurrence and a stationary probability measure therefore separate on this infinite state space.

With nonzero bias, $F'(1) = 4pq/|p - q|$ is finite and equals the unnormalized finite-return first moment $\sum n f_n$. The unconditional return time still has an atom at infinity and infinite extended expectation. Conditional on eventual return, its mean is $F'(1)/r$; for $p = 0.6$, it is $4.8/0.8 = 6$ steps. A finite mean conditional on success does not make the state positive recurrent, because the return probability is below one. This distinction matters when simulation averages are computed only over returning paths.

21.6 Computational laboratory: solve before fitting

The laboratory constructs gasket edges recursively using exact integer coordinates, counts unique shared vertices, builds neighbor lists, and solves the sparse backward linear system. No fitted exponent is used in computing the mean times. It verifies $T_n/5^n = 1$ to numerical tolerance for generations zero through six. The largest graph has 1,095 vertices. It also evaluates first-return masses by logarithmic factorials, checks one hundred renewal convolutions against exact return probabilities, and sums 10,000 even-time coefficients.

```
side=1: vertices=3; mean exit=1.000000 steps
side=2: vertices=6; mean exit=5.000000 steps
side=4: vertices=15; mean exit=25.000000 steps
side=8: vertices=42; mean exit=125.000000 steps
side=16: vertices=123; mean exit=625.000000 steps
side=32: vertices=366; mean exit=3125.000000 steps
side=64: vertices=1095; mean exit=15625.000000 steps
walk dimension=2.32192809
p=0.5: return=1.000000; partial mass=0.99435817; partial moment=112.836506
p=0.6: return=0.800000; partial mass=0.80000000; partial moment=4.800000
```

The unbiased partial mass is close to one while its partial first moment is already large and continues to grow. The biased partial mass converges rapidly to 0.8; renormalizing it to one would conceal transience. Exact absorbing-chain or renewal calculations make these effects visible without waiting for rare long simulated excursions. For much larger gaskets, sparse factorization and storage become material costs; an iterative solve or exact decimation may be preferable.

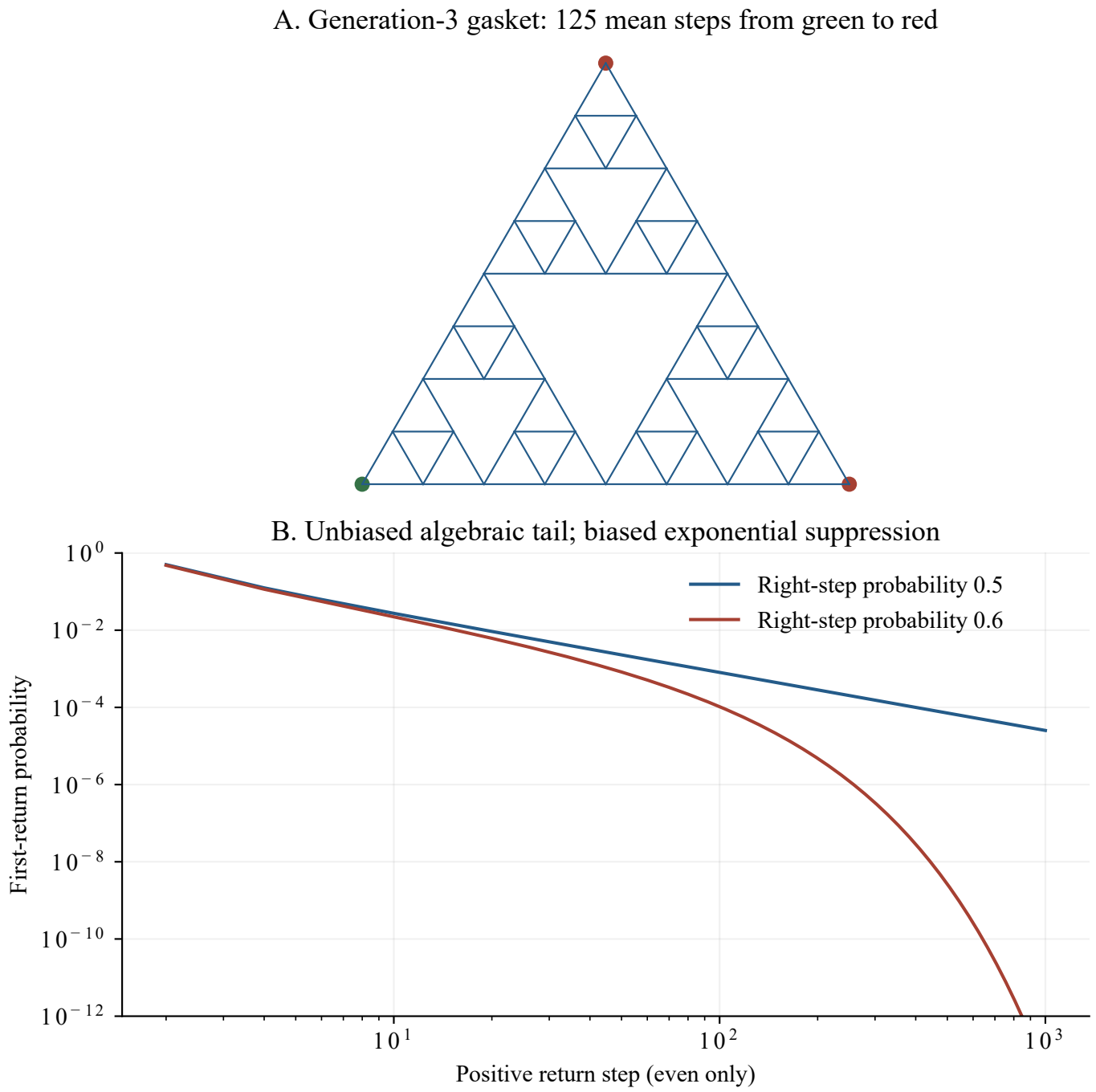


Figure 21.1. The finite gasket illustrates the explicitly specified exit geometry; its mean time is 125 steps. The line-walk plot concerns a different graph and shows exact even-time first-return masses. The generator reproduces these panels from the retained data.

```
import math


def first_return_even(m, right_probability):
    if m < 1 or not 0 < right_probability < 1:
        raise ValueError("require positive m and probability in (0, 1)")
    p = right_probability
    return math.comb(2*m, m)*(p*(1-p))**m/(2*m-1)
```

```
print(first_return_even(1, 0.5))
print(first_return_even(2, 0.5))
```

The snippet prints 0.5 and 0.125 and changes no files. It is intended for modest integer m; the full lab uses logarithmic factorials for a stable long tail. The full program writes output and JSON, makes no network calls, and overwrites these products on rerun. Run it and then the plot generator with the declared dependencies.

21.7 Keep the geometric and probabilistic limits distinct

Finite connected gasket graphs with reachable traps have finite mean exit times. The infinite graph limit can change recurrence and normalizability; it is not established by labeling the finite graph ergodic. The exact exponent here belongs to nearest-neighbor uniform discrete-time jumps with the stated family of targets. Holding times, long jumps, random conductances, or disconnected pieces can alter it. On the line, transience concerns the atom at infinite return time, while null recurrence concerns the moment of an almost surely finite positive return. The next chapter tests how dimension changes the return criterion and how a separate physical-time clock changes transport.

21.8 Exercises with worked solutions

Exercise 21.1 — A time-zero mistake (conceptual)

A return law is defined to include $f_0 = 1$. Why does applying (21.2) to it fail?

Worked solution

The initial visit occurs on every path and is not a first positive return. If it is included as the first return, the resulting time is identically zero; its generating function is one. Then $1/(1 - F)$ is undefined, and a decomposition by later unique first returns has lost its content. Keep $u_0 = 1$ as the initial renewal and start f_n at $n = 1$. For a chain with a holding probability, a one-step return is valid and should not be discarded; only the time-zero visit is excluded.

Exercise 21.2 — Transience hidden by conditioning (conceptual)

A biased walk has a measured six-step average among eventually returning paths. Does this imply positive recurrence?

Worked solution

No. For $p = 0.6$, the calculation gives conditional mean six but eventual return probability 0.8. One fifth of paths never return. Positive recurrence requires probability one of return as well as finite mean. The extended unconditional return time is infinite on that one-fifth event, hence has infinite expectation. A data collection that retains only completed returns cannot estimate the missing probability without separately tracking failed or censored trials.

Exercise 21.3 — Solve the first refined gasket (derive)

Use the three symmetry equations to compute the mean values at the starting corner, its adjacent midpoints, and the opposite midpoint. Check each equation after substitution.

Worked solution

From $a = 1 + u$ and $v = 1 + u/2$, the middle equation becomes $4u = 4 + (1 + u) + u + (1 + u/2) = 6 + (5/2)u$. Hence $u = 4$, $a = 5$, and $v = 3$. The corner equation reads $5 = 1 + 4$; the adjacent-midpoint equation is $4 = 1 + (5 + 4 + 3)/4$; the opposite-midpoint equation is $3 = 1 + 4/2$. The constant one in every equation is elapsed time for the first step, including a step that goes immediately to a trap. Omitting it for trap-directed jumps understates the time.

Exercise 21.4 — Recover the return tail (derive)

Starting from the explicit unbiased first-return masses, derive their large-m asymptotic and the growth of the truncated first moment through step $2M$.

Worked solution

Stirling gives $\binom{2m}{m} 4^{-m} \sim (\pi m)^{-1/2}$. Dividing by $2m - 1 \sim 2m$ gives $f_{2m} \sim [2\sqrt{\pi}\, m^{3/2}]^{-1}$. Therefore $\sum_{m=1}^{M} 2m f_{2m} \sim \pi^{-1/2} \sum_{m=1}^{M} m^{-1/2} \sim 2\sqrt{M/\pi}$. The last asymptotic follows by upper and lower integral bounds for the decreasing function $x^{-1/2}$. The missing probability mass decays like $1/\sqrt{\pi M}$, so apparent normalization can improve while the truncated mean diverges. This is the discrete counterpart of the half-line Brownian tail.

Exercise 21.5 — Test an exponent without assuming it (compute/design)

Describe how to distinguish a measured walk exponent from the exact gasket scaling in a new graph family. Which details of the laboratory prevent a circular check?

Worked solution

Construct the actual graph and targets at several scales, use its neighbor probabilities to solve the backward equations, then estimate $\log[T(2R)/T(R)]/\log 2$ with numerical error estimates. Do not set time values equal to a hypothesized power before fitting. Here edges are recursively generated, shared vertices are merged, and the sparse linear equations depend only on adjacency and absorbing vertices. The factor five is checked afterward. Exact decimation supplies an independent proof for this graph family; a stable finite-size ratio on an unrelated graph would remain numerical evidence. Changing target shape between scales can create an apparent exponent that reflects changed experiments.

Exercise 21.6 — Quantify a biased conditional return (compute/design)

For $p = 0.6$, compute total return probability, finite-return first moment, and conditional mean. Predict the same three quantities when p is replaced by 0.4.

Worked solution

The return probability is $r = 1 - |2p - 1| = 0.8$. Differentiating F gives $F'(1) = 4p(1-p)/|2p-1| = 4.8$. Dividing by r gives conditional mean six steps. Reflection of every path exchanges right and left, so replacing 0.6 by 0.4 leaves all three values unchanged. It does reverse the drift and changes which one-sided target is easy to reach; return to the origin is reflection invariant. The lab's sum over 10,000 even times reproduces the finite-return moment 4.8, while its unnormalized probabilities sum to 0.8.

21.9 Sources and attribution

- V. Balakrishnan, IIT Madras, lecture 25, NPTEL 115106089: gasket first-step construction, return renewal, line return probability and null recurrence.
- NIST DLMF 5.11: classical gamma and factorial asymptotics underlying the stated Stirling limit. The renewal and finite-graph proofs are supplied directly here.

The sparse solves, exact convolution checks, plots, and six solutions are original teaching constructions. The exact gasket claim is limited to the specified finite graph family and stopping rule; no general fractal heat-kernel theorem is inferred from the numerical calculation.

Recurrence by dimension and random walk clocks

A recorded search trajectory can look trapped for two different reasons: its spatial graph may keep returning it to familiar sites, or its clock may impose long pauses between otherwise ordinary jumps. To distinguish these, first analyze a walk by its jump count, then attach elapsed waiting times. This separation preserves the geometry while exposing the information that a nonexponential clock adds to the state.

> The full caption track of lecture 26 was read. Its sequence supplies lattice recurrence, Fourier Green functions, walk rescaling, and renewal clocks. One frame at 1375 seconds was inspected; the small-wave-number denominator is visible, while some equations are occluded. The exact parity factors, clock-age model, primary-source continuum distinction, numerical checks, and exercises are developed here. No independent audio review is claimed.

22.1 Fourier space retains the exact jump geometry L26 2:54

Let Y_n be a walk on the integer lattice $\mathbb{Z}^d$, where $d \geq 1$ is an integer and n counts steps. Start at $Y_0 = 0$. Each independent jump chooses one of the $2d$ vectors $\pm e_j$ with equal probability $1/(2d)$; e_j is the unit vector in coordinate direction j. Coordinates here are dimensionless lattice units. A physical spacing can be restored by multiplying position by ℓ.

For Fourier variable $q = (q_1, \ldots, q_d)$ with components in $[-\pi, \pi]$, the one-step characteristic function is $\varphi(q) = d^{-1} \sum_{j=1}^{d} \cos q_j$. Independent sums give characteristic function $\varphi(q)^n$. Discrete Fourier inversion therefore gives the exact mass at the starting site, $u_n = (2\pi)^{-d} \int_{[-\pi,\pi]^d} \varphi(q)^n \, d^d q$. This is a lattice probability, not a continuum density at a point. The finite integration domain is a fundamental period of the lattice transform.

22.2 The low-wave-number singularity determines recurrence L26 18:17

Define the return Green function $U(z) = \sum_{n \geq 0} u_n z^n$ for real $0 \leq z < 1$. Summing the uniformly convergent geometric series inside the Fourier integral yields $U(z) = (2\pi)^{-d} \int [1 - z\varphi(q)]^{-1} \, d^d q$. As in chapter 21, recurrence is equivalent to $U(z) \to \infty$ as $z \uparrow 1$.

Proposition 22.1 — Symmetric lattice recurrence threshold. The specified nearest-neighbor walk is recurrent in dimensions one and two and transient in every dimension $d \geq 3$.

Derivation.

The only point with $\varphi(q) = 1$ on the periodic integration torus is $q = 0$. Away from a neighborhood of it the $z \uparrow 1$ denominator is bounded away from zero; the negative-φ portion is uniformly bounded as well. Near zero,

the cosine expansion gives $1-\varphi(q)=|q|^2/(2d)+O(|q|^4)$. Thus positive constants bound $1-\varphi(q)$ above and below by multiples of $|q|^2$ in a sufficiently small ball.

The possible divergence of the limiting Green integral is therefore that of $\int_0^\epsilon r^{d-1}r^{-2}\,dr=\int_0^\epsilon r^{d-3}\,dr$. It diverges for $d=1$ or two and converges for $d\geq 3$. On the positive-φ neighborhood monotone convergence justifies the $z\uparrow 1$ integral limit; on its complement bounded convergence applies. The return renewal criterion then proves the assertion. ∎

The divergence is strongest in one dimension and logarithmic in two. In three dimensions a finite Green function gives return probability $1-1/U(1)<1$. Its exact value depends on the lattice geometry and transition rule; the threshold proved here does not say every graph embedded in three-dimensional space is transient. A graph confined to a line remains a line even if its coordinates are stored in three dimensions. Nondegenerate homogeneous Euclidean lattice walks and arbitrary networks require different hypotheses.

22.3 A local Gaussian approximation needs the allowed lattice cell L26 26:59

Each step reverses the parity of the sum of lattice coordinates, so $u_n=0$ at odd n. At even time $2m$, the leading return mass is $u_{2m}\sim 2[d/(4\pi m)]^{d/2}$. The factor two is essential. To derive it, note that $|\varphi|=1$ at two points on the Fourier torus: zero and the point with all coordinates equal to π. Near zero, $\varphi(q)^{2m}\sim e^{-m|q|^2/d}$; near the second point the sign reverses but its even power has the same Gaussian contribution. The rest of the torus decays exponentially with m after neighborhoods of these points are removed. Each saddle contributes $(2\pi)^{-d}\int_{\mathbb{R}^d}e^{-m|q|^2/d}\,d^dq=[d/(4\pi m)]^{d/2}$.

The two saddles are the Fourier expression of a spatial support constraint. At even times only one half of the integer lattice is accessible; its effective cell volume is two rather than one. A continuum Gaussian of covariance $(2m/d)I$ evaluated at zero without this correction gives half the lattice mass. In one dimension, the result reduces to $u_{2m}\sim 1/\sqrt{\pi m}$; in two it is $1/(\pi m)$. Summing these tails reproduces the recurrence threshold. This local asymptotic is stronger than the central limit theorem, which concerns probabilities over scaled regions and does not by itself determine individual lattice masses.

22.4 A lattice site is not a continuum point

In planar Brownian motion, every neighborhood of a fixed point is visited repeatedly, but a specified point distinct from the start is hit with probability zero. Points are called *polar* when their hitting probability vanishes in this sense. Mörters–Peres, Corollary 2.26, printed page 48, establishes the point statement; their Theorem 3.20, printed page 73, gives neighborhood recurrence. These are cited results, not consequences obtained by inserting $d=2$ in the lattice mass formula.

The distinction can be understood through the probability to reach a small circle of radius ϵ before leaving a circle of radius R. In a planar annulus this harmonic probability is $\log(R/r)/\log(R/\epsilon)$ for starting radius r. At fixed outer radius R it tends to zero as $\epsilon\downarrow 0$. Shrinking the target to a point and removing the outer boundary are different limiting operations. A path can be dense in the sense of visiting all rationally specified open disks

while avoiding any one fixed point almost surely. A two-dimensional path dimension also does not assert literal filling of the plane.

22.5 Time-change the same spatial path L26 44:02

Let $J_1, J_2, \dots$ be independent identically distributed spatial jumps, and let $W_1, W_2, \dots$ be independent identically distributed strictly positive waiting times, independent of all jumps. Define renewal epochs $S_n = W_1 + \cdots + W_n$, with $S_0 = 0$, and the renewal count $N(t) = \max\{n : S_n \le t\}$. The physical-time position is $X(t) = \sum_{j=1}^{N(t)} J_j$. The process begins with a fresh waiting interval at zero. This preparation is part of its definition.

Proposition 22.2 — Independent clock mixture. If $p_n(x)$ is the mass or density after n spatial jumps and $w_n(t) = \Pr(N(t) = n)$, then

$$\Pr(X(t) \in dx) = \sum_{n \ge 0} w_n(t)\, p_n(dx). \tag{22.1}$$

Exponential waits of rate λ give $w_n(t) = e^{-\lambda t}(\lambda t)^n / n!$ and a time-homogeneous Markov walk.

Derivation.

Condition on the independent count $N(t)$. Given that count, the spatial sum has exactly its n-jump law, giving the mixture. Exponential waiting times have the memoryless survival $e^{-\lambda t}$; successive independent waits generate a Poisson counting process, so the displayed weights follow. Its independent future increments make position a Markov compound-Poisson process with generator $\mathcal{L}g(x) = \lambda \mathbb{E}[g(x + J) - g(x)]$. ∎

In this renewal model, nonexponential waits generally make position alone non-Markov: the current waiting age matters. This is not a universal claim that every non-Poisson one-time count law forbids every Markov model. A time-dependent deterministic hazard can produce a time-inhomogeneous Markov process; degenerate observations can hide timing altogether. Conversely, specifying normalizable count weights separately at each time is not enough to construct consistent joint-time laws. Independent identically distributed positive waits provide that construction and ensure $S_n \to \infty$ almost surely, so there are finitely many renewals in finite time.

22.6 Waiting age is information about the future L26 1:00:01

Let $\psi(w)$ be the waiting-time density, with units inverse seconds, and let $\Psi(w) = \Pr(W > w) = \int_w^\infty \psi(u)\, du$ be its survival. These two functions are different objects. If the present waiting interval has already lasted age a, its remaining survival for a further time t is a conditional probability:

Proposition 22.3 — Residual waiting law and hazard. Whenever $\Psi(a) > 0$,

$$\Pr(W > a + t \mid W > a) = \frac{\Psi(a+t)}{\Psi(a)}, \qquad \eta(a) = \frac{\psi(a)}{\Psi(a)}. \tag{22.2}$$

Here η is the instantaneous hazard rate when the density is regular at a. It is constant for an exponential waiting law.

Derivation.

The event $W > a + t$ is contained in $W > a$, so conditional probability gives the survival ratio. The probability of ending within a short additional interval h is $[\Psi(a) - \Psi(a+h)]/\Psi(a) = \psi(a)h/\Psi(a) + o(h)$, proving the hazard. If $\eta(a) = \lambda$ at all ages and $\Psi(0) = 1$, the differential equation $\Psi' = -\lambda\Psi$ gives $\Psi(a) = e^{-\lambda a}$. Conversely this exponential makes (22.2) equal to $e^{-\lambda t}$ at every age. ∎

For a Lomax waiting law $\Psi(w) = (1 + w/\tau_0)^{-\alpha}$, with scale $\tau_0 > 0$ seconds and exponent $0 < \alpha < 1$, the mean wait is infinite and the hazard is $\alpha/(\tau_0 + a)$. A long observed pause predicts a smaller immediate jump rate. At $\alpha = 1/2$, $\tau_0 = 1$ second, the chance of no jump during the next second is 0.70710678 at age zero, 0.81649658 at age one second, and 0.95742711 at age ten seconds. Position alone does not encode those different forecasts.

Adding the age to position restores a Markov description for this specified renewal construction: between events, age increases at unit rate; at hazard $\eta(a)$, position jumps by the independent jump law and age resets to zero. For a smooth test function $g(x, a)$ its generator is $\mathcal{L}g = \partial_a g + \eta(a)\mathbb{E}[g(x + J, 0) - g(x, a)]$ on a suitable domain. This is a piecewise deterministic state model. The density-based hazard formula presumes absolutely continuous waiting laws; deterministic or atomic waits need boundary or measure-valued event rules rather than pretending they have an ordinary density.

An event epoch density is not a jump-count probability

The time S_n of the nth renewal has density $g_n = \psi^{*n}$, a convolution of n waiting densities. But the occupancy weight is $w_n(t) = \Pr(S_n \le t < S_{n+1})$, which includes survival after that nth arrival. For $n \ge 1$, $w_n = g_n * \Psi$, and $w_0 = \Psi$. In Laplace space $\tilde{g}_n = \tilde{\psi}^n$, while $\tilde{w}_n(s) = \tilde{\psi}(s)^n[1 - \tilde{\psi}(s)]/s$. The extra survival factor makes $\sum_n \tilde{w}_n = 1/s$, as normalization requires. Omitting it turns an arrival rate density into a probability weight and breaks the position law. This distinction supplies the starting point for chapter 23's Fourier–Laplace solution.

22.7 Computational laboratory: geometry first, clocks second

The lab computes exact even-time return masses in dimensions one, two, and three from multinomial counts. In three dimensions, group the numbers of paired steps in two coordinates: the required combinatorial sum is $u_{2m} = \binom{2m}{m} 6^{-2m} \sum_{k=0}^{m} \binom{m}{k}^2 \binom{2k}{k}$. Logarithmic factorials and a logarithmic sum avoid overflow. The code checks the two-step value $1/(2d)$ and the parity-correct asymptotic at step 2,000. Its partial Green sums include time zero.

```
d=1: G_200=11.32604428; G_2000=35.69586130; parity-correct ratio=0.99987501
d=2: G_200=2.53452726; G_2000=3.26532131; parity-correct ratio=0.99975003
d=3: G_200=1.46990210; G_2000=1.50163700; parity-correct ratio=0.99962510
Residual survival at ages 0,1,10: 0.70710678, 0.81649658, 0.95742711
```

The finite sums do not prove divergence or convergence; the Green-integral argument does. The ratios test the local normalization that a naive continuum density would miss by a factor two. A second experiment attaches exponential and Lomax waits to the same sequence of 200 unbiased spatial jumps using seed 2026110826. The heavy-tailed clock can leave the walker paused for most of the displayed interval even though its jump sequence has not changed.

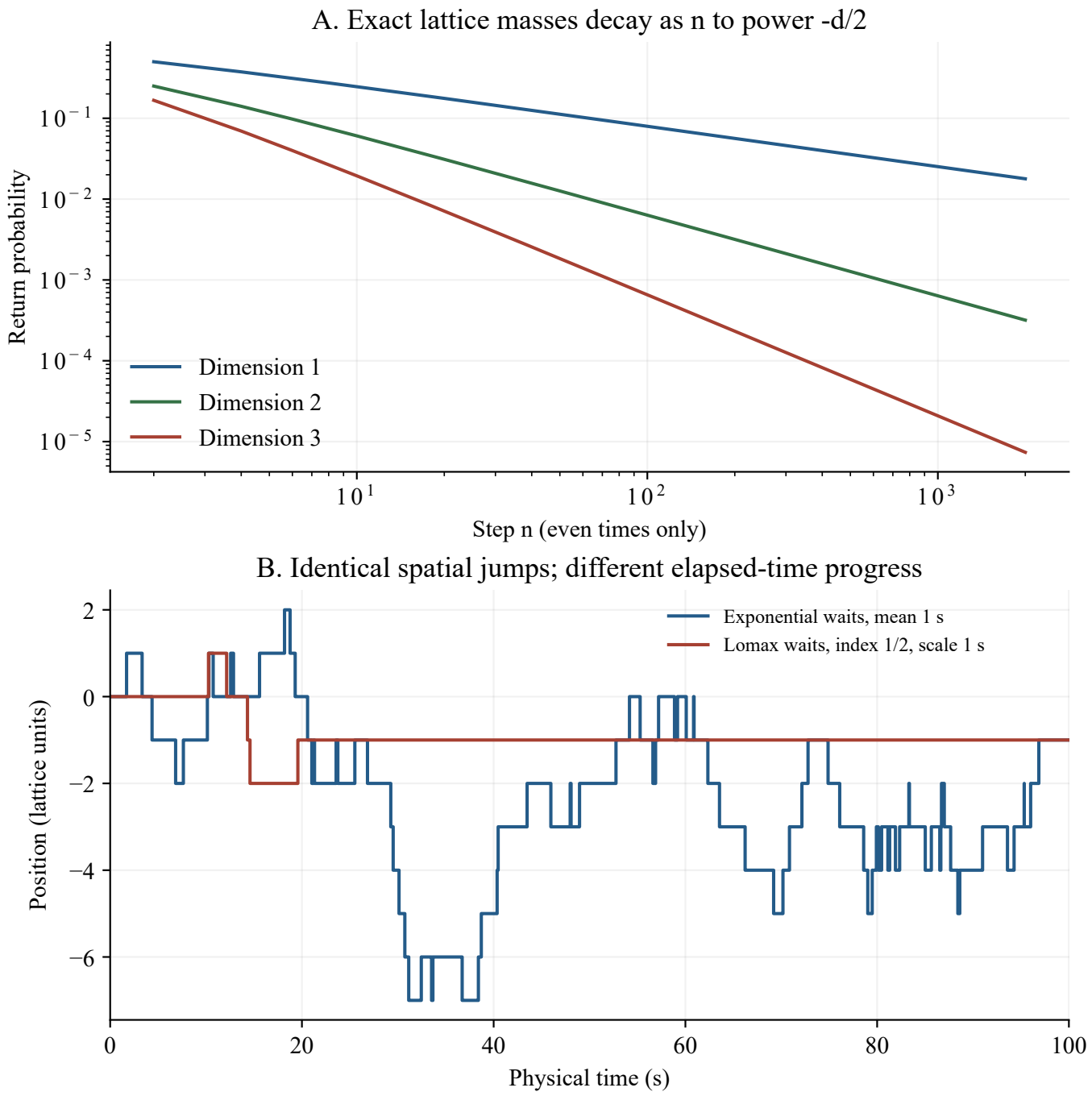


Figure 22.1. Return decay depends on spatial dimension; physical-time progress also depends on the clock. Only even-time lattice masses are plotted. The lower panel is one synthetic trajectory under each independent clock, not an ensemble exponent estimate. The generator uses the retained data.

```
def lomax_residual(age, extra, scale=1.0, alpha=0.5):
    if min(age, extra) < 0 or scale <= 0 or alpha <= 0:
        raise ValueError("invalid age, extra time, scale, or exponent")
    return ((scale+age+extra)/(scale+age))**(-alpha)


for waiting_age in (0.0, 1.0, 10.0):
    print(f"{lomax_residual(waiting_age, 1.0):.8f}")
```

This function has no file effect and computes (22.2) directly. The full lab writes JSON and the report; rerunning overwrites them and does not use the network. Generate the figure after the lab with the declared packages. A rare very large Lomax wait is an intended property, not an overflow to be silently clipped.

22.8 What a clock preserves

Independent strictly positive finite waits preserve the ordered sequence of visited spatial states over infinite time because infinitely many renewals eventually occur. They can change physical-time moments, finite-time observation probabilities, and aging. A time-zero renewal is generally not an equilibrium age preparation. Exponential waits uniquely erase age within this homogeneous renewal model, while general density-based hazards provide a larger Markov state. Lattice recurrence concerns repeated visits to sites; planar Brownian neighborhood recurrence concerns open targets and must not be stated as repeated hits of every point.

22.9 Exercises with worked solutions

Exercise 22.1 — A missing factor two (conceptual)

A continuum Gaussian approximation gives $u_{2m} \sim (4\pi m)^{-1/2}$ for the one-dimensional line walk. Diagnose the error.

Worked solution

At step $2m$, position is an even integer, so the accessible cell spacing is two. A unit-normalized continuous density must be multiplied by that spacing to approximate a single accessible mass. Thus the answer is $2/\sqrt{4\pi m} = 1/\sqrt{\pi m}$. The exact binomial coefficient confirms it. Assigning the same mass approximation at odd sites also violates support: those probabilities are exactly zero. A local limit theorem must encode both the Gaussian scale and lattice periodicity.

Exercise 22.2 — Recurrence under a slower clock (conceptual)

Does replacing exponential waits by positive independent Lomax waits make a recurrent spatial walk transient?

Worked solution

No, provided each wait is finite almost surely and the renewal sum diverges. Every finite spatial jump index is eventually reached, so the same infinite sequence of sites is visited in the same order. Eventual return probabilities are preserved. The distribution of elapsed return time changes, and an infinite-mean wait can make a previously finite physical-time mean diverge. A finite observation window can show many walkers still in their first pause; that is not a change in the jump-chain recurrence classification.

Exercise 22.3 — Derive the occupancy survival factor (derive)

Derive $\widetilde{w}_n(s)$ from the event-epoch density and verify normalization. What goes wrong if $\widetilde{\psi}^n$ alone is used as the count weight?

Worked solution

For $n \geq 1$, the nth renewal occurs at some $u \leq t$ with density $g_n(u)$; no next renewal occurs in the remaining time with probability $\Psi(t-u)$. Hence $w_n(t) = \int_0^t g_n(u)\Psi(t-u)\,du$. For zero renewals $w_0(t) = \Psi(t)$. Integration by parts gives $\widetilde{\Psi} = (1-\widetilde{\psi})/s$, and convolution gives $\widetilde{w}_n = \widetilde{\psi}^n(1-\widetilde{\psi})/s$ including $n = 0$. Summing the geometric series yields $1/s$, the transform of one. Using $\widetilde{\psi}^n$ alone instead gives $1/(1-\widetilde{\psi})$ and has the wrong time units and normalization.

Exercise 22.4 — Obtain the age-dependent generator (derive)

Use a short-time argument to derive the generator for the position-age state given in Section 22.6.

Worked solution

During h, with probability $1 - \eta(a)h + o(h)$ there is no event: position remains x and age advances to $a + h$. With probability $\eta(a)h + o(h)$ there is one renewal: position becomes $x + J$ and age is zero to leading order. Thus the expected change of g is $h\partial_a g + \eta(a)h\mathbb{E}[g(x+J,0) - g(x,a)] + o(h)$. Divide by h and let it vanish. The expansion assumes a regular finite hazard at the current age and negligible multiple events at that order; singular or atomic waiting laws need their own event construction.

Exercise 22.5 — Interpret partial Green sums (compute/design)

The dimension-three sum through step 2,000 is about 1.501637. Can this alone certify transience, or give an exact return probability?

Worked solution

A finite partial sum cannot certify convergence of an infinite series; even the recurrent one- and two-dimensional sums are finite at every cutoff. The Green-integral proof establishes transience. Once convergence is known, the partial sum is a lower bound on $U(1)$, so $1 - 1/U_{2000}$ is a lower bound on the true return probability because this function increases with U. An asymptotic tail estimate or rigorous tail bound is needed for a controlled numerical probability. The lab uses finite sums as checks and deliberately does not label their ratio an exact eventual-return probability.

Exercise 22.6 — An aging detector (compute/design)

A walker has already waited ten seconds under the chapter's Lomax law. Compute its hazard and probability of waiting a further second. Contrast an exponential mean-one-second clock and describe an experiment that distinguishes them.

Worked solution

The Lomax hazard is $0.5/(1+10) = 1/22$ per second. The one-second residual survival is $(12/11)^{-1/2} = 0.95742711$. An exponential mean-one-second clock retains hazard one per second at every age and residual survival $e^{-1} = 0.36787944$. Group observed intervals by current age and compare their subsequent event frequencies, handling right censoring and using independent intervals rather than repeated correlated snapshots as independent trials. A decreasing age-conditioned event rate supports this renewal-aging mechanism more directly than a broad position histogram.

22.10 Sources and attribution

- V. Balakrishnan, IIT Madras, lecture 26, NPTEL 115106089: lattice Green function, recurrence threshold, rescaling, and renewal clock motivation.
- P. Mörters and Y. Peres, Brownian Motion, author-hosted manuscript, Corollary 2.26, printed p. 48, and Theorem 3.20, printed p. 73: planar point polarity versus neighborhood recurrence.

The Green singularity proof, parity normalization, age-state formulation, occupancy weights, exact coefficient laboratory, figure, and exercises are developed here. The random-clock mixture presumes independent spatial jumps and waiting intervals; correlated or coupled alternatives require a different renewal equation.

Renewal memory and anomalous diffusion

A transport record has a displacement variance growing slower than time. Is the jump geometry unusual, or does the walker spend most of its time waiting? A renewal clock with an infinite mean wait produces subdiffusion on an ordinary lattice. Adding a small directional bias creates another complication: random clock dispersion can make the centered variance grow with a different power from the unbiased variance. The mean, centered variance, and uncentered mean square displacement must be calculated separately.

> The full caption track of lecture 27 was read. Its sequence supplies the spatial/temporal factorization, the essential final-wait survival factor, transform solution, and long-time moments. One still at 1630 seconds was inspected, with part of the transform obscured. The exact half-stable clock, biased-moment audit, primary Tauberian qualification, numerical experiment, and exercises are additions. Independent audio was not reviewed.

23.1 Specify a clock independently of the jumps L27 1:33

Let the iid real spatial jumps J_j have characteristic function $\phi(k) = \mathbb{E}[e^{ikJ_j}]$, mean μ_J, and finite variance σ_J^2. Their units are length and length squared; k has inverse-length units. Let iid strictly positive waits W_j be independent of the jump sequence, with density ψ, survival Ψ, and Laplace transform $\tilde{\psi}(s) = \mathbb{E}[e^{-sW}]$ for $s > 0$ in inverse seconds. Start at the origin at a fresh renewal. As in chapter 22, $S_n = \sum_{j=1}^{n} W_j$, $N(t) = \max\{n : S_n \le t\}$, and $X(t) = \sum_{j=1}^{N(t)} J_j$.

On a hypercubic lattice in d dimensions, one jump can have probabilities p_i^+ and p_i^- for $\pm \ell e_i$, with $\sum_i (p_i^+ + p_i^-) = 1$. Its transform is $\sum_i [p_i^+ e^{ik_i\ell} + p_i^- e^{-ik_i\ell}]$. Individual directional probabilities are fractions of the total jump choice; they need not sum to one separately in each coordinate. The scalar argument below applies to any one coordinate or to general independent scalar jumps. Persistence in the jump directions cannot be hidden inside the scalar clock law while retaining the iid spatial factorization.

23.2 The final waiting interval enforces normalization L27 16:51

Write $w_n(t) = \Pr(N(t) = n)$. The nth arrival density is ψ^{*n}, but occupancy at n jumps requires no additional arrival during the remaining interval. Hence $w_n = \psi^{*n} * \Psi$, with $w_0 = \Psi$. The waiting density need not decrease with time: a sharply peaked positive waiting law is valid too. Nonnegativity, unit integral, independence, and the positive-wait construction are the assumptions used here.

Proposition 23.1 — Fourier–Laplace renewal solution. Define $\hat{\tilde{P}}(k,s) = \int_0^\infty e^{-st}\mathbb{E}[e^{ikX(t)}]\,dt$. Then

$$\hat{\tilde{P}}(k,s) = \frac{1 - \tilde{\psi}(s)}{s[1 - \tilde{\psi}(s)\phi(k)]}. \tag{23.1}$$

Derivation.

Conditional on $N(t) = n$, independence gives position transform $\phi(k)^n$. The occupancy convolution gives $\widetilde{w}_n = \widetilde{\psi}^n \widetilde{\Psi}$, with $\widetilde{\Psi} = (1 - \widetilde{\psi})/s$ by integration of survival. Summing over $n \geq 0$ is an absolutely convergent geometric series because $|\phi(k)| \leq 1$ and $0 \leq \widetilde{\psi}(s) < 1$ for positive waits and positive s. This yields (23.1). At $k = 0$, $\phi(0) = 1$ and the result becomes $1/s$, proving probability normalization. ∎

This is the decoupled Montroll–Weiss formula; Meerschaert–Scheffler, Section 2, equation (4), PDF page 3, gives the same occupancy derivation. The numerator includes the probability of still waiting after the final observed jump. Omitting it would make the zero-wave-number transform $1/(1 - \widetilde{\psi})$ rather than $1/s$, exposing the normalization failure before any inverse transform is attempted.

For exponential waits of rate λ, $\widetilde{\psi} = \lambda/(s + \lambda)$ reduces (23.1) to $[s + \lambda(1 - \phi(k))]^{-1}$. Inverting in time gives $\mathbb{E}[e^{ikX(t)}] = \exp[\lambda t(\phi(k) - 1)]$, the compound-Poisson transform. For nearest-neighbor jumps it recovers the lattice laws of chapters 6–7. The same check also fixes the normalization of the waiting-rate parameter.

23.3 Bias couples position scatter to clock scatter L27 32:49

Let $m(t) = \mathbb{E}[N(t)]$ and $v_N(t) = \mathrm{Var}(N(t))$. Conditional on a finite count n, the sum has mean $n\mu_J$ and variance $n\sigma_J^2$. At fixed finite time these count moments are finite for the stated iid strictly positive wait construction. For example, choose $\epsilon > 0$ with $\Pr(W \geq \epsilon) > 0$; the event of arbitrarily many renewals by t requires an atypically small number of those positive-length waits, giving an exponentially decaying count tail.

Proposition 23.2 — Exact random-sum moments.

$$\mathbb{E}[X(t)] = \mu_J m(t), \quad \mathrm{Var}(X(t)) = \sigma_J^2 m(t) + \mu_J^2 v_N(t), \tag{23.2}$$

$$\mathbb{E}[X(t)^2] = \mathrm{Var}(X(t)) + \mu_J^2 m(t)^2. \tag{23.3}$$

Derivation.

Total expectation averages the conditional mean $N\mu_J$. For the variance, expand $X - \mathbb{E}X$ into $X - \mathbb{E}[X \mid N]$ plus $\mathbb{E}[X \mid N] - \mathbb{E}X$. The cross term has zero conditional expectation. The first squared term averages to $\sigma_J^2 \mathbb{E}N$, and the second to $\mu_J^2 \operatorname{Var} N$. Adding the squared mean gives the uncentered second moment. ∎

When $\mu_J = 0$, position variance measures the expected number of jumps. When $\mu_J \neq 0$, walkers with different jump counts accumulate different mean displacements; their clock scatter creates extra centered variance. Under a Poisson clock $m = v_N = \lambda t$, giving $\operatorname{Var} X = \lambda t \mathbb{E}[J^2]$. The uncentered second moment also contains a ballistic $\mu_J^2 \lambda^2 t^2$ term. Calling that entire quantity diffusion without subtracting drift would misidentify the mechanism.

To extract the count moments from transforms, define $G(z, t) = \mathbb{E}[z^{N(t)}]$. Equation (23.1) with ϕ replaced by z gives its time transform. Differentiating at $z = 1$ yields $\widetilde{m}(s) = \widetilde{\psi}/[s(1 - \widetilde{\psi})]$ and

$\mathcal{L}\{\mathbb{E}[N(N-1)]\} = 2\widetilde{\psi}^2/[s(1-\widetilde{\psi})^2]$. The second derivative generates a factorial moment, not N^2; add $m(t)$ before computing $v_N(t)$. This is the same ordinary-versus-factorial distinction encountered for counting laws in chapter 1.

23.4 A small-transform power needs a Tauberian condition L27 37:59

Assume $1-\widetilde{\psi}(s) \sim (\tau_0 s)^\alpha$ as $s \downarrow 0$, with $0 < \alpha < 1$ and time scale $\tau_0 > 0$. This nonanalytic correction signals an infinite mean waiting time. For comparison, any finite positive mean μ_W gives $1-\widetilde{\psi}(s) \sim \mu_W s$ by dominated convergence applied to $(1-e^{-sW})/s$. Full analyticity at zero is not required for that finite-mean relation.

The inference from transform behavior to a long-time function is a *Tauberian* implication: it needs restrictions that prevent arbitrary oscillatory counterexamples. The form used here concerns a nondecreasing right-continuous function U whose Laplace–Stieltjes transform is finite for positive arguments. If $\int e^{-st}dU(t) \sim cs^{-\beta}$, for $c > 0$ and $\beta \geq 0$, then $U(t) \sim ct^\beta/\Gamma(1+\beta)$. This is the constant-slow-factor case of Meerschaert–Scheffler, Section 3, equations (3.1)–(3.2). The gamma function is $\Gamma(z) = \int_0^\infty u^{z-1}e^{-u}du$ for positive z. The theorem is cited here; applying it to monotone renewal moments is the argument supplied next.

Proposition 23.3 — Heavy-clock ensemble scaling. Under the stated independent renewal construction and transform asymptotic,

$$m(t) \sim \frac{(t/\tau_0)^\alpha}{\Gamma(1+\alpha)}, \quad v_N(t) \sim C_\alpha (t/\tau_0)^{2\alpha}, \tag{23.4}$$

where $C_\alpha = 2/\Gamma(1+2\alpha) - 1/\Gamma(1+\alpha)^2$. For zero-mean jumps, $\operatorname{Var} X = \sigma_J^2 m(t)$ has sublinear power t^α.

Derivation.

The exact moment transforms give $s\widetilde{m}(s) \sim \tau_0^{-\alpha}s^{-\alpha}$ and $s\mathcal{L}\{\mathbb{E}[N(N-1)]\} \sim 2\tau_0^{-2\alpha}s^{-2\alpha}$. Both moments are nonnegative, nondecreasing, and right-continuous in time. They vanish initially and have finite positive-argument transforms, so these are their Laplace–Stieltjes transforms. The cited Tauberian theorem gives the first formula and factorial moment $2(t/\tau_0)^{2\alpha}/\Gamma(1+2\alpha)$. Adding m and subtracting m^2 gives the variance asymptotic. The coefficient is positive for $0 < \alpha < 1$; it represents nonvanishing relative clock dispersion in this limit. Finally apply (23.2) with $\mu_J = 0$. ∎

At nonzero bias, the $\mu_J^2 v_N$ contribution has power $t^{2\alpha}$ and eventually dominates $\sigma_J^2 m$. Thus the same clock gives an unbiased centered variance with exponent α and a biased centered variance with exponent 2α. The uncentered second moment adds another term of order $t^{2\alpha}$. At $\alpha = 1/2$, unbiased variance is proportional to $\sqrt{t}$, while biased variance is asymptotically linear in t. The clock has not become memoryless. A fitted exponent of one under bias does not establish ordinary Markov diffusion.

The positivity used in the variance asymptotic can also be checked directly. Define $B(\alpha,\alpha) = \int_0^1 u^{\alpha-1}(1-u)^{\alpha-1}du = \Gamma(\alpha)^2/\Gamma(2\alpha)$. For $0 < \alpha < 1$, its integrand exceeds $u^{\alpha-1}$, so

$\alpha B(\alpha,\alpha) > 1$. Hence $\Gamma(1+2\alpha)/\Gamma(1+\alpha)^2 = 2/[\alpha B(\alpha,\alpha)] < 2$, precisely the inequality $C_\alpha > 0$. The beta–gamma identity follows by substituting total radius and fraction in the product of two gamma integrals.

With a finite mean wait, the same monotone transform argument gives $m(t) \sim t/\mu_W$, hence ordinary linear ensemble variance for zero-mean finite-variance jumps, even if the waiting law is nonexponential and the position process has memory. Biased variance additionally depends on clock fluctuations; finite mean alone does not determine those. Long-memory timing and a transport exponent are distinct properties.

23.5 A clock with an exact count-tail formula

For a concrete synthetic clock, measure time in seconds and set $W = 1/(2Z^2)$ seconds, where Z is standard Gaussian. Chapter 4's Lévy transform gives $\widetilde{\psi}(s) = e^{-\sqrt{s}}$ in these numerical units. Sums of n waits have the law of $n^2 W$, so $\Pr(N(t) \geq n) = \Pr(S_n \leq t) = \operatorname{erfc}[n/(2\sqrt{t})]$ for positive integers n. This is exact at every positive time, not just asymptotic. In a dimensional expression replace t by t/τ_0 with $\tau_0 = 1$ second.

The integer identities $N = \sum_{n\geq 1} \mathbf{1}_{N\geq n}$ and $N^2 = \sum_{n\geq 1}(2n-1)\mathbf{1}_{N\geq n}$ give rapidly convergent series for its mean and variance. They supply independent numerical targets for the renewal Monte Carlo. At long time $m(t) \sim 2\sqrt{t/\pi}$ and $v_N(t) \sim (2-4/\pi)t$ in these units. The exact mean at time one is 0.675802, far below its leading asymptotic 1.128379. A long-time exponent can be right while a finite-time prefactor estimate remains poor.

23.6 Computational laboratory: compare three moments and two averages

The lab simulates 20,000 independent Lévy renewal clocks at four observation times with seed 2026110827. It generates each positive wait from the reciprocal Gaussian square, preserving jumps at their actual epochs. The exact count-tail series gives the independent mean and variance targets. Conditional binomial draws implement either unbiased signs or right-step probability 0.65, with jump mean 0.3 and variance 0.91 in lattice units. Thus the random-sum predictions can be checked separately from the clock simulation.

```
t=1: E[N]=0.675802; sample=0.679050; biased variance=0.678054; biased MSD=0.719157
t=10: E[N]=3.083128; sample=3.062350; biased variance=3.457654; biased MSD=4.313165
t=100: E[N]=10.788494; sample=10.674900; biased variance=16.356321; biased MSD=26.831565
t=1000: E[N]=35.183969; sample=35.491250; biased variance=97.423803; biased MSD=208.835855
Lag10 ensemble MSD=3.083128; time-average mean=0.353841, across-path SD=0.319048
```

The final line compares the ensemble displacement over the first ten seconds of fresh clocks with a different statistic: squared ten-second increments averaged along each of 64 paths of duration 1,000 seconds, sampled every second. The latter intervals mostly begin at a positive waiting age and overlap in time. Their across-path standard deviation is large. This finite-horizon experiment demonstrates preparation and averaging effects under a specified protocol; it is not a proof of a general ergodic theorem or a universal relation for time averages. The exact renewal analysis supplies the ensemble scaling.

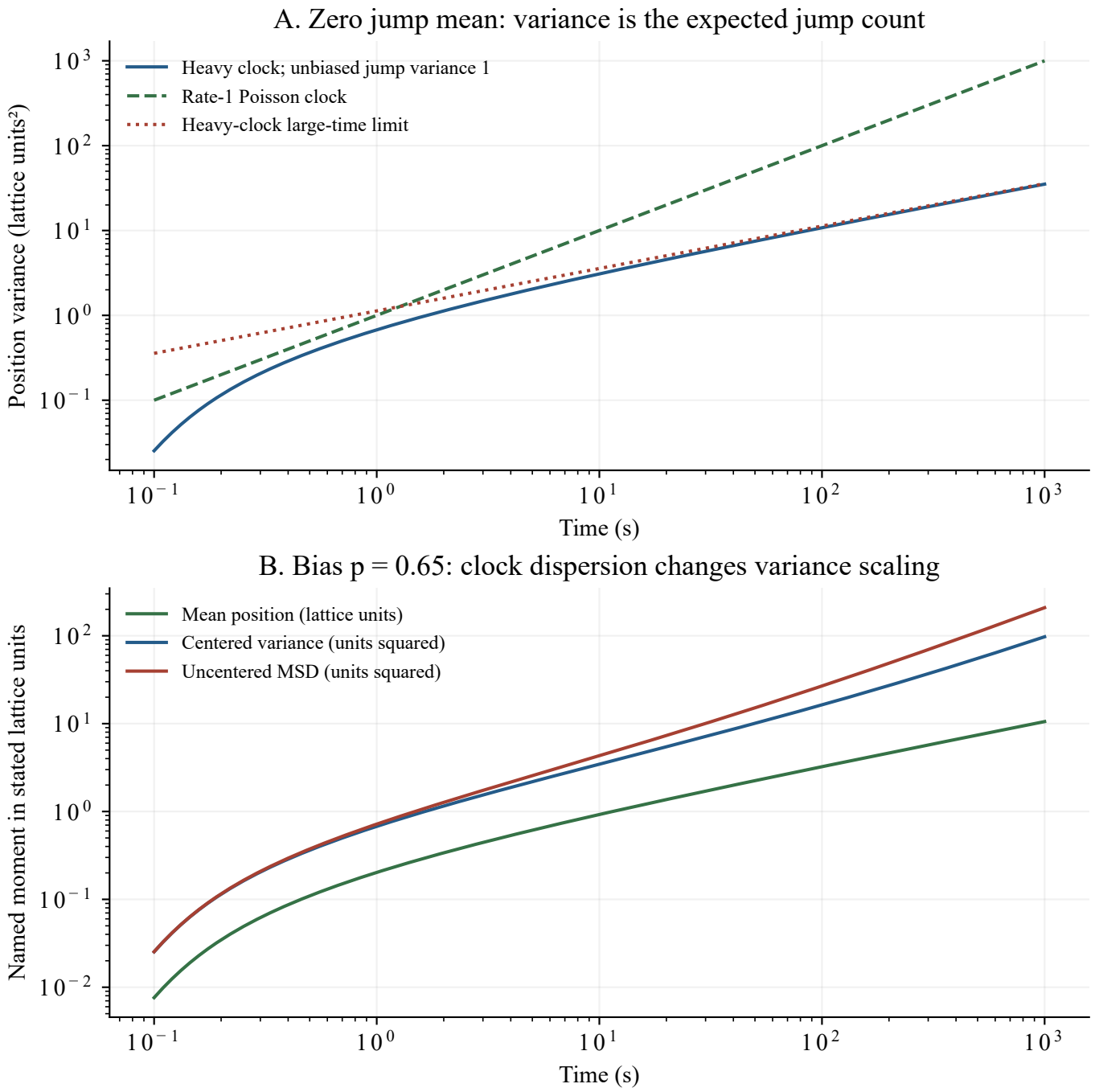


Figure 23.1. Exact count-tail sums determine these theoretical curves. The lower panel labels each moment and its power of lattice units explicitly; comparing their numerical heights does not equate their dimensions. Bias makes clock-count fluctuations contribute to centered position scatter. The generator reproduces the curves from the same clock law checked in the laboratory records.

```
def random_sum_moments(jump_mean, jump_variance, count_mean, count_var):
    mean = jump_mean*count_mean
    variance = jump_variance*count_mean+jump_mean**2*count_var
    return mean, variance, variance+mean**2


print(random_sum_moments(0.3, 0.91, 10.0, 10.0))
```

For a Poisson clock of mean ten this prints mean three, centered variance ten, and uncentered second moment nineteen, up to floating-point representation. It accepts supplied valid finite moments and has no file effects. The full lab writes JSON and the report, overwrites them on rerun, and makes no network calls. Run the lab before the figure generator with the declared dependencies. The broad Monte Carlo checks allow sampling scatter; the exact series, transform normalization, and limiting cases are the more precise implementation targets.

23.7 An anomalous exponent does not identify one mechanism

The derivation assumes independent iid waits, independent iid finite-variance jumps, and a fresh renewal at time zero. Coupled jump/wait laws, persistent jump directions, aging preparation, finite propagation speed, and heavy spatial jumps change the transform or its moments. A memory integral equation can still exist even when a local-in-time Markov master equation for position alone does not. Subdiffusion can also arise from geometry, as in chapter 21; the same displacement exponent does not identify a unique microscopic cause. Return probabilities of the jump chain, physical-time return distributions, and ensemble versus time averaging must be measured separately.

23.8 Exercises with worked solutions

Exercise 23.1 — An arrival density is not an occupation probability (conceptual)

A proposed transport transform is $[1-\tilde{\psi}(s)\phi(k)]^{-1}$. Diagnose it without performing an inverse transform.

Worked solution

At $k=0$ the result becomes $1/(1-\tilde{\psi})$, whereas the time transform of total probability one must be $1/s$. It also has no time units: each transform of an arrival density is dimensionless, while the Laplace transform of an occupation probability has units of time. Arrival at the last epoch must be followed by survival with no new event. Multiplication by $\tilde{\Psi}=(1-\tilde{\psi})/s$ restores both units and normalization. This test holds for arbitrary positive iid waits, independent of a diffusive approximation.

Exercise 23.2 — Linear variance under a non-Markov clock (conceptual)

In the half-order clock, an experiment with biased unit jumps finds centered position variance proportional to time. Does this establish an exponential waiting law?

Worked solution

No. Equation (23.2) contains the clock contribution $\mu_J^2 \operatorname{Var} N$. For $\alpha=1/2$, that contribution is asymptotic to $\mu_J^2(2-4/\pi)t$, while the intrinsic jump contribution is of order $\sqrt{t}$. The resulting leading linear variance coexists with mean displacement of order $\sqrt{t}$ and nonexponential waiting ages. One should compare a zero-bias run, count statistics, and age-conditioned survival. Linear centered variance alone neither identifies the count law nor establishes a position-only Markov description. Subtracting the mean before measuring scatter is necessary but does not remove clock dispersion.

Exercise 23.3 — Differentiate the count-generating transform (derive)

Derive the two count-moment transforms and recover $\operatorname{Var} N=\lambda t$ for an exponential clock.

Worked solution

Write $q = \widetilde{\psi}(s)$, fixed for positive s. The transform is $(1-q)/[s(1-zq)]$. Its first derivative at one is $q/[s(1-q)]$; its second derivative is $2q^2/[s(1-q)^2]$. These generate $\mathbb{E}N$ and $\mathbb{E}[N(N-1)]$ respectively because differentiating z^n twice gives $n(n-1)$. With $q = \lambda/(s+\lambda)$, the transforms reduce to λ/s^2 and $2\lambda^2/s^3$. Inversion gives $m = \lambda t$ and factorial moment $\lambda^2 t^2$. Therefore $\mathbb{E}N^2 = \lambda^2 t^2 + \lambda t$, and subtracting m^2 leaves λt. Confusing the second derivative with the ordinary second moment would incorrectly remove the Poisson variance.

Exercise 23.4 — Finite mean without an analytic transform (derive)

Let the waiting survival be $\Psi(t) = (1 + t/\tau)^{-3/2}$ for $\tau > 0$. Find its mean, explain why its transform is not analytic at zero, and derive the unbiased long-time position variance.

Worked solution

Integrating survival gives $\mu_W = \int_0^\infty (1+t/\tau)^{-3/2} dt = 2\tau$. The second moment equals $2\int_0^\infty t\Psi(t)dt$, whose integrand has tail proportional to $t^{-1/2}$, so it diverges. A finite second derivative of the Laplace transform at zero would equal that moment by monotone convergence; therefore no ordinary analytic Taylor expansion exists there. Nevertheless $0 \le (1 - e^{-sW})/s \le W$, and dominated convergence yields $1 - \widetilde{\psi}(s) \sim 2\tau s$. Thus $s\widetilde{m}(s) \sim 1/(2\tau s)$ and the monotone Tauberian theorem gives $m(t) \sim t/(2\tau)$. For independent zero-mean jumps of variance σ_J^2, the exact random-sum identity gives $\operatorname{Var} X(t) \sim \sigma_J^2 t/(2\tau)$. This proves linear ensemble variance despite a nonexponential, nonanalytic clock; it does not determine the biased variance from the waiting mean alone.

Exercise 23.5 — Separate drift from clock scatter (compute/design)

At time 1,000 in the lab, $m = 35.1839691055$ and $v_N = 726.7376822080$. With right-step probability 0.65, compute the position mean, centered variance, and uncentered second moment. Which variance contribution dominates?

Worked solution

The jump mean is $0.65 - 0.35 = 0.3$ and variance $1 - 0.3^2 = 0.91$. Hence mean position is $0.3m = 10.5551907317$. Intrinsic jump scatter contributes $0.91m = 32.0174118860$, while clock scatter contributes $0.09 v_N = 65.4063913987$. Their sum is the centered variance 97.4238032848. The squared mean is 111.4120513818, so the uncentered second moment is 208.8358546665. Clock dispersion already exceeds intrinsic jump scatter at this observation time. Reporting the last number as centered diffusion would additionally include the larger squared drift. Repeating at zero bias would isolate the count mean, giving variance 35.1839691055.

Exercise 23.6 — Compare averaging protocols (compute/design)

Design a defensible comparison between the lab's lag-ten ensemble MSD and its lag-ten time-averaged squared increments. What conclusions do the reported 3.083128 and 0.353841 support?

Worked solution

Generate many independent paths from a fresh renewal at zero. For the ensemble statistic, square each displacement between zero and ten and average across paths. For the time average, retain each complete path to time 1,000, compute the 991 overlapping increments $X(j+10) - X(j)$ for $j = 0, \ldots, 990$, square them, and average within that path. Then report the distribution of these path averages, not 991 correlated increments as independent replicates. Reuse the same spatial and clock laws, units, and observation grid; independently vary the total duration, lag, and number of paths. The reported values show that these finite-horizon protocols give markedly different averages in the simulated aging clock. The across-path standard deviation 0.319048 indicates substantial path variation for 64 retained paths. They do not prove a limiting distribution, a universal ratio, or the failure of every possible ergodic theorem. Comparing intervals conditioned on their waiting age and using fresh-clock controls identifies the preparation effect directly.

23.9 Sources and attribution

- V. Balakrishnan, IIT Madras, lecture 27, NPTEL 115106089: general renewal clock, Fourier–Laplace transport transform, bias, and anomalous ensemble scaling.
- M. M. Meerschaert and H.-P. Scheffler, Continuous time random walks and space-time fractional differential equations, author-hosted review, Section 2, equation (4), PDF p. 3: decoupled renewal transform.
- M. M. Meerschaert and H.-P. Scheffler, Tauberian theorems for matrix regular variation, Transactions of the AMS 365 (2013), 2207–2221, Section 3, equations (3.1)–(3.2): the scalar Laplace–Stieltjes Tauberian result used here.

The moment proofs, finite-mean qualification, exact half-order count-tail sums, simulation, original figure, and worked exercises are developed here. The independent-jump and independent-clock assumptions remain essential throughout; no universal transport law is asserted for coupled or persistent alternatives.

Deterministic maps and invariant probability

A deterministic map fixes every future state once the initial point is known. An ensemble of uncertain initial points nevertheless has an evolving probability law. The doubling map makes the distinction exact: it transports a nonuniform density toward uniform, and its half-interval observations are independent fair bits under uniform initialization. A different partition of the same map retains memory. The probability law, the observation, and the arithmetic used to compute an orbit all matter.

> The full caption track of lecture 28 was read sequentially. One low-resolution still at 31:22 was inspected; the invariant-density equation is partly occluded. The lecture's flow, coarse-graining, invariant-density and recurrence sequence supplies the spine. Explicit preimage calculations, symbolic counterexamples and the precision experiment are developed here. The pointwise ergodic theorem is cited with its measure-theoretic hypotheses; recurrence and the mean-return identity are distinguished in chapter 25.

24.1 Sample a flow without adding external noise L28 1:43

An autonomous ordinary differential equation $\dot{x} = f(x)$ specifies a state vector in phase space. Under local Lipschitz regularity, its initial-value solution is locally unique; restrict attention to an invariant domain on which solutions exist for all required times. Write $\Phi_t(x)$ for the state after elapsed time t. Uniqueness yields $\Phi_{s+t} = \Phi_s \circ \Phi_t$. Sampling every $\Delta t > 0$ produces the deterministic map $T = \Phi_{\Delta t}$ and sequence $x_n = T^n x_0$. For a complete reversible flow each sampled map is invertible, but general discrete maps need not be. The doubling map below is a discrete model in its own right, not the full-state time map of such an invertible flow.

For a harmonic oscillator, $H = p^2/(2m) + m\omega^2 q^2/2$ is constant and its nonzero-energy trajectories are ellipses of period $2\pi/\omega$. For two excited oscillators with an irrational frequency ratio, returning both phases exactly would require two integer periods with that ratio, which is impossible. Approximate simultaneous returns can still occur. Projected trajectories may cross in a position plot without violating uniqueness in full phase space. A recurrence question must specify a full state or a finite-resolution target and a sampling interval.

24.2 Probability follows inverse images L28 24:02

Let Ω be the state space, $\mathcal{F}$ its measurable sets, μ an initial probability measure, and $T : \Omega \to \Omega$ a measurable map. A target set A is occupied after one step precisely when the initial point belongs to its inverse image $T^{-1}A = \{x : T(x) \in A\}$. Thus the transported or *pushforward measure* is

$$(T_*\mu)(A) = \mu(T^{-1}A). \tag{24.1}$$

The symbol $T^{-1}A$ denotes a set of preimages and does not require an inverse function. Countable additivity and total mass one follow because inverse images preserve disjoint unions and $T^{-1}\Omega = \Omega$.

A measure is *invariant* if $T_*\mu = \mu$, meaning $\mu(T^{-1}A) = \mu(A)$ for every measurable A. It gives stationary ensemble observations under that initialization. Invariance does not assert convergence to that measure from other initial laws, uniqueness, or that every trajectory samples it. A *density* ρ relative to ordinary length satisfies $\mu(A) = \int_A \rho(x)dx$; some invariant measures have no such density.

Our running map is $T(x) = 2x \pmod 1$ on $[0,1)$, with normalized Lebesgue length measure μ. For an interval $A = [a,b)$ its inverse image is $[a/2, b/2) \cup [(a+1)/2, (b+1)/2)$, two disjoint intervals of total length $b-a$. This proves length invariance on intervals and hence on their generated Borel sets. Yet the point mass at zero is also invariant because $T(0) = 0$. It is singular relative to length. A stationary-density equation cannot detect every possible invariant measure.

24.3 Stretch each branch and count every preimage L28 27:14

Proposition 24.1 — Density transfer by regular branches. Suppose a one-dimensional map has finitely many smooth monotone branches, each with nonzero derivative in its interior. For a nonnegative integrable input density, the output density is, at regular target values,

$$(\mathcal{P}\rho)(y) = \sum_{x:T(x)=y} \frac{\rho(x)}{|T'(x)|}. \tag{24.2}$$

Branch endpoints and any exceptional target values are handled up to sets of zero length. Positivity and total mass are preserved.

Derivation.

For any bounded measurable test function g, the transported expectation is $\int g(T(x))\rho(x)dx$. Split the integral into the monotone branches. On each branch change variable $y = T(x)$; the magnitude of its inverse Jacobian is $1/|T'(x)|$. Summing the resulting integrals gives $\int g(y)(\mathcal{P}\rho)(y)dy$. Taking indicator functions identifies the density, and taking $g = 1$ proves mass conservation. Nonnegativity is immediate from the formula. ∎

The equivalent distributional expression is $\rho_{n+1}(y) = \int \rho_n(x)\delta(y - T(x))dx$, with a Dirac delta. Its branch interpretation is essential: retaining just one inverse point would lose mass. An invariant density solves $\mathcal{P}\rho = \rho$ together with $\rho \geq 0$ and $\int \rho = 1$. The fixed-point equation alone neither guarantees existence nor proves attraction.

For doubling, $T' = 2$ on both branches, so

$$(\mathcal{P}\rho)(y) = \tfrac{1}{2}[\rho(y/2) + \rho((y+1)/2)]. \tag{24.3}$$

The uniform density is fixed. Starting instead from $\rho_0(x) = 2x$, substitution gives $\rho_1(y) = y + 1/2$ and, by induction,

$$\rho_n(y) = 1 + 2^{-n}(2y - 1). \tag{24.4}$$

Its distance from uniform in total variation is $\frac{1}{2}\int_0^1 |\rho_n - 1|dy = 2^{-n}/4$. Every finite-step density is normalized and nonnegative. This explicit convergence concerns that initial density; it cannot describe the invariant point mass at zero.

24.4 A deterministic orbit can reveal independent bits L28 22:53

Except for the countable ambiguity at dyadic rationals, write $x = 0.b_1b_2b_3\ldots$ in binary. Doubling discards the first digit: $T(x) = 0.b_2b_3\ldots$. Define the observation $Y_n = 0$ in $[0, 1/2)$ and $Y_n = 1$ in $[1/2, 1)$. Then $Y_n = b_{n+1}$. The initial point fixes this entire sequence, so there are no fresh random kicks. Randomness enters through the initial measure.

Proposition 24.2 — Exact cylinder probabilities. Under uniform initial measure, every specified word of m successive observed bits has probability 2^{-m}. Consequently these observations are iid fair bits.

Derivation.

A word $c_1\ldots c_m$ fixes an integer $j = \sum_{r=1}^{m} c_r 2^{m-r}$ and the initial interval $[j/2^m, (j+1)/2^m)$, apart from irrelevant endpoint representations. Its measure is 2^{-m}, the product of m fair-bit probabilities. Marginalizing unspecified digits proves independence for any finite collection of bit positions. ∎

Knowing only the observed past has revealed the discarded prefix, while the unknown tail remains uniformly distributed. Knowing the exact real initial point determines every future bit. These are different information conditions. The point $1/7$ has repeating binary digits and the exact period-three orbit $1/7, 2/7, 4/7$; that exceptional orbit does not disprove an almost-everywhere statistical statement under length measure.

24.5 Separate invariance, ergodicity, and mixing L28 35:45

For an invariant probability measure, a measurable set A is invariant modulo null sets when $T^{-1}A$ differs from A by a set of measure zero. The system is *ergodic* when every such set has measure zero or one. The pair of map and measure matters. The identity map preserves every probability measure but is not ergodic for a uniform continuum ensemble: every proper interval is invariant.

Cited theorem — Pointwise ergodic theorem, ergodic case. For a measurable measure-preserving map on a probability space, if the system is ergodic and g is integrable, then

$$\frac{1}{N}\sum_{n=0}^{N-1} g(T^n x) \longrightarrow \int g\,d\mu \quad \text{for } \mu\text{-almost every } x. \tag{24.5}$$

The null exceptional set may depend on g. See Tao, 254A Lecture 9, Theorems 2 and 3. In the general nonergodic case the limit is the conditional expectation on the invariant sigma-algebra, meaning the collection of invariant measurable sets; it need not be the global ensemble mean.

This theorem is not proved by displaying a stationary histogram. For the identity map and $g(x) = x$ under uniform measure, the time average is exactly x for every duration, while the ensemble mean is $1/2$. More generally, disconnected invariant components can preserve a stationary global mixture even though an individual orbit never samples another component.

Mixing is the stronger decorrelation property $\mu(A \cap T^{-n}B) \to \mu(A)\mu(B)$ for measurable A, B. Doubling with length measure is mixing, as can be proved from the binary construction. First take A and B to be unions of finitely many dyadic intervals. Membership in each depends on a finite initial block of digits. For sufficiently large n, membership in $T^{-n}B$ depends on a disjoint later block. Proposition 24.2 gives exact factorization. For arbitrary measurable A, B, approximate each in measure by such finite unions. Invariance bounds the shifted approximation error by the original one; intersection and product errors together tend to zero with those approximation errors. This extends the factorization limit to all measurable sets.

Mixing implies ergodicity: for an invariant set A, $\mu(A \cap T^{-n}A) = \mu(A)$ at every n, so mixing gives $\mu(A) = \mu(A)^2$. The converse fails. A two-point space with uniform measure and map swapping the points is ergodic, since its only invariant subsets are empty and full. Its correlations alternate forever. The time average on either orbit is nevertheless the uniform two-point mean. Neither chaos nor mixing is required for the ergodic time-average conclusion.

24.6 A coarse partition can retain predictive memory

Now observe the same doubling map through $A = [0, 3/4)$ and $B = [3/4, 1)$. Under stationary uniform initialization,

$$\Pr(Y_{n+1} = B \mid Y_n = A) = \frac{1/8}{3/4} = \frac{1}{6}. \tag{24.6}$$

The numerator is the length of $A \cap T^{-1}B = [3/8, 1/2)$. But conditioning also on $Y_{n-1} = B$ forces the intermediate coordinate into $[1/2, 3/4)$; its next image lies in $[0, 1/2)$, inside A. Therefore

$$\Pr(Y_{n+1} = B \mid Y_n = A, Y_{n-1} = B) = 0. \tag{24.7}$$

The conditioning history has positive probability $1/8$. The observation process is stationary but is not a first-order Markov chain. Deterministic evolution in the full state does not make arbitrary coarse cell labels sufficient predictive states.

One can always form one-step stationary conditional probabilities between cells. Multiplying that finite matrix repeatedly is an additional closure assumption unless the partition and conditional distributions justify it. In particular, replacing a density by its cell averages erases variations inside each cell. Under a map, those variations affect how much mass crosses each future boundary. A finite matrix transfer of cell-uniform approximations need not equal the exact cell masses of an evolving nonuniform density. The laboratory therefore evaluates actual inverse branches for (24.4), rather than silently imposing cellwise uniformity after each step.

24.7 Computational laboratory: ensemble transfer and finite precision

The lab computes the 2^n preimages of each of 64 cell midpoints and averages $\rho_0 = 2x$ over them with the exact Jacobian weights. The result is compared with (24.4). Midpoint integration is exact for these linear densities, so normalization and total variation can be tested to roundoff. Separately, all 4,096 equally spaced dyadic starting points are enumerated for three-bit words and the history-conditioned coarse probabilities. These finite indicator integrals are exact because every event boundary falls on the enumeration grid.

```
step=0: total variation=0.25000000
step=1: total variation=0.12500000
step=2: total variation=0.06250000
step=3: total variation=0.03125000
step=4: total variation=0.01562500
step=5: total variation=0.00781250
step=6: total variation=0.00390625
Three-bit cylinder counts (4096 initial points): [512, 512, 512, 512, 512, 512, 512, 512]
P(next B | A)=0.16666667; P(next B | previous B, A)=0.00000000
Binary float starting at0.1 reaches zero at step55.
Exact rational orbit: ['1/7', '2/7', '4/7'], period3.
```

Every stored binary floating-point number in $[0, 1)$ is a dyadic rational with a finite binary expansion. Doubling shifts away one digit at a time until zero remains. This calculation uses multiplication by two and subtraction of an integer, so this digit-exhaustion effect is particularly direct. The stored approximation to decimal 0.1 reaches zero at step 55. Exact decimal $1/10$, by contrast, eventually cycles and does not reach zero, while the exact rational $1/7$ cycles from the start. Uniform real initial measure places probability zero on all rational points. Increasing run duration after a fixed precision has exhausted its digits gives a worse representation of a typical continuum orbit.

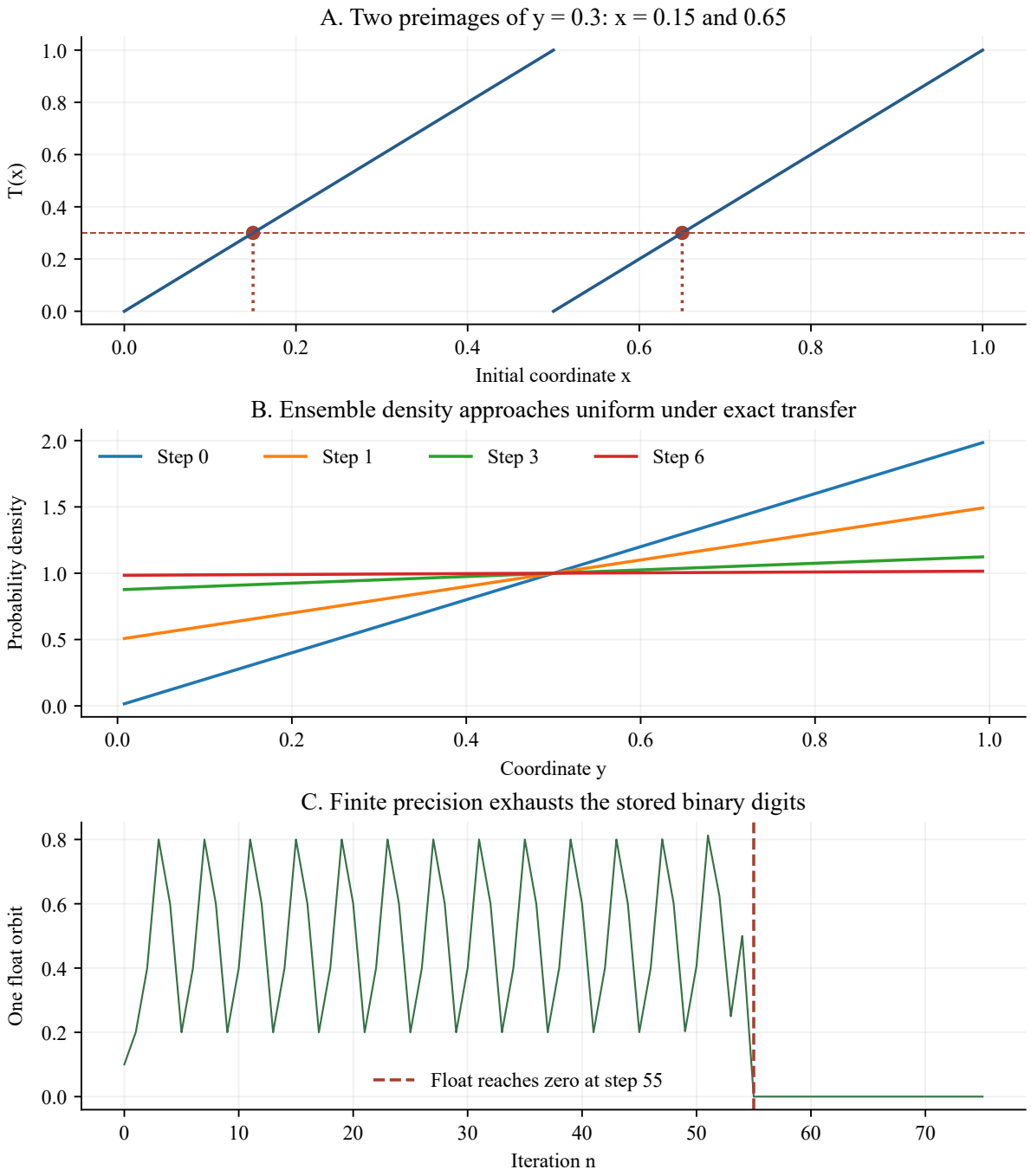


Figure 24.1. The preimage geometry determines ensemble evolution. Independent density transfer approaches uniform while one computed float orbit exhausts its stored digits. The lower panel is a precision diagnostic for the specified arithmetic, not a physical attractor of the ideal map under length measure. Figure generator; numerical records.

```
def doubling_transfer(rho, y):
    return 0.5*(rho(y/2)+rho((y+1)/2))


print(doubling_transfer(lambda x: 2*x, 0.3))
```

This function evaluates one regular density transfer and prints 0.8, up to floating-point representation; it has no file effects. Its input must be an evaluable density on the unit interval. The full lab writes JSON and its report, overwriting them on rerun, without network access. Run it before the generator with the declared packages. No long finite-precision orbit is used to prove ergodicity or mixing.

24.8 Formulate escape and recurrence before averaging L28 39:59

For a measurable target C with $\mu(C) > 0$, initialize with conditional invariant measure $\mu_C(A) = \mu(A \cap C)/\mu(C)$. The first exit is $\tau_{\text{out}} = \inf\{n \geq 1 : T^n x \notin C\}$; the first positive return is $\tau_C^+ = \inf\{n \geq 1 : T^n x \in C\}$. Infima of empty sets are infinity. Return at step one counts even if the path has not first exited. A return after departure is a different stopping event. The survival probability $\Pr(\tau_{\text{out}} > n \mid x \in C)$ is a decreasing tail, not a probability mass to normalize by summing over n.

In a finite measure-preserving system, almost every point of C returns to C; this Poincaré recurrence statement requires no ergodicity. Under ergodicity, the conditional mean positive return is $1/\mu(C)$, the Kac identity proved in the next chapter. Neither statement asserts that a trajectory hits every exact point or that return intervals are independent. A zero-measure target lies outside that conditional formula; measure zero alone does not prove absence of periodic or other exact returns. A stationary invariant density also need not be a long-time limit for every initial ensemble.

24.9 Exercises with worked solutions

Exercise 24.1 — A stationary atom (conceptual)

Does the normalized solution $\rho = 1$ of the doubling-map density equation establish a unique invariant probability law?

Worked solution

No. The fixed point zero gives invariant point mass δ_0, and each periodic orbit gives the uniform probability on that orbit. These are singular relative to length and have no ordinary integrable length density. The density equation describes only absolutely continuous laws, and its normalization does not exclude singular measures. Convex mixtures of invariant measures are invariant as well, by linearity of the pushforward. The proved attraction of $2x$ to uniform is a statement about a specified initial ensemble and cannot turn an atom at a fixed point into a density.

Exercise 24.2 — Independent symbols without random kicks (conceptual)

Explain why the fair-bit cylinder calculation and determinism do not contradict each other. Would the result remain true with an initial point chosen from $\{1/7, 2/7, 4/7\}$?

Worked solution

The cylinder law averages over uncertain initial digits under uniform continuum measure. Once all digits are fixed, their future order is fixed too; observing earlier digits does not reveal later independent digits under that ensemble. Uniform choice among the three rational points instead gives a stationary periodic symbolic process. A word such as three successive zeros has probability zero there, whereas the length-measure ensemble assigns probability $1/8$. The map is the same, but the initial invariant measure changes. Probabilistic independence is a

property of joint laws under specified information, not a statement that the dynamical update contains external noise.

Exercise 24.3 — Generalize the branch calculation (derive)

For $T(x) = mx \pmod 1$ with integer $m \geq 2$, derive density transfer and evolve $\rho_0 = 2x$. Find the total-variation distance after n steps.

Worked solution

The m preimages are $(y + j)/m$, $j = 0, \ldots, m - 1$, each with derivative magnitude m. Thus $\mathcal{P}\rho(y) = m^{-1} \sum_j \rho((y + j)/m)$. Substituting $2x$ yields $2y/m + (m - 1)/m = 1 + (2y - 1)/m$, using $\sum j = m(m - 1)/2$. The constant density is fixed, while the mean-zero linear perturbation is divided by m. Induction gives $\rho_n = 1 + m^{-n}(2y - 1)$. Consequently total variation is $m^{-n} \int_0^1 |2y - 1| dy/2 = m^{-n}/4$. Normalization remains one and the minimum density is $1 - m^{-n} \geq 0$. The contraction factor concerns this linear perturbation, not an assertion that every singular initial law converges.

Exercise 24.4 — Prove a transfer contraction (derive)

For integrable densities ρ, σ, prove $\|\mathcal{P}\rho - \mathcal{P}\sigma\|_1 \leq \|\rho - \sigma\|_1$ for doubling. Does this inequality alone prove convergence to uniform?

Worked solution

Put $h = \rho - \sigma$. By the triangle inequality, the integral of $|\mathcal{P}h|$ is at most $\frac{1}{2} \int_0^1 |h(y/2)| dy + \frac{1}{2} \int_0^1 |h((y + 1)/2)| dy$. Changing variables turns these into the integrals over $[0, 1/2)$ and $[1/2, 1)$, whose sum is $\|h\|_1$. This is a nonexpansion bound with constant one, not a strict uniform contraction less than one. It alone supplies neither a convergence rate nor a uniqueness proof. Additional structure, such as explicit evolution for the chosen density or approximation by dyadic step functions, is needed. The latter functions become constant after sufficiently many exact transfers, and the same nonexpansion then extends convergence to general length-integrable densities by approximation; it still says nothing about singular measures.

Exercise 24.5 — Test a coarse Markov fit (compute/design)

A fitted two-cell chain uses the partition $A = [0, 3/4)$, $B = [3/4, 1)$ and one-step stationary frequencies. Compute its predicted probability for B, A, B and compare with the map.

Worked solution

The initial probability of B is $1/4$. Given B, its image is uniform over $[1/2, 1)$, half of which lies in A, so $\Pr(A \mid B) = 1/2$. The one-step fit has $\Pr(B \mid A) = 1/6$. Its Markov product predicts $(1/4)(1/2)(1/6) = 1/48$. In the actual map the event B, A restricts the original point to $[3/4, 7/8)$; after two steps the image is $[0, 1/2)$,

so the triple probability is zero. The discrepancy is exact, not a small-sample effect. A validation set should test history-conditioned transition probabilities and word frequencies, rather than checking only the stationary cell histogram and one-step matrix.

Exercise 24.6 — Diagnose a spurious attractor (compute/design)

A simulation of doubling reports that every orbit tends to zero. Design checks that distinguish numerical digit exhaustion from a property of the invariant length measure.

Worked solution

Record the exact stored starting value as a rational with denominator a power of two and count how many binary shifts erase it. The zero time should track the available finite digits. Repeat at greater arithmetic precision and compare with exact rational examples of odd denominator, such as the period-three $1/7$ orbit. Independently evolve an ensemble density using inverse branches and verify mass conservation and the analytic formula (24.4). For finite word statistics, integrate cylinder sets exactly or generate a sufficiently long random digit prefix, while labeling that method as sampling the initial measure rather than adding noise to each update. A finite grid remains a singular ensemble and cannot validate arbitrary infinite-time limits. The agreement of precision-dependent collapse with binary digit counts identifies the artifact; it does not deny that zero is a genuine exceptional fixed point.

24.10 Sources and attribution

- V. Balakrishnan, IIT Madras, lecture 28, NPTEL 115106089: flows, symbolic observations, invariant density, Frobenius–Perron evolution, and the recurrence problem.
- T. Tao, 254A Lecture 9: Ergodicity, 4 February 2008, Theorems 2 and 3: pointwise ergodic theorem and its ergodic specialization.

The pushforward and branch proofs, exact nonuniform density, binary-cylinder and mixing arguments, coarse-history counterexample, precision laboratory, original figure, and exercises are developed here. Neither the general pointwise ergodic theorem nor arbitrary nonlinear dynamics is claimed to follow from the laboratory. All return statements refer to a specified measure, target, and positive sampling time.

Sojourn, escape, and dynamical recurrence

An observer starts a stopwatch when a trajectory enters a cell. Another starts at an arbitrary occupied snapshot. A third asks for the next positive return, counting continued occupancy at the next sample. These observers measure different random variables. Once their conditioning and endpoints are fixed, stationary block probabilities yield exact escape and return laws. Finite invariant measure guarantees recurrence; ergodicity gives a reciprocal-measure mean without making visits independent.

> The full caption track of lecture 29 was read sequentially. One low-resolution frame at 47:57 was inspected for departure conditioning; the denominator is partly visible and partly occluded. The discrete block definitions were checked against the complete spoken sequence and independently rederived below. Poincaré recurrence and Kac's mean identity are named separately. The continuous-time discussion states an explicit crossing-rate hypothesis rather than assuming a universal sampling limit.

25.1 Specify the target, preparation, and endpoints L29 1:46

Let T preserve a probability measure μ on a measurable state space Ω. Fix a measurable cell C of measure $q = \mu(C)$ with $0 < q < 1$, and write $B = \Omega \setminus C$. Start the ensemble with $x \sim \mu$, so the binary indicator $I_n = \mathbf{1}_C(T^n x)$ is stationary. Conditional initialization in C uses $\mu_C(A) = \mu(A \cap C)/q$. A sampling step has physical duration $\Delta > 0$; all integer lengths below are in steps until multiplied by Δ.

The first positive return is $\tau_C^+ = \inf\{n \geq 1 : I_n = 1\}$ for $I_0 = 1$. An immediate repeat at step one counts. The first exit from B is $E_B = \inf\{n \geq 1 : I_n = 1\}$ for $I_0 = 0$. Although the formulas look similar, their conditional initial measures differ. Infinite values are allowed until return or escape is established. Residence at an entrance counts all occupied samples starting with that entrance and excludes the first outside sample. A run of ones at indices 2,3,4,5 has length four and exits at index 6. From observation index 4 its forward residual length is two.

25.2 Escape is a first difference; return is a second L29 3:11

Define the joint outside-block probabilities

$$W_0 = 1, \qquad W_n = \Pr(I_0 = \cdots = I_{n-1} = 0), \quad n \geq 1. \tag{25.1}$$

Thus $W_1 = 1 - q$. The sequence decreases to a limit $W_\infty \geq 0$. Decrease alone does not make that limit zero. The conditional outside survival is $H_n = \Pr(E_B > n \mid I_0 = 0) = W_{n+1}/(1-q)$ for $n \geq 0$. These are tail probabilities of nested events, not masses of mutually exclusive outcomes.

Proposition 25.1 — Stationary block identities. For $n \geq 1$,

$$\Pr(E_B = n \mid I_0 = 0) = \frac{W_n - W_{n+1}}{1-q}, \tag{25.2}$$

$$R_n := \Pr(\tau_C^+ = n \mid I_0 = 1) = \frac{W_{n-1} - 2W_n + W_{n+1}}{q}. \tag{25.3}$$

Moreover $\Pr(\tau_C^+ > n \mid I_0 = 1) = (W_n - W_{n+1})/q$ for $n \geq 0$. No independence or Markov assumption is used.

Derivation.

Outside escape at n is survival through $n-1$ minus survival through n, yielding (25.2). For the return tail with $n \geq 1$, subtract the event that indices $0, \ldots, n$ are all outside from the event that $1, \ldots, n$ are outside. By stationarity their probabilities are W_{n+1} and W_n. The difference forces $I_0 = 1$ while excluding all positive returns through n. Dividing by q gives the tail formula; at $n = 0$ it is $(W_0 - W_1)/q = 1$. Differencing successive tails gives (25.3), including $n = 1$. Equivalently, for $n \geq 2$ expand the endpoint factors in $\mathbb{E}[I_0 \prod_{j=1}^{n-1}(1 - I_j) I_n]$ and use stationarity for the shifted blocks. ∎

At $n = 1$, the numerator is $1 - 2(1-q) + \Pr(I_0 = I_1 = 0) = \Pr(I_0 = I_1 = 1)$. This endpoint check prevents losing immediate returns. Since every $R_n \geq 0$, the first differences $D_n = W_n - W_{n+1}$ form a nonincreasing nonnegative sequence. Thus a stationary outside-block sequence is constrained more strongly than mere monotonicity: its second differences are nonnegative. An arbitrary decreasing list cannot automatically be used as these block probabilities.

If $W_\infty = 0$, the outside escape law is normalized and its mean is

$$\mathbb{E}[E_B \mid I_0 = 0] = \frac{1}{1-q} \sum_{n=1}^{\infty} W_n, \tag{25.4}$$

by summing the survival tail. This sum may diverge. A sequence tending to zero need not have a finite sum. The return mean will involve a different telescoping series and can remain finite.

25.3 Finite invariant measure forbids a positive wandering part

Theorem 25.2 — Poincaré recurrence. Let T be measurable and preserve a finite measure. For every measurable cell C, almost every point of C returns to C infinitely often. Invertibility, ergodicity, mixing, and independent visits are not required.

Derivation.

Let $F = \{x \in C : T^n x \notin C \text{ for every } n \geq 1\}$ be the nonreturning part. The inverse images $T^{-j}F$, $j \geq 0$, are pairwise disjoint. Indeed, if x lay in both $T^{-i}F$ and $T^{-j}F$ with $i < j$, then $T^i x \in F$ would return to $F \subset C$ after $j - i$ steps, contradicting its definition. Invariance gives each inverse image measure $\mu(F)$. Finite total measure therefore forces $\mu(F) = 0$.

A point of C with finitely many future visits has a last visit, after which it belongs to F. All such initial points lie in the countable union $\bigcup_{j \geq 0} T^{-j}F$, a null set by invariance. The remaining points return infinitely often. ∎

This is a guarantee of eventual repeated visits to the specified measurable set, not an upper bound on waiting time or an exact return to every point. Translation $T(x) = x + 1$ on the real line preserves length but has infinite total measure; an interval of length less than one never receives a positive return from its own points. The finiteness hypothesis is consequential. See Bruin, Section 7, Theorem 12, printed p. 22, for the finite-probability recurrence statement.

25.4 Kac's mean identity is a telescoping tail sum L29 36:08

Theorem 25.3 — Conditional mean positive return. If T is ergodic and measure-preserving on a probability space and $q = \mu(C) > 0$, then

$$\mathbb{E}_{\mu_C}[\tau_C^+] = \frac{1}{q}, \qquad \int_C \tau_C^+ \, d\mu = 1. \tag{25.5}$$

The return starts from invariant measure conditioned on C. This is Kac's identity, rather than the finite-measure recurrence assertion alone.

Derivation.

The event $G = \bigcap_{j \geq 0} T^{-j} B$ consists of points never visiting C, and has measure W_∞ by continuity for decreasing measurable sets. We have $G \subset T^{-1} G$; invariance gives equal measures, so they agree modulo a null set. Ergodicity forces $\mu(G)$ to be zero or one. Because $G \subset B$ with $\mu(B) = 1 - q < 1$, it must be zero.

For a positive integer-valued time, including a possible infinite value, its expectation is the sum of its survival tails by monotone convergence. Proposition 25.1 gives the exact finite partial sum

$$\mathbb{E}_{\mu_C}[\min(\tau_C^+, N)] = \sum_{n=0}^{N-1} \frac{W_n - W_{n+1}}{q} = \frac{1 - W_N}{q}.$$

Let $N \to \infty$. Since $W_N \to 0$, the mean equals $1/q$ and is finite. Multiplying by q yields the unconditioned integral over C. No cancellation of conditionally convergent signed series or assumption of independent excursions is needed. ∎

The conditional normalization matters: $\int_C \tau_C^+ d\mu$ equals one, while the conditional mean equals $1/q$. Without ergodicity the same tail proof yields $(1 - W_\infty)/q$, which can be smaller than $1/q$. For the identity map, every point of C returns at step one and $W_\infty = 1 - q$, giving mean one. Finite-measure recurrence still holds. Bruin, Section 7, Lemma 4, printed p. 23, states the ergodic Kac formula; the stationary-tail proof above also covers noninvertible maps.

If a finite ergodic system has M equally weighted configurations, the mean return to a specified configuration is M samples. For a model of N independent binary degrees of freedom with uniform invariant weights, $M = 2^N$. This illustrates how a certain eventual recurrence can have an enormous mean. It is a finite-state model under explicit equilibrium weights, not a claim that every physical gas is uniformly distributed over a finite list of configurations.

25.5 Entrance runs and arbitrary-time intervals have different laws

Consider an iid binary stationary record indexed by all integers, with occupied probability q. An *entrance* at index zero means $I_{-1} = 0, I_0 = 1$. Let $L \geq 1$ be the occupied-run length, so $I_0, \dots, I_{L-1} = 1$ and $I_L = 0$. Conditioning on the entrance leaves future symbols independent, hence

$$\Pr(L = n \mid \text{entrance}) = (1-q)q^{n-1}, \qquad \mathbb{E}L = \frac{1}{1-q}. \tag{25.6}$$

This two-sided record is an observation ensemble; defining its past does not require an invertible one-sided doubling map. It can be realized by a two-sided Bernoulli shift, or by observing a sufficiently long stationary bit record away from its boundaries.

Now select an arbitrary occupied sample and ask for the *total* length L^* of its containing run, including the part before the observation. A run of length n offers n possible occupied observation sites, so its law is length-biased:

$$\Pr(L^* = n \mid I_0 = 1) = \frac{n \Pr(L = n)}{\mathbb{E}L} = n(1-q)^2 q^{n-1}. \tag{25.7}$$

The geometric sums $\sum n q^{n-1} = (1-q)^{-2}$ and $\sum n^2 q^{n-1} = (1+q)/(1-q)^3$ give

$$\mathbb{E}[L^*] = \frac{\mathbb{E}[L^2]}{\mathbb{E}L} = \frac{1+q}{1-q}. \tag{25.8}$$

It exceeds the entrance mean by $q/(1-q)$. Repeatedly sampling occupied snapshots overrepresents long intervals even though the underlying symbols are independent.

The forward residual R counts occupied samples from the observation through the end of the run, including the current one. For iid symbols, $\Pr(R = n \mid I_0 = 1) = (1-q)q^{n-1}$, so $\mathbb{E}R = 1/(1-q)$, equal to the entrance mean. This memoryless equality does not remove the length bias of the total interval. For a general stationary collection of finite runs with entrance law L and finite $\mathbb{E}L$, counting the residuals $1, \dots, n$ in a run of length n gives

$$\mathbb{E}R = \frac{\mathbb{E}[L(L+1)]}{2\mathbb{E}L}. \tag{25.9}$$

The residual can have infinite mean when the entrance mean is finite but the second moment diverges.

This resolves the apparent conflict between a finite mean return and an infinite mean escape from an arbitrary outside snapshot. For example, construct a stationary alternating renewal record with every occupied run of length one and outside-run lengths L having finite mean but infinite second moment. One may choose a positive-integer law proportional to $n^{-1-\beta}$, $1 < \beta < 2$. A stationary preparation weights cycles by their lengths and places the observation uniformly within the selected cycle; finite mean cycle length permits its normalization. Returns from occupied samples have mean $1 + \mathbb{E}L$, while an arbitrary outside sample has forward residual mean (25.9), which is infinite. The two experiments average different distributions, rather than conditioning one on an event of probability zero.

25.6 Exclude immediate stays by conditioning on a departure L29 45:30

For finer sampling of a continuous trajectory, most successive observations inside a cell remain inside it. Kac's positive-return time legitimately counts those immediate stays; multiplying its mean by Δ gives Δ/q. To measure an excursion after leaving, change the stopping event and preparation instead of taking that quantity as a continuous-time excursion duration.

Let $D = \{I_0 = 1, I_1 = 0\}$ be an observed departure, and set $d = \Pr(D)$. Stationarity and the outside-block definitions give $d = W_1 - W_2$. For $0 < q < 1$ in an ergodic system, $d > 0$: if it vanished, the cell would be invariant modulo null sets. Conditional on D, let $U = \tau_C^+ - 1 \geq 1$ be the number of outside samples starting at index one, until return at index $U + 1$. Then

$$\Pr(U = n \mid D) = \frac{qR_{n+1}}{d} = \frac{W_n - 2W_{n+1} + W_{n+2}}{d}, \quad n \geq 1. \tag{25.10}$$

Its normalization is $\sum_{n\geq 1} qR_{n+1}/d = q(1 - R_1)/d = 1$, because $q(1 - R_1) = d$. The endpoint shift is essential: n measures from the first outside sample, while the return occurs $n + 1$ steps after the last occupied sample.

Under the ergodic hypotheses of Theorem 25.3,

$$\mathbb{E}[U \mid D] = \frac{q}{d} \sum_{m\geq 2} (m - 1)R_m = \frac{1 - q}{d}. \tag{25.11}$$

The elapsed steps from the last occupied sample at zero to re-entry are instead $U + 1$, with mean $1 + (1 - q)/d$. A full entrance-to-next-entrance cycle includes both occupied and outside runs and has mean $1/d$ in an ergodic stationary two-sided record with positive transition rate. These quantities must not share an unlabeled name such as recurrence time.

This is the departure-conditioned modification discussed in the lecture in connection with Smoluchowski. A useful continuous-time limit requires a dynamical crossing assumption. If observed departure probability satisfies $d(\Delta) \sim J\Delta$ for a finite positive crossing rate J per unit physical time, then the mean outside excursion approaches $(1 - q)/J$ and the full entrance cycle approaches $1/J$. Such scaling may fail when arbitrarily rapid recrossings dominate. The published Balakrishnan–Nicolis–Nicolis study, Physical Review E 61, 2490 (2000), abstract, specifically contrasts observation-resolution limits in deterministic and stochastic dynamics. Its abstract motivates this boundary; formulas (25.10)–(25.11) follow from the discrete definitions here, without relying on an uninspected continuous-time formula.

25.7 Correlated visits can keep the same reciprocal mean L29 53:51

In the natural half-interval observation of the doubling map, visits are iid with $q = 1/2$. The outside blocks have $W_n = (1 - q)^n$, so (25.3) gives $R_n = q(1 - q)^{n-1}$, the geometric law of mean $1/q$. This agrees with Kac but uses much more information than Kac's mean identity.

Instead take the cylinder cell $C = [3/4, 1)$, corresponding to two successive binary digits 11. Its measure is $1/4$. Successive observations overlap in one digit, so the visits are correlated. Given an initial 11, the next visit occurs immediately if the next digit is one, with probability $1/2$. If that next digit is zero, a visit at step two is impossible because its first digit is zero. Thus $R_1 = 1/2$ and $R_2 = 0$, already excluding the geometric law of parameter $1/4$.

For the entire return law, track whether the last digit is zero or one before completing a new 11. In the row-state convention used just in this calculation, the substochastic transition matrix and success vector are

$$K_{\mathrm{surv}} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 0 \end{pmatrix}, \qquad r = \begin{pmatrix} 0 \\ 1/2 \end{pmatrix}.$$

A missing row mass is absorption on completing 11. With row $e_1 = (0, 1)$, because the initial occurrence ends in one, $R_n = e_1 K_{\mathrm{surv}}^{n-1} r$. The first six masses are $1/2, 0, 1/8, 1/16, 1/16, 3/64$. Row notation is explicit here; no column convention for the chapter 5 state probabilities is being silently transposed.

Let h_0, h_1 be expected additional bits to completion from trailing digit zero or one. Conditioning on the next bit gives $h_0 = 1 + (h_0 + h_1)/2$ and $h_1 = 1 + h_0/2$. Solving yields $h_0 = 6, h_1 = 4$. Finiteness is justified independently: in every two newly generated bits there is probability $1/4$ of 11, so failure probability over disjoint two-bit blocks is at most $(3/4)^m$. The return mean four agrees with $1/\mu(C)$, despite a strongly nongeometric distribution. A stationary measure fixes this mean under ergodicity but does not fix all return statistics.

25.8 Computational laboratory: use the right sampling unit

The lab generates one million iid occupancy symbols with $q = 0.7$ and seed 2026110829. It extracts complete occupied runs, excluding runs truncated by either record boundary. Each complete run contributes once to the entrance average, contributes its length to occupied sampling weights, and contributes residuals $1, \dots, L$ to the residual average. These weighted sums avoid treating repeated snapshots of one run as separate independent intervals.

A separate exact experiment enumerates all 2^{16} binary words, conditions on an initial 11, and records the first subsequent 11 through step 14. Every probability is compared with the matrix formula $e_1 K_{\mathrm{surv}}^{n-1} r$. A backward linear solve independently checks the exact mean four. The unreturned fraction at step 14 is retained; the finite list is not renormalized into a complete law and its truncated mean is not labeled the infinite-horizon mean.

```
Complete occupied runs: 210221
Entrance mean: sample=3.328112; exact=3.333333
Occupied-time total-run mean: sample=5.647062; exact=5.666667
Occupied-time residual mean: sample=3.323531; exact=3.333333
Cylinder11 return masses n=1..6: [0.5, 0.0, 0.125, 0.0625, 0.0625, 0.046875]
Cylinder11 mean from backward equations: 4.000000
Cylinder11 unreturned at step14: 0.03723145
```

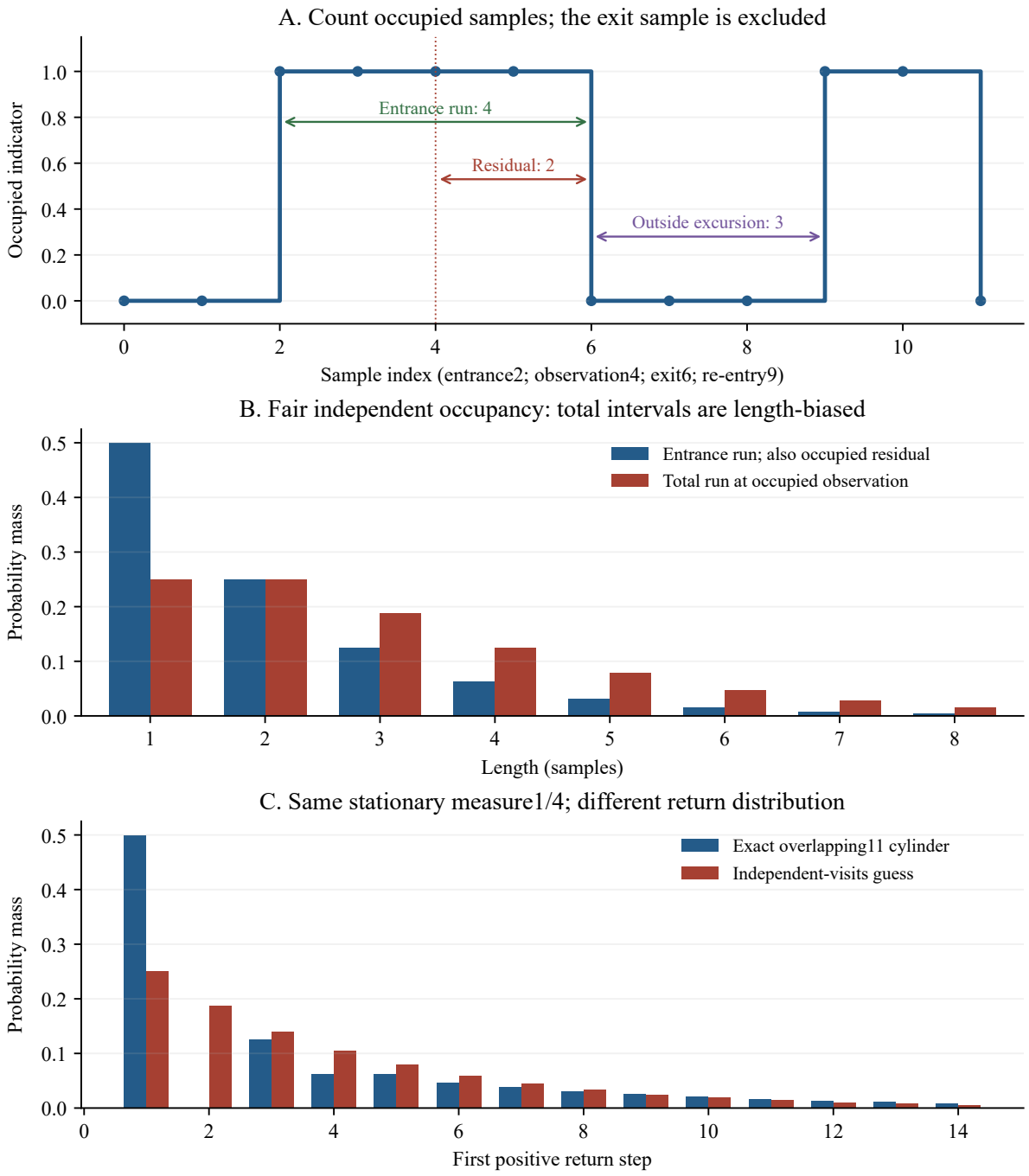


Figure 25.1. The top record fixes endpoints before any difference is taken. In fair iid occupancy the entrance and residual distributions agree, while occupied-time total runs are length-biased. The lower comparison shows a correlated cell of measure one quarter whose return mean is four but whose probability masses differ sharply from independent visits. Generator; numerical records.

```
def return_mass_from_blocks(previous, current, following, cell_measure):
    return (previous-2*current+following)/cell_measure


print(return_mass_from_blocks(1.0, 0.5, 0.25, 0.5))
```

The function evaluates (25.3) from valid stationary block inputs and prints 0.5; it has no file effects. It does not validate arbitrary supplied numbers as realizable block probabilities. The full lab writes JSON and its report, overwrites them on rerun, and uses no network. Run it before the generator with the declared dependencies. Its simulated symbols represent the invariant binary ensemble, not a long orbit computed in fixed floating-point precision.

25.9 What the recurrence identities do and do not determine

The finite-measure recurrence proof needs measure preservation but no independent visits. Kac's reciprocal mean additionally uses ergodicity, or an equivalent full-saturation condition for the chosen target. Stationary block differences determine escape and return laws only when the correct block sequence is known. A target's measure alone generally gives no return distribution. Arbitrary-time residual residence and entrance-conditioned runs differ, and reducing the physical sampling interval changes which immediate stays are counted. In an experiment, record the target, initialization, direction of crossing, sampling interval, endpoint convention, and censoring rule before comparing a fitted distribution with these identities.

25.10 Exercises with worked solutions

Exercise 25.1 — Certain recurrence without the reciprocal mean (conceptual)

On a uniform probability interval let T be the identity and let $\mu(C) = 0.2$. Which recurrence conclusions hold, and why is the mean positive return not five?

Worked solution

Every initial point of C is in C at all subsequent samples, so it returns infinitely often and $\tau_C^+ = 1$ identically. Poincaré recurrence holds exactly. The map is not ergodic: C itself is an invariant set of measure strictly between zero and one. The outside blocks satisfy $W_0 = 1$ and $W_n = 0.8$ for every $n \geq 1$. Thus the tail proof gives mean $(1 - W_\infty)/q = 0.2/0.2 = 1$, consistent with the direct orbit. Dividing by the global measure without checking ergodicity would average in an inaccessible outside component. There are no observed departures, so departure-conditioned excursion statistics are undefined.

Exercise 25.2 — Memoryless residuals and length bias (conceptual)

For fair iid occupancy, the forward residual and entrance length both have mean two. Does it follow that the total interval containing an occupied snapshot also has mean two?

Worked solution

No. Residual length looks only forward from the observation, while total length includes its past. Each run of length n offers n occupied observation sites, giving total-length law $n2^{-(n+1)}$. Its mean is three, rather than two. Conditional on total length, the occupied observation site is uniform among its n positions; its average forward residual is $(n + 1)/2$. Averaging that quantity under the length-biased law gives $(3 + 1)/2 = 2$, reconciling the results. Memorylessness of future iid symbols is fully compatible with preferential sampling of long containing intervals.

Exercise 25.3 — Return law for a persistent binary chain (derive)

Let an irreducible stationary two-state Markov chain satisfy $\Pr(1 \mid 1) = a$ and $\Pr(1 \mid 0) = b$, with $0 < a, b < 1$. Find its occupied measure and positive-return law, and verify Kac's mean.

Worked solution

Stationary balance gives $q = qa + (1-q)b$, hence $q = b/(1-a+b)$. Starting in one, immediate return has probability $R_1 = a$. A later first return at $n \geq 2$ requires transition to zero, then $n-2$ continued zeros, then transition to one. Thus $R_n = (1-a)(1-b)^{n-2}b$. Its mass is $a + (1-a)b\sum_{j\geq 0}(1-b)^j = 1$. Conditional on leaving at the first step, the remaining outside length is geometric with mean $1/b$, so the unconditional positive-return mean is $a + (1-a)(1+1/b) = 1 + (1-a)/b = 1/q$. This also follows by summing nR_n. The entrance occupied-run mean is instead $1/(1-a)$, showing why residence and positive-return means cannot be interchanged. The outside blocks $W_n = (1-q)(1-b)^{n-1}$ for $n \geq 1$, with the special value $W_0 = 1$, reproduce the same law through (25.3).

Exercise 25.4 — A finite entrance mean and infinite residual mean (derive)

Suppose positive-integer entrance lengths obey $\Pr(L = n) = cn^{-1-\beta}$ with $1 < \beta < 2$, where c normalizes the sum. Determine the moment behavior of entrance, occupied-time total, and residual lengths.

Worked solution

The normalizing series $\sum n^{-1-\beta}$ converges. The entrance mean is $c\sum n^{-\beta}$, finite because $\beta > 1$. Its second moment is $c\sum n^{1-\beta}$, divergent because $1-\beta > -1$. Length-biased total-run probabilities are $n\Pr(L=n)/\mathbb{E}L$, a normalized law since the denominator is finite. Their mean is $\mathbb{E}[L^2]/\mathbb{E}L = \infty$. Summing residuals $1, \ldots, n$ over the n possible observation sites gives $\mathbb{E}R = \mathbb{E}[L(L+1)]/(2\mathbb{E}L) = \infty$. Thus a finite mean of entrance-sampled intervals does not control snapshot residuals. Finite records right-censor their longest intervals; deleting all incomplete runs can strongly understate these tails and must be distinguished from a proof of a finite population moment.

Exercise 25.5 — Check overlapping returns without simulation (compute/design)

For the doubling-map cell 11, compute R_1, R_2, R_3 directly from new bits and recover the mean with a backward calculation. Explain why the finite enumeration must retain an unreturned mass.

Worked solution

After an initial 11, a new one gives immediate return, so $R_1 = 1/2$. If the new bit is zero, a return one step later cannot begin with one, hence $R_2 = 0$. First return at step three requires the new sequence 011, probability $1/8$. For the mean, use trailing-digit states: $h_0 = 1 + (h_0 + h_1)/2$ and $h_1 = 1 + h_0/2$. The second equation includes immediate success with its one elapsed step. Solving gives $h_0 = 6, h_1 = 4$, the required initial-state mean. The all-failure probability has a geometric upper bound from independent disjoint pairs, ensuring finite expectation. At the enumeration horizon 14 the exact unreturned mass is 0.03723145; normalizing only the observed returns conditions on unusually short events and alters the target law. A validation should compare each exact finite-horizon mass and the remaining survival probability separately.

Exercise 25.6 — Restore physical units after conditioning (compute/design)

A smooth-flow record has occupied measure $q = 0.2$, sample spacing $\Delta = 0.005$ seconds, and departure probability $d = 0.001$ per sample. Estimate outside excursion and full entrance-cycle means under $d(\Delta) \sim J\Delta$. Compare with the sampled positive-return mean.

Worked solution

The inferred crossing rate is $J = d/\Delta = 0.2$ per second. The outside excursion mean from the first outside sample is $\Delta(1-q)/d = 0.005(0.8)/0.001 = 4$ seconds. A full entrance cycle has mean $\Delta/d = 5$ seconds; its mean occupied portion is $q/J = 1$ second, so the portions add consistently. The sampled positive-return mean is $\Delta/q = 0.025$ seconds, dominated by immediate stays. From the last occupied sample to re-entry, the discrete endpoint convention adds one sample, giving 4.005 seconds. Repeat the measurement at several decreasing spacings and test whether d/Δ stabilizes; also check missed crossings, boundary grazing, and finite-window censoring. Without the finite positive crossing-rate scaling, these continuum extrapolations are not justified by Kac's identity alone.

25.11 Sources and attribution

- V. Balakrishnan, IIT Madras, lecture 29, NPTEL 115106089: outside sojourn blocks, first and second differences, mean recurrence, departure conditioning, and the Bernoulli shift.
- H. Bruin, Notes on Ergodic Theory, author-hosted version dated 8 May 2023, Section 7, Theorem 12, printed p. 22, and Lemma 4, printed p. 23: finite-probability recurrence and the ergodic Kac identity.
- V. Balakrishnan, G. Nicolis, and C. Nicolis, Recurrence time statistics in deterministic and stochastic dynamical systems in continuous time: A comparison, Physical Review E 61 (2000), 2490–2499, doi:10.1103/PhysRevE.61.2490. The inspected abstract concerns sampling-resolution limits; no detailed formula is attributed to unseen full text.

The disjoint-preimage recurrence proof, stationary-tail proof of the mean identity, explicit departure endpoint formulas, run-sampling derivations, overlapping-cylinder calculation, numerical laboratory, original figure, and six worked solutions are developed here. The course closes with the same discipline used for its first counting models: state the probability law, its conditioning, the measured observable, and the assumptions before interpreting a computed number.